

**HHL**LEIPZIG
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OF MANAGEMENT

Dissertation

Development Aid - A Perspective on the World Bank Performance

Calculating the Social Return on Investment for the Least Developed Countries

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Abstract:

This doctoral thesis focuses on the evaluation of the World Bank (WB) performance in delivering development aid to the Least Developed Countries (LDCs). For this purpose, an extensive research was performed to analyze a set of 790 Implementation Completion and Results reports for key economic and financial indicators. Results of this research provide various insights for the appraisal and the results stage of project delivery of the LDCs in different continents. In the final part of the economic and financial analysis the minimum Social Return on Investment (SROI) of the LDCs including all project costs was calculated. This SROI ratio outcome of 1 and 1.06 in the weighted and 1.3 and 1.72 in the unweighted case indicate that projects delivered by the WB have a positive effect on the poor countries. In the second part of this research project the data set of the ICR reports was qualitatively researched for negative ratings according to 3 core assessment categories for the overall project performance: Sustainability, bank performance and borrower performance. As a result the most critical categories respectively risks were outlined. In conclusion, the research analyses and findings support the general demand to provide even more development assistance to poor countries.



Development Aid - A Perspective on the World Bank Performance

Calculating the Social Return on Investment for the Least Developed Countries

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Summary

This doctoral thesis focuses on the evaluation of the World Bank (WB) performance in delivering development aid to the Least Developed Countries (LDCs). For this purpose, an extensive research was performed to analyze a set of 790 Implementation Completion and Results (ICR) reports for key economic and financial indicators such as the Net Present Value (NPV) and the Economic Rate of Return (ERR). Results of this research are presented in the economic and financial analysis part of this paper and provide various insights for the appraisal and the results stage of project delivery of the LDCs in different continents: The project appraisal portfolio NPV (91 projects) for LDCs lay between 17,667 and 20,067 Mio USD with total project costs of 13,981 Mio USD. When the 3 NPVs above 1,000 Mio USD were excluded the NPV value decreased to 4,678 in the minimum and 4,713 Mio USD in the maximum case with total project costs of 8,021 Mio USD. The portfolio NPV at the result stage of the LDCs (168 projects) remained positive with NPV values ranging from 42,059 to 50,779 Mio USD (33,506 Mio USD total project costs) and from 6,188 to 7,799 Mio USD (20,280 Mio USD total project costs) excluding the 7 outlier projects with abnormally “high-value” NPVs. Overall, the portfolio NPV and the related NPV/capital cost ratio of the Asian continent were slightly higher than those of Africa. The direct comparison of the appraisal vs. the result values showed that project outcomes developed as planned. The results of the minimum and maximum arithmetic means of the ERRs of the non-NPV projects were 18.91 and 31.35 percent. Even though lower than the ERRs of the NPV projects, they were still noticeably higher than the corresponding discount rate of 11 percent. In the final part of the economic and financial analysis the theoretic minimum Social Return on Investment (SROI) of the LDCs was calculated. The SROI ratio outcome of 1 and 1.06 in the weighted and 1.3 and 1.72 in the unweighted case indicate that projects delivered by the WB have a positive effect on the poor countries. In the second part of this research project the data set of the ICR reports was qualitatively researched according to 3 core assessment categories for the overall project performance: Sustainability, bank performance and borrower

performance. The objective of the qualitative data analysis was to determine the major reasons for negative category ratings. It was found that fiduciary risks were the primary risks relating to 54.8 percent of the negative sustainability projects of the LDCs. In the bank performance category, implementation framework was the major issue affecting 96 percent of all negative bank performance projects within the LDCs, whereat the borrower performance suffered mostly from inadequate borrower competence affecting every fifth project researched within this paper. In conclusion, the research analyses and findings support the general demand to provide even more development assistance to poor countries. The paper closes with a critical acclaim and recommendations for future development aid project execution as well as an outlook and suggestions for future research within the area of development aid.

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List of Abbreviations

BCR	Benefit-Cost-Ratio
CAS	Country Assistance Strategy
CBA	Cost-Benefit-Analysis
CSR	Corporate Social Responsibility
DAC	Development Assistance Committee
EIRR	Economic Internal Rate of Return
ERR	Economic Rate of Return
EVI	Economic Vulnerability Index
FNPV	Financial Net Present Value
FRR	Financial Rate of Return
GDP	Gross Domestic Product
GNI	Gross National Income
GNP	Gross National Product
GPRS	Ghana's Poverty Reduction Strategy
HAI	Human Assets Index
HIPC	Heavily Indebted Poor Countries
IBRD	International Bank for Reconstruction and Development
ICR	Implementation Completion and Results (Report)
IDA	International Development Association
IEG	Independent Evaluation Group
IMF	International Monetary Fund
IRR	Internal Rate of Return
LDCs	Least Developed Countries
MDGs	Millennium Development Goals
LA	Latin America
NGO	Non-Governmental Organization
NPV	Net Present Value
NPVCR	NPV-Cost-Ratio
ODA	Official Development Assistance

OECD	Organization of Economic Cooperation
PDO	Project Development Objective
PRSP	Poverty Reduction Strategy Paper
PV	Present Value
RR	Rate of Return
SA	Sensitivity Analysis
SROI	Social Return on Investment
ROI	Return on Investment
UN	United Nations
UNDP	United Nations Development Programme
UNICEF	United Nations Children's Fund
VBA	Visual Basic for Applications
WB	World Bank
WBG	World Bank Group
WFP	World Food Programme
WHO	World Health Organization

1 Introduction

1.1 Introduction to the Topic

Over the course of the last 2 centuries, global prosperity has accelerated and each generation has been encouraged to meet new challenges to “make the world better” by lifting up human well-being. Nevertheless, the picture does not look promising everywhere. Still today, about 3 Mio people live on less than 2 dollars a day. More than 8 hundred Mio do not have enough to eat, 10 Mio children die every year from diseases which could be easily prevented, and AIDS is killing close to 3 Mio people a year. 1 billion lack access to clean water and some 2 billion people lack access to sanitation. Furthermore, roughly 1 billion adults are still illiterate and up to 25 percent of the children in poor countries do not finish primary school.¹

According to experts, there are 10 significant challenges within the global poverty context: air pollution, conflict, disease, global warming, education, sanitation and water, malnutrition and hunger, trade barriers and subsidies, women, and development and terrorism.² To address global poverty problems and help the poorest billion improve their situations, many development aid organizations and so called human aid institutions have arisen, particularly in the last few decades.³

The new millennium has also offered prospective hope in solving global prosperity problems through emerging technologies as part of the ongoing IT boom and the continued economic progress in China, India, and Russia. Although Africa is still in a miserable crisis, a spread of democracy throughout the continent has given hope to the possibility of activating processes to use new technologies to fight different diseases. The most vivid reflection of this was the Millennium Assembly which took place at the United Nations (UN) in New York. It was the largest assembly of world leaders in history with 147 heads of state and government attending. For this occasion, UN Secretary-General Kofi Annan presented the document “We the Peoples:

¹ Cf. Easterley (2006), p. 7.; Sachs (2005), p. 360.

² Cf. Lomborg (2009), p. 2.

³ Cf. Easterley (2008), unpag.

The Role of the United Nations in the 21st Century”, laying out a critical view of the global challenges of our time, such as extreme poverty, environmental damage, major disease problems, civil conflicts and war. This document became the basis for the Millennium Declaration which sets forth a series of time-bound and quantified goals, the Millennium Development Goals (MDGs).⁴

1.2 Assessing Poverty Problems and Achieving Economic Growth

The problem of how to end poverty in our world has been widely discussed throughout literature, whereat most of all the research indicates that those countries affected are caught in a "poverty trap." A combination of poor geography, health care and infrastructure prevent some societies from generating any economic surplus (this is especially the case for Sub-Saharan Africa). To help such countries make the first step on the economic ladder of development, far more aid assistance from “rich world countries”, debt forgiveness, better trade terms, and access to good technologies need to be ensured. This is generally referred as the “top-down” approach for economic assistance. Jeffery D. Sachs can be regarded as a main driver of this approach, and there are also other important supporters, such as Paul Collier.⁵

There are, however, opponents who do not believe in any “utopian” top-down approach. William Easterly is a well-known proponent among those who believe that helping the poor is only possible through simple and cost-effective means of foreign aid such as dietary supplements (e.g. vitamins, infant formula, and iodine), fertilizer subsidies, education in sexual practices (usage of condoms) and urban water provision.⁶

Beyond these competing opinions of how foreign aid assistance should be applied, a new “contra foreign aid” opinion had also emerged. This was first advocated by

⁴ Cf. Sachs (2005), p. 210 et seq.

⁵ Cf. Collier (2008), unpag.; Sachs (2005), p. 242 et seq.

⁶ Cf. Easterly (2006), p. 327 et seq.

Dambisa Moyo with her book “Dead Aid”, arguing that economic growth and a significant decline in poverty can be achieved without reliance on foreign aid or aid-related assistance.⁷

Nevertheless, there are some common ideas and agreements on how to start immediately with foreign aid solutions, regardless of the opposing opinions on how to apply Official Development Assistance (ODA) most efficiently:⁸

- Promotion of understanding ODA as a subsidy
- Grants instead of loans⁹
- Differentiated diagnoses according to the country specific needs by shifting from supply to demand focus
- Competitive advantages and accountabilities of aid agencies

1.3 Millennium Development Goals

The MDGs were unanimously agreed by the 191 UN member states that culminated in the signing of the United Nations Millennium Declaration in 2002. In principle, these goals stand for the main objectives of our endeavors to solve world poverty problems and gain global prosperity. The MDGs consist of the following 8 goals:¹⁰

- Goal 1: Eradicate extreme poverty and hunger
- Goal 2: Achieve universal primary education
- Goal 3: Promote gender equality and empower women
- Goal 4: Reduce child mortality
- Goal 5: Improve maternal health
- Goal 6: Combat HIV/AIDS, malaria and other diseases

⁷ Cf. Moyo (2009), unpag.

⁸ Cf. Easterly (2006), unpag.; Sachs (2005), unpag.; Schabbel (2006), p. 281 et seq.

⁹ Author's Note: Grants are funds given as subventions to recipient countries, whereas this money normally does not need to be paid back. In contrast, loans are borrowed money which the borrower generally must pay back, usually with interest costs.

¹⁰ Cf. Easterly (2006), p. 8; Sachs (2005), p. 25; United Nations Development Programme (2011): Millennium Development Goals.

- Goal 7: Ensure environmental sustainability
- Goal 8: Develop a Global Partnership for Development

Today, the 8 MDGs are broken further down into 21 quantifiable targets that are measured by 60 indicators.¹¹ To ensure that the MDGs can be achieved and to put a realistic plan into place, the UN Millennium Project was founded in 2002. The project was engaged by 250 central global expert participants who represented each part of the entire UN system (WHO, UNICEF, the Food and Agriculture Organization, The United Nations Environment Program, etc.). In order to estimate the total amount of foreign aid available through the MDGs, each country must first offer a detailed costing plan based on the Millennium Project methodology. This has been outlined by a minimum amount of 135 to 195 billion USD per year for the period of 2005 through 2015 (this is about 0.44 to 0.54 percent of the rich-world GNP each year). Based on the official calculations from 2005/06 and the MDGs Summit outcome in 2010, this means that ODA would need to be more than doubled for the majority of the Least Developed Countries (LDCs) to reach the MDGs and for poverty to be halved by 2015.¹²

The most disappointing results in solving extreme poverty can be observed in Africa. While some of the North African states will be able to halve poverty on time, most

¹¹ Author's Note: Find the full list including all targets and indicators in *Appendix A*.

¹² Cf. Sachs (2005), p. 223 et sqq.; United Nations Development Programme (2011): The 2010 MDG Summit Outcome.

of the Sub-Saharan African countries (known as Black Africa) will not succeed in doing so, despite an average economic per capita growth rate¹³ of 3.3 percent.¹⁴

A high-level forum was convened by the Secretary-General on the 23 September 2013. The forum was used to catalyze and accelerate further action to achieve the MDGs and to enhance the counselling of the General Assembly. Using concrete examples of scaling up success and identifying further opportunities was a key objective as well. Looking at the fact sheets of the single MDGs, the progress towards reaching them is good, but not sufficient in order to completely achieve the full scope for each one of them on time.¹⁵

Days later on 25 September 2013, the president of the UN General Assembly hosted a special event in order to follow up on the efforts made towards achieving the MDGs.¹⁶ At this special event UN Secretary-General Ban Ki-moon presented to member states his report entitled “A Life of Dignity for All”. This document was adopted by the member states, and herein the world leaders renewed their commitment to meet the MDG’s targets and goals. In order to do so a high-level summit will be scheduled for September 2015. The aim of the summit is to adopt a new set of goals which will build upon the achievements of the MDGs.

In this context, questions such as these have been introduced into the global poverty discussion:¹⁷

¹³ Cf. Besley/Burgess (2003), unpag.; Lopez (2004), p. 2 et sqq.; Schabbel (2006), p. 192 et seq., p. 213: A substantial number of studies found that there is a positive relationship between growth and poverty, which is a finding that reaches a relative high standard of consensus among development researchers. Although empirical observations support the view that growth is pro-poor, there is still a very uneven extend to which growth reduces poverty (at a given growth rate, poverty reduction is very uneven across countries). Among 71 studies about the relationship between growth and poverty, only 1 study outlined a statistically negative relationship and 31 studies reported inconclusive results (statistically insignificant), which is due to the fact that there is a lack of a (strong) analytical framework for evaluating the causal relationships in various studies because aid is given in many different forms and for a lot of different purposes. Moreover, 40 studies showed a statistically positive impact of aid on growth. This shows that the majority of case studies have proven that foreign economic development assistance can affect economic results positively.

¹⁴ Cf. Calderisi (2006), p. 2 et sqq.; Collier (2008), p. 3 et sqq.; Schabbel (2006), p. 114; Wolff (2005), unpag.

¹⁵ Cf. UN (2015); Cf. *Appendix B* provides an overview of the current progress towards the MDGs.

¹⁶ Author’s Note: The president of the UN General Assembly is appointed every year. In 2013 the president of the UN General Assembly was John William Ashe from Antigua and Barbuda.

¹⁷ Cf. Ahrens (2005), unpag.; Easterley (2006), p. 24; Köhler et al. (1996), unpag.

- Does Western-aid have a positive effect on the LDCs from a financial point of view?
- How effective are development aid projects at a country or continent (portfolio) level?
- Why are so many countries still failing to achieve economic success?
- How sustainable are the outcomes of development assistance projects?
- What are the problems of development aid projects?

With the aim of unearthing answers to these questions, this doctoral thesis will focus on analyzing development aid effectiveness, bank and borrower performance, as well as the sustainability of development assistance projects lead and conducted by the WB within the LDCs.

1.4 Development Aid

Development aid has multiple names in the vernacular, such as development assistance, international aid, foreign aid, technical assistance, ODA. In principle, all of these terms are based on the same understanding: Financial aid provided by governments and other agencies in order to support and address the challenges of developing countries. The challenges to be addressed can vary by country and nature, e.g. economic, social, environmental and political development deficits. However, at the very core, providing or delivering development assistance should always follow the humanitarian aspect –focusing on alleviating poverty in the long term.¹⁸

Generally, each low-income country which wants to receive development assistance from most major donors and lenders is asked to define country specific poverty reduction strategy paper (PRSP). A PRSP defines the country's biggest needs and areas where development assistance is required. PRSPs are most of all required by the International Monetary Fund (IMF) and the WB before a country can be consid-

¹⁸ Cf. Georgeou (2012), p. 23 et sqq.; Kanbur (2006), unpag.

ered for debt relief. Besides the country specific needs a PRSP should be comprehensive, aim for partnership relations, result-oriented and based on a long-term perspective. Partnership-oriented relations between the donors and its recipients have gained more importance during the last decades in order to move away from the traditional situation which was dominated by the wealth and knowledge leadership of one side. Traditionally, development aid comes from the Western industrialized countries whereat nowadays poorer countries also do their best to contribute. There are different dimensions of how development aid can be provided:¹⁹

1. Multilateral: Given by a donor country to international or supranational “acting” development aid organizations, such as the WB or the UN agencies (World Health Organization (WHO), World Food Programme (WFP), United Nations Children's Fund (UNICEF), United Nations Development Programme (UNDP) and International Monetary Fund (IMF).
2. Bilateral: Given from one country directly to another; also referred as nation-state development aid, e.g. the ODA of a government.
3. Non-state (mostly associations): Provided by non-governmental organizations (NGO) such as Oxfam or Welthungerhilfe.

Approx. 80 percent of development assistance is provided by governmental sources as ODA whereat the proportion is currently about 70 percent bilateral and 30 percent multilateral. The remaining 20 percent of delivered aid comes from private sources such as NGOs or other development foundations.²⁰

¹⁹ Cf. Georgeou (2012), p. 23 et sqq.; Kanbur (2006), unpag.

²⁰ Cf. Georgeou (2012), unpag.; Kanbur (2006), unpag.; Sachs (2005), p. 18 et sqq.

2 Research Approach

In this chapter, the research objective and relevant terms, definitions, and methods are explained in detail. Furthermore, the structure of this doctoral thesis, the data set used as a research basis, and the conclusive findings are also presented and discussed.

2.1 Objective

The purpose of this research project is to establish an evaluation approach for assessing the (economic) performance of development aid projects and to understand the gaps for successful delivery. In order to do so, key financial figures such as the NPV and the ERR are evaluated. Comparison of NPV appraisal values to actual results at project completion will provide insights to understanding how projects develop in general. Reviewing the outlined ERRs against overall project performance ratings will serve as an approach to assessing the potential ERR outcomes for projects which have not disclosed any economic or financial figures. The data set upon which these analyses shall be based is sourced from the Implementation Completion and Results (ICR) Reports of the WB. As a central element the Social Return on Investment (SROI) concept will be discussed within this research paper. The idea is to calculate the SROI ratio of the LDCs as well as to compare its continental outcomes based on development aid's direct market values. Additionally, causes for low sustainability of the outcomes and bad performances of the bank and borrower will be qualitatively assessed and recommendations for future project deliveries outlined. Overall, this paper aims to contribute to the ongoing development aid discussions driven by the WB and the UN.

2.2 Structure

In principal this doctoral thesis consists of 2 main parts: A financial and a qualitative data analysis based on the data set of the ICR reports of the WB. In order to lead the reader to the central topic of the economic and financial analysis the first chapter

deals with the challenges to fight global poverty. In the second chapter the research approach itself is presented to set the basic understanding for this paper, and the objective and structure of this paper are further defined. Furthermore, important facts and definitions about the LDCs, WB, the data access and relevance of it and the term performance are given. The chapter is concluded by specifying the study and research questions and outlining the challenges and contribution of this thesis. Chapter 3 focuses on the theoretical foundations of the economic and financial analysis. Here, the concept of the SROI, its process and impact map, and the fundamentals of the Cost-Benefit-Analysis (CBA) as well as the proper calculation of the SROI are explained and outlined. The second part of Chapter 3 then specifies the financial figures and identified NPV-Cost-Ratios (NPVCRs) of the SROI calculation according to the background of the ICR data set. The final part of the chapter introduces additional economic and financial indicators which are disclosed within the ICR reports and could eventually be used to evaluate the WB performance further. In the fourth chapter the economic and financial analysis itself is conducted. In this context, the appraisal and result NPV values of the LDCs and its continents are presented and compared. Additionally, the results of the disclosed ERRs and the potential outcome for the non-financial figure disclosure projects are pointed out. Last but not least the results of the additional financial figures are presented. The final subchapter deals with the evaluation of the proper SROI ratio of the LDCs and its continents under consideration of different calculation approaches.

In chapter 5 the foundations for the qualitative data analysis are set, before the results of low sustainability and the causes for bad bank and borrower performances are evaluated and outlined. This chapter concludes the findings of this thesis and also presents recommendations for future development aid project deliveries of the WB. In this final chapter a critical acclaim to the outlined approaches and an outlook to further researches within the context of development aid delivered by the WB are provided to round off this dissertation project.

2.3 Least Developed Countries

The categorization of LDCs²¹ was created by the UN Economic and Social Council through its Committee for Development Policy in 1971. Currently 49²² developing countries from Africa (34 countries), Asia (14 countries) and Latin America (LA; 1 country) are classified as least developed by having a low level of per capita income and human resource development and a high degree of economic vulnerability:²³

Count	Africa	Asia	Latin America
1	Angola	Afghanistan	Haiti
2	Benin	Bangladesh	
3	Burkina Faso	Bhutan	
4	Burundi	Cambodia	
5	Central African Republic	East Timor	
6	Chad	Kiribati	
7	Comoros	Laos	
8	Democratic Republic of the Congo	Myanmar	
9	Djibouti	Nepal	
10	Equatorial Guinea	Samoa	
11	Eritrea	Solomon Islands	
12	Ethiopia	Tuvalu	
13	Gambia	Vanuatu	
14	Guinea	Yemen	
15	Guinea-Bissau		
16	Lesotho		
17	Liberia		
18	Madagascar		
19	Malawi		
20	Mali		
21	Mauritania		
22	Mozambique		
23	Niger		
24	Rwanda		
25	São Tomé and Príncipe		
26	Senegal		
27	Sierra Leone		
28	Somalia		
29	Sudan		
30	South Sudan		
31	Togo		
32	Tanzania		
33	Uganda		
34	Zambia		

Table 1: Overview of the LDCs²⁴

²¹ Author's Note: In principal, there have been 3 country groups identified by the UN which face specific development challenges: a) the LDCs itself, b) Land-locked Developing Countries (LLDCs) and c) Small Island Developing States (SIDS). However, subject to this paper are the LDCs only.

²² Author's Note: South Sudan became a LDC in 2012.

²³ Cf. Least Developed Countries (2014); UNCTAD (2014): Data on Least Developed Countries.

²⁴ Cf. UNCTAD (2014): UN list of Least Developed Countries.

The following criteria, which were revisited by the Committee in 2012 during the triennial review, apply to all LDCs:²⁵

1. Low-income criterion: based on a 3-year average estimate of the gross national income (GNI) per capita
2. Human Asset Index (HAI) – based on indicators of
 - a) nutrition – percentage of population undernourished
 - b) health – mortality rate for children aged 5 years or under
 - c) education – the gross secondary school enrolment ratio and
 - d) adult literacy rate
3. Economic Vulnerability Index (EV) – based on indicators of
 - a) population size
 - b) remoteness
 - c) merchandise export concentration
 - d) share of agriculture, forestry and fisheries in gross domestic product
 - e) share of population living in low elevated coastal zones
 - f) instability of exports of goods and services
 - g) victims of natural disasters and
 - h) instability of agricultural production

In order for a country to be defined as a LDC all of the following criteria have to be met:

- GNI Per Capita: 992 USD or less
- HAI: 60 or less
- EVI: 36 or more

As per an evaluation by the UN, the annual gross domestic product (GDP) per capita for LDCs amounted to 684 USD on average.²⁶

²⁵ Cf. Least Developed Countries (2014); UNCTAD (2014): Data on Least Developed Countries.

²⁶ Cf. Least Developed Countries (2014); UNCTAD (2014): Data on Least Developed Countries.

In order for a country to graduated out of the LDC status, 2 of the following 3 criteria must be met:²⁷

- GNI Per Capita: 1,190 USD or more
- HAI: 66 or more
- EVI: 32 or less

All LDCs have adopted special support measures in order to recognize and monitor their particular problems. Meanwhile, their development challenges are on the agenda of major international UN conferences and assemblies. The Fourth UN conference on the LDCs, held in 2011 in Istanbul (Turkey), took stock of the actual economic and social situation in the LDCs and adopted a set of development actions and strategies for these countries for the period of 2011-2020. As a result, the following 8 items have been defined as priority areas of action:

1. Productive capacity
2. Agriculture, food security and rural development
3. Trade
4. Commodities
5. Human and social development
6. Multiple crises and other emerging challenges
7. Mobilizing financial resources for development and capacity-building and
8. Good governance at all levels

The LDCs benefit from differential treatment in international economic and financial relations, such as market access preferences and priorities in terms of technical assistance by donor nations. All measures are designed to overcome structural disadvantages, support physical infrastructure development, develop human resources, and strengthen institutional capacities. The overarching goal is to

²⁷ Cf. Least Developed Countries (2014): A country also qualifies for graduation if its GNI per capita is 2,380 USD or more, independent of its HAI and EVI scores.

eradicate poverty and achieve the agreed development goals in order to move out of the LDC category.²⁸

2.4 World Bank

The WB is one of the largest global financial institutions that serves as a vital source of financial and technical assistance providing funds (most of all loans respectively credits) to developing countries for capital programs. It is not a bank in the ordinary sense but a unique partnership and global union to reduce poverty and support development. The WB was created at the Bretton Woods Conference in New Hampshire (US) in 1944 and became operational on the 27th of December in 1945. Along with it, IMF was also established. Since July 1st in 2012 Jim Yong Kim is the 12th president leading the WB.

The underlying objective of the WB and its operations is to increase productivity, incomes and welfare (“standard of living”). Furthermore, wages and employment should be raised and the working conditions, such as the conditions of labor, in the territories of member countries should be improved.²⁹

The WB itself must not be confused with the World Bank Group (WBG) which is a member of the UN. The headquarters of WBG is in Washington, D.C., employing more than 10,000 employees in around 120 locations worldwide. The Group consists of 5 international organizations which all supply funds to the LDCs:³⁰

- International Bank for Reconstruction and Development (IBRD)
- International Development Association (IDA)
- International Finance Corporation (IFC)
- Multilateral Investment Guarantee Agency (MIGA)
- International Centre for Settlement of Investment Disputes (ICSID)

²⁸ Cf. UNCTAD (2014): Data on Least Developed Countries; Author’s Note: So far, the following 3 countries have graduated from LDC status: Botswana in 1994, Cape Verde in 2007 and Maldives in 2011.

²⁹ Cf. WB (2014): About; WB (2015): IEG – Cost-benefit Analysis in World Bank Projects; WB (2014): What we do.

³⁰ Cf. WB (2014): About; WB (2014): What we do.

The WB comprises only the first 2 institutions mentioned, namely the IBRD and the IDA. Today, 188 countries are part of the IBRD and 172 of the IDA institutions.³¹ In fiscal 2012, the WBG committed 52.6 billion USD in loans, grants, equity investments, and guarantees to its members and to private businesses, of which IBRD and IDA took the lion share:³²

- IBRD commitments totaled 20.6 billion USD (compared with 26.7 billion USD in 2011).
- IDA (Bank's fund for the poorest countries) made commitments of 14.8 billion USD (compared with 16.3 billion USD in 2011).

In general, the WBG is aiming to achieve the following 2 goals for the world by the year 2030:³³

- End extreme poverty by decreasing the percentage of people living on less than 1.25 USD a day to no more than 3 percent.
- Promote shared prosperity by fostering the income growth of the bottom 40 percent for every country.

2.5 Data Access and Relevance

This chapter deals with the data set used for this research project and thus provides an overview of the data access and details on the collection of the ICR reports. Afterwards the ICR reports and their content are highlighted before further details on the various project types available conclude this chapter.

³¹ Cf. WB (2014): About.

³² Cf. WB (2014): About; WB (2014): What we do.

³³ WB (2014): What we do.

2.5.1 Data Basis

Research data used for this paper was obtained from ICR documents published by the WB on the bank's homepage. Audiences of this data set are governments, beneficiaries amongst individual countries and around the globe as well as the general public.³⁴ As explained in the previous chapter, the WB and its institutions are key supporters for addressing and solving the world's poverty problems. For this reason and considering the importance of data quality, WB projects were chosen to be the subject of this research project. The initial data load was conducted in September 2012 with documents (type: spreadsheet) pertaining to 49 LDCs respectively. These spreadsheets contain links to around 30,500 documents³⁵ –about 2,000 of which being ICR documents– of roughly 5,500 projects. To maintain focus on data with required research relevance, this paper only references ICR documents released after the MDGs were officially passed (9/8/2000).³⁶ Therefore, this paper is based on researching 790 ICR documents among all of the LDCs.³⁷

2.5.2 Implementation Completion and Results Reports

The ICR report is one of the main instruments of self-evaluation and serves as an integral part to increase development effectiveness of the WB. Reports are prepared by the WB at the close of every IDA or IBRD funded operation containing major financial figures. The project's NPV and the ERR³⁸ are determined and additionally the ICR evaluates the degree to which Project Development Objectives (PDO) have been achieved by providing outcome ratings in different project categories, such as

³⁴ Cf. WB (2012).

³⁵ Author's Note: The roughly 30,500 documents consisted of various document types, such as Project Appraisal, Staff Appraisal, Procurement Program, Project Plan Country Assistance Strategy, Credit Agreement, Environmental Assessment, Sectorial Review (e.g. energy, infrastructure), Annual and Board as well as Social Analysis documents respectively reports.

³⁶ Author's Note: Projects might have been started and closed before that date already.

³⁷ Cf. WB (2012).

³⁸ Author's Note: Any economic and financial indicator will be explained in *chapter 3 Economic and Financial Analysis*.

bank performance³⁹, borrower performance, sustainability respectively risk to development outcome. Furthermore, the ICR represents a continuous process of self-evaluation, lessons learned, knowledge sharing and being accountable for results. The following summarizes the main intention of the ICR and its system:

- Provide a complete account of the performance and results of each project and operation.
- Capture and dispose experience from previous projects in order to:
 - a) improve future interventions to achieve the goals of the Country Assistance Strategy (CAS),
 - b) improve the design and implementation of up-coming operations through lessons learned, and
 - c) ensure a greater development impact and sustainability for these future operations.
- Provide accountability and transparency at the project level while considering the bank, borrower and involved stakeholders.
- Provide an instrument for realistic self-evaluation of performance by the bank and borrowers (government and implementation agency).
- Contribute to databases for analysis and reporting, especially by the Quality Assurance Group and the Independent Evaluation Group (IEG)⁴⁰ on the effectiveness of development assistance in contributing to development strategies at the various levels (sector, country, and global).

The audience for the ICR is both internal (e.g. board members, bank managers and staff) and external (governments and their agencies, stakeholders, and beneficiaries

³⁹ Author's Note: In this research project the term "bank performance" is used for the evaluation of the performance of the WB.

⁴⁰ Author's Note: The WB projects are being "cross-checked" by the IEG (please refer to <http://ieg.worldbank.org/>). The focus of the IEG lies on reevaluating the project outcome ratings. The IEG data evaluation catalog can be accessed via <http://data.worldbank.org/data-catalog/IEG> (call on 2014/12/30).

in partner countries, as well as the general public). In general, the final ICR is publicly disclosed at the time it is submitted within the WB and to the board.⁴¹

Although there are “old format style” and “new format style” ICR documents, each ICR report follows a certain chapter structure. The old format style ICR reports have been disclosed before mid-2005⁴², the new ones after this date. Besides some minor differences, such as project type disclosure, controversial sustainability rating categories and a different chapter setup, the information content of both format styles is the same. The generic structure of an ICR report can be presented as follows:

No	Chapter
1.	Project Context, Development Objectives and Design
2.	Key Factors Affecting Implementation and Outcomes
3.	Assessment of Outcomes
4.	Assessment of Risk to Development Outcome / Sustainability
5.	Assessment of Bank and Borrower Performance
6.	Lessons Learned
7.	Comments on Issues Raised by Borrower/Implementing Agencies/Partners
Annex 1	Project Costs and Financing
Annex 2	Outputs by Component
Annex 3	Economic and Financial Analysis

Table 2: ICR Report Chapter Structure⁴³

The relevant chapters for this research project can be grouped into 2 different areas:

1. Chapters relevant for the economic and financial analysis are (part 1):
 - a. “Annex 3 – Economic and Financial Analysis”

⁴¹ Cf. WB (2013), p. 3 et seq.; World Press (2006), p. 1 et sqq.

⁴² Author’s Note: The exact date can’t be determined as there are both: ICR documents which were disclosed before 2005 based on the new format style structure as well as ICR documents which were disclosed after 2005 still being based on the old format style structure.

⁴³ Source: Own illustration; Author’s Note: Going forward any table, figure and equation created by the author is not referenced as such.

- b. “Annex 1 – Project Costs and Financing”
- 2. Chapters relevant for the qualitative data analysis (part 2):
 - a. “2. – Key Factors Affecting Implementation and Outcomes”
 - b. “4. – Assessment of Risk to Development Outcome / Sustainability”
 - c. “5. – Assessment of Bank and Borrower Performance”

The chapter “Project Context, Development Objectives and Design” was used to get a better understanding of the project setup, its design as well as the project types and sectors. Additionally, chapter “3. – Assessment of Outcomes” and “6. – Lessons Learned” were examined to ensure that reasonable conclusions can be made to the economic and financial and the qualitative data analysis.

2.5.3 Project Types

In order to have a common basis for comparing project outcomes in Africa and Asia, a study on the available project types needed to be conducted. The classification of project types is based and defined as “Sector and Theme Codes” within the ICR documents. Each project can consist of multiple “sector codes” (up to 5 in total) that determine the project type for the project funding provided by the WB. As illustrated in the following table, for each sector type a percentage is used to indicate how much of WB funding allocated to the project had been planned at the appraisal stage (“Original” column) and actually disbursed at project completion (“Actual” column):⁴⁴

⁴⁴ Author’s Note: Very often the WB is not the only project donor whereas the sector and theme codes distribution is only available for the WB funds but not for other donor funding. As outlined later in this thesis the total WB funding only makes up a minor portion of the total funding for the whole of the researched projects.

D. Sector and Theme Codes		
	Original	Actual
Sector Code (as % of total Bank financing)		
Central government administration	36	36
General education sector	4	4
General public administration sector	9	9
Health	40	40
Other social services	11	11

Table 3: Example of a Project Type Definition⁴⁵

In this example the planned WB financing per sector code has been estimated as actually disbursed.

2.6 Term “Performance”

Generally, performance can be defined as the fulfillment of a given task measured against a pre-defined standard of accurateness, cost, and schedule.⁴⁶ However, in this research project performance is referred as the positive or negative outcomes of the researched data set. Hereby several factors need to be considered and answers to the following questions will help formulate a judgment on development aid delivered by the WB and the potential value-add for the developing countries:

- How can the disclosed financial indicators for the WB projects be interpreted and is their outcome positive?
- In general, do projects run positively or negatively?
- Are the deliverables of the WB sustainable?

2.7 Study and Research Questions

In order to evaluate the economic performance and outcome of development aid provided by the WB, the NPV, ERR and other financial key figures available in the

⁴⁵ Cf. Afghanistan: AF_Completion_ICR1263.

⁴⁶ Cf. BusinessDictionary (2015).

ICR documents were gathered and analyzed. In the economic and financial analysis chapter the disclosed project NPVs are used to calculate SROI per continent and for the LDCs as a whole. Furthermore, a method for calculating the potential ERR outcomes for non-financial indicator (disclosure) projects is presented. The economic and financial analysis focuses on the following study questions:

- Are there continental differences for the NPV and cost disclosures?
- Do projects actually perform as planned?
- Can the SROI ratio be calculated for development aid projects?
- What is the average SROI of the LDCs?
- Are there additional financial indicators which help to evaluate project outcomes?

In the second part of this doctoral research project several qualitative data analyses are performed to discover causes for bad bank and borrower performances and negative sustainability ratings⁴⁷. Based on these research results the following questions should be answered:

- What are the reasons for bad project performance within the LDCs?
- How does the WB perform and what are the reasons for bad performance of the LDCs and across continents?
- How does the development aid recipient perform and what are the reasons for bad performance within the LDCs and across Africa and Asia?
- Are the WB projects sustainable? What are the reasons for negative sustainability ratings within the LDCs and across continents?

Additionally, based on the research findings, recommendations to avoid future bad performances shall be given.

⁴⁷ Author's Note: The definition of a negative sustainability rating will be provided in *chapter 5.2.1 Sustainability Rating Definition*.

Ultimately, the goal of answering the above study questions is to get a detailed understanding of the effectiveness of development aid projects. This shall in turn provide an answer to the fundamental question raised by this paper:

- Do WB development aid projects have a positive outcome for the LDCs and how can they be improved?

2.8 Challenges of this Doctoral Thesis

In general, the common challenges of writing and facilitating a dissertation applied to this doctoral research project. These challenges included not only the need for development and deepening of professional expertise, but also the creation of a realistic work and time concept, and the execution of such concept through thorough research and effective self-management.

Additionally, specific to this research project, challenges directly related to the nature of the research data set include:

- The data relevant for this research project is only available in pdf format, as there is no database which offers public access to the portfolio data required for this research project.⁴⁸ Thus, all economic and financial figures, project performance ratings as well as text sections relevant for the qualitative data analysis had to be captured manually through each of the 790 ICR documents.
- Even though this is a doctoral research project, the volume of data which was researched for the economic, financial and qualitative data analysis was enormous. The average number of pages per ICR document is 53. Nearly all of the reports were disclosed in the English language, with some exceptions

⁴⁸ Author's Note: The only structured database for viewing ICR relevant project data (project rating categories) on a portfolio level is currently created by the IEG (<http://data.worldbank.org/data-catalog/IEG>). However, this database is still not comprehensive (call on 2012/12/26; confirmed via mail (2012/12/27) from Jeanette Smith – WB contact (Internal Documents Unit)).

in Spanish. In order to perform economic and financial analysis each of document was:

- a) Searched by specific and multiple key words and abbreviations⁴⁹ in order to ensure different naming conventions and expressions used in the ICR documents are covered and
 - b) Examined across all relevant sections to ensure no details were missed.
- In many ICR documents project costs are disclosed across different sections which made it difficult to capture the total project costs. Further, different basis for disclaiming project costs (discounted vs. undiscounted costs) were used and thus several iterations of matching and validation processes to determine the correct project costs were required.
 - The SROI can only be calculated where the NPV and other relevant key figures are disclosed. However, there are no mandatory guidelines which enforce even the attempt to calculate and disclose a project NPV. In many ICR reports the following figures and information were difficult to capture –often not even disclosed– and therefore needed to be calculated based on a standard approach:
 - a) Total project life span⁵⁰
 - b) Rate of Return (RR)⁵¹ used for the NPV and ERR calculation
 - c) NPV calculation understanding: At the initial research stage it was not clear if NPV calculations at the result stage are a mixture of accumulation and discounting compared to NPV calculations at the appraisal stage where values are only discounted due to pure future projections.⁵²

⁴⁹ Cf. *Chapter 4.1 Analysis Approach and Setup*.

⁵⁰ Cf. *Chapter 3.2.2.4 NPV-horizon*.

⁵¹ Cf. *Chapter 3.2.2.5 Discount Rate*.

⁵² Author's Note: As understood during the course of the research conducted any of the outlined NPV values within the ICR reports always refers to the start date of the project, regardless if the NPV is based on the appraisal (here it is the planned project start date) or the result calculation (actual project start date). This guarantees a common basis of comparison as well as a more precise evaluation of the project components and their outcomes during the result stage.

- d) Costs of the NPV calculation: Capital and/or recurring operational and maintenance costs⁵³
- e) Discounted vs. undiscounted project costs: There are no common rules for the detailed NPV calculation tables within the ICR reports. Often the costs and benefits were already outlined as discounted values for each of the project periods, whereat sometimes this was not the case.
- In order to facilitate the content analysis, relevant sections of each “negative” ICR rating category (sustainability and bank and borrower performance) had to be placed into a separate word file which could then be further used for using qualitative data research tools (e.g. ATLAS.ti⁵⁴).

2.9 Contribution of this Thesis

One of the key objectives of this research project is to contribute to the overall discussion of development aid by determining the effectiveness of the WB projects using the SROI concept. The research analysis on NPV and additional relevant economic and financial indicators (such as costs, RR, project lifetime of the NPV calculation) in defined ICR reports lays the foundation for applying the SROI concept. The focus predominantly being to find the average SROI across all the LDCs. Additionally, a comparison of the 2 major continents –Africa and Asia– is conducted. Quantitative assessment of bad performance projects and the subsequent qualitative analysis are performed to identify detailed reasons for negative sustainability ratings and negative bank and borrower performance. Based on the findings, potential solutions and recommendations to overcome such hurdles are made. In the end, captured data from the ICR reports is placed in a structured database which can be kept alive by adding future ICR data to it.⁵⁵ A Visual Basic for Applications (VBA) macro

⁵³ Author’s Note: In this thesis “recurring operational and maintenance costs” are referred as “recurring costs”.

⁵⁴ Cf. <http://atlasti.com/>.

⁵⁵ Author’s Note: The database is spreadsheet-based. Please reach out to the author (schaefer.dominik@yahoo.de) for further information.

is created to evaluate data fed into the database. Utilizing both tools –the database and the VBA program– any set of ICR data can be quickly accessed, analyzed and compared (e.g. Africa vs. Asia).

3 Economic and Financial Analysis

This chapter deals with the economic and financial analysis of ICR reports used in this research paper. The economic and financial analysis represents one of the 2 major parts of this doctoral thesis.

The SROI concept – which represents the central element of the economic and financial analysis – will be described below in further detail. After providing the official definition of SROI and having a closer look at the concept and its calculation as described throughout literature, the second chapter will highlight how the generic SROI approach can be applied to the economic and financial data provided by the WB ICR reports by looking specifically at key figures provided within the closing documents. In the last chapter the additional economic and financial indicators disclosed within ICR documents are presented.

3.1 SROI Concept

In the following sections, the SROI concept is explained. Firstly, the proper definition, approach and principles as well as processes of carrying out the SROI analysis is compiled and provided. Then the CBA as the context providing method combining many of the indicators relevant for SROI calculation is presented. Finally, calculation of the SROI value is discussed.

3.1.1 SROI Definition

The SROI is an approach which puts a monetary value on different social investments. The SROI concept was developed by the Roberts Enterprise Development Fund (REDF), a philanthropy organization in California. In general, the approach can be used by a range of different organization types, such as non-profit (or voluntary) and social enterprises across public and private sectors, independent of the

company size and maturity. Specifically, non-profit organizations and social enterprises use the SROI as a management tool to improve their performance and outline added value.⁵⁶

Using the SROI approach, social, economic and environmental outcomes of a project, organization or policy can be better understood and managed. The objective is to reveal the value of outcomes which does not have direct market values, and thus the approach uses well-founded financial assumptions based on stakeholders' objectives.

There are 2 types of SROIs, the evaluative and the forecast SROI. Evaluative SROIs are based on past outcomes, whereat forecast SROIs predict the social value of the future. The approach is similar to the Return on Investment (ROI) method which businesses use to value return on investments (also known as discounted cash flow method). The ROI approach takes the cash flows which an investment is expected to generate over its lifetime and then "discounts" the value of these future cash flows to today's value. A key factor to note about the SROI, is that the calculated ratio should not be evaluated as a single number on its own. Instead, the SROI concept is a framework which includes the values of people telling a whole story, so that the value calculated can be better grasped and managed. The approach is based on 7 principles:⁵⁷

1. Involve all stakeholder groups to understand the way in which the organization creates change
2. Understand what changes through acknowledging and articulating all values, objectives and stakeholders to define the organization's scope
3. Value those things which are excluded from the markets in the same terms as used in the markets by using financial proxies for indicators
4. Be precise in how activities create change and evaluate this based on the evidence gathered

⁵⁶ Cf. The SROI Network (2014): The Guide in English 2012 Edition, p. 11 et sqq.

⁵⁷ Cf. Compass Partnership – Management Consultants (2011): Social Return on Investment; nef – economics as if people and the planet mattered (2011): Social Return on Investment; The SROI Primer (2011): Measuring Social Impact: The Foundation of Social Return on Investment [SROI]; The SROI Network (2014): What is SROI?.

5. Ensure performance is not overestimated through comparing impact with benchmarks, targets and existing external standards
6. Demonstrate the basis for the findings as well as by the use of appropriate reporting based on discussions with the stakeholders to ensure transparency
7. The results should be verified with an independent verification method

The SROI approach is also related to other approaches and concepts such as the CBA⁵⁸, Social accounting⁵⁹, and Sustainability reporting⁶⁰ as well as to different outcome approaches and methods and techniques of economic appraisal⁶¹ and the Environmental Impact Assessment (EIA)^{62, 63}

3.1.2 SROI Process and Impact Map

In order to carry out a SROI analysis, 6 stages need to be followed:⁶⁴

1. Establish scope and identify key stakeholder: This defines the necessary boundaries about what the SROI analysis will cover as well as who will be involved and how is important to nail down.
2. Map outcomes: One of the key elements of the SROI analysis is the so called “impact map” (also called a theory of change or logic model). The impact map should be established through stakeholder engagement and outlines the relationship between inputs, outputs and outcomes.

⁵⁸ Cf. *Chapter 3.1.3 Cost-Benefit-Analysis*.

⁵⁹ wiseGEEK (2014): “Social accounting is a type of accounting that a business performs to place a value on the influence its operations have on society. It requires that enterprises look closely at all that it does and what kind of impact its activities have on people, places, and the environment.”

⁶⁰ Global Reporting Initiative (2014): “A sustainability report is a type of corporate or organizational report. A sustainability report conveys sustainability-related information in a way that is comparable with financial reporting.”

⁶¹ HealthKnowledge (2014): “Economic appraisal and economic evaluation are general names for a set of techniques that weigh up the costs of an action against the benefits that it provides.”

⁶² European Commission (2014): “Environmental assessment is a procedure that ensures that the environmental implications of decisions are taken into account before the decisions are made.”

⁶³ Cf. The SROI Network (2014): SROI Guide - How SROI relates to other approaches.

⁶⁴ Cf. The SROI Network (2014): The Guide in English 2012 Edition; The SROI Network (2014): SROI Guide - Impact Map for worked example.

3. Evidence outcomes and put a value on them: This stage requires finding data to show whether outcomes have happened followed by giving them a monetary value.
4. Establish impact: The SROI analysis requires that those aspects of change that resulted as a matter of fact and would have happened anyway or are a result of other factors are taken out.
5. Calculate the SROI: In this stage all benefits are added up before any negatives are subtracted and finally the result is compared with the investment. This is also the stage where the sensitivity of the results can be subject to tests.
6. Report, share and embed: This critical last step involves the verification of the report created, sharing the findings with stakeholders using an interactive approach, and embedding the processes which generated good outcomes.

As previously mentioned, a key element of the SROI analysis is the impact map which illustrates the chain of economic value creation. During the process of establishing one, stakeholders, inputs and outputs as well as calculations and financial proxies for the changes are defined, and the SROI specific calculation variables for the identified activities such as the deadweight⁶⁵, displacement⁶⁶, attribution⁶⁷ and drop-off⁶⁸ (e.g. by using benchmark indicators) are determined. The following illustrations provide an example based on “MDG 6: Combat HIV/AIDS, malaria and other

⁶⁵ The SROI Network (2014): The Guide in English 2012 Edition, p. 56: “Deadweight is a measure of the amount of outcome that would have happened even if the activity had not taken place. It is calculated as a percentage.”

⁶⁶ The SROI Network (2014): The Guide in English 2012 Edition, p. 84: “An assessment of how much of the outcome has displaced other outcomes.”

⁶⁷ The SROI Network (2014): The Guide in English 2012 Edition, p. 59: “Attribution is an assessment of how much of the outcome was caused by the contribution of other organisations or people. Attribution is calculated as a percentage (ie the proportion of the outcome that is attributable to your organisation).”

⁶⁸ The SROI Network (2014): The Guide in English 2012 Edition, p. 61: “In future years, the amount of outcome is likely to be less or, if the same, will be more likely to be influenced by other factors, so attribution to your organisation is lower. Drop-off is used to account for this and is only calculated for outcomes that last more than 1 year.”

diseases” to demonstrate the SROI concept and how an impact map should be set up.⁶⁹

Social Return on Investment – MDGs Impact Map - 1					
Organization	United Nations				
Objective	Goal 6: Combat HIV/AIDS, malaria and other diseases: • Target 6.A: Have halted by 2015 and begun to reverse the spread of HIV/AIDS				
Scope	Activity	One example of <u>many</u> activities for this goal could be: • Invest in HIV/AIDS prophylaxis projects		Objective of Activity	...
	Contract/Funding/ Part of Organization	...		Purpose of Analysis	...
Stage 1			Stage 2		
Stakeholders	Intended/ unintended changes	Inputs		Outputs	The Outcomes (what changes)
Who do we have an effect on?	What do you think will change for them?	What do they invest	Value [monetary unit]	Summary of activity in numbers	Description
Who has an effect on us?					How would you describe the change?
e.g. developing countries	e.g. people use health services less	e.g. time	-	e.g. - „x“ prophylaxis medications provided per year - „y“ experts working around the globe	e.g. people feel healthier, less complaints
e.g. „rich“ countries governments	e.g. provide expert personnel, malaria prophylaxis medications	e.g. money	e.g. aid assistance amount		e.g. material outcomes for developing countries only. All outcomes for this stakeholder already considered above.
...	...	...	...		...

Figure 1.1: MDGs Impact Map⁷⁰

⁶⁹ Cf. The SROI Network (2014): The Guide in English 2012 Edition; The SROI Network (2014): SROI Guide - Impact Map for worked example.

⁷⁰ Source: Own illustration, following The SROI Network (2014): SROI Guide - Impact Map for worked example.

Social Return on Investment – MDGs Impact Map - 2						
Organization	United Nations					
Objective	Goal 6: Combat HIV/AIDS, malaria and other diseases: • Target 6.A: Have halted by 2015 and begun to reverse the spread of HIV/AIDS					
Scope	Activity	One example of <u>many</u> activities for this goal could be: • Invest in HIV/AIDS prophylaxis projects	Time Period		2002 - 2015	
			Evaluation & Forecast		Evaluation & Forecast	
Stage 3						
The Outcomes (what changes)						
Indicator	Source	Quantity	Duration	Financial proxy	Value [monetary unit]	Source
How would you measure it?	Where did you get the information from?	How much change was there?	How long does it last?	What proxy would you use to value the change?	What is the value of the change?	Where did you get the information from?
e.g. decrease of malaria infection rate	e.g. Progress Reports, World Bank homepage, IMF homepage	e.g. „z“ less malaria infections	e.g. „w“ years	e.g. „hospital accommodation costs“	e.g. „hospital accommodation costs amount“	e.g. World Bank homepage, IMF homepage
-	-	-	-	-	-	-
...	...	...	...	...	...	...

Figure 1.2: MDGs Impact Map⁷¹

Social Return on Investment – MDGs Impact Map - 3							
Organization	United Nations						
Objective	Goal 6: Combat HIV/AIDS, malaria and other diseases: • Target 6.A: Have halted by 2015 and begun to reverse the spread of HIV/AIDS						
Scope	Activity	One example of <u>many</u> activities for this goal could be: • Invest in HIV/AIDS prophylaxis projects					
Stage 4					Stage 5		
Deadweight [%]	Displacement [%]	Attribution [%]	Drop Off [%]	Impact	Calculating Social Return on Investment		
What would have happened without the activity?	What activity did you displace	Who else contributed to the change?	Does the outcome drop off in future years?	Calculation: (quantity * financial proxy) - deadweight, - displacement, - attribution.	Rate of Return per year [%]		
					Year 1	...	Year X
[%]; e.g. Benchmark indicator	[%]; e.g. Benchmark indicator	[%]; e.g. Benchmark indicator	[%]; e.g. Benchmark indicator	[monetary amount]	[monetary amount]	[monetary amount]	[monetary amount]
-	-	-	-	-	-	-	-
...	...	...	...	...	...	...	...
					[Sum from above columns activities' amounts]	[Sum from above columns activities' amounts]	[Sum from above columns activities' amounts]
					[Present Value]		
					[Total Present Value]		
					[Net Present Value]		
					[Social Return]		

⁷¹ Source: Own illustration, following The SROI Network (2014): SROI Guide - Impact Map for worked example.

Figure 1.3: MDGs Impact Map⁷²

As indicated by the example in the above figures, the setup of the SROI analysis for the identified MDGs' activities consists of 5 stages. Based on the activities relevant for calculating the respective NPV/SROI, the stakeholders and the changes for each of them will be evaluated in the first stage. In the second stage the stakeholders' investments (inputs) and the outcomes of the same will be described. Stage 3 then examines the indicators for measuring the outcomes, the sources used to determine the changes as well as the financial proxies utilized and the concrete values of the changes. The fourth stage then determines the measures and assessments for calculating the impacts of the changes. Finally, stage 5 calculates the PVs based on the RR for future years to determine the SROI, before in stage 6 the results are verified and shared with stakeholders.⁷³

3.1.3 Cost-Benefit-Analysis

The CBA –sometimes called Benefit-Cost-Analysis– is often used as an umbrella term for different analysis in order to compare costs and benefits, or rather whether the present value (PV) of benefits of a given project exceeds the PV of costs. The CBA as a term refers to 2 things:⁷⁴

1. Discipline which is used to help assess and appraise the case for a project or a proposal (project appraisal).
2. Informal approach to any kind of decision making.

Regardless of the definition the process requires the weighting of the expected total costs against the total benefits of the different options available. It is an economic tool to aid social decision-making often used by governments to assess the impacts and attractiveness of a given intervention in markets and different types of cross-regional programs. The aim is to evaluate the efficiency of the planned intervention

⁷² Source: Own illustration, following The SROI Network (2014): SROI Guide - Impact Map for worked example.

⁷³ Cf. The SROI Network (2014): The Guide in English 2012 Edition, p. 9. et seq.

⁷⁴ Cf. Ferrara (2010), unpag.; Gabler Wirtschaftslexikon (2014): Kosten-Nutzen-Analyse.

project relative to the status quo. The related costs and benefits of the intervention impacts are assessed and rated in terms of the general willingness of the public to pay for them (benefits) or willingness to pay to avoid them (costs). The principle idea of the CBA is to consider all parties and stakeholders affected by the intervention and put a monetary value of the change on their “welfare” as it would be valued by them. Inputs are normally measured in terms of opportunity costs, whereat it is often difficult to construct meaningful measures of the costs and benefits of specific tasks, actions, and outcomes. In practice, the estimation of costs and benefits is generally based on survey methods and historical and benchmark data, whereat the CBA attempts to have a common temporal basis of comparison for all costs and benefits. Therefore, the CBA is normally based on a time value of money formula which basically means that any future expected cash flows of costs and benefits are discounted to the PV. During the course of a CBA, monetary values are often assigned to less tangible effects, for instance to various risks such as market penetration, long-term strategies, environmental and political changes that could negatively affect certain project components or the project as a whole. This normally comes into play when governments use methods or techniques in order to decide whether to introduce new business regulations, offer new drugs or build new roads. In such a case a value must be put onto human life and the environment which obviously comes along with controversial discussions. The main purpose of the usage for CBA, but not the exclusive one, is to assess the monetary amount of (very) large private as well as public sector projects and programs. This is because those types of projects tend to include both scenarios where costs and benefits are either less flexible to be expressed in financial terms (e.g. environmental changes) or the opposite. The practice of CBA generally differs between countries and amongst sectors within countries (e.g. health and transport). Methods and indicators often used within the context of the CBA, such as the Benefit-Cost-Ratio (BCR), NPV, ERR,

Net Benefits⁷⁵ are also used in the ICR evaluation of the WB and thus forms a part of this research project.⁷⁶

3.1.4 SROI Calculation

The first step in calculating the SROI ratio requires the projection of all outcomes – based on their persistence– into the future. This sets the basis for calculating the PV and NPV. Based on the outcomes’ persistence, the values of their impact are set –normally for 1 time period (usually 1 year). The value for each outcome is then applied across the number of periods it will last, before any drop-offs are subtracted for each of the future periods after the first period. In the second step the NPV needs to be calculated. In order to do so the costs paid and the benefits received in different periods need to be added up. To ensure that the costs and benefits are compared on an equal basis, discounting is used (this is generally referred as “time value of money”). In order to appropriately value future cash flows, the determination of the appropriate RR is key. Obviously, this is a controversial area with ongoing discussion and research since short-termism is encouraged by discounting the future. The most commonly applied model of the PV can be described as follows:⁷⁷

$$PV = \sum_{t=1}^n \frac{\text{Value of Impact}_t}{(1+i)^t}$$

where

Value of Impact = benefit during each period

n = analytic horizon

i = effective periodic interest rate or rate of return

t = time period

Equation 1: Present Value⁷⁸

⁷⁵ Author’s Note: all of the mentioned methods and concepts will be explained in detail at the appropriate place during the course of this research paper.

⁷⁶ Cf. Boardman/Greenberg/Vining/Weimer (2010), unpag.; Ferrara (2010), unpag.; Gabler Wirtschaftslexikon (2014): Kosten-Nutzen-Analyse.

⁷⁷ Cf. Investopedia (2014): NPV; Projektmagazin (2014); Investopedia (2014): PV; The SROI Network (2014): The Guide in English 2012 Edition, p. 66. et seq.

⁷⁸ Cf. Own illustration, following The SROI Network (2014): The Guide in English 2012 Edition, p. 66. et seq.

The PV is defined as the current worth of a future sum of money of cash flows given a specified RR. In principle, future cash flows are discounted at a specific RR, whereat the higher the RR, the lower the PV of the future cash flows. The NPV can be defined as the difference between the PV of cash inflows and the PV of cash outflows. The following formula shows the calculation of the NPV. The only difference comparing to the formula of the PV is that instead of the “Value of Impact” the total cash flows (cash in and out) per period are applied to the numerator:⁷⁹

$$NPV = \sum_{t=1}^n \frac{Cash\ Flow_t}{(1+i)^t}$$

where

Cash Flow = cash flow (cash in and out) during each period (including initial investment in t=0)

n = analytic horizon

i = effective periodic interest rate or rate of return

t = time period

Equation 2: Net Present Value⁸⁰

A NPV greater than 0 indicates the investment as value adding. A NPV less than 0 indicates that the investment would have a negative outcome in terms of adding value for the investors and thus is not recommended. In the case of NPV equal to 0, the investment provides neither a positive nor a negative value add.⁸¹

In order to calculate the SROI ratio the discounted value of benefits (PV) needs to be divided by the discounted total respective investment costs:

$$SROI\ Ratio = \frac{PV}{Total\ Costs} = \frac{NPV + Total\ Costs}{Total\ Costs}$$

Equation 3: Social Return on Investment Ratio⁸²

⁷⁹ Cf. Investopedia (2014): NPV; Projektmagazin (2014); Investopedia (2014): PV; The SROI Network (2014): The Guide in English 2012 Edition, p. 66. et seq.

⁸⁰ Cf. Own illustration, following The SROI Network (2014): The Guide in English 2012 Edition, p. 66. et seq.

⁸¹ Cf. Investopedia (2014): NPV; Investopedia (2014): PV; The SROI Network (2014): The Guide in English 2012 Edition, p. 66. et seq.

⁸² Cf. Own illustration, following The SROI Network (2014): The Guide in English 2012 Edition, p. 68. et seq.

An alternative calculation is the Net SROI ratio, whereat here the NPV is divided by the discounted total costs:

$$\text{Net SROI Ratio} = \frac{NPV}{Total\ Costs}$$

Equation 4: Net Social Return on Investment Ratio⁸³

There is one thing which needs to be noted when looking at official documents or publications such as the SROI Network (2014) or Többen (2011): In general the calculated SROI ratio is based on a one-time investment in period $t = 0$ –there are no recurring costs for the periods after $t = 0$. Since normally cash flows do vary from period to period (Hannerer (2011) provides a better example with varying, recurring costs per period), this in general does not reflect reality. Therefore, recurring costs based on further investments, resource and maintenance costs need to be considered.⁸⁴ As discussed in more detail in *chapter 3.2.2.2 Capital and Recurring Costs* recurring costs do apply for most of the WB projects.

3.2 SROI of World Bank Projects

In this chapter details on the calculation of the SROI using the ICR reports of the WB are discussed. In the following the purpose of the CBA and the indicators which are relevant for the SROI calculation, specifically within the context of the WB database used for this research paper, will be analyzed. Afterwards the different types of NPVCRs identified during the course of this research are introduced, before the approach for calculating the proper SROI ratio of the LDCs is presented. The chapter closes with outlining the understanding of the portfolio and sensitivity analysis of this thesis.

⁸³ Cf. Own illustration, following The SROI Network (2014): The Guide in English 2012 Edition, p. 68. et seq.

⁸⁴ Cf. The SROI Network (2014): The Guide in English 2012 Edition, p. 64 et sqq.; Többen (2011), p. 29 et sqq.; Hannerer (2011), p. 48 et seq.

3.2.1 Purpose of the Cost-Benefit-Analysis

The CBA is a technique which has been used by the WB since the early 1970s. In principle, it serves the WB as a self-evaluation and verification tool for stakeholders outside the WB. Further, it is one of the core in-house reporting instruments to inform the Board of Directors that the WB's operations indeed have a net positive effect on the welfare of the member countries. Since the CBA normally produces both a NPV and an ERR calculation, the WB's policy directs it to help the borrowing countries to select the highest-NPV project. Further on this, a country would as well be instructed to "do nothing" in case even the best alternative discloses a negative project NPV. Generally, a negative project NPV implies that through the borrowing process at a certain interest rate the country as a whole becomes poorer. However, in regards to IDA funds which usually contain large grant elements, a negative project NPV does not necessarily mean that the country would get poorer. Nevertheless, such negative NPV projects waste funds from donor countries when possibly better alternatives could have been pursued and obviously global resources get wasted. Interestingly, neither of the described issues is empirically significant, whereat projects with a CBA and NPV calculation almost always report an ERR as well. Still, since the ERR says nothing about the quality of the analysis, the NPV method is the main criterion for the WB in order to evaluate project outcomes.⁸⁵

3.2.2 Indicators of the SROI Calculation

In this chapter the relevant indicators of the ICR report for the SROI calculation of the LDCs are presented and discussed.

3.2.2.1 Net Present Value

The NPV disclosure within the ICR reports builds the foundation for the proper SROI calculation of the LDCs of this research project. Not only is NPV the central concept

⁸⁵ Cf. Baneth, Jean (1996), p. 29 et sqq.; Pearce/Giles/Susana (2006), p. 70 et seq.; WB (2015): IEG – Cost-benefit Analysis in World Bank Projects, p. 2.

of the SROI calculation in general and that it defines if an investment is meaningful or not, it is further the preferred method of project evaluation of the WB. The NPV disclosed within a report is also commonly referred to as the Economic-NPV or E-NPV⁸⁶. A positive NPV contributes to the fundamental objective underlying the WB's policy which is the strengthening of a country's economy, and more generally, the increase in welfare of poorer countries. Thus, a positive effect on a single country's economy has a positive ripple effect on the world's economy itself or –most positively spoken– the country itself does require less or none funds in the future.⁸⁷ In principle, there are 2 types of NPVs which can be identified for WB projects:

1. Overall-NPV: The sum of NPVs of each existing project component. Generally, all project costs (all capital/investment and recurring costs) have been considered in order to calculate the overall-NPV.⁸⁸ Depending if the project consists of 1 or multiple components and if in the latter case the ICR document discloses each component NPV separately or the sum of the overall project NPV itself, it is necessary to distinguish between 2 overall-NPVs:
 - a. Manually calculated overall-NPV: The sum of the NPVs of each existing project component which has been manual calculated by the author.
 - b. Overall-disclosed NPV: The disclosed NPV of the whole project. Independent of whether the project consists of 1 or multiple project components, the overall-disclosed NPV is the accumulated NPV generated from all project components. In the ICR documents the overall-disclosed NPV is generally referred as the “NPV of the whole project” or the “overall-NPV of the project”.

⁸⁶ Author's Note: Within this research paper the abbreviation “NPV” is used for the E-NPV.

⁸⁷ Cf. Pearce/Giles/Susana (2006), p. 70.

⁸⁸ Author's Note: Please refer to the NPV calculation of any country ICR document publicly available under the WB homepage (<http://documents.worldbank.org/curated/en/country>). However, there are some overall-NPVs where not all invested capital costs of the project have been considered for the NPV calculation.

2. Partial-NPV: The NPV of a project component or of multiple project components, but never refers to all project components. A partial-NPV is only declared in case the ICR document does not disclose an overall-NPV. To calculate a partial-NPV only related project costs (capital and recurring costs) of the corresponding project components have been considered.⁸⁹ There are 2 types of partial-NPVs which can be distinguished:⁹⁰
 - a. Component-NPV: Based on at least 1 project component, but can also be the NPV of multiple components. In case of multiple components the component-NPV is mostly manually calculated.
 - b. Sample-NPV: Refers to neither a whole project nor multiple project components. A sample-NPV normally showcases based on a few project specific activities or tasks how valuable the investment would be.

For all of the above mentioned types and sub-types of NPVs there are again 2 differentiations:

1. Standalone-NPV: This NPV is outlined as a single value. That is the usual disclosure of a NPV within the ICR reports.
2. Scenario-NPV: This NPV is disclosed as a range of multiple values, whereat there are at least 2 values. The 2 values outlining the most negative and most positive value are defined as the minimum and the maximum NPV of the scenario-NPV. For this research project only the outlined minimum and maximum NPV values are taken into account.⁹¹

Furthermore, every NPV is categorized into 1 of 3 groups, depending on its valuation:

⁸⁹ The question why the ICR department of the WB doesn't take all project components into account when calculating the partial-NPVs' is discussed in *chapter 3.2.4 Calculation of the proper SROI Ratio*. However, it is an open question if the amount of the recurring costs for partial-NPVs would actually be higher in case all project components would have been considered.

⁹⁰ Author's Note: Please refer to the NPV calculation of any country ICR document publicly available under the WB homepage (<http://documents.worldbank.org/curated/en/country>).

⁹¹ Author's Note: The reason for the existence of scenario-NPVs are sensitivity analysis using different parameters for e.g. the RR, project lifetime, cash flows in order to account for unpredictable changes in the future.

1. Positive NPV: Described by a positive value which is greater than 0. In case of a scenario-NPV, the overall- or partial-NPV of the project is defined as positive if the minimum and maximum NPVs are both positive.
2. Neutral NPV: Exists only for a scenario-NPV and is described by the fact that the minimum NPV value is negative and the maximum NPV value is positive.
3. Negative NPV: Described by a negative value which is less than 0. In case of a scenario-NPV, the overall- or partial-NPV of the project is defined as negative if the minimum and maximum NPVs are both negative.

3.2.2.2 Capital and Recurring Costs

In order to calculate SROI based on disclosed NPVs the corresponding cost information needs to be gathered based on the NPV-disclosing ICR document. In principle, there are 2 different types of costs which need to be distinguished:

1. Investment or capital costs: Capital costs are costs of funds used for financing a business.⁹² Cost of capital depends on the mode of financing used, whereat in the context of development aid funds from the donor countries – or more specific from the WB– it is provided to the government of the fund-receiving country. The agreed funding⁹³ is then used to ensure that the project objectives are fulfilled within the foreseen project timeframe. Therefore, capital costs can include expenses on works, goods and equipment, services, consulting, training, resources as well as contingencies. In general, the funding provided does mirror or cover the project required capital costs. Depending on whether it is a partial- or overall-NPV, the costs to be applied may vary.

⁹² Cf. Investopedia (2014): Cost of Capital.

⁹³ Author's Note: Only in exceptional cases the funding amount is higher/lower than the capital costs required for the project. In case of a higher funding amount the remaining budget might need to be given back or used to cover some of the recurring costs (case-to-case decision of the WB responsible in charge). In case the funding amount is lower than the capital costs the country itself may have been requested to cover the additional capital expenses.

In case of a partial-NPV only related capital costs are considered, whereas in general all capital costs are taken into account for an overall-NPV.⁹⁴

2. Recurring operational and maintenance costs: Regular costs which incur repeatedly or for each item produced or each service performed.⁹⁵ In general, recurring costs are not part of the funding provided to the receiving country and therefore normally provided by the country itself. Examples of recurring operation and maintenance costs include continuous street maintenance and necessary reparations during and after the delivery of an infrastructure project or ongoing energy costs for running a power plant, hospital, or other welfare institutions. Only in few project cases there are no recurring costs existing.⁹⁶ The issue with the ICR documents is that in nearly all cases where a NPV value is outlined, there is no detailed calculation available⁹⁷. Further, very often the amount of recurring costs is not even outlined despite their existence. Although recurring costs have already been considered during the NPV calculation they need to be taken into account again in order to establish the project SROI ratio.

For both types of costs the amount applied may be different in the case of a scenario-NPV. In general, the differentiation for minimum and maximum NPV is based on the regulation of the applied RR, project lifetime as well as varying benefits per period across the lifetime.

⁹⁴ Author's Note: If or not disclosed partial-NPVs must actually be lower since nor the remaining capital neither potential recurring costs of the remaining capital costs have been taken into account is discussed in *chapter 3.2.4 Calculation of the proper SROI Ratio*.

⁹⁵ Cf. BusinessDictionary (2014): Recurring Cost.

⁹⁶ Cf. Niger – Report 26930, p.23: “The investment costs used in the analysis were based on the actual costs of rehabilitation, including design and supervision, but excluding taxes. There was little or no annual routine maintenance on the 6 roads analyzed, and the annual routine maintenance costs are therefore estimated to be zero.”; Zambia – Report 24019, p.24: see NPV calculation table.

⁹⁷ Author's Note: Based on the evaluation of the disclosed NPVs at the result stage, only about 10 percent (17 out of 168) of the NPV projects are not based on recurring costs. For another 13.1 percent of the NPV projects no information on the fact if recurring costs exist is given.

Furthermore, to provide a common basis for comparison, both cost types need to be discounted before the SROI ratio calculation.⁹⁸ Normally, only cases where a detailed NPV calculation is available can the discounted values (benefits and costs) can be gathered from ICR documents. In a standard case NPV related capital costs are disclosed as a lump sum for the total project duration and on an undiscounted basis in “Annex 1 – Project Costs and Financing” of the ICR document (refer to *chapter 2.5.2 Implementation Completion and Results Reports*). In comparison recurring operation and maintenance costs are normally disclosed within the detailed NPV calculation in “Annex 3 – Economic and Financial Analysis”, whereat they are either disclosed on an undiscounted or discounted basis. In case the ICR document discloses neither discounted capital and recurring costs nor a detailed calculation on the NPV, capital costs as well as recurring costs are distributed and discounted evenly across the project duration.⁹⁹

3.2.2.3 Project Dates and Duration

Project duration is defined by project start and project end date, which delineate the time period of project implementation duration. For both dates there is a planned and an actual date. As per the ICR documents, the following definitions apply:¹⁰⁰

- Planned Start date: “Effectiveness – Original Date”
- Planned Finish date: “Closing – Original Date”
- Actual Start date: “Effectiveness – Revised / Actual Date”
- Actual Finish date: “Closing – Revised / Actual Date”

The planned and actual duration of the project are not disclosed in the ICR reports and therefore need to be calculated in order to conduct the discounting. The actual

⁹⁸ Cf. The SROI Network (2014): The Guide in English 2012 Edition, p. 64 et sqq.

⁹⁹ Author’s Note: The decision to distribute and discount evenly across the project duration respectively the NPV-horizon period can be reinforced by the fact that the projects containing a detailed NPV calculation do generally examine an evenly spending of both cost types across the corresponding period. Furthermore, an uneven distribution, for instance applying all costs at the beginning respectively end of the period, would be too pessimistic respectively optimistic to be justified generally.

¹⁰⁰ Cf. Project Management Lexicon (2014).

project duration is the time period used for discounting the NPV relevant capital costs in order to establish the SROI ratio.¹⁰¹

3.2.2.4 NPV-horizon

NPV-horizon is defined as the time period which is used in order to discount the recurring costs. It is generally referred to as the total lifespan of the respective project during which the whole costs occur and benefits should be generated. A NPV-horizon begins with the project start period. In general, NPV calculation is based on a time period which exceeds the project implementation duration due to the assumptions that NPV calculations of development aid projects are based on outlook beyond project implementation time frame. Therefore, this extended period needs to be considered to ensure a common basis of comparison for all benefits and costs of the SROI ratio. In case no detailed NPV calculation is outlined in the ICR document for recurring costs, a standard discounting approach of distributing and discounting the recurring costs evenly per period across the NPV-horizon using the project RR is used.¹⁰²

3.2.2.5 Discount Rate

Discount rate can be defined as the periodic interest rate or RR. It is the multiplier that converts anticipated or expected returns from an investment to their current market value (PV). Often, the discount rate used for PV calculations is a RR taken from the corresponding economic or financial markets or branches. Sometimes this can be very controversial as for instance a high discount rate normally implies a (very) low value on the welfare of future generations that might then have a huge impact on the desirability of interventions to help the environment. The discount rate is also referred to as the “hurdle level” or “percentage of capital cost” which is the

¹⁰¹ Cf. Project Management Lexicon (2014).

¹⁰² Author's Note: One could argue that the discounting of the recurring costs for the periods starting from $t = 0$ is not meaningful as in general project outcomes need to be generated first before maintenance costs apply. However, for the majority of ICR reports outlining a detailed NPV calculation the discounting of recurring costs is conducted from the project start on ($t = 0$).

minimum project outcome target. The WB conventions for the usage of a certain discount rate are however quite specific. For instance, a discount rate of 12 percent needs to be used for the evaluation of bank-financed transport projects. Generally, the discount rate used is not necessarily an accurate reflection of the opportunity cost of capital of the borrower's country. Instead it must be rather seen as the rationing tool for the WB funds. If the appraisal requirements of a country request for a different discount rate than 12 percent, then both the WB's specific discount rate and the country specific one need to be tested. Suggesting a discount rate lower than 10 percent is unlikely to be approved, since recent researches have proven that the cost of capital are higher than this in developing countries. In any case the project manager will need to justify and demonstrate any alternative rate.¹⁰³

3.2.3 Types of NPV-Cost-Ratios

One main focus of this doctoral thesis is to filter, evaluate and make use of the disclosed NPVs of the WB projects in order to calculate the proper SROI ratio for the LDCs. In order to do so, the corresponding cost information for each disclosed NPV needs to be filtered from the ICR documents. The coefficient calculated for each of the projects is based on the principle of the SROI ratio (PV/Value of Inputs) instead of the Net SROI ratio. This guarantees a direct assessment if a value-add exists based on the sum of all costs –discounted in the period they accrued– added to both nominator and denominator. However, due to variations in the disclosure of recurring costs across the entirety of ICR reports and the existence of different NPV types (partial- and overall-NPV), there are 4 different NPVCR types which can be identified.¹⁰⁴ Each of the NPVCRs follows the same principle which is similar to the one of the proper SROI ratio:

¹⁰³ Cf. Baneth, Jean (1996), p. 29 et sqq.; Boardman/Greenberg/Vining/Weimer (2010), unpag.; BusinessDictionary (2014): Discount Rate; Laos – Report 22325, p. 6 et seq.; Nepal – Report ICR1197, p. 38 et sqq. and Report ICR1311, p. 28; Malawi – Report ICR764, p. 32; Senegal – Report 1415, p. 14 et seq.; WB (2015): Transport Notes, p.8.

¹⁰⁴ Cf. The SROI Network (2014); WB (2003).

$$NPVCR = \frac{NPV + Costs}{Costs}$$

Equation 5: NPVCR

Through the 4 types of NPVCRs a common basis of project comparison is given whereat their results are used to calculate the proper SROI ratio of the LDCs (refer to the next chapter).

For each of the identified ratio types, consideration is taken from stakeholders' perspective with pros and cons noted in each case. Furthermore, there are 2 versions for each NPVCR defined:

1. Discounted NPVCR: The discounted ratio reflects the proper NPV-cost-relation, since all parameters of the ratio are compared on a discounted basis. It therefore provides a dynamic view on the monetary outcomes achieved.
2. Undiscounted NPVCR: Due to missing discounted cost values in the ICR reports, undiscounted cost types are more readily available than discounted ones. Using the undiscounted costs and the disclosed NPVs (always discounted) in order to calculate the NPVCRs of course does not provide a correct view on the cost-benefit relationship. However, those NPVCRs describe ratios which can be built directly out of the information available within the ICR reports without any modification. Additionally, many of the discounted costs within the ICR reports are based on an even cost distribution. Therefore, the undiscounted ratio can be regarded as the "most negative" discounted ratios, i.e. all costs are assumed to incur in period $t = 0$.

In the following the 4 different NPVCRs are presented and discussed.

3.2.3.1 Pro-Rata-Capital-Costs Ratio

The pro-rata-capital-costs ratio is based on NPV related capital costs only. No recurring costs are used to construct this NPVCR, even in case the NPV is based on this cost type. Therefore, the total project capital costs are considered for overall-

NPVs, but only the related partial capital costs for partial-NPVs. For any overall-NPV project which is not based on recurring costs the discounted version of the ratio already reflects the proper SROI ratio. From a stakeholder's perspective the funds supplied by the donors and moreover their perspective and interests are the central element of this ratio, since the monetary evolution of the provided support is shown. The pros and cons of this ratio can be described as follows:

- + For all projects outlining a NPV value this ratio can be built without the need to calculate standard recurring costs.
- + It only considers the NPV related capital costs and not the total capital costs of the project. Therefore, the ratio is
 1. focusing on the benefit generating components only and
 2. in line with the capital costs considered for both NPV types.
- Since only the related capital costs of the partial-NPVs are considered, the remaining capital costs for those projects are not part of this NPVCR.
- Recurring costs have not been considered in the NPVCR itself –although the NPV might be based on this cost type.
- The partial-NPVs itself are disclosed too positive due to the fact that only related capital costs instead of the total project capital costs are part of their calculation. Thus, by default the ratio might be too positive.¹⁰⁵

3.2.3.2 Total-Capital-Costs Ratio

The total-capital-costs ratio is constructed using the total capital investment of the project but no recurring costs. For any overall-NPV project which is not based on recurring costs the discounted version of the ratio already reflects the proper SROI

¹⁰⁵ Author's Note: This is a general weakness of all 4 NPVCRs. A solution for this weakness will be described in *chapter 3.2.4 Calculation of the proper SROI Ratio*. The results of the calculation of the proper SROI ratio of the LDCs will be presented in *chapter 4.8.3 Calculating the Minimum SROI Ratio*.

ratio. The consideration is based purely on the donors'¹⁰⁶ funds and the value generated out of them. It basically shows what value-ratio is generated with the funding provided. Following pros and cons need to be noted for this NPVCR:

- + For all projects outlining a NPV value this ratio can be built without the need to calculate standard recurring costs.
- + The stakeholders are provided with a complete picture of the project's monetary outcome from the total capital investment, regardless of which project components were responsible for producing it.
- For partial-NPV projects there is a mismatch between the costs used to calculate the NPV and the ones used to construct the ratio.
- Recurring costs have not been considered in the NPVCR itself –although the NPV might be based on such cost types.
- The partial-NPVs itself are disclosed too positive due to the fact that only related capital costs instead of the total project capital costs are part of their calculation. Thus, by default the ratio might be too positive.

3.2.3.3 Pro-Rata-Capital plus Recurring-Costs Ratio

The pro-rata-capital plus recurring-costs ratio is based on the NPV related capital and the corresponding recurring costs. In case of overall-NPVs the total project capital costs are taken into account whereat only the corresponding partial capital costs are considered for partial-NPVs. However, for both cases the NPV related recurring costs are included in the ratio. Since only quantifiable project components have been used to calculate the NPV and the costs considered in the NPVCR are NPV corresponding, the discounted version of this ratio can be considered as the SROI ratio as the WB would define it.¹⁰⁷ The stakeholders can best be described as the

¹⁰⁶ Author's Note: Usually, the WB is the donor which supplies the largest portion of the total fund provided. Still, many projects are supported by multiple donors.

¹⁰⁷ Cf. The SROI Network (2014); WB (2003).

country's economy itself and its supported people and the achievement of the donors as a whole, such as the WB including the UN. The pros and cons of this ratio can be described as the following:

- + The ratio only considers the NPV related capital costs and not the total capital costs of the project and is thus focusing on the direct relationship of investment and outcome.
- The partial-NPVs itself are disclosed too positive due to the fact that only related capital costs instead of the total project capital costs are part of their calculation. Thus, by default the ratio might be too positive as well.

3.2.3.4 Total-Capital plus Recurring-Costs Ratio

The total-capital plus recurring-cost ratio takes the total capital investment of the project including the NPV related recurring costs into account. In terms of partial-NPVs the ratio-applied costs do not match the costs used for calculating the NPV as the full projects costs are applied in the ratio. Still, this NPVCR illustrates the total BCR of all project activities – even if some of the activities were not evaluated (partial-NPVs). The stakeholder group is similar to the one of the pro-rata-capital plus recurring-costs ratio, mainly the country's economy itself as well as all fund providing donors. The following are the pros and cons of this ratio:

- + The ratio considers all project-related costs, including recurring costs and thus provides an overall picture of the cost-benefits of the whole project.
- For partial-NPV projects there is a mismatch between the costs used to calculate the NPV and the ones used to construct the ratio.
- The partial-NPVs itself are disclosed too positive due to the fact that only related capital costs instead of the total project capital costs are part of their calculation. Thus, by default the ratio might be too positive.

3.2.4 Calculation of the proper SROI Ratio

All of the above described ratios have a common weakness: Due to the fact that the disclosed partial-NPVs only consider corresponding projects costs, each ratio involving partial-NPVs misses the remaining project costs. Therefore, a basis of comparison between ratios based on the total project costs and the partial-NPVs is not warranted. On the other hand the ratios based on pro-rata costs do not consider all project components costs, nor does the NPV calculation itself. The reasons why often partial-NPVs and no overall-NPVs are disclosed are unknown and not mentioned within the ICR documents, but a timely known and broadly discussed issue. However, the following considerations might help explain this circumstance:¹⁰⁸

- Whether or not all of the project components respectively project costs and benefits are quantifiable is depending on the type of project.
- Besides the dependence on the project type, following reasons might as well favor the fact that some project costs and benefits have not been quantified:
 - At the time of project evaluation, the ICR team did not have sufficient data available to conduct an economic analysis for the remaining project components.
 - The ICR team decided to only focus on the most valuable project components within the analysis.
 - Some of the unevaluated project components are too difficult to evaluate with the NPV method.
 - For certain project costs and benefits an economic analysis is simply too circumstantial and not yielding for results.
 - Some of the unevaluated project components are too small and thus an economic evaluation was not cost-effective enough.
 - The project components that are not considered balance themselves out, resulting in no benefit or cost overrun.

¹⁰⁸ Cf. Baneth, Jean (1996), p. 28 et sqq.; Squire/van der Tak (1975); p. 21 et sqq.; WB (2015): Transport Notes, p.8.

- As a matter of fact it needs to be noted that an assessment of a variety of unevaluated, remaining project components showed that many of those components were part of an economic analysis some projects only.

The focus and goal of this research paper is to calculate a minimum SROI ratio which considers all capital and recurring costs in order to give a clear picture of the cost-benefit relationship of the development assistance provided by the WB. Therefore, the remaining project costs of the partial-NPV projects need to be considered as part of the NPV calculation itself. However, the following challenges need to be considered when trying to manipulate the partial-NPVs disclosed by the ICR department of the WB:¹⁰⁹

- Are there benefits related to the remaining project costs from unevaluated project components?
- Besides the remaining capital costs of the project, are there also additional recurring costs for those components which need to be considered?

In order to disclose a minimum SROI for the LDCs, no benefits for the remaining components of all partial-NPV projects will be assumed. This ensures that the calculated SROI ratio is not valued too high due to inaccurate assumptions on potential benefits. The calculation for a minimum SROI will follow a two-step approach, in which adjustment need to happen on a project level first before the SROI values can be aggregated onto a portfolio level (e.g. Asia, Africa, all LDCs; refer to the following chapter):

1. For the first SROI ratio –referred as capital SROI ratio– the partial-NPV value will be adjusted by applying the discounted remaining capital costs to its calculation. Herewith, all known costs have then been considered.
2. In order to also account for potential recurring costs of the remaining capital costs of the partial-NPVs a standard approach for calculating them will be

¹⁰⁹ Cf. Baneth, Jean (1996), p. 28 et sqq.; Pearce/Giles/Susana (2006), p. 86 et seq.; Squire/van der Tak (1975); p. 21 et sqq.

applied. Those costs will then be applied to the NPV calculation in the first step and be taken into account when calculating the final, minimum SROI ratio.

3.2.5 Portfolio Analysis

Portfolio analysis is commonly defined as an important tool of strategic management. The idea behind it is to determine specific strategies for individual business areas and divisions. To that end, the products or services of the company are categorized.¹¹⁰ However, for the economic and financial analysis of this research paper a different definition for the portfolio analysis needs to be applied: a detailed assessment and evaluation of the ICR reports of the LDCs, specifically looking at the project cost-benefits in order to aggregate the outcomes to different portfolio levels, e.g. continental and LDCs. Therefore, each single ICR report is analyzed for meaningful economic and financial key figures using specific key words as well as key abbreviations (refer to *chapter 2.8 Challenges of this Doctoral Thesis*), before the descriptive analysis of the LDCs as a whole and the comparison analysis for each continent (Africa and Asia)¹¹¹ can be conducted.

3.2.6 Sensitivity Analysis

The Sensitivity Analysis (SA) is a tool which aims to determine how model uncertainty depending upon the informational input, its structure and the framing assumptions applied to build the model. The information gained through the SA can be invaluable for obtaining the acceptance of different types of stakeholders. In other words the SA can be seen as “study of how the variation in the output of a model

¹¹⁰ Cf. CONTROLLING-Portal.de (2014); Gabler Wirtschaftslexikon (2014): Portfolio-Analyse.

¹¹¹ Author's Note: None of the Haitian ICR reports discloses a NPV or any other financial values. Therefore -as the only LDC belonging to LA – it can be disregarded for the financial comparison analysis.

(numerical or otherwise) can be apportioned, qualitatively or quantitatively, to different sources of variation”¹¹², as defined by the European Commission Joint Research Centre in 2011.¹¹³ In this research paper different kinds of SA are used. As described in *chapter 4.8.2 Standardized NPV-Cost-Ratios* standard values will be applied for missing discount rates, NPV-horizons and recurring costs. Due to potential continental differences, standard values will be calculated based on average values of the LDCs as a whole and on continental averages only. The second kind of SA is the minimum SROI ratio calculation itself by taking the remaining capital costs of partial-NPV projects into account and by considering potential recurring costs of the remaining project components.

3.3 Additional Economic and Financial Indicators

This chapter introduces additional economic and financial figures, methods and tools. All of them are disclosed within the ICR reports. The purpose is to use them for an overview of their portfolio values in order to compare projects with NPV with those without. Considering project performance ratings of such grouped projects will then allow better understanding and evaluation of outcomes of all 790 ICR reports.

3.3.1 Economic Rate of Return

ERR –also often called Economic Internal Rate of Return (EIRR) or simply Internal Rate of Return (IRR)– is defined as the interest rate at which the NPVs of cash flows (both positive and negative) from all periods in a project or investment add up to 0. The concept behind ERR is similar to that of NPV: To evaluate the attractiveness of a project or investment. If the ERR of a planned project exceeds the agreed RR, then the project is deemed desirable. When the ERR is lower, the investment is not considered profitable from a monetary standpoint of view. ERR is calculated as:¹¹⁴

¹¹² European Commission Joint Research Centre (2011): Sensitivity Analysis.

¹¹³ Cf. European Commission Joint Research Centre (2011): Sensitivity Analysis; Saltelli et al. (2008), p. 1.

¹¹⁴ Cf. InvestingAnswers (2014); The Comoros: KO_Completion_ICR2054, p. 35 et seq.

$$Initial\ Investment_{t=0} = \sum_{t=1}^n \frac{(CashIn_t - CashOut_t)}{(1 + ERR)^t}$$

where

ERR = Economic Rate of Return

CashIn/CashOut = cash flow during each period

n = analytic horizon

t = time period

Equation 6: Initial Investment¹¹⁵

A general rule of thumb is that ERR cannot be derived analytically. Instead, ERR value must be found by “trial-and-error”¹¹⁶ to derive the appropriate rate.¹¹⁷ Furthermore, the disclosed ERRs can be categorized into overall- and partial-ERRs as well as standalone- and scenario-ERRs, similar to rules used in the NPV concept (refer to *chapter 3.2.2.1 Net Present Value*). The analytic horizon of the ERR formula is referred as “ERR-horizon” within this thesis.

3.3.2 Benefit-Cost-Ratio

BCR is the indicator formally used in the context of the CBA. It is expressed in the form of a quotient and summarizes the monetary value of a proposal, project or an intervention in general. Therefore, BCR is the ratio of benefits gained from intervention relative to its costs, both expressed in monetary terms and in discounted present values. In other words BCR compares the financial gains realized by performing the intervention against costs from intervention execution. The higher the BCR the more profitable the intervention. The general rule of thumb is that if the benefits of an intervention are higher than its costs, the investment is rentable.¹¹⁸ Obviously, in order to calculate BCR a discount rate as well as an analytic horizon (referred as

¹¹⁵ Cf. Own illustration, following The Comoros: KO_Completion_ICR2054, p. 35 et seq.

¹¹⁶ Author’s Note: The “trial-and-error” approach is a fundamental method of solving problems characterized by repeated, varied attempts which are generally continued until success has been reached.

¹¹⁷ Cf. InvestingAnswers (2014).

¹¹⁸ Cf. Boardman/Greenberg/Vining/Weimer (2010), unpag.; Ferrara (2010), unpag.; Gabler Wirtschaftslexikon (2014): Kosten-Nutzen-Analyse.

“BCR-horizon”) is needed. BCR can be categorized as an overall- or a partial-BCR, using similar rules applied to NPV concept (refer to *chapter 3.2.2.1 Net Present Value*).

3.3.3 Net Benefit

The Net (economic) Benefit –also referred as the total Net Benefit of the project– of an investment or a project can be defined as the total benefits minus the total costs of the project. The steps to calculate the Net Benefit are rather simple and easy to follow:¹¹⁹

1. Determine all direct and indirect costs associated with the project.
2. Sum up the direct and indirect costs to arrive the total costs of the project.
3. Subtract the total costs from the total revenue/turnover to arrive at the gross benefit of the project.¹²⁰

The Net Benefit of a project can also be obtained through the accumulation of the NPV via its net cash flows across the single time periods.¹²¹ In order to calculate the Net Benefit a discount rate and an analytic horizon (referred as “Net Benefit-horizon”) are used. The Net Benefit can be distinguished between an overall- and a partial-Net Benefit, whereat the same differentiation rules as for the NPV concept do apply (refer to *chapter 3.2.2.1 Net Present Value*).

3.3.4 Financial Net Present Value

The Financial Net Present Value (FNPV) can be defined as the NPV of a specific object of consideration. In comparison to the E-NPV it does not consider the whole economy; instead it focuses on a specific enterprise such as the provider of a power

¹¹⁹ Cf. Campbell/Brown (2003), unpag.; Friedag/Schmidt (2004), unpag.

¹²⁰ Author's Note: As an additional last step the deduction of taxes, depreciation and legal reserves from the gross benefit to obtain the Net Benefit of the investment is often mentioned. However, this step refers to the production and sale of a product including car pool and fixed assets only.

¹²¹ Cf. Campbell/Brown (2003), unpag.; Friedag/Schmidt (2004), unpag.

plant or the road implementing company, the WB or the implementation agency of the project.¹²² The investment is rentable for the object of consideration when FNPV is greater than 0 and not recommended in cases where FNPV less than 0.

3.3.5 Financial Rate of Return

The Financial Rate of Return (FRR) is comparable to the ERR (refer to *chapter 3.3.1 Economic Rate of Return*). However, instead of considering the whole economy, it is the RR of the object of consideration in the FNPV. In other words, there is no difference in FRR calculations from that of ERR other than it scrutinizes only a specific object of consideration as mentioned in the previous chapter (refer to *chapter 3.3.4 Financial Net Present Value*).

¹²² Cf. Laos – Report 32004, p. 9 sqq.; Tanzania – Report 32225, p. 15 sqq.

4 Results of the Economic and Financial Analysis

The following chapters further evaluate the results of the financial analysis of this doctoral thesis. Firstly, the setup of the financial analysis will be explained. Subsequently, the concept and approach used for data aggregation on a country, continental level and the LDCs as a whole will be explained in the first chapter. The next chapter then analyzes the outcomes of the appraisal values as well as the outcome comparison of the African and Asian continents. Afterwards an overview on ICRs providing result indicators for the LDCs and the 2 continents will be given. In order to have a comparison between the appraisal and final results, the chapter 4.4 analyzes the appraisal vs. result outcomes. The following chapter then deals with the results of the ERR. Additional economic and financial figures such as the BCR, Net Benefit, FNPV and FRR for both –NPV as well as for non-NPV projects– will be disclosed in chapter 4.6. Using the overall project performance ratings an approach for calculating potential ERR outcomes of non-financial indicator (disclosure) projects will be presented before the calculation of the NPVCRs and the minimum SROI ratio is outlined in the penultimate chapter. The final chapter then provides a summary of the outcomes and a conclusion.

4.1 Analysis Approach and Setup

Based on the initial data load conducted in September of 2012, 790 ICR documents serve as the data source for the whole economic and financial analysis. Data available within these documents are regarded as raw data. In order to be able to access and analyze the economic and financial indicators as provided in ICR documents, each ICR had to be examined by browsing through relevant sections to locate specific financial key words and the abbreviations of them. This ensures that no relevant details get missed and that the different naming conventions and expressions used in the ICR documents are covered.¹²³ The ICR documents vary in size from 9 to 154

¹²³ Author's Note: Common key words and there abbreviations are: Net Present Value (NPV), Net Present, Present Value (PV), Economic Rate of Return (ERR), Economic Internal Rate of Return (EIRR), Internal Rate of Return (IRR), Rate of Return (RR), Financial Net Present Value (FNPV),

pages. The average document size is 53 pages. With a total of 790 ICR documents, it amounts to more than 42,000 pages. The following table provides an overview of the ICR document counts per continent:

	Total No of "Implementation Completion & Results Reports"
TOTAL	790
Africa	574
Asia	206
Latin America (Haiti)	10

Table 4: Total Number of ICR Reports per Continent

As shown in the table above, most of the ICR reports are focused on Africa (574 reports) followed by Asia as a remote second (206 reports). None of the Haitian ICR reports discloses a NPV or any other economic or financial values. As it is the only LDC in LA, LA is therefore disregarded and not part of the analysis.

In order to evaluate the available data in each ICR report and aggregate outcomes to different portfolio level, the following steps were taken:

1. Download relevant ICR documents via the country's section of the WB homepage: <http://documents.worldbank.org/curated/en/country>.
2. Save each ICR document using the following naming convention: "ICR_Completion_ReportNumber".
3. Access and capture the data available within each ICR document and enter it in the corresponding country spreadsheet:
 - Rows: ICR reports
 - Columns:
 - Project type(s)
 - Planned and actual project dates

Financial Rate of Return (FRR), Rate(s) of Return, Present Value Cost (PVC), Present Value Benefit (PVB), Net Benefit, Total Benefit.

- Planned and actual funding amount (including the share of the WB)
 - Project appraisal and result indicators on NPV, ERR, FNPV, FRR, CBR, Net Benefit (including the corresponding reference information, such as costs, time horizon and discount rate)
4. Aggregate available data on a continent level and then LDCs as a whole (setup of summary spreadsheets).
 5. Define and establish a summary portfolio analysis using VBA.¹²⁴
 6. Apply the VBA analysis, collate results, interpret on a LDC level and perform a continental comparison of Africa and Asia via appropriate indicators.

Additionally, each of the country spreadsheets has been further enhanced in order to examine the different NPVCRs and SROI ratios and to provide information required for qualitative data analysis in the second part of this thesis.

4.2 NPV Outcomes at the Appraisal Stage

Figures and calculations examined and visualized in this chapter are based on appraisal estimates of the ICR reports, which are in turn determined by evaluations and analysis done in the appraisal stage of a project. Generally, the appraisal estimates are verified, confirmed or adjusted during the respective closing stage of the project.¹²⁵ In the following chapters the relevant appraisal values will be outlined, before the next chapter goes into details of the result value outcomes.¹²⁶

¹²⁴ Author's Note: The VBA program analyzes and summarizes the data available on a country, continent and LDCs level and provide the portfolio indicators including the corresponding reference information. Please reach out to the author (schaefer.dominik@yahoo.de) for further information.

¹²⁵ Author's Note: The result values can still be regarded as "estimates" since they are based on future predictions on cash flows, e.g. for the recurring costs.

¹²⁶ Author's Note: Other than the result the appraisal values are less covered and within the ICR documents, since there are separate appraisal documents for each project. However, appraisal documents are not subject to this research paper.

4.2.1 Appraisal NPVs of the LDCs

In this chapter the appraisal NPVs and their corresponding values for the LDCs as a whole are presented and interpreted. In order to set the basis for understanding the appraisal analysis, the following table below outlines the counts and types of appraisal NPVs included in the ICR reports:

		LDCs NPV Counts	
Total NPVs	Standalone-NPVs	91	86
	Scenario-NPVs		5
Overall-NPVs	Standalone-NPVs	38	35
	Scenario-NPVs		3
Partial-NPVs	Standalone-NPVs	53	51
	Scenario-NPVs		2

Table 5: Appraisal NPV Counts of the LDCs

Out of the 790 ICR reports, 11.5 percent disclose NPVs based on appraisal calculations (91 NPVs in total). All of the NPVs are categorized as positive. As per the definitions presented in *chapter 3.2.2.1 Net Present Value*, out of the 38 overall-NPVs, only 2 were manually calculated and are standalone-NPVs. 36 of the overall-NPVs are overall-disclosed NPVs, including 3 scenario-overall-disclosed NPVs. The partial-NPVs are standalone-sample-NPVs. Thus, 49 of the partial-NPVs are standalone-component-NPVs and only 2 are scenario-component-NPVs.

The following table shows the outcome of the economic and financial analysis of the appraisal values disclosed within all ICR documents. It provides cumulative values based on the total sum [Cum Sum] as well as the arithmetic mean [$\bar{x}$] of the appraisal NPVs and their corresponding cost values:

Values in Mio USD		LDCs					
		Total NPVs		Overall-NPVs		Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	17,667	194	3,539	93	14,128	267
	Maximum	20,067	221	5,921	156	14,145	267
Total NPV Capital Costs (planned)	Minimum	11,539	127	7,092	187	4,447	84
	Maximum	11,547	127	7,092	187	4,455	84
Total Project Costs (planned)		13,981	154	7,092	187	6,889	130
Total World Bank Costs (planned)		5,237	58	1,537	40	3,700	70
Discount Rate [%]		11.19		11.43		11.05	
Planned Project Duration [years]		5.33		5.42		5.26	
NPV-horizon [years]		19.68		17.01		21.06	

Table 6: Appraisal NPV and Cost Values of the LDCs¹²⁷

The NPVs and their corresponding cost values in the above table are expressed in Mio USD.¹²⁸ The “minimum” case is based on the minimum NPVs, resulting from the existence of scenario-NPVs. It always considers the standalone-NPVs and the minimum NPVs of the scenario-NPVs, whereas the “maximum” case is based on the maximum NPVs considering the standalone-NPVs and the maximum NPVs of the scenario-NPVs. Accordingly, the minimum respectively maximum case of the total NPV capital costs refer to the minimum respectively maximum NPV related capital costs case.¹²⁹

The cumulative sum of the minimum NPV appraisal values of the 91 ICR reports is approx. 17,667 Mio USD and could increase to more than 20,067 Mio USD depending on each single project’s outcome variation. Those values are based on a discount rate of 11.19 percent, planned project duration of 5.33 years and on a NPV-horizon of approx. 19.68 years on average. Interesting to see is that the sum of the

¹²⁷ Author’s Note: For each of the following tables the total project funding provided by the WB will be referred as “Total World Bank Costs”.

¹²⁸ Author’s Note: All values are rounded to 1 Mio USD. The currency of the ICR documents is usually USD. In case the fund provided is based on a different currency, the document specific conversion factor for USD was used, whereas the conversion factor can be different for the appraisal and the result values.

¹²⁹ Author’s Note: Another point to mention is that the cumulative sums and arithmetic means are the aggregations and quotients of the whole of the appraisal NPVs disclosed since the MDGs have been officially passed on the 8th of September 2000. All of the disclosed values, financial figures and currency conversion rates have been taken as outlined within the ICR reports, regardless of varying disclosure dates amongst the reports.

disclosed total NPV values is at least 53 percent higher than that of the related capital costs in the minimum case¹³⁰ and more than 26 percent higher (minimum case) than that of the total cumulative project costs. Roughly 38 percent of total funding for those projects has been provided by the WB¹³¹. The rest is contributed by other donors, primarily the respective country's government but also bilateral agencies or embassy and ministries or other donor organizations. Another interesting point to mention is that the cumulative minimum NPV (14,128 Mio USD) based on related capital costs of the 53 partial-NPVs (roughly 4,447 Mio USD) is much higher than the cumulative maximum NPV (5,921 Mio USD) based on the capital costs of 38 overall-NPVs (7,092 Mio USD). Even from an average arithmetic mean perspective, partial-NPV projects seem to be more rentable with higher average NPV values (267 Mio USD per NPV project in average) but lower related capital costs (84 Mio USD compared to 187 Mio USD related capital costs of an overall-NPV project). However, it should be noted that there are 3 out of the 91 appraisal NPV projects with disclosed NPV values in the 1- to low 2-digit billion range.¹³² In comparison, the remainder of the 88 projects shows on average NPV values within the 2- to 3-digit Mio range.¹³³ The "Third Road Rehabilitation and Maintenance Project (RRMP III)" outlines the highest of all project NPV values of 11,179 Mio USD at the appraisal stage.¹³⁴ Excluding those "high-value" NPV projects from the appraisal NPV project

¹³⁰ Author's Note: For the appraisal NPVs no recurring costs have been disclosed in the ICR documents. In nearly all cases only the NPV values without the corresponding calculations have been disclosed. Still, the 91 appraisal NPV projects have been researched whether the NPV calculations are based on recurring costs or not. For 70 projects this is the case, whereas 6 NPV calculations are not based on recurring costs and for 12 projects it could not be identified if their NPV calculations are based on recurring costs or not.

¹³¹ Author's Note: Nearly all funds respectively credits of the WB are provided by the IDA; only few are provided by the IBRD.

¹³² Cf. *Appendix C* provides an overview of the appraisal projects which disclose NPVs of more than 1,000 Mio USD.

¹³³ Cf. Author's Note: The NPV values of the remainder of the 88 projects do have a range from 30 USD till 416 Mio USD with an average arithmetic mean of 53,163,698 USD in the minimum and 53,552,011 USD in the maximum case. Leaving out the 2 sample-NPVs the NPV range ranges from 12,040 USD till 416 Mio USD, with an average arithmetic mean of 54,400,062 USD in the minimum and 54,797,405 USD in the maximum case.

¹³⁴ Author's Note: Since bolter values usually distort the picture of a data set it is in the sense of making meaning to consider a cleaned-up data set. Therefore, specifically high bolter values embellishing the data set will be excluded for each of the analyses whereas the exclusion will always be mentioned within the running text.

data set would decrease the total NPV drastically to 4,678 Mio USD with an arithmetic mean of 53 Mio USD in the minimum case and to 4,713 Mio USD with an arithmetic mean of 54 Mio USD. The total cost values would decrease to 5,748 Mio USD in the minimum case and to 5,756 Mio USD in the maximum case, both with an arithmetic mean of about 65 Mio.¹³⁵ The following table provides an overview of the “high-value” NPVs adjusted for appraisal values:

Values in Mio USD		LDCs					
		Total NPVs		Overall-NPVs		Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	4,678	53	1,729	48	2,949	57
	Maximum	4,713	54	1,746	49	2,966	57
Total NPV Capital Costs (planned)	Minimum	5,748	65	1,660	46	4,088	79
	Maximum	5,756	65	1,660	46	4,096	79
Total Project Costs (planned)		8,021	91	1,660	46	6,360	122
Total World Bank Costs (planned)		4,619	52	1,192	33	3,427	66
Discount Rate [%]		11.21		11.58		11.03	
Planned Project Duration [years]		5.31		5.39		5.25	
NPV-horizon [years]		19.72		16.73		21.1	

Table 7: Appraisal NPV and Cost Values of the LDCs excluding “high-value” NPVs

4.2.2 Appraisal NPV Continent Comparison

This chapter deals with the comparison of the appraisal values between Africa and Asia. In order to provide the user with further understanding of the cumulative and arithmetic mean NPV and related cost values, table 8 shows the distribution of the NPV counts of the African and the Asian continent:

¹³⁵ Author’s Note: the NPV and cost values outlined for any of the “high-value” NPV projects have been double-checked by as well considering other documents than the ICR reports, such as the project appraisal documents as well as other project-related financial disclosures.

		Africa NPV Counts		Asia NPV Counts	
Total NPVs	Standalone-NPVs	62	60	29	26
	Scenario-NPVs		2		3
Overall-NPVs	Standalone-NPVs	26	25	12	10
	Scenario-NPVs		1		2
Partial-NPVs	Standalone-NPVs	36	35	17	16
	Scenario-NPVs		1		1

Table 8: Appraisal NPV Counts of Africa and Asia

Even though Africa has close to 3 times more ICR documents than Asia (574 compared to 206 ICR documents), its ICR reports disclose only twice as many appraisal NPVs as that of Asia (62 NPVs in Africa vs. 29 NPVs in Asia). Therefore, the disclosure rate of the appraisal NPVs is 10.8 percent for Africa and 14.1 percent for Asia. The following table compares the outcomes of the NPVs and the corresponding cost values for all NPV projects in Africa and Asia based on cumulative sum [Cum Sum] and arithmetic mean [Ø]:

Values in Mio USD		Africa Total NPVs		Asia Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	5,009	81	12,658	436
	Maximum	5,027	81	15,040	519
Total NPV Capital Costs (planned)	Minimum	5,809	94	5,730	198
	Maximum	5,817	94	5,730	198
Total Project Costs (planned)		7,583	122	6,399	220
Total World Bank Costs (planned)		3,516	57	1,721	59
Discount Rate [%]		11.44		10.61	
Planned Project Duration [years]		5.31		5.38	
NPV-horizon [years]		19.69		19.64	

Table 9: Appraisal NPV and Cost Values Comparison of Africa and Asia¹³⁶

¹³⁶ Author's Note: The NPV and its corresponding cost values are expressed in Mio USD, whereas all values are as well rounded to 1 Mio USD. Furthermore, *appendices D* and *E* provide an overview of the outcomes of the NPVs and corresponding cost values for the African and Asian overall- and partial-NPVs.

Even though Asia discloses less NPVs the cumulative sum of the NPVs is much higher than in Africa (12,658 compared to 5,009 Mio USD), whereat the corresponding capital costs (roughly 5,810 compared to 5,730 Mio USD), average discount rates, project duration as well as NPV-horizon periods are similar. The main difference is due to the “high-values” of the 17 Asian partial-NPVs which account for 11,577 Mio USD of the cumulative sum of 12,658 of the minimum NPV and for 11,594 Mio USD of the cumulative sum of 15,040 Mio USD of the maximum NPV of Asia. To identify other potential reasons to help explain the differences between the 2 continents, corresponding project types of the African and Asian continents are compared. However, due to a significant portion of appraisal NPV projects use old format style (16 out of the 29 in Asia and 32 out of 62 in Africa), there is no meaningful sector code analysis in order to compare project types possible. This is because the old format style documents only disclose actual sector codes and their corresponding percentages, but without planned versions. Furthermore, the total WB financing (“Total World Bank Costs (planned)”) only makes up about 46 percent of the total funding (“Total Project Costs (planned)”) in Africa and even less, namely about 27 percent, in Asia.

Excluding the 3 “high-value” NPV projects, the comparison between the continents shows a completely different picture as demonstrated in the following table:

Values in Mio USD		Africa Total NPVs		Asia Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	3,809	62	869	32
	Maximum	3,827	63	886	33
Total NPV Capital Costs (planned)	Minimum	4,677	77	1,071	40
	Maximum	4,685	77	1,071	40
Total Project Costs (planned)		6,451	106	1,570	58
Total World Bank Costs (planned)		3,471	57	1,148	43
Discount Rate [%]		11.48		10.56	
Planned Project Duration [years]		5.3		5.33	
NPV-horizon [years]		19.3		20.63	

Table 10: Appraisal Comparison of Africa and Asia excluding “high-value” NPVs

Noticeably, the NPV and cost values of Asia drop drastically once the 2 Bangladeshi “high-value” NPV projects are excluded. In this case the African continent is disclosing higher cumulative sums for the minimum and maximum NPV as well as capital cost values. Furthermore, the NPV/capital cost ratio of Asia (minimum 0.81 and maximum 0.83) is still slightly higher than that of Africa (minimum 0.81 and maximum 0.82).

4.3 NPV Outcomes of the Result Stage

This chapter examines and describes the figures and calculations for the result values of the ICR reports. The result values outlined in this chapter are based on the disclosed result values within the ICR documents. Additional calculations on the missing cost values and NPVCRs are described in *chapter 4.8 NPV-Cost-Ratios and SROI Calculation* and not part of this chapter.

4.3.1 Result NPVs of the LDCs

In this chapter the result NPVs and their corresponding values for the LDCs as a whole are presented. In order to provide readers with a better understanding of the subsequent analysis, the following table outlines the counts and types of result NPVs outlined within the 790 ICR reports:

		LDCs							
		NPV Counts		Positive NPV Counts		Neutral NPV Counts		Negative NPV Counts	
Total NPVs	Standalone-NPVs	168	120	154	113	7	0	7	7
	Scenario-NPVs		48		41		7		0
Overall-NPVs	Standalone-NPVs	60	42	52	38	4	0	4	4
	Scenario-NPVs		18		14		4		0
Partial-NPVs	Standalone-NPVs	108	78	102	75	3	0	3	3
	Scenario-NPVs		30		27		3		0

Table 11: Result NPV Counts of the LDCs

In total, 168 NPVs based on results calculations are disclosed which represents a disclosure rate of about 21.3 percent. Out of the 168 NPVs, 154 are positive,

whereat approx. 8 percent of the disclosing reports show a neutral or negative NPV. In total there are 60 overall- and 108 partial-NPVs. The overall-NPVs consist of 42 standalone-NPVs whereat 54 out of the 60 are overall-disclosed NPVs. The partial-NPVs consist of 78 standalone-NPVs whereat 98 out of the 108 are component-NPVs. The positive overall- and partial-NPVs display similar distributions. Furthermore, it can be noted that the neutral and negative NPVs only contain overall-disclosed and component-NPVs for the overall- respectively partial-NPVs.

The following table shows outcomes of the economic analysis of the result NPV and cost values based on the cumulative sum and the arithmetic mean:

Values in Mio USD		LDCs					
		Total NPVs (168)		Overall-NPVs (60)		Partial-NPVs (108)	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	42,059	250	6,779	113	35,280	327
	Maximum	50,779	302	14,931	249	35,847	332
Total NPV Capital Costs (actual)	Minimum	23,408	139	15,648	261	7,760	72
	Maximum	23,578	140	15,641	261	7,937	73
Total Project Costs (actual)		33,506	199	15,810	263	17,696	164
Total World Bank Costs (actual)		9,110	54	2,582	43	6,528	60
Discount Rate [%]		11.21		11.39		11.11	
Actual Project Duration [years]		7.02		7.1		6.98	
NPV-horizon [years]		19.33		18.15		19.95	

Table 12: Result NPV and Cost Values of the LDCs

The cumulative sum of the minimum NPV result values from the 168 ICR reports is approx. 42,059 Mio USD at minimum and could increase to more than 50,000 Mio USD based on the maximum values disclosed. The average NPV value per project ranges between 250 and 302 Mio USD. Therefore, it can be concluded that the NPV disclosing projects are generally profitable. These values are based on an average 11.2 percent discount rate, 7 years project duration, and an average 19 years NPV-horizon. The cumulative sum of the NPVs is approx. 80 percent higher than the related capital costs in the minimum case and 115 percent higher in the maximum case. Consequently, it can be concluded that the result NPV disclosing projects generate values twice as much as their corresponding NPV capital costs.¹³⁷ There

¹³⁷ Author's Note: Out of the 168 result NPVs 129 are based on recurring costs, whereas for 75 out of the 129 projects the value of the recurring costs have not been disclosed. 17 NPVs are not based

are no noteworthy differences found between the minimum and maximum cumulative sum of the NPV related capital costs. The difference of approx. 8,000 Mio USD between the minimum and maximum cumulative sum of the total NPVs is due to the difference between the minimum and the maximum cumulative sum of the overall-NPVs¹³⁸ (approx. 6,779 Mio USD vs. 14,391 Mio USD). In the case of minimum cumulative sum of the overall-NPVs, it is less than half of the cumulative sum of the related capital costs and also lower than the related capital costs in the case of maximum sum. Another point to mention is that the cumulative maximum amount of the capital costs of the overall-NPVs is lower than the cumulative minimum amount. Here the reason lies in a project neutral NPV calculation, since for the negative minimum overall-NPV calculation all project costs but for the positive maximum overall-NPV only specific project costs have been considered.¹³⁹ Taking a closer look at the partial-NPVs it can be noticed that those are the ones making up more than 80 percent of the total cumulative minimum NPV sum and even more than 70 percent of the total cumulative maximum NPV sum. The related capital costs only account for a share of approx. 22 percent of the cumulative sum of the NPV. The partial-NPV capital costs are in general only half as high as the overall-NPV capital costs, whereat the total project costs of the partial-NPV projects actually exceed the ones of the overall-NPV projects by roughly 2,000 Mio USD. The fact that the partial-NPV related total project costs are actually 2.3 times higher than the related capital costs is not surprising, since it lies in the nature of a partial-NPV and hence only NPV related components and costs are considered. Another point to mention is that the WB financing only makes up about 27 percent of the total project costs. Therefore, it can be concluded that also for the result values the disclosed sector codes of the

on recurring costs and for 22 projects the information on whether recurring costs do exist or not could not be gathered. Please refer to *chapter 4.8.2 Standardized NPV-Cost-Ratios* to understand the principle of the calculation of missing recurring costs.

¹³⁸ Author's Note: Taking a closer look at the costs of the overall-NPV projects there is a slight difference of the overall-NPV related capital costs which are in this case a bit lower than the total project costs. This discrepancy is due to the fact that there are projects where funding is used for operational and maintenance works or mismatches between the outlined total project costs in the Appendices of the ICR documents (refer to *table 2: ICR Report Chapter Structure: Annex 1 - Project Costs and Financing*) and the costs disclosed in the detailed NPV calculation tables (if available).

¹³⁹ Cf. Yemen: JE_Completion_22117 p. 5 et sqq.

WB financing of a project are not representative for the project type as a whole. However, the sector types defining the usage of WB spending can at least indicate a possible direction for what the total project financing has been used. In total the 168 NPV projects are defined by 61 sector codes, whereat each project accounts for 1 percentage point (“%-Points”). The following table provides an overview of the 12 major sector codes of the WB financing:

Sector Code	Project Counts	%-Points	%-Share	World Bank Costs [Mio USD]
Roads and highways	55	34.47	20.52	2,849
Water supply	28	16.83	10.01	849
Central government administration	94	16.21	9.65	703
Power	18	13.31	7.92	876
General agriculture, fishing and forestry sector	14	6.28	3.74	178
Irrigation and drainage	14	5.95	3.54	167
Other social services	31	5.62	3.35	239
Sub-national government administration	36	5.57	3.32	320
Agricultural extension and research	22	4.94	2.94	157
Health	18	4.94	2.94	381
General water, sanitation and flood protection sector	13	3.16	1.88	182
General public administration sector	13	3.14	1.87	157
Total		120.41	71.67	7,058

Table 13: Major Sector Codes of the NPV Projects of the LDCs¹⁴⁰

The “%-Share” column shows the total percentage share of a sector code in relation to all 61 sector codes. In total the 12 major sector codes account for roughly 120 %-points (out of 168 %-points in total) which is a %-share of about 72 percent. The total WB costs of the 12 major sector codes account for about 77 percent of the total WB costs of all NPV projects (7,058 Mio USD out of 9,110 Mio USD; refer to *table 12: Result NPV and Cost Values of the LDCs*). The “Project Counts” column shows how many projects disclose the sector code, whereat different sector codes might belong to the same projects. The ‘Roads and highways’ sector code has the highest %-points (34.47) and owns the largest financing share of more than 20 percent of the WB (2,849 Mio USD) projects. The ‘Water Supply’ sector code is the second largest sector code with a %-share of roughly 10 percent followed by the ‘Central government administration’ (%-share of 9.65 percent) and the ‘Power’ (%-share of 7.92 percent) sector codes. Interestingly, the ‘Central government administration’ sector code ranks the fourth highest in terms of WB costs of 703 Mio USD but still belongs to 94 projects. As the research has shown, al-though this sector code is the

¹⁴⁰ Cf. *Appendix F* provides an overview of the full list of sector codes and the corresponding %-points as well as the WB costs.

core sector code for a number of projects it is as well often just running “aside” with other major sector codes. This makes perfect sense, since central governmental activities and administration define the core elements of delivering development aid to foreign countries. For none of the NPV projects which are based on 2 or more sector codes the project types are identical, since even for matching sector codes the corresponding %-points per sector code are different.¹⁴¹ In order to better understand for which type of projects the WB financing is used, following project type categories can be identified:

Project Type Categories	LDCs		
	%-Share	% of total WB Costs	WB Costs [Mio USD]
Transportation Development	27.43	37.17	3,386
Agriculture	15.09	8.89	810
Water & Sanitation	13.57	11.45	1,043
Governmental Administration	13.36	11.94	1,088
Power	10.53	13.22	1,204
Other	5.73	4.93	449
Sectorial Development & Reforming	3.75	2.4	219
Health	3.47	4.52	412
Education	2.73	2.32	211
Natural Resources	2.38	1.46	133
Financial Development	1.46	1.61	147
IT & Communication	0.5	0.09	8
Total	100	100	9,110

Table 14: Project Type Categories of the NPV Projects of the LDCs¹⁴²

‘Transportation Development’ (e.g. setup and establishment of rural and urban roads, railways, ports, aviation) is the largest category with a %-share of more than 27 percent and a 37 percent share of the total WB costs of 3,386 Mio USD out of 9,110 Mio USD total WB costs for the 168 NPV projects. The ‘Agriculture’ category shows the second highest %-share of 15 percent, whereas the WB costs of 810 Mio USD only account for the fifth highest share of the total WB costs. The ‘Power’,

¹⁴¹ Cf. *Appendix G* provides an overview of the projects and the shared sector codes amongst these.

¹⁴² Cf. *Appendix H* provides an overview of the project type categories and the assigned sector codes.

‘Governmental Administration’ and ‘Water and Sanitation’ project type categories with 1,204 Mio USD, 1,088 Mio USD and 1,043 Mio USD WB costs respectively, hold the second, third and fourth largest share of the total WB financing.

Looking at the 154 positive NPV projects, the picture and facts as described for the whole of the LDCs stays the same as following table outlines:

Values in Mio USD		LDCs					
		Total Positive NPVs		Positive Overall-NPVs		Positive Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	42,109	273	6,793	131	35,315	346
	Maximum	50,758	330	14,903	287	35,855	352
Total NPV Capital Costs (actual)	Minimum	22,910	149	15,501	298	7,410	73
	Maximum	23,090	150	15,504	298	7,586	74
Total Project Costs (actual)		32,630	212	15,668	301	16,961	166
Total World Bank Costs (actual)		8,616	56	2,457	47	6,159	60
Discount Rate [%]		11.21		11.35		11.14	
Actual Project Duration [years]		7.01		7.1		6.97	
NPV-horizon [years]		19.27		18.28		19.75	

Table 15: Positive Result NPV and Cost Values of the LDCs

The main difference lies in higher cumulative sums for any of the NPV figures, purely because of the fact that any neutral or negative NPV values have been removed in the positive NPV presentation above. For the same reason why NPV values have been increasing, the decrease in cost values is simply due to neutral and negative projects having been removed from the data set.

The only noteworthy fact to mention when looking at the 7 neutral NPVs –containing 3 partial- and 4 overall-NPVs and by definition all scenario-NPVs– is that the maximum cumulative sum of the NPVs reaches the double absolute amount as a positive figure than the minimum cumulative sum of the NPVs is negative. The following table provides an overview of the total neutral and negative NPV and cost values of the LDCs. In order to show the low impact of the neutral and negative NPV projects in regards to all LDC NPV projects, the “Total NPVs” column has been included at the beginning of the table:

Values in Mio USD		LDCs					
		Total NPVs		Total Neutral NPVs		Total Negative NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	42,059	250	-23	-3	-27	-4
	Maximum	50,779	302	47	7	-27	-4
Total NPV Capital Costs (actual)	Minimum	23,408	139	296	42	202	29
	Maximum	23,578	140	285	41	202	29
Total Project Costs (actual)		33,506	199	533	76	344	49
Total World Bank Costs (actual)		9,110	54	294	42	200	29
Discount Rate [%]		11.21		11.17		11.2	
Actual Project Duration [years]		7.02		7.43		6.86	
NPV-horizon [years]		19.33		22.6		23.75	

Table 16: Neutral and Negative Result NPV and Cost Values of the LDCs¹⁴³

The 7 negative standalone-NPV projects (3 partial- and 4 overall-NPVs) account for a negative amount of about 27 Mio USD in total and thus only have a negligible impact on the total cumulative sum of the NPVs of the LDCs.

Furthermore, it should be noted that there are 7 out of the 168 result NPV projects which disclose NPV values within the 1- to 2-digit billion range.¹⁴⁴ In comparison, the NPV values of the remaining 161 result NPV projects are located in the 2- to 3-digit Mio range.¹⁴⁵ Again, the “Third Road Rehabilitation and Maintenance Project (RRMP III)” is the project outlining the highest result NPV value of 11,179 Mio USD at the appraisal stage. The following table shows the outcome, when all “high-value” NPV projects data are excluded:

¹⁴³ Cf. *Appendices I* and *J* provide an overview of the overall- and partial-NPVs of the neutral and negative result NPV and cost values of the LDCs.

¹⁴⁴ Cf. *Appendices K* and *L* provide an overview of the result “high-value” NPV projects with NPVs higher than 1,000 Mio USD and the sector codes of the WB financing of those projects.

¹⁴⁵ Cf. Author’s Note: The NPV values of the remainder of the 161 projects do have a range from -13.451.752 USD till 423 Mio USD with an average arithmetic mean of 38.431.762 Mio USD in the minimum and 48.440.645 Mio USD in the maximum case. Leaving out the 10 sample-NPVs the NPV range stays, but the average arithmetic mean changes to 40.136.324 Mio USD in the minimum and 49.787.447 Mio USD in the maximum case.

Values in Mio USD		LDCs					
		Total NPVs		Overall-NPVs		Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	6,188	38	2,226	57	3,961	38
	Maximum	7,799	48	3,270	58	4,529	43
Total NPV Capital Costs (actual)	Minimum	10,458	65	3,506	63	6,952	66
	Maximum	10,627	66	3,499	62	7,129	68
Total Project Costs (actual)		20,280	126	3,668	65	16,612	158
Total World Bank Costs (actual)		7,845	49	1,934	35	5,911	56
Discount Rate [%]		11.2		11.42		11.08	
Actual Project Duration [years]		7		7.04		6.98	
NPV-horizon [years]		19.74		18.94		20.13	

Table 17: Result NPV and Cost Values of the LDCs excluding “high-value” NPVs

Excluding the 7 “high-value” NPV projects results in a fully reduced and “slimmed” values picture. The resulting NPV now only lies at between 6,188 and 7,799 Mio USD which is only about 15 percent of the NPV of the total LDCs. Therefore, the resulting arithmetic mean of the NPVs as well drops to less than a sixth of the original value. In contrast, the NPV related capital costs decrease as well, but only by a little over 50 percent to 10,458 respectively 10,627 Mio USD. There are no noteworthy changes to the average values of the discount rate, actual project duration or to the NPV-horizon. The notable difference between the NPV and cost values of the overall- and the partial-NPVs diminishes as well. Still, the partial-NPVs disclose higher NPV values compared with the overall-NPVs. Taking a closer look by comparing the overall- vs. the partial-NPVs, the NPV/capital cost ratio of the overall-NPVs (0.63 in the minimum and 0.93 in the maximum case) is better than the one of the partial-NPVs (0.57 in the minimum and 0.64 in the maximum case). This difference between the overall- and partial-NPV/capital cost ratios would even be more significant where all project related capital costs have been taken into account for the partial-NPV projects. Another interesting point to mention when excluding the “high-value” projects is that the share of the total WB costs of the total project costs increases to 39 percent compared to 27 percent. With the exception of decreasing WB cost amounts per project type category there are no additional noteworthy changes to the ranking of the project type categories or the %-shares, as shown in the following table:

Project Type Categories	LDCs		
	%-Share	% of total WB Costs	WB Costs [Mio USD]
Transportation Development	27.8	36.59	2,871
Agriculture	15.16	9.78	767
Water & Sanitation	14.69	13.29	1,043
Governmental Administration	12.4	9.57	751
Power	10.58	14.16	1,111
Other	5.79	5.31	417
Sectorial Development & Reforming	3.77	2.46	193
Health	3.26	3.76	295
Natural Resources	2.38	1.7	132
Education	2.33	1.4	110
Financial Development	1.42	1.87	147
IT & Communication	0.42	0.11	8
Total	100	100	7,845

Table 18: Project Type Categories of the LDCs excluding “high-value” NPVs¹⁴⁶

The most interesting change is the adjusted amount of the WB costs of the ‘Governmental Administration’ project type category, which decreases by more than 30 percent from 1,088 to 751 Mio USD. Another interesting point to note is the decrease of the WB cost amount for the ‘Transportation Development’ project type category from 3,386 to 2,871 Mio USD. This basically shows that most of the WB financing of the 7 “high-value” NPV projects was used to support governmental activities as well as transportation developments.

To follow up on this analysis a general question to be answered is ‘how positive are the NPV projects at all?’ In order to find out if the total of partial- and overall-NPV disclosing projects are positive, chapter 4.8 *NPV-Cost-Ratios and SROI Calculation* will outline and discuss an approach for the calculation of the presented types of NPVCRs (refer to chapter 3.2.3 *Types of NPV-Cost-Ratios*), as well as on the minimum SROI ratio of the LDCs.

¹⁴⁶ Cf. *Appendix M* and *N* provide an overview of the sector codes and the corresponding percentages as well as the assigned sector codes per project type category for the LDCs when “high-value” NPV projects are excluded.

4.3.2 Result NPV Continent Comparison

This chapter deals with the comparison of the result NPV values between the African and Asian continents. In order to set the basis of comparison the following table provides an overview of the NPV counts for Africa and Asia, whilst the matrix distinguishes between total, overall- and partial-NPVs per row and the total number and between positive, neutral and negative NPV counts as per the columns:

		Africa							
		NPV Counts		Positive NPV Counts		Neutral NPV Counts		Negative NPV Counts	
Total NPVs	Standalone-NPVs	109	80	104	76	1	0	4	4
	Scenario-NPVs		29		28		1		0
Overall-NPVs	Standalone-NPVs	37	29	35	27	0	0	2	2
	Scenario-NPVs		8		8		0		0
Partial-NPVs	Standalone-NPVs	72	51	69	49	1	0	2	2
	Scenario-NPVs		21		20		1		0

		Asia							
		NPV Counts		Positive NPV Counts		Neutral NPV Counts		Negative NPV Counts	
Total NPVs	Standalone-NPVs	59	40	50	37	6	0	3	3
	Scenario-NPVs		19		13		6		0
Overall-NPVs	Standalone-NPVs	23	13	17	11	4	0	2	2
	Scenario-NPVs		10		6		4		0
Partial-NPVs	Standalone-NPVs	36	27	33	26	2	0	1	1
	Scenario-NPVs		9		7		2		0

Table 19: Result NPV Counts of Africa and Asia

Overall, Africa outnumbers Asia with 109 compared to 59 NPVs respectively. Still, when looking at the number of ICR reports researched per continent –Africa 574 and Asia 206 ICR documents– the ratio of NPV disclosures of 29 percent in Asia is higher compared to 19 percent in Africa. In total, there are 37 African and 23 Asian overall-NPVs, whereas the partial-NPV count is twice as much in Africa (72) with as in Asia (36). What is interesting to note is that the number of negative NPVs is nearly equal with 3 negative NPV counts in Asia and only 1 more in Africa. Most surprisingly in terms of the 3 valence groups is the difference within the neutral NPV counts,

since there is only 1 in Africa, but 6 in Asia. So in total, Asia counts 9 neutral and negative out of 59 NPVs, whereas Africa only counts 5 neutral and negative out of a total of 109 NPVs.

The following table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for all NPV projects based on cumulative sum [Cum Sum] and arithmetic mean [Ø] values:

Values in Mio USD		Africa Total NPVs		Asia Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	6,458	59	35,602	603
	Maximum	8,981	82	41,797	708
Total NPV Capital Costs (actual)	Minimum	8,575	79	14,833	251
	Maximum	8,578	79	14,999	254
Total Project Costs (actual)		17,666	162	15,841	268
Total World Bank Costs (actual)		6,127	56	2,983	51
Discount Rate [%]		11.18		11.27	
Actual Project Duration [years]		7.34		6.44	
NPV-horizon [years]		19.76		18.62	

Table 20: Result Total NPV and Cost Values Comparison of Africa and Asia¹⁴⁷

Even though Asia discloses only about half of the NPVs of Africa, and moreover a higher count of neutral and negative NPVs, the cumulative sum of the NPVs and its arithmetic mean values are about 6 and 10 times higher in the minimum and close to 5 and 9 times higher in the maximum case for those summary financial figures. Therefore, Asian NPVs make up about 85 percent of the total cumulative sum of NPVs of the whole LDCs in the minimum and 82 percent in the maximum case. Although the NPV related capital costs as a total amount are higher in Asia than in Africa, the NPV/capital cost ratio is still significantly higher as well. The African minimum case discloses the lowest ratio of 0.75 followed by its maximum case of 1.05. In contrast the ratios in Asia are 2.4 in the minimum and 2.79 in the maximum case.

¹⁴⁷ Cf. *Appendices O* and *P* provide an overview on the outcomes of the NPVs and corresponding cost values of the African and Asian overall- and partial-NPVs.

This basically underlines the appraisal results of the continent comparison, namely that in case all NPVs are taken into account, Asian projects generally have a better input/output-ratio than Africa.¹⁴⁸ Noteworthy is the fact the higher NPV values in Asia are achieved with shorter project durations by about a year, with average actual project durations of 6.44 years in Asia and 7.34 years in Africa, as well as with a shorter NPV-horizon of 18.62 years compared to 19.76 years in Africa. Since the amount of the corresponding total project costs is similar for both continents (17,666 Mio USD for the African and 15,841 Mio USD for the Asian project) the low share of 27 percent of the WB financing of the total project costs mainly results from the low bank financing amount in Asia of about 2,983 Mio USD compared to 6,127 Mio USD in Africa. In order to provide the user with a better understanding on the continental project types, the following table compares the category mapping of the project types based on the costs covered by the WB:

Project Type Categories	Africa			Asia		
	%-Share	% of total WB Costs	WB Costs [Mio USD]	%-Share	% of total WB Costs	WB Costs [Mio USD]
Transportation Development	27.38	37.73	2,313	27.48	35.98	1,073
Water & Sanitation	14.91	13.15	806	11.09	7.95	237
Agriculture	14.63	9.76	598	15.68	7.11	212
Governmental Administration	12.6	10.42	639	12.96	15.06	449
Power	11.29	15.76	966	9.13	7.95	237
Other	4.99	4.58	281	7.05	5.67	169
Sectorial Development & Reforming	4.27	2.72	167	2.82	1.74	52
Health	3.16	3.96	242	4.01	5.7	170
Natural Resources	2.75	0.88	53	1.68	2.65	79
Education	1.87	0.58	34	4.32	5.9	176
Financial Development	0.92	0.43	26	2.47	4.09	123
IT & Communication	0.05	0.03	2	1.31	0.2	6
Total	100	100	6,127	100	100	2,983

Table 21: Continental Project Type Category Comparison of the NPV Projects¹⁴⁹

¹⁴⁸ Author's Note: This statement does not consider that the projects disclosing NPVs in the appraisal stage might not be the ones disclosing NPVs in the result stage. Therefore, a specific comparison analysis will be conducted in *chapter 4.4 Appraisal vs. Result NPVs*.

¹⁴⁹ Cf. *Appendices Q, R and S* provide an overview of the sector codes and the corresponding percentages as well as the assigned sector codes per project type category per continent.

For both continents the ‘Transportation Development’ project type category shows a very similar %-share of approx. 27 percent as well as a similar share of the total WB costs of approx. 38 percent for Africa and 26 percent for Asia. Aside from these similarities, differences can be found for all the other categories. For instance, ‘Water and Sanitation’ shows a difference of more than 5 percent for the share of the total WB costs with 13.15 percent in Africa and only 7.95 percent in Asia. Furthermore, although this category type has the second highest %-share within Africa, it only accounts for the third highest %-share in Asia sharing it with the ‘Power’ project type category. The ‘Power’ project type category actually outlines the highest difference for the share of the total WB costs with 15.76 percent in Africa and only 7.95 percent in Asia. The ‘Governmental Administration’ project type category accounts for the second highest share of the total WB costs in Asia with 15.06 percent, whereat it only accounts for the fourth highest share of 10.42 percent in Africa.

The picture for the positive NPV projects looks similar to the one of the total NPVs of the 2 continents¹⁵⁰, since the amounts for the neutral and negative NPVs per continent are similarly low and do not have a noteworthy impact on the total values as the following table shows:¹⁵¹

Values in Mio USD		Africa				Asia			
		Total Neutral NPVs		Total Negative NPVs		Total Neutral NPVs		Total Negative NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	-13	-13	-17	-4	-9	-2	-10	-3
	Maximum	7	7	-17	-4	40	7	-10	-3
Total NPV Capital Costs (actual)	Minimum	210	210	91	23	86	14	111	37
	Maximum	210	210	91	23	75	13	111	37
Total Project Costs (actual)		392	392	233	58	141	23	111	37
Total World Bank Costs (actual)		196	196	158	39	98	16	42	14
Discount Rate [%]		11		11.33		11.2		11	
Actual Project Duration [years]		8		8		7.33		5.33	
NPV-horizon [years]		22.6		23.33		18.25		25	

Table 22: Neutral and Negative Result NPVs and Costs Comparison of Africa and Asia

¹⁵⁰ Cf. *Appendices T, U, and V* provide an overview of the outcomes of the NPVs and corresponding cost values of the African and Asian positive total, overall- and partial-NPVs.

¹⁵¹ Cf. *Appendices W, X, Y and Z* provide an overview of the outcomes of the NPVs and corresponding cost values of the African and Asian neutral and negative overall- and partial-NPVs.

Looking at the total neutral NPVs the single 1 in Africa actually discloses a lower NPV value with -13 Mio USD compared against the 6 Asian ones with a combined -9 Mio USD in the minimum case. A vast difference can be found when looking at the NPV related capital costs which are more than 2.5 times higher for the African neutral project than for the Asian projects.

When comparing the continental total negative NPVs there are no noteworthy differences within the NPV and the corresponding cost values. However, a difference is noted in the average actual project duration between the 2 continents, with a 2.66 years shorter duration in Asia than in Africa. Furthermore, for both continents the NPV-horizon is longer compared with the corresponding continental total NPVs, namely 3.4 years for the African and even 6.08 years for the Asian projects. However, despite some noted differences, no generalized conclusions can be derived as the number of projects used for the comparative analysis is too few.

Should the 7 “high-value” NPV projects be excluded however, the continent comparison has a totally different outlook as shown in the table below. Of those 7 projects, 3 are African and 4 Asian (all Bangladeshi projects) including ICR report 35449 (Bangladesh) disclosing a NPV of 28,918 Mio USD which accounts for more than half of the cumulative NPV sum of all 168 NPV projects:

Values in Mio USD		Africa Total NPVs		Asia Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	4,054	38	2,134	39
	Maximum	5,091	48	2,708	49
Total NPV Capital Costs (actual)	Minimum	8,117	77	2,341	43
	Maximum	8,120	77	2,508	46
Total Project Costs (actual)		17,015	161	3,264	59
Total World Bank Costs (actual)		5,709	54	2,136	39
Discount Rate [%]		11.15		11.3	
Actual Project Duration [years]		7.32		6.38	
NPV-horizon [years]		20.06		19.17	

Table 23: Result Comparison of Africa and Asia excluding “high-value” NPVs

Although the African cumulative NPV sum drops by 2,404 Mio USD from 6,458 to 4,054 Mio USD in the minimum case and by 3,890 Mio USD from 8,981 to 5,091 Mio USD in the maximum case, the most impacting factor is the drop of the Asian NPVs. In the minimum case the cumulative NPV drops from 35,602 to 2,134 Mio USD in the minimum and from 41,797 to 2,708 Mio USD, however, there are no noteworthy changes to the corresponding discount rates, actual project durations or the NPV-horizons. Comparing the dropped outcomes with the revised NPV counts per continent (106 NPV counts in Africa and 55 NPV counts in Asia) the average arithmetic mean of the NPVs shows that the average NPV outcomes of both components are similar. However, the NPV/capital cost ratio of Asia is still higher than the one of Africa with 0.91 and 1.08 compared to 0.5 in the minimum and 0.63 in the African maximum case. This basically means that projects conducted within the Asian continent seem to be more capital cost effective than the ones in Africa. Taking a closer look by comparing the overall- vs. the partial-NPVs the result for both continents looks similar: The NPV/capital cost ratio of the overall-NPVs is again higher respectively better for both continents than the one for the partial-NPVs (refer to *Appendix AA* for the results of the continental overall-NPV comparison and to *Appendix AB* for the ones of the partial-NPV comparison). However, looking at the total project costs it needs to be noted that analysis still includes an African “high-value” total project costs project which discloses total project costs of 3,985 Mio USD and a partial-NPV of 25 Mio USD.¹⁵² In order to be able to compare the adjusted project type categories and the corresponding WB costs, the following table outlines the revised shares per category:

¹⁵² Cf. Tanzania – Report ICR1511: Total WB costs of 170 Mio USD.

Project Type Categories	Africa			Asia		
	%-Share	% of total WB Costs	WB Costs [Mio USD]	%-Share	% of total WB Costs	WB Costs [Mio USD]
Transportation Development	26.94	34.94	1,996	27.93	41.03	876
Water & Sanitation	15.33	14.11	806	11.91	11.1	237
Agriculture	14.54	9.7	554	16.64	9.93	212
Governmental Administration	13.1	10.2	583	11.46	7.92	169
Power	11.6	16.91	966	8.92	6.74	144
Other	5.14	4.92	281	7.38	6.42	137
Sectorial Development & Reforming	4.39	2.92	166	2.87	1.17	25
Health	3.24	4.24	242	3.57	2.44	52
Natural Resources	2.82	0.95	53	1.82	3.7	79
Education	1.91	0.61	35	3.42	3.56	76
Financial Development	0.94	0.46	25	2.66	5.71	123
IT & Communication	0.05	0.04	2	1.42	0.28	6
Total	100	100	5,709	100	100	2,136

Table 24: Continental Project Type Category Comparison excluding “high-value”

NPVs¹⁵³

Due to the 4 Bangladeshi “high-value” NPV projects the decrease of the total amount of the Asian WB costs by roughly 29 percent from 2,983 to 2,136 Mio USD is a lot higher than the decrease of the WB costs in Africa by roughly 7 percent from 6,127 to 5,709 Mio USD. Therefore, even though there is a higher decrease of the ‘Transportation Development’ project type according to the amount in Africa than in Asia, the share of the total WB costs decreases in Africa to 34.94 from 37.73 percent, whereat it actually increases to 41.03 from 35.98 percent in Asia. Besides some minor changes to the %-shares in both continents the noteworthy difference found for the share of the total WB costs of the ‘Power’ project type category further increases from 7.81 percent (in the case where all projects are considered) to 10.17 percent where the 7 “high-value” NPV projects are excluded. Another interesting change in Asia is the drop in the share of the total WB costs for the ‘Governmental Administration’ project type category from 15.06 to 7.92 percent.

Although differences can be found by comparing the continental sector codes and their respective project types, no conclusion can be made since the outlined sector codes of the ICR reports only refer to the WB financing part of a project. In the total

¹⁵³ Cf. *Appendices AC, AD and AE* provide an overview of the sector codes and the corresponding percentages as well as the assigned sector codes per project type category per continent when excluding “high-value” NPV projects for both continents.

case of the 168 NPV projects, the WB share is only about 27 percent (9,110 Mio USD out of 33,506 Mio USD; refer to *table 12: Result NPV and Cost Values of the LDCs*), whereat it increases to about 39 percent (7,845 Mio USD out of 20,280 Mio USD; refer to *table 17: Result NPV and Cost Values of the LDCs excluding “high-value” NPVs*) in the case where the 7 “high-value” NPV projects are excluded. Unfortunately, the data basis provided within the ICR documents does not allow for further analysis to examine the continental differences. Therefore, it must be concluded that the continental differences could be due to project type differences, but as well to other factors, such as the project handling by the implementation agency, country specific differences, governmental processes, project teams.

4.4 Appraisal vs. Result NPVs

This chapter outlines the differences between the appraisal and the result values. To assure that there is a common basis of comparison only those NPV projects outlining an NPV value for the appraisal as well as for the result stage are considered. Furthermore, the following points need to be highlighted in order to better understand the appraisal vs. result comparison:

- To ensure a common basis of comparison the NPV of the appraisal and the result stage must be of the same type, meaning for example, that the overall-NPV of the appraisal stage as well requires the disclosure of the overall-NPV of the result stage. In order to ensure the common comparison basis for the partial-NPVs, each relevant partial-NPV project is researched for the components the partial-NPV is based on.
- The calculation of the NPV values of the appraisal stage is based on the planned project start date, whereas the calculation for the result stage is based on the actual project start date. The advantage of the result NPV calculation vs. the appraisal NPV calculation is that the approximations for future cash flows are more precise since a validation of in- and outflows has already happened during the course of the project implementation respectively duration period.

- Due to this approach the benefits and costs as well as the applied project duration, discount rate and NPV-horizon can vary between the appraisal and the result stage of a project.

4.4.1 Results of the LDCs

In this chapter the cumulative sums and arithmetic means of the NPVs and their related capital costs, as well as the corresponding average discount rates, project durations and NPV-horizons at the appraisal and the result stage are presented and compared. In order to do so, the following table provides the reader with the NPV counts of the same type which are outlined for the appraisal and the result stage within an ICR report:

		LDCs									
		Appraisal NPV Counts		Result NPV Counts		Result Positive NPV Counts		Result Neutral NPV Counts		Result Negative NPV Counts	
Total NPVs	Standalone-NPVs	67	64	67	51	63	50	3	0	1	1
	Scenario-NPVs		3		16		13		3		0
Overall-NPVs	Standalone-NPVs	29	28	29	22	26	21	2	0	1	1
	Scenario-NPVs		1		7		5		2		0
Partial-NPVs	Standalone-NPVs	38	36	38	29	37	29	1	0	0	0
	Scenario-NPVs		2		9		8		1		0

Table 25: NPV Counts of the LDCs at the Appraisal and Result Stage

Obviously, the number of NPV counts is the same for the appraisal and the result stage for the 67 total, 29 overall- and 38 partial-NPVs. However, differences can be found in the sub-types, since 13 standalone-NPVs of the appraisal stage have been replaced by scenario-NPVs at the result. Generally, the conversion from standalone- to scenario-NPVs happens for both the overall- and the partial-NPVs.¹⁵⁴ Furthermore, all of the appraisal NPVs are positive NPVs whereas out of the 67 NPVs 3 are neutral and 1 is negative at the result stage.

¹⁵⁴ Author's Note: Both, the appraisal and the result NPV counts contain 2 manually calculated overall-NPVs and 1 sample-NPV, whereas at the appraisal stage the manually calculated overall-NPVs and the sample-NPV are outlined as standalone-NPVs, but only 1 manually calculated overall-NPV is outlined as standalone-NPV at the result stage, the other 2 have been converted to scenario-NPVs.

The following table shows the outcome of the economic comparison analysis of the appraisal and result values for the total NPVs based on the cumulative sum and the arithmetic mean values:

Values in Mio USD		LDCs Appraisal - Total NPVs		LDCs Result - Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	16,047	240	33,648	502
	Maximum	18,429	275	36,343	542
Total NPV Capital Costs	Minimum	10,169	152	10,002	149
	Maximum	10,178	152	9,991	149
Total Project Costs		11,936	178	12,141	181
Total World Bank Costs		3,923	59	3,857	58
Discount Rate [%]		11.14		11.28	
Project Duration [years]		5.34		6.87	
NPV-horizon [years]		19.35		20.91	

Table 26: Total NPV and Cost Values of the LDCs at the Appraisal and Result Stage

At a glance it can be said that for the evaluated 67 projects, the NPV values in the minimum as well as in the maximum case actually double from the appraisal towards the result stage, whereat the costs slightly decrease. This essentially means that the initially expected outcome was expected to be only half as it was reevaluated at project completion. Even though the planned average project duration increases by 1.52 years from 5.34 to 6.87 years as well as the NPV-horizon by 1.56 years from 19.35 to 20.91 years, the higher average discount rate of the result stage, namely 11.28 compared to 11.14 percent, compensates the longer NPV-horizon period in case of positive cash flows. Although the total project costs for the 67 projects increases by roughly 205 Mio USD, the share of the costs covered or funded by the WB decreases by 66 Mio USD. Interestingly, the doubling of the NPV cumulative sum values is due to the high increase of the partial-NPVs. Those increase from 13,054 to 31,990 Mio USD in the minimum case and from 13,072 to 32,014 Mio USD in the maximum case from the appraisal towards the result stage. In comparison the related capital costs only slightly increase by 419 Mio USD in the minimum and by 410 Mio USD in the maximum case –both to 3,909 Mio USD. However, the

overall-NPV values decrease from 2,992 to 1,658 Mio USD in the minimum and from 5,357 to 4,329 Mio USD in the maximum case. In line with this, the overall-NPV related capital costs as well decrease by 587 Mio USD to 6,092 Mio USD in the minimum and by 597 Mio USD to 6,082 Mio USD in the maximum case.¹⁵⁵ As a matter of fact it needs to be noted that those values contain 4 “high-value” NPVs (2 partial- and 2 overall-NPVs), whereat the decisive project is again the Bangladeshi project “35449” disclosing a NPV of 11,179 Mio USD at the appraisal and 28,918 Mio USD at the result stage.¹⁵⁶ Should the 4 “high-value” NPV projects be excluded, the results will appear as follows:

Values in Mio USD		LDCs Appraisal - Total NPVs		LDCs Result - Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	2,805	45	2,680	43
	Maximum	2,822	45	3,010	48
Total NPV Capital Costs	Minimum	3,990	63	4,321	69
	Maximum	3,998	63	4,311	68
Total Project Costs		5,437	86	6,210	99
Total World Bank Costs		2,996	48	2,980	47
Discount Rate [%]		11.15		11.31	
Project Duration [years]		5.3		6.84	
NPV-horizon [years]		19.36		20.73	

Table 27: Total NPV and Cost Values of the LDCs at the Appraisal and Result Stage excluding “high-value” NPVs

In the minimum case the cumulative sum of the NPVs slightly decreases by 125 Mio USD for the 63 projects, whereat it increases by 188 Mio USD in the maximum case. Therefore, it can be noted that in principle the appraised NPV values do develop as expected during the course of the project implementation. However, looking at re-

¹⁵⁵ Cf. *Appendices AF* and *AG* provide an overview of the overall- and partial-NPV and cost values of the LDCs at the appraisal and result stage.

¹⁵⁶ Cf. *Appendix AH* shows the “high-value” NPV projects disclosing a NPV at the appraisal and the result stage.

lated capital costs it needs to be noted that there is an increase of at least 7.8 percent in average in order to achieve the anticipated NPVs. Furthermore, the total project costs also increase by 14.2 percent during the course of the project implementation, whereat the share of the WB costs again slightly decreases by 0.5 percent. Looking at the overall- and partial-NPVs separately the picture as described above when considering the “high-value” NPV projects is similar: The result overall-NPV values and the related capital costs decrease compared to the appraised ones, whereat the partial-NPV values and the related capital costs increase at the result stage compared to the appraisal values (refer to *Appendices AI* and *AJ*). Furthermore, it can be noted that the NPV/capital cost ratio of the overall-NPVs (0.94 (appraisal) and 0.70 (result) in the minimum and 0.94 (appraisal) and 0.97 (result) in the maximum case) is again higher respectively better than the one for the partial-NPVs (0.59 (appraisal) and 0.59 (result) in the minimum and 0.6 (appraisal) and 0.6 (result) in the maximum case) in both the appraisal and the result stages. As a conclusion of the appraisal vs. result comparison it can be noted that expected outcomes of the appraisal stage are generally achieved and projects develop as initially planned, although both –the related capital costs and the total project costs– do slightly increase during the course of the project implementation phase.

4.4.2 Continent Comparison

This chapter deals with the comparison of the appraisal and the result values between the African and Asian continents. In order to set the basis of comparison the following table provides an overview of the appraisal and result NPV counts for both continents, whereat the matrix distinguishes between total, overall- and partial-NPVs per row, and the total number and between positive, neutral and negative NPV counts as per the columns:

		Africa				Asia			
		Appraisal NPV Counts		Result NPV Counts		Appraisal NPV Counts		Result NPV Counts	
Total NPVs	Standalone-NPVs	45	44	45	38	22	20	22	13
	Scenario-NPVs		1		7		2		9
Overall-NPVs	Standalone-NPVs	20	20	20	18	9	8	9	4
	Scenario-NPVs		0		2		1		5
Partial-NPVs	Standalone-NPVs	25	24	25	20	13	12	13	9
	Scenario-NPVs		1		5		1		4

Table 28: NPV Counts of Africa and Asia at the Appraisal and Result Stage

The distribution of the NPV counts per continent is almost exactly two-thirds of the NPV reports in Africa (45) and one-third (22) in Asia, whereat all 3 neutral and the negative NPV of the result stage belong to the Asian continent. However, the disclosure rate for ICR reports disclosing an appraisal as well as result NPV is 9.7 percent in Asia which is higher than the 7.8 percent in Africa. For both continents the share of the overall-NPVs is lower with 20 and 9 overall-NPVs compared to 25 and 13 partial-NPVs in Africa and Asia.¹⁵⁷ Interestingly, for all of the continental overall- and partial-NPVs the share of the scenario-NPVs at the result stage is higher than at the appraisal stage. A possible interpretation for this could be that the single NPV values estimated at project appraisal turn out to be dependent on varying factors influencing future cash flows, so that it is more appropriate to provide a range for the result NPV estimations at project completion.

The following table compares the outcomes of the total NPVs and the corresponding cost values of Africa and Asia at the appraisal and the result stages based on cumulative sum and arithmetic mean values:

¹⁵⁷ Author's Note: In Africa there are 2 manually calculated overall- and 1 sample-NPV at the appraisal and the result stage, whereas in Asia there is neither a manually calculated overall- nor a sample-NPV.

Values in Mio USD		Africa Appraisal - Total NPVs		Africa Result - Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	3,525	78	3,578	80
	Maximum	3,525	78	3,664	81
Total NPV Capital Costs	Minimum	4,788	106	4,618	103
	Maximum	4,796	107	4,618	103
Total Project Costs		6,137	136	6,434	143
Total World Bank Costs		2,612	58	2,612	58
Discount Rate [%]		11.32		11.4	
Project Duration [years]		5.42		7.2	
NPV-horizon [years]		19.27		21.47	

Values in Mio USD		Asia Appraisal - Total NPVs		Asia Result - Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	12,522	569	30,070	1,367
	Maximum	14,904	677	32,679	1,485
Total NPV Capital Costs	Minimum	5,381	245	5,383	245
	Maximum	5,381	245	5,373	244
Total Project Costs		5,799	264	5,707	259
Total World Bank Costs		1,311	60	1,245	57
Discount Rate [%]		10.69		11.06	
Project Duration [years]		5.18		6.18	
NPV-horizon [years]		19.5		19.75	

Table 29: Total NPV and Cost Values of Africa and Asia at the Appraisal and Result Stage

Obviously, the high increase of the cumulative sum of the NPVs at the result stage is due to the Asian continent and its “high-value” projects. At the appraisal stage Asia makes up 78 percent in the minimum case and 81 percent in the maximum case of the cumulative sum of the NPVs of the LDCs, whereas its share actually increases to 89 in the minimum case and 90 percent in the maximum case at the result stage. In Africa, the NPV values for the minimum and the maximum case only slightly increase during the project implementation phase together with a slight decrease of the related capital costs. In Asia however, the cumulative sum of the NPVs

risers about 140 percent to about 30,070 Mio USD in the minimum case and about 219 percent to 32,679 Mio USD in the maximum case, whereas there is actually no change to the related capital costs. The NPV/capital cost ratio is below 1 in Africa for the appraisal and result stages, whereas the Asian ratios are all above 1 with 2.33 for the appraisal minimum and 2.77 for the maximum case, rising to 5.16 for the minimum and 6.08 for the maximum case at the result stage. Africa's share of the total project costs is slightly larger compared against Asia with 6,137 Mio USD compared 5,799 Mio USD at the appraisal stage. The difference actually increases when looking at the result stage, whereas comparing it continental wise there is an increase to 6,434 Mio USD in Africa and even a slight decrease to 5,707 Mio USD in Asia. For the African countries, the share of the WB financing of the total project costs decreases whereas the amount of 2,612 Mio USD stays. In Asia there is even a slight decrease for the WB financing portion from 1,311 Mio USD at the appraisal to 1,245 Mio USD at the result stage.¹⁵⁸

Excluding the 4 "high-value" NPV projects basically reduces the African minimum and maximum NPVs by 1,453 Mio USD at the appraisal and by 1,441 Mio USD at the result stage. On the other hand the Asian NPVs are reduced by 11,789 Mio USD in the appraisal minimum and by 14,154 Mio USD in the maximum case and even more drastically by 29,528 Mio USD in the minimum and by 31,893 Mio USD in the maximum case at the result stage as following table highlights:

¹⁵⁸ Author's Note: *Appendices AK and AL* provide an overview over the overall- and partial-NPV and cost values of Africa and Asia at the appraisal and result stage.

Values in Mio USD		Africa Appraisal - Total NPVs		Africa Result - Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	2,073	48	2,138	50
	Maximum	2,073	48	2,224	52
Total NPV Capital Costs	Minimum	3,267	76	3,621	84
	Maximum	3,275	76	3,621	84
Total Project Costs		4,467	104	5,271	123
Total World Bank Costs		2,258	53	2,257	52
Discount Rate [%]		11.34		11.42	
Project Duration [years]		5.40		7.19	
NPV-horizon [years]		18.735		21.20	

Values in Mio USD		Asia Appraisal - Total NPVs		Asia Result - Total NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	733	37	542	27
	Maximum	750	37	786	39
Total NPV Capital Costs	Minimum	723	36	699	35
	Maximum	723	36	689	34
Total Project Costs		970	49	939	47
Total World Bank Costs		738	37	723	36
Discount Rate [%]		10.64		11.06	
Project Duration [years]		5.1		6.1	
NPV-horizon [years]		20.94		19.73	

Table 30: Total NPV and Cost Values of Africa and Asia at the Appraisal and Result Stage excluding “high-value” NPVs

The picture shown above is totally different than the one described for the continental appraisal and result comparison including “high-value” NPVs: Now Africa holds the higher share of the cumulative sum of the NPVs of the LDCs with about 74 percent in the appraisal minimum and 73 percent in the maximum case. At the result stage the share is even higher with about 80 percent in the minimum and 74 percent in the maximum case. For both cases –minimum and maximum– the cumulative sum of the NPVs slightly increases in the African case. However, for the remainder of the 20 Asian projects the cumulative NPV value drops from 733 to 542 Mio USD

in the minimum and only increases by 36 Mio USD to 786 Mio USD in the maximum case. What is interesting to note is that in Africa all costs (related capital, total project and WB costs) increase at the result stage, whereas in Asia all of the cost types have dropped. The NPV/capital cost ratio is below 1 for any of the African cases for both appraisal and result stages, whereas the ratios actually decrease further at project completion (appraisal: 0.63 (min) and 0.63 (max); result: 0.59 (min) and 0.61 (max)). In Asia a ratio above 1 exists for both appraisal cases (1.01 (min) and 1.04 (max)), but below 1 for the minimum case (0.78) of the result stage. Only in the Asian maximum case the ratio is again above 1 (1.14). Even though Asia has the possibility to achieve a better NPV/cost ratio than appraised, the result ratios deviate more from the appraisal ones than it is the case within Africa.¹⁵⁹

4.5 Economic Rate of Return Result Values

In this chapter the analysis of the ERR is presented. In order to be able to evaluate and compare the outcome of portfolio ERRs of non-NPV projects, the following chapters will share the results of the ERR analysis conducted based on the same 790 ICR data basis which was used for the NPV analysis. Firstly, the results of the LDCs are outlined and sensitized by excluding “high-value” ERR projects. In the subsequent chapter a continent comparison for all ERRs and for the ones excluding “high-values” is conducted.

4.5.1 Results of the LDCs

This chapter outlines the results of the disclosed ERRs of the result stage within the ICR reports. In order to set the basis for a common understanding, the following table provides an overview of the total ERR counts of the LDCs in comparison to the ERR counts for projects outlining a NPV and for those which do not:

¹⁵⁹ Author's Note: *Appendices AM* and *AN* provide an overview of the overall- and partial-NPV and cost values of Africa and Asia at the appraisal and result stage excluding the 4 “high-value” projects.

LDCs		ERR Counts		ERR Counts of NPV Projects		ERR Counts of Non-NPV Projects	
Total ERRs	Overall-ERRs	218	81	159	55	59	26
	Partial-ERRs		137		104		33

Table 31: Result ERR Counts of the LDCs

In total there are 218 ERRs disclosed within the ICR report data set. The intersection for the reports disclosing an ERR as well as a NPV at the result stage is 159 counts (in total there have been 168 result NPVs disclosed). On top of that there are 59 ERR disclosures which are based on non-NPV projects. For each of the ERR counts the partial-ERRs do prevail as it is the case for the NPV counts of the LDCs (refer to *table 11: Result NPV Counts of the LDCs*).¹⁶⁰

The table below shows the outcome of the ERR analysis of the result stage. The arithmetic mean ($\emptyset$) and range of the ERRs are expressed in percentage, whereat there is always a minimum and a maximum value due to disclosure of ERR-scenarios. Further to this, there is a “Min” and a “Max Scenario” for the ERR-range:

Values in Percentage		Total ERRs		Total ERRs of NPV Projects		Total ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean ($\emptyset$)		29.98	48.24	26.7	44.93	38.8	57.17
ERR-Range	Min Scenario	-8	1,105.65	-4	137.55	-8	1,105.65
	Max Scenario	-8	1,105.65	3.3	311	-8	1,105.65
Actual Total Project Costs [Mio USD]		37,192		32,732		4,460	
Actual Total World Bank Costs [Mio USD]		10,812		8,770		2,042	
Discount Rate [%]		11.19		11.25		10.72	
Actual Project Duration [years]		6.91		6.98		6.73	
ERR-horizon [years]		19.3		19.64		17.55	

Table 32: Result ERRs of the LDCs

The arithmetic mean of the 218 ERRs lies between 29.98 and 48.24 percent. This is at least 3 times as much as the corresponding discount rate of 11.19 percent. The

¹⁶⁰ Author's Note: Except a few cases, the disclosed ERR type (overall or partial) is of the same type as the disclosed NPV. See *Appendices AO* and *AP* for the overall- and partial-ERR results of the LDCs.

minimum and maximum value for both ERR-range scenarios are -8 and 1,105.65¹⁶¹. Furthermore, the total costs of 37,192 Mio USD of the total ERRs are 3,686 Mio USD higher than the ones for the result total NPVs of 33,506 Mio USD (refer to *table 12: Result NPV and Cost Values of the LDCs*). Accordingly, the WB costs are also higher with 10,812 Mio USD, namely about 1,072 Mio USD which is approx. 19 percent, compared to the total WB costs of the result total NPV projects of the LDCs with about 9,110 Mio USD. Looking at the 159 intersection projects it can be noted that the percentages are slightly lower and the ERR-ranges are narrower respectively showing less extreme values than the ones for the total ERR projects. On the other hand the percentage values of the total ERRs for non-NPV projects are generally higher with a minimum percentage of 38.8 and a maximum of 57.17, whereas the ranges of the minimum and maximum scenario are defining the ranges of the total 218 ERRs.

When excluding “high-value” projects the ERR counts are as follows:

LDCs		ERR Counts		ERR Counts of NPV Projects		ERR Counts of Non-NPV Projects	
Total ERRs	Overall-ERRs	198	75	143	51	55	24
	Partial-ERRs		123		92		31

Table 33: Result ERR Counts of the LDCs excluding “high-value” ERR projects

In total, 20 projects are defined as “high-value” ERR projects¹⁶² disclosing ERR percentages above 100. 15 projects of them disclose ERRs with percentage values between 101 and 200 and 5 projects disclose ERR values above 200 percent. The table below shows the adjusted outcome for the ERR counts excluding the “high-value” ERR projects:

¹⁶¹ Author’s Note: The ERR-range is based on the whole data set including “high-value” ERRs, whereas ERRs of 100 or higher percentages are defined as “high”.

¹⁶² Cf. *Appendix AQ* provides an overview of the result ERRs disclosing values above 100 percent.

Values in Percentage		Total ERRs		Total ERRs of NPV Projects		Total ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		22.21	32.8	23.48	33.36	18.91	31.35
ERR-Range	Min Scenario	-8	99	-4	84	-8	99
	Max Scenario	-8	99	3.3	95	-8	99
Actual Total Project Costs [Mio USD]		35,994		31,724		4,270	
Actual Total World Bank Costs [Mio USD]		10,132		8,221		1,911	
Discount Rate [%]		11.17		11.21		10.83	
Actual Project Duration [years]		6.97		7.06		6.74	
ERR-horizon [years]		19.58		20.09		17.05	

Table 34: Result ERRs of the LDCs excluding “high-value” ERR projects¹⁶³

Generally, it can be noted that the ERR values decrease for each of the researched areas. The highest decrease of the arithmetic mean can be noted for total ERRs of the non-NPV projects from 38.8 to 18.91 percent in the minimum and even more drastically from 57.17 to 31.35 percent in the max case.¹⁶⁴ The decrease of the ERRs of the non-NPV projects is as well reflected by the fact that the average discount rate increases from 10.72 to 10.82 percent while the ERR-horizon drops by half a year. Furthermore, the total ERRs of the NPV projects now show higher minimum and maximum values for the arithmetic mean than the total ERRs of the non-NPV projects. Obviously, the exclusion of projects comes along with a reduction in the cost values, whereat the total project costs of the total ERRs decrease from 37,192 to 35,994 Mio USD and the WB costs from 10,812 to 10,132 Mio USD.

As an result of this analysis it needs to be noted that besides the positive outcome of the 168 result total NPV projects there are additionally 59 result total ERR projects which do outline higher and respectively better ERR values than the intersectional 159 NPV projects outlining ERRs. The reason for the higher ERR arithmetic means of the non-NPV projects lies in their overall-ERRs, whereby the minimum arithmetic mean of the overall-ERRs of the non-NPV projects is still about 23 percent higher than the maximum arithmetic mean of the overall-ERRs of the NPV projects (refer

¹⁶³ Author's note: See *Appendices AR* and *AS* for the overall- and partial-ERR results of the LDCs excluding “high-value” projects.

¹⁶⁴ Author's Note: The decrease of ERR values of the non-NPV projects is mainly due to the exclusion of the Guinea-Bissau ICR1347 report which outlines an overall-ERR of 1,105.65 percent.

to *Appendix AO*). Although the partial-to-overall-ERR counts ratio is higher for the NPV-projects (1.89; 104 partial-ERRs compared to 55 overall-ERRs) than for the non-NPV projects (1.27; 33 partial-ERRs compared to 26 overall-ERRs), the partial-ERRs of the non-NPV projects do account for about two-third of the actual total project costs with about 2,917 Mio USD compared to 1,543 Mio USD total project costs of the overall-ERRs (refer to *Appendix AP*). On the other hand the total project costs of the partial-ERRs of the NPV projects of 16,605 Mio USD are more or less equal to the total project costs of the overall-ERRs of 16,127 Mio USD. Still, looking at the minimum and maximum arithmetic means of the partial-ERRs, they are comparably high for both groups with a 5 percent higher value in the minimum and a 4 percent higher value in the maximum case of the ERRs of the NPV projects. In the case where all “high-value” ERR projects are excluded, the intersectional average ERRs of the 143 projects are slightly higher than the ones from the non-NPV projects, whereat here the partial-ERRs of the non-NPV projects disclose a higher max arithmetic mean with 36.55 percent than the one from the NPV projects with 35.45 percent (refer to *Appendix AS*). In contrast, the overall-ERRs of the non-NPV projects do disclose slightly lower values for the minimum and the maximum case than the overall-ERRs of the NPV projects (refer to *Appendix AR*). However, even in the “high-value” exclusion case the average arithmetic mean of the ERRs of the non-NPV projects lies between 18.91 and 31.35 percent compared to its average discount rate of 10.83 percent.

4.5.2 Continent Comparison

This chapter deals with the comparison of the result values between the African and Asian continents. In order to set the basis of comparison the following table provides an overview of the NPV counts for Africa and Asia, whereat the matrix distinguishes between total, overall- and partial-NPVs per row and the total number and between the total number of ERR counts, the intersection of ERR counts and NPV projects as well as the ERR counts of non-NPV projects:

Africa		ERR Counts		ERR Counts of NPV Projects		ERR Counts of Non-NPV Projects	
Total ERRs	Overall-ERRs	142	45	103	34	39	11
	Partial-ERRs		97		69		28
Asia		ERR Counts		ERR Counts of NPV Projects		ERR Counts of Non-NPV Projects	
Total ERRs	Overall-ERRs	76	36	56	21	20	15
	Partial-ERRs		40		35		5

Table 35: Result ERR Counts of Africa and Asia

Overall, Africa outnumbers Asia with 142 compared to 76 ERRs. However, Asia still shows a higher ERR disclosure ratio of 37 percent (76 out of 206 ERRs) compared to that of Africa which reports 25 percent (142 out of 574 ERRs). For both continents the total partial-ERRs again prevail. In Asia the partial-ERRs count 40 compared to 36 overall-ERRs whereas in Africa there are 97 partial- and 45 overall-ERRs. Interestingly, the reverse is evident in the Asian ERR counts of non-NPV projects, where the overall-ERRs amount to 3 times (15) the number of partial-ERRs (5).

The following table compares the outcomes of the continental ERRs of the result stage, based on the minimum and maximum values of the arithmetic mean and the ERR ranges:

Africa Values in Percentage		Total ERRs		Total ERRs of NPV Projects		Total ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		33	53.18	27.03	45.22	48.79	74.19
ERR-Range	Min Scenario	-8	1,105.65	-4	137.55	-8	1,105.65
	Max Scenario	-8	1,105.65	3.3	202	-8	1,105.65
Actual Total Project Costs [Mio USD]		20,637		16,913		3,724	
Actual Total World Bank Costs [Mio USD]		7,305		5,805		1,500	
Discount Rate [%]		11.14		11.25		10.5	
Actual Project Duration [years]		7.18		7.28		6.9	
ERR-horizon [years]		19.34		19.76		17.21	
Asia Values in Percentage		Total ERRs		Total ERRs of NPV Projects		Total ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		24.32	39.02	26.11	44.39	19.31	23.97
ERR-Range	Min Scenario	3	120	5	120	3	51.5
	Max Scenario	9	311	9	311	11	51.5
Actual Total Project Costs [Mio USD]		16,554		15,819		736	
Actual Total World Bank Costs [Mio USD]		3,507		2,964		542	
Discount Rate [%]		11.27		11.26		11.5	
Actual Project Duration [years]		6.42		6.43		6.4	
ERR-horizon [years]		19.22		19.43		18.13	

Table 36: Result Total ERR Values Comparison of Africa and Asia¹⁶⁵

In general, the arithmetic mean of the African total ERRs is higher with a minimum of 33 and a maximum of 53.18 percent compared to the Asian total ERRs arithmetic mean minimum value of 24.32 and a maximum value of 39.02 percent. The actual total project costs of 20,637 Mio USD of Africa are about 25 percent higher than the 16,554 Mio USD in Asia, whereby the related total WB costs are more than twice as high in Africa with 7,305 Mio USD as in Asia with 3,507 Mio USD. Interestingly, the total ERRs of the NPV projects of both continents do show similarly “high-values” for all ERR and cost values. However, the continental difference can be found in the area of total ERRs of non-NPV projects. Here, the African minimum arithmetic mean is more than twice as high as the Asian maximum arithmetic mean (48.79 percent compared to 23.97 percent). Furthermore, the African maximum arithmetic mean

¹⁶⁵ Cf. *Appendices AT* and *AU* provide an overview of the overall- and partial-ERR outcomes of Africa and Asia.

reaches more than 74 percent, which is more than 7 times higher than the corresponding discount rate of 10.5 percent.

In case the 20 “high-value” ERR projects are excluded the continental ERR counts look as per following table:

Africa		ERR Counts		ERR Counts of NPV Projects		ERR Counts of Non-NPV Projects	
Total ERRs	Overall-ERRs	126	41	91	32	35	9
	Partial-ERRs		85		59		26
Asia		ERR Counts		ERR Counts of NPV Projects		ERR Counts of Non-NPV Projects	
Total ERRs	Overall-ERRs	72	34	52	19	20	15
	Partial-ERRs		38		33		5

Table 37: Result ERR Counts of Africa and Asia excluding “high-value” ERRs

Out of the 200 “high-value” ERR projects, 16 can be found in Africa. Additionally, 4 of the above 200 ERRs also belong to Africa. Obviously, the percentage difference within the disclosure rate of the 2 continents further decreases, namely to 36 percent in Asia (72 out of 202) and to 23 percent in Africa (126 out of 558). Interestingly, there are no changes to the ERR counts of the non-NPV projects in Asia, whereas 4 out of the 16 African “high-value” ERRs belong to this area. The following table compares the outcomes of the continental ERRs based on the minimum and maximum values of the arithmetic mean and the ERR ranges, and based on the exclusion of the 20 “high-value” ERRs:

Africa Values in Percentage		Total ERRs		Total ERRs of NPV Projects		Total ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		22.37	33.43	23.79	32.6	18.69	35.57
ERR-Range	Min Scenario	-8	99	-4	84	-8	99
	Max Scenario	-8	99	3.3	85	-8	99
Actual Total Project Costs [Mio USD]		19,436		15,902		3,534	
Actual Total World Bank Costs [Mio USD]		6,591		5,223		1,368	
Discount Rate [%]		11.13		11.21		10.59	
Actual Project Duration [years]		7.27		7.4		6.94	
ERR-horizon [years]		19.57		20.2		16.33	
Asia Values in Percentage		Total ERRs		Total ERRs of NPV Projects		Total ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		21.94	31.71	22.94	34.69	19.31	23.97
ERR-Range	Min Scenario	3	67.34	5	67.34	3	51.5
	Max Scenario	9	95	9	95	11	51.5
Actual Total Project Costs [Mio USD]		16,470		15,734		736	
Actual Total World Bank Costs [Mio USD]		3,457		2,914		542	
Discount Rate [%]		11.26		11.23		11.5	
Actual Project Duration [years]		6.46		6.48		6.4	
ERR-horizon [years]		19.59		19.89		18.13	

Table 38: Result Total ERRs of Africa and Asia excluding “high-value” ERRs¹⁶⁶

Removing the “high-value” ERRs sheds a different light on the continental comparison. Firstly, the arithmetic means of the 2 continents are now similarly high with 22.37 to 33.43 percent for Africa and 21.94 to 31.71 percent for Asia. For both continents the minimum arithmetic mean is at least twice as high as the corresponding discount rate and approx. 3 times higher in the maximum case. What is interesting to note is the fact that the African minimum and maximum arithmetic means of the ERRs of the NPV projects of 23.79 and 32.6 percent lie between the minimum (18.69 percent) and maximum (35.57 percent) arithmetic means of the ERRs of the non-NPV projects. On the other hand, the minimum (19.31 percent) and maximum (23.97 percent) arithmetic means of the non-NPV projects in Asia are worse than the minimum (22.94 percent) and maximum (34.69 percent) of the NPV projects. However, shrinking differences can be found when comparing these results to the

¹⁶⁶ Cf. *Appendices AV* and *AW* provide an overview of the overall- and partial-ERR outcomes of Africa and Asia when “high-value” ERR projects are excluded.

minimum and maximum arithmetic means of 26.11 and 44.39 percent of the ERRs of NPV projects (refer to *table 36: Result Total ERR Values Comparison of Africa and Asia*).

In terms of total project costs, the 19,436 Mio USD total costs for Africa are 2,966 Mio USD higher than the 16,470 Mio USD total project costs for Asia. Furthermore, the total WB costs of 6,591 Mio USD in Africa are twice as high as the 3,457 Mio USD WB costs in Asia. Therefore, the share of the WB costs of the total project costs is higher in Africa with a share of approx. 34 percent compared to 21 percent in Asia. However, it needs to be noted that this costs analysis includes 3 “high-value” total project costs projects with total project costs above 1,000 Mio USD (2 in Asia and 1 in Africa) where the disclosed ERRs are below 100 percent.¹⁶⁷

4.6 Additional Economic and Financial Indicator Result Values

This chapter deals with the result description of the indicator analysis other than the NPV and the ERR. As described in *chapter 3.1 Additional Economic and Financial Indicators*, additional economic and financial indicators other than NPV and ERR which are also outlined within the ICR reports include the BCR, Net Benefit, FNPV as well as FRR. In order to determine and evaluate the ICR documents on the broadest financial basis, the outcomes of each of these additional economic and financial indicators for various portfolio groups need to also be considered. The first portfolio group provides the results on the total values of the additional economic and financial indicators. The second group then examines the results of the intersectional portfolio values which are shared with the projects either outlining a NPV or an ERR value. In the third group, the portfolio values of the economic and financial indicators which are outlined in non-NPV and non-ERR reports are examined. The fourth and the fifth group are sub-groups of the third portfolio group which are based on the outcome of the following pre-study:

¹⁶⁷ Author's Note: Africa: Tanzania – Report ICR1511: partial-ERR of 8 percent, total project costs of 3,985 Mio USD; Asia: Bangladesh – Report ICR317: overall-ERR of 21 and 51 percent, total project costs of 4,300 Mio USD; Asia: Bangladesh – Report ICR2036: overall-ERR of 16 and 23 percent, total project costs of 7,508 Mio USD.

1. Within the third group there are intersections between the BCRs and the Net Benefits as well as between the FNPVs and the FRRs.
2. Within the third group there are no intersections between the BCRs/Net Benefits and the FNPVs/FRRs.
3. All of the additional economic and financial indicators within the third group are intersectional values (as described under 1.) or single values which represent the only economic or financial indicator within an ICR document.

The fourth group therefore outlines the outcome of the intersectional values for the third group, whilst the fifth group provides an overview of the single value results of the third group for any of the additional economic or financial indicators.¹⁶⁸

4.6.1 Benefit-Cost-Ratio and Net Benefit

This chapter outlines the results of the BCRs and Net Benefits of the result stage within the researched ICR reports. The following table provides an overview of the BCR and Net Benefit counts of the LDCs according to the 5 portfolio groupings mentioned:

		Counts		Counts of NPV/ERR Projects		Counts of Non-NPV/ERR Projects			
BCRs	Overall	26	12	20	9	6	3	Intersection	1
	Partial		14		11		3		1
Net Benefits	Overall	42	18	39	17	3	1	2	1
	Partial		24		22		2		1

Table 39: Result BCR and Net Benefit Counts of the LDCs¹⁶⁹

Even though the total number of Net Benefits is much higher (42) than the total number of BCRs (26), the counts of the non-NPV/ERR projects is higher for the BCRs than for the Net Benefits with a ratio of 2:1. This is actually no surprise given

¹⁶⁸ Author's Note: Due to the limited counts of the additional economic and financial indicators the following analysis will focus on the results of the LDCs only, since a continent comparison is not meaningful.

¹⁶⁹ Author's Note: No analysis on appraisal values of the BCRs and the Net Benefits was conducted.

that in order to calculate the NPV of a project, the Net Benefit must also be calculated as well. However, the disclosure rate of the Net Benefits of the NPV projects is only about 18 percent (30 Net Benefits out of 168 NPVs).

In total there are 39 Net Benefits and 20 BCR counts for the NPV/ERR projects. For each of the indicators, a distinction is made between overall- and partial-values. With the exception of the BCRs of the non-NPV/ERR projects, the partial-values always outnumber the overall-values. Looking at the 2 intersectional values of the non-NPV/ERR projects it can be concluded that there are 4 single BCRs and only 1 single Net Benefit.

The table below shows the outcome of analysis for the total BCRs as well as for the BCRs of the NPV/ERR projects. The arithmetic mean ($\emptyset$) of the BCR is expressed in percentage, whereat the figures for the total project costs as well as for the total WB costs are expressed in Mio USD. Furthermore, a cumulative sum (Cum Sum) underneath the arithmetic mean for both total costs types is given. For all BCR groups and their respective figures, the total, overall and partial-BCRs are provided, whereat the corresponding discount rates, actual project durations and BCR-horizons are only outlined in cases where they have been disclosed within the ICR reports:

		Total BCRs		BCRs of NPV/ERR Projects	
		Ø (Cum Sum)		Ø (Cum Sum)	
BCR	Overall	3.41	2.45	3.8	2.6
	Partial		4.23		4.78
Actual Total Project Costs [Mio USD]	Overall	133 (3,446)	53 (641)	106 (2,110)	41 (367)
	Partial		200 (2,804)		158 (1,743)
Actual Total World Bank Costs [Mio USD]	Overall	57 (1,471)	38 (452)	49 (973)	21 (193)
	Partial		73 (1,019)		71 (781)
Discount Rate [%]	Overall	10.25	11.71	10.68	11.71
	Partial		9.32		9.95
Actual Project Duration [years]	Overall	5.96	5.58	6.45	6.22
	Partial		6.29		6.64
BCR-horizon [years]	Overall	18.54	15.94	19.65	17.64
	Partial		22		22

Table 40: Result BCRs of the LDCs – 1

Overall, the arithmetic mean for all 26 disclosed BCRs is 3.41, demonstrating a strong positive outcome for those projects. Looking at the BCRs of the NPV/ERR projects – which according to the total project costs make up approx. 61 percent (2,110 Mio USD out of 3,446 Mio USD) of the comparison basis– the ratio actually increases to 3.8. The values of the arithmetic mean for the total BCRs range from 1.01 to 10.6 (respectively from 1.01 to 5.85 for the overall-BCRs and from 1.39 to 10.6 for the partial-BCRs). Therefore, the arithmetic mean of the partial-BCRs is a lot higher than that of the corresponding overall-BCRs. Referring back to *chapter 4.3.1 Result NPVs of the LDCs*, this is also reflected in the results for the partial-NPVs when comparing them to the overall-NPVs. Whereat in case where the 7 “high-value” NPV projects have been excluded, the outcome was reversed with slightly lower NPV/capital cost ratios for the overall-NPVs. The following table provides an overview of the remaining 3 groups of the BCR grouping, starting with the third group from the left:

		BCRs of Non-NPV/ERR Projects		BCRs of Non-NPV/ERR Projects – Net Benefit Intersection		BCRs of Non-NPV/ERR Projects – Single	
		Ø (Cum Sum)		Ø (Cum Sum)		Ø (Cum Sum)	
BCR	Overall	2.11	2	3	3.78	1.67	1.11
	Partial		2.23		2.21		2.23
Actual Total Project Costs [Mio USD]	Overall	223 (1,335)	91 (274)	135 (269)	244 (244)	266 (1,066)	15 (30)
	Partial		354 (1,062)		25 (25)		518 (1,036)
Actual Total World Bank Costs [Mio USD]	Overall	83 (498)	86 (259)	130 (260)	234 (234)	59 (238)	12 (24)
	Partial		80 (239)		25 (25)		107 (214)
Discount Rate [%]	Overall	3	n.a.	3	n.a.	n.a.	n.a.
	Partial		3		3		n.a.
Actual Project Duration [years]	Overall	4.33	3.67	3.5	2	4.75	4.5
	Partial		5		5		5
BCR-horizon [years]	Overall	4	4	n.a.	n.a.	4	4
	Partial		n.a.		n.a.		n.a.

Table 41: Result BCRs of the LDCs – 2

In summary, the arithmetic mean of 2.11 for the 6 BCRs of the non-NPV/ERR projects is slightly more than half of the 3.8 arithmetic mean of the NPV/ERR projects. The arithmetic mean of 2.23 for the partial-BCRs is only slightly higher than the one of the overall-BCRs of 2. Out of the 6 BCRs of the non-NPV/ERR, the 2 intersectional ones disclose an average arithmetic mean of 3, whereat in this case the partial-BCR shows a lower arithmetic mean of 2.21 compared against the overall-BCR of 3.78. The arithmetic mean of the 4 single BCRs is even lower with 1.67, whereat the arithmetic mean of the 2 partial-BCRs of 2.23 is twice as high as the one of the 2 overall-BCRs.

In order to ensure that there is a common basis of comparison for the BCRs of the non-NPV/ERR projects with the ones from the NPV/ERR projects, the total project costs need to be considered.¹⁷⁰ The average total projects costs of the overall-BCRs of the NPV/ERR projects is approx. 41 Mio USD (refer to *table 40: Result BCRs of*

¹⁷⁰ Author's Note: In general, the relevant and corresponding project costs need to be considered. Unfortunately, only for a few partial-BCRs the cost basis was provided within the ICR reports. As an alternative the total project costs will be considered.

the LDCs – 1), compared to the total project costs of 244 Mio USD of the 1 intersectional overall-BCR project, and compared to the average total project costs of 15 Mio USD of the 2 single Overall-BCR projects. On the other hand the average total projects costs of the partial-BCRs of the NPV/ERR projects is approx. 158 Mio USD (refer to *table 40: Result BCRs of the LDCs – 1*), compared to the total project costs of 25 Mio USD of the 1 intersectional partial-BCR project and compared to the average total project costs of 518 Mio USD of the 2 single partial-BCR projects. Thus, it needs to be concluded that due to different cost basis –which might be caused by the fact that the quantity of the non-NPV/ERR BCRs is limited only– no general judgment for the non-NPV/ERR project BCRs in comparison to the ones of the NPV/ERR project BCRs can be made.

The following table shows the outcome of the Net Benefit analysis for all of the 5 portfolio groups. The cumulative sum and the arithmetic mean ($\emptyset$) are provided for all groups, whereat the Net Benefit and all cost types are expressed in Mio USD:

Values in Mio USD	Total Net Benefits	Net Benefits of NPV/ERR Projects	Net Benefits of Non-NPV/ERR Projects	Net Benefits of Non-NPV/ERR Projects – BCR Intersection	Net Benefits of Non-NPV/ERR Projects – Single
	Cum Sum ($\emptyset$)	Cum Sum ($\emptyset$)	Cum Sum ($\emptyset$)	Cum Sum ($\emptyset$)	Cum Sum ($\emptyset$)
Net Benefit	17,581 (419)	16,893 (433)	688 (229)	672 (336)	16 (16)
Total Net Benefit Costs	6,721 (160)	6,450 (165)	271 (90)	269 (135)	2 (2)
Actual Total Project Costs	4,685 (112)	4,399 (113)	286 (95)	269 (135)	17 (17)
Actual Total World Bank Costs	1,875 (45)	1,601 (41)	273 (91)	260 (130)	13 (13)
Discount Rate [%]	10.44	10.73	3	3	n.a.
Actual Project Duration [years]	7.1	7.33	4	3.5	5
Net Benefit-horizon [years]	24.87	24.87	n.a.	n.a.	n.a.

Table 42: Result Net Benefits of the LDCs

In total, the 42 disclosed Net Benefits account for a cumulative sum of 17,581 Mio USD with an arithmetic mean of 419 Mio USD. The total Net Benefit costs include

both –capital as well as recurring costs.¹⁷¹ For many disclosed Net Benefits, there is no separation of the 2 cost types provided within the ICR reports and thus no distinction between the overall- and partial-Net Benefits can be made as no proper comparison basis exists. This also explains that in general the disclosed Net Benefit costs are higher than actual total project costs. Obviously, the 39 Net Benefits of the NPV/ERR projects disclose similar values than the total of 42 Net Benefits across all indicators. When comparing the Net Benefits and the Net Benefit costs of the NPV/ERR and non-NPV/ERR project groups, linearity can be recognized since both arithmetic means of the NPV/ERR project group outline values which are nearly twice as high as the ones from the non-NPV/ERR project group. The picture looks completely different when looking at the overall and partial-Net Benefits separately as no linearity can be recognized anymore.¹⁷² However, since there are only 3 non-NPV/ERR Net Benefits compared to 39 NPV/ERR Net Benefits the basis for the group comparison is even worse than the one of the BCRs. Even with the exclusion of 5 of the “high-value” Net Benefits¹⁷³, no linearity or basis of comparison can be found.¹⁷⁴

As a summary of this subchapter, the following points need to be highlighted:

- For the non-NPV/ERR projects, there were 6 BCR projects: 2 Net Benefit intersectional projects and 4 single projects.
- For the intersectional and the single BCRs of the non-NPV/ERR group the outcome was positive, although not as positive as the one of the NPV/ERR projects. Additionally, no direct connection was found between those 2 groups as there was no basis provided for the comparison of total project costs.

¹⁷¹ Author’s Note: For more than half of the disclosed Net Benefits the recurring costs have not been outlined and thus no consideration of them was possible.

¹⁷² Cf. *Appendices AX* and *AY* provide an overview of the total, overall- and partial-Net Benefits.

¹⁷³ Cf. *Appendix AZ* provides an overview of the “high-value” Net Benefit projects.

¹⁷⁴ Cf. *Appendices BA* and *BB* provide an overview of the total, overall- and partial-Net Benefits when “high-value” Net Benefits are excluded.

- For the non-NPV/ERR projects, there were an additional 3 projects outlining Net Benefits: 2 BCR intersectional ones and 1 single project.
- For the intersectional and the single Net Benefits of the non-NPV/ERR group the Net Benefits were higher than the corresponding Net Benefit costs. However due to a lack of comparison basis, no direct connection to the outcomes of the NPV/ERR project Net Benefits could be made.

4.6.2 Financial Net Present Value and Financial Rate of Return

This chapter outlines the results of the FNPVs and FRRs of the result stage within the researched ICR documents. The table below provides an overview of the FNPV and the FRR counts of the LDCs:

	Counts	Counts of NPV/ERR Projects	Counts of Non-NPV/ERR Projects	
FNPVs	38	32	6	Intersection
FRRs	57	49	8	5

Table 43: Result FNPVs and FRR Counts of the LDCs¹⁷⁵

Overall, the total FRRs (57) outnumber the total FNPVs (38). For both financial indicators, NPV/ERR projects comprise approx. 85 percent of the total project counts (i.e. 32 out of 38 for the FNPVs and 49 out of 57 for the FRRs). In total there are 6 FNPVs and 8 FRRs within the non-NPV/ERR project group, with an intersection of 5 projects. Within the 38 counts of FNPVs, there is 1 neutral and 5 negative FNPVs which all belong to the NPV/ERR projects. Additionally, the counts of the non-NPV/ERR projects include 1 “high-value” FNPV project.¹⁷⁶ Since the FNPV and the FRR always refer to a certain object of consideration, there is no differentiation between overall- and partial-values. However, for both financial indicators there are standalone- and scenario-FNPVs and FRRs.

¹⁷⁵ Author’s Note: No analysis on appraisal values of the BCRs and the Net Benefits has been conducted.

¹⁷⁶ Cf. *Appendix BC* provides an overview of the single “high-value” FNPV project.

In order to compare the FNPVs of the 5 portfolio groups, the following table provides an overview of the results of the LDCs. This data set excludes the single identified “high-value” FNPV project belonging to the NPV/ERR project group.¹⁷⁷ When analyzing the FNPV values, the key issue encountered was the missing costs for the objects of consideration of the FNPVs, as for the majority of the FNPV disclosing ICR reports the corresponding costs were not outlined.¹⁷⁸ However, as an approximation for the comparison, the total project costs for similar objects of considerations have been taken into consideration. The following table provides an overview of the FNPVs of the LDCs whereat the same format structure as for the NPVs applies:

Values in Mio USD		Total FNPVs		FNPVs of NPV/ERR Projects		FNPVs of Non-NPV/ERR Projects		FNPVs of Non-NPV/ERR Projects – FRR Intersection		FNPVs of Non-NPV/ERR Projects – Single	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
FNPV	Minimum	469	13	461	14	8	2	8	2	n.a.	n.a.
	Maximum	510	14	502	16	8	2	8	2	n.a.	n.a.
Actual Total Project Costs		2,378	64	2,280	71	99	20	99	20	n.a.	n.a.
Actual Total World Bank Costs		1,548	42	1,455	45	93	19	93	19	n.a.	n.a.
Discount Rate [%]		10.89		10.87		11		11		n.a.	
Actual Project Duration [years]		7.05		7.19		6.2		6.2		n.a.	
FNPV-horizon [years]		16.7		17.93		8.5		8.5		n.a.	

Table 44: Result FNPVs of the LDCs excluding the single “high-value” FNPV Project¹⁷⁹

When comparing the 5 non-NPV/ERR project group FNPV values with the NPV/ERR project group values there are 2 things to be noted:

1. The “high-value” FNPV project is the 1 single FNPV of the non-NPV/ERR project group. Therefore, all of the non-NPV/ERR project group FNPVs consist of an intersection with the FRR non-NPV/ERR project group.

¹⁷⁷ Cf. *Appendices BD* and *BE* provide an overview of the total result FNPV as well as a comparison of the total, positive, neutral and negative FNPVs.

¹⁷⁸ Cf. *Appendix BF* provides an overview of the objects of consideration of the total FNPVs.

¹⁷⁹ Cf. *Appendix BG* provides a comparison of the total, positive, neutral and negative FNPVs when the single “high-value” FNPV project is excluded.

2. The 5 intersectional FNPVs disclose a positive arithmetic mean for the FNPVs, whereas the comparison based on the total project costs is not meaningful due to different objects of considerations (different cost objects).

Below table provides an overview of the FRRs of the LDCs whereat the same format structure as for the ERRs applies:

Values in Percentage		Total FRRs		FRRs of NPV/ERR Projects		FRRs of Non-NPV/ERR Projects		FRRs of Non-NPV/ERR Projects – FNPV Intersection		FRRs of Non-NPV/ERR Projects – Single	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
FRR-Arithmetic Mean ($\emptyset$)		22.77	53.69	20.53	54.64	36.5	47.88	34	51.8	40.67	41.33
FRR-Range	Min Scenario	-7.66	116	-7.66	116	12	77	12	77	13	66
	Max Scenario	-7.66	847	-7.66	847	13	100	14	100	13	68
Actual Total Project Costs [Mio USD]		4,782		4,627		155		99		56	
Actual Total World Bank Costs [Mio USD]		2,481		2,346		136		93		43	
Discount Rate [%]		10.9		10.92		10.8		11		10	
Actual Project Duration [years]		7.16		7.33		6.13		6.2		6	
FRR-horizon [years]		17.96		19.15		8.5		8.5		n.a.	

Table 45: Result FRRs of the LDCs

As indicated in the above illustration, the FRRs of the NPV/ERR projects show similar arithmetic means and FRR ranges compared to the total FRRs. 8 of the FRRs pertaining to the non-NPV/ERR projects show a 16 percent higher minimum arithmetic mean compared to the FRRs of the NPV/ERR projects, and is thus at least 3 times higher than the corresponding discount rate. On the other hand their maximum arithmetic mean of 47.88 percent is approx. 7 percent lower than one of the NPV/ERR projects. Looking at the intersectional FRRs as well as at the single ones, they also disclose minimum arithmetic means which are at least 3 to 4 times higher than the corresponding discount rates. However, the total project costs of 155 Mio USD of the 8 non-NPV/ERR FRRs have no correlation to the total project costs of 4,627 Mio USD of the 49 NPV/ERR FRRs. This is largely due to the fact that in contrast to the ERR, the FRR is based on a specific object of consideration which may either be a minor or major stakeholder, or comprise multiple stakeholders of the project at the same time. Unfortunately, for the majority of the outlined FRRs the costs of the objects of consideration could not be identified within the corresponding

ICR reports.¹⁸⁰ Of the 57 FRRs, there are 3 projects which disclosed FRR values above 100 percent.¹⁸¹ All of those FRRs belong to the NPV/ERR project group. In case the “high-value” FRR projects are excluded, the difference of the arithmetic means between the NPV/ERR and the non-NPV/ERR project group increases, whereat the maximum arithmetic mean of 26.4 percent of the NPV/ERR project group is around 10 percent lower than the minimum arithmetic mean of 36.5 percent of the non-NPV/ERR project group.¹⁸² However, there are still no noteworthy changes to the total project costs.¹⁸³

As an outcome of this subchapter the following points need to be captured:

- For the non-NPV/ERR project group FNPVs, there were in total 6 projects: 5 FRR intersectional ones and 1 single project, whereat the single FNPV project needed to be handled as a “high-value” FNPV.
- For the non-NPV/ERR project group FRRs, there were an additional 8 projects outlining FRRs: 5 FNPV intersectional ones and 3 single projects.
- For both financial indicators, disclosure details such as the total costs of the object of consideration were only sporadically provided within the ICR reports, and thus no direct comparison of the outcomes of the NPV/ERR project FNPVs/FRRs and the ones from the non-NPV/ERR projects was possible.
- However, the analysis showed that for the additional 9 projects (5 intersectional projects, a single “high-value” FNPV and 3 single FRR projects) positive outcomes for certain project objectives were achieved.

¹⁸⁰ Cf. *Appendix BF* provides an overview of the objects of consideration of the total FRRs.

¹⁸¹ Cf. *Appendix BH* provides an overview of the 3 “high-value” FRR projects.

¹⁸² Author’s Note: The data set excluding “high-value” FRRs still includes 3 projects which disclose FRR values of 100 percent exactly, with 2 of them belonging to the NPV/ERR project group and 1 to the non-NPV/ERR project group.

¹⁸³ *Appendix BI* provides an overview of the result FRRs when the “high-value” FRRs are excluded.

4.7 Overall Project Performance

This chapter deals with the overall project performance of the 790 ICR reports. In order to set the basis for a common understanding the following chapter provides the definition and visualizes the components determining the overall performance of a project. Following this, the project performance for the relevant economic and financial groups (as defined in the previous chapter) as well as for non-financial indicator projects will be evaluate and discussed. This analysis is conducted for the purpose of finding commonalities or linkages between the economic and financial indicator outcomes and the overall project performances. In the last subchapter a calculation approach for assessing the possible outcomes of all “non-financial indicator” projects –those are all projects which do not outline any economic or financial indicator– is provided. For this assessment, the NPV and the ERR serve as the key and determining elements.

4.7.1 Definition

The overall project performance defines the overall outcome of a project. The performance itself is described by the overall outcome rating which is defined as “the extent to which the operation's major relevant objectives were achieved, or are expected to be achieved, efficiently.”¹⁸⁴ Therefore, the overall outcome rating serves as the measurement of the project's overall performance, whereat it is derived from prior assessments of the relevance of objectives and design, and the degree to which each of the project objectives have been achieved and efficiency. Since the WB is defined as an objective-based institution, the measurement of achievements of the project development objectives (PDOs) has priority. In order to follow a common evaluation approach, each project's objectives encompass both –PDOs and key associated outcome targets. In cases where the PDOs are vaguely worded, the key associated outcome targets will be used for the detailed project objectives as-

¹⁸⁴ WB (2013), p. 30.

assessment. The goal of the outcome rating is to assess which of the defined objectives have been achieved or are to be expected to be achieved, efficiently. In principle, the guidance for the evaluators is based on the assessment of shortcomings¹⁸⁵:

- Shortcomings in the achievement of objectives: Some of the objectives have not been achieved/are not expected to be achieved, or where some of the objectives have been achieved/are expected to be achieved to a certain extent only.
- Shortcomings in efficiency: The project fails to achieve/is not expected to achieve a return which exceeds the opportunity cost of capital and/or is not the least cost alternative.
- Shortcomings in relevance: The extent to which the project's objectives, design or implementation are inconsistent with the country's development plans and the sectorial assistance strategies and aligned corporate goals with the WB.

The rating scale for the overall project performance consists of 6 classifications, which are described as per the following definition:¹⁸⁶

	Rating Category		Definition
Positive Ratings	Highly Satisfactory	●●●	No project shortcomings in the achievement of its objectives, in its efficiency, or in its relevance.
	Satisfactory	●●	Minor project shortcomings in the achievement of its objectives, in its efficiency, or in its relevance.
	Moderately Satisfactory	●	Moderate project shortcomings in the achievement of its objectives, in its efficiency, or in its relevance.
Negative Ratings	Moderately Unsatisfactory	●	Significant project shortcomings in the achievement of its objectives, in its efficiency, or in its relevance.
	Unsatisfactory	●●	Major project shortcomings in the achievement of its objectives, in its efficiency, or in its relevance.
	Highly Unsatisfactory	●●●	Severe project shortcomings in the achievement of its objectives, in its efficiency, or in its relevance.

Table 46: Overall Project Performance Rating Categories¹⁸⁷

¹⁸⁵ Cf. WB (2013), p. 30 et sqq.

¹⁸⁶ Cf. WB (2013), p. 31 et seq.

¹⁸⁷ Author's Note: The content analysis will only focus on the negative ratings in order to provide the reasons for bad performances within each area.

Besides the overall project performance there are 3 additional key performance areas which are evaluated during the course of the project implementation period. All of these additional performance areas will be described and handled in more detail during the course of the content analysis (refer to *chapter 5 Qualitative Data Analysis*):

- Sustainability (Risk to Development Outcome) Rating: Refer to *chapter 5.2 Sustainability*
- Bank Performance: Refer to *chapter 5.3 Bank Performance*
 - Quality at Entry (beginning after the design phase with project start)
 - Quality of Supervision (of the WB)
- Borrower Performance: Refer to *chapter 5.4 Borrower Performance*
 - Government Performance
 - Implementation Agency Performance

Each of these performance areas is evaluated separately. As shown above, the bank and borrower performances both contain sub-areas of performance. The same rating scale used for the overall project performance applies, with the exception for sustainability which is based on other rating categories and requires specific handling. The following figure shows the performance framework of the WB evaluation process:

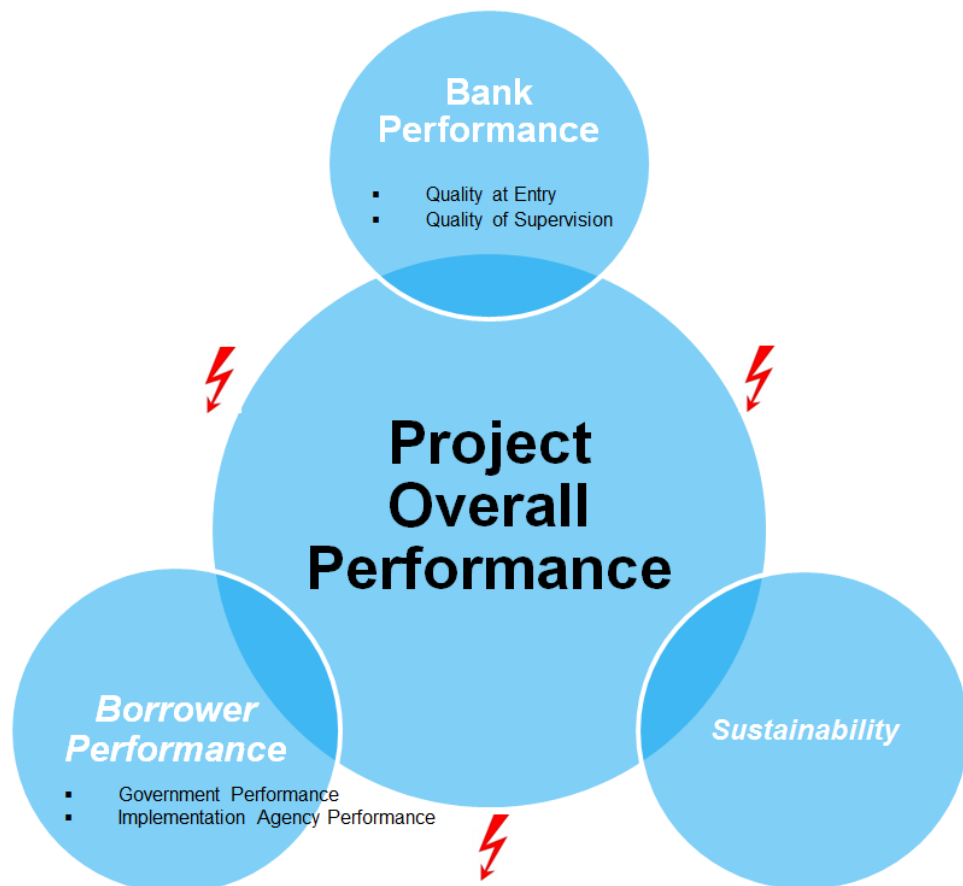


Figure 2: Overall Project Performance Categories

As shown above, each of the additional key performance areas has the potential to affect the overall project performance. Even though each of the additional performance areas is based on a separate assessment, a weak performance in one of those areas generally leads to a downgrading of the overall project performance. There may be cases where a negative borrower performance does not necessarily affect the overall project performance, since the PDOs are not affected. However, usually negative performances of the additional performance areas do affect the overall outcome of a project. An example of such would be the case of natural disasters which affect the sustainability of a project, and may potentially lead to a limited achievement of a certain PDO.

4.7.2 Overall Project Performance Ratings

This chapter outlines the overall project performance for different groups which have already been presented and discussed in the previous chapters. The principle idea is to see if there are some parallels between the overall project performance and the outcome of the economic and financial analysis. Therefore, the ratio of the negative projects in comparison to total number of projects defines the negative share of each group. For each group the total negative share as well as the shares per rating category are given. The following table provides an overview of the 3 major groups, namely the “all projects” group containing all 790 researched ICR reports, the “all financial indicator projects” group which consists of all projects outlining at least one of the presented economic or financial indicators and the “non-financial indicator projects” group which comprises all projects that do not outline any economic or financial indicator:

		All Projects		All Financial Indicator Projects		Non-Financial Indicator Projects	
		Counts: 789		Counts: 243		Counts: 546	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	19.01	150	13.58	33	21.43	117
“Moderately Unsatisfactory” Share	negative	7.35	58	5.76	14	8.06	44
“Unsatisfactory” Share	negative	11.41	90	7.82	19	13	71
“Highly Unsatisfactory” Share	negative	0.25	2	0	0	0.37	2

Table 47: Negative Project Shares – Group Comparison 1¹⁸⁸

In total there are 150 negative performance ratings out of 789 ICR reports, which lead to a total negative share of 19.01 percent. The all financial indicator group containing only 33 negative performance projects out of a total of 243 projects therefore shows a lower but respectively better total negative share of 13.58 percent compared against the 21.43 percent of the non-financial indicator group. In total there

¹⁸⁸ Author’s Note: The difference of 1 count for the 789 counts of the all projects group is due to the ICR report 36569 of Chad which does not outline any performance rating.

are only 2 “Highly Unsatisfactory” rated projects. Interestingly, for all groups the majority of negative ratings can be found in the “Unsatisfactory” category.¹⁸⁹ The following table illustrates the outcome of the different financial groups:

		Total NPV Projects		Total ERR Projects		Other Financial Indicator Projects	
		Counts: 168		Counts: 218		Counts: 113	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	12.5	21	13.3	29	14.16	16
“Moderately Unsatisfactory” Share	negative	5.36	9	5.96	13	6.19	7
“Unsatisfactory” Share	negative	7.14	12	7.34	16	7.96	9
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

Table 48: Negative Project Shares – Group Comparison 2¹⁹⁰

The outcome of the total negative shares for the 3 groups shows that the group with the total NPVs leads the financial groups comparison with 12.5 percent, followed by the total ERR projects with 13.3 percent and the other financial indicators projects¹⁹¹ with 14.16 percent. In order to compare the ERR outcomes presented in *chapter 4.5 Economic Rate of Return Result Values, table 32: Result ERRs of the LDCs* the ERRs of the NPV projects and the non-NPV projects need to be considered as well. The following table examines the results of the ERRs of NPV projects as well as of non-NPV projects:

¹⁸⁹ Cf. *Appendix BJ* shows the results for the 3 groups when all “high-value” projects of all financial indicators are excluded, whereas there are no noteworthy changes.

¹⁹⁰ *Appendix BK* shows the results for the groups comparison when all “high-value” projects of all financial indicators are excluded, whereas there are no noteworthy changes.

¹⁹¹ Author’s Note: This group consists of all projects outlining any economic or financial indicator other than the NPV or ERR, whereas those projects might as well outline a NPV or ERR value.

		ERRs of NPV Projects		ERRs of non-NPV Projects	
		Counts: 159		Counts: 59	
		%	Counts	%	Counts
Total Negative Share	negative	11.95	19	16.95	10
"Moderately Unsatisfactory" Share	negative	5.66	9	6.78	4
"Unsatisfactory" Share	negative	6.29	10	10.17	6
"Highly Unsatisfactory" Share	negative	0	0	0	0

Table 49: Negative Project Shares – Group Comparison 3

When considering the total negative share of 11.95 percent of the 159 ERR disclosing NPV projects the difference is even larger compared to the negative share of 13.3 percent of the total ERR projects. Therefore, it needs to be noted that there is no exact linearity found between the overall project performance and the ERR arithmetic mean. However, in case where the “high-values” are excluded from the ERR outcomes as presented in *chapter 4.5 Economic Rate of Return Result Values, table 34: Result ERRs of the LDCs excluding “high-value” ERR projects*, both the minimum and maximum arithmetic mean of ERRs of the NPV projects are higher than those of the total ERRs and the ERRs of the non-NPV projects. This result is in line with the outcome of the above table, namely that higher ERR arithmetic means are linked to a better overall project performance. This relation is further endorsed in case all “high-values” or only the “high-value” ERRs are excluded when calculating the negative share of the overall project performance (refer to *Appendices BL and BM*).

Still, no concrete linearity can be determined, since the minimal difference within the arithmetic means does not explain the larger difference between the overall performances of the groups. In support of more evidence for this linkage, the other financial indicator groups could serve as a control group. However, even though the negative shares for both groups –the total financial indicator group (refer to *table 49: Negative Project Shares – Group Comparison 2*) and the one of the non-NPV/ERR

projects (refer to *table 49: Negative Project Shares – Group Comparison 3*)– underline the found linkage, it is difficult to make a final conclusion due to following reasons:

1. There are 4 different types of additional economic and financial indicators, which due to the lack of disclosure quantity could not be compared based on the ERR basis.
2. For the FNPV and FRR, the related capital costs for the objects of consideration could not be gathered from the ICR reports, as some recurring costs for the Net Benefit consideration appear to be absent.

In order to still substantiate the relation between the overall project performance and the ERR arithmetic mean, the table below presents the results of the negative and neutral NPV/ERR as well as the ones from the positive NPV and ERR projects:

		Negative & Neutral NPV/ERR Projects		Positive NPV/ERR Projects	
		Counts: 19		Counts: 191	
		%	Counts	%	Counts
Total Negative Share	negative	42.11	8	8.9	17
“Moderately Unsatisfactory” Share	negative	15.79	3	3.66	7
“Unsatisfactory” Share	negative	26.31	5	5.24	10
“Highly Unsatisfactory” Share	negative	0	0	0	0

Table 50: Negative Project Shares – Group Comparison 4

For the negative and neutral NPVs and ERRs only projects which disclose either a negative or a neutral ERR¹⁹² or NPV or both financial indicators are considered. The reason to include neutral values as well is to increase the quantity of the counts.

¹⁹² Author’s Note: An ERR value below 10 percent can be defined as neutral and below 0 percent as negative.

While the 19 counts of this group might not be representative, the total negative share of 42.11 percent favors the hypothesis that the lower the ERRs, the higher the negative share of the overall project performance. The positive NPV/ERR projects include only positive ERRs and/or positive NPVs, whereat the 191 counts represent an appropriate share of the total ICR reports. Their low negative share of 8.9 percent again favors the hypothesis.

When excluding all “high-values” the results are even more distinct with a negative share of 47.06 percent for the negative and neutral NPV/ERR projects and a negative share of 8.07 percent for the positive NPVs and ERRs.¹⁹³ Therefore, the results of this analysis which need to be captured are:

- There are no “Highly Unsatisfactory” ratings for positive NPV or ERR projects.¹⁹⁴
- Positive NPV and ERR projects disclose negative overall performance ratings. The reasons behind negative overall outcomes of positive NPV and ERR projects can be diverse. Examples may include instances where a financial perspective has not been defined within the PDOs, or the project was never supposed to reach a positive NPV (e.g. financial support project).
- Even though there is no proven evidence for a direct linearity between the overall project performance and the ERR arithmetic mean the analysis based on the exclusion of “high-values” has shown that the higher the ERR arithmetic the lower the negative share of the overall project performance.

4.7.3 Outcome Calculation for Non-Financial Indicator Projects

As outlined in the previous chapter, the negative share of the overall project performance generally corresponds with the ERR arithmetic mean. Therefore, this chapter aims to make use of this linkage by iteratively calculating a possible ERR arithmetic

¹⁹³ Cf. *Appendix BN* shows the results for the groups comparison when all “high-value” projects of all financial indicators are excluded.

¹⁹⁴ Author’s Note: This has been verified throughout all NPV and ERR projects.

mean for the non-financial indicator projects. In order to do so the following points need to be taken into consideration:

- The 2 main ERR groups, namely the ERRs of the NPV projects and the ERRs of the non-NPV projects serve as the central element to determine the possible ERR arithmetic mean of the non-financial indicator projects, given that all disclosed ERRs pertain to either one of them.
- Aside from the 2 mentioned groups, the negative and neutral NPV/ERR project group and the positive NPV/ERR project group are also considered so as to account for the most extreme negative/neutral and positive ERR arithmetic means (refer to *table 50: Negative Project Shares – Group Comparison 4*).
- There is no differentiation between overall- and partial-ERRs or NPVs due to the following reasons:
 1. Not all overall- and partial-NPVs relate to overall- and partial-ERRs and as such, the counts of the overall- and partial-NPV/ERR sub-groups is too limited to serve as a calculation basis.
 2. Partial-NPV and ERR projects generally disclose lower negative shares compared to overall-NPV and ERR projects. Thus, a high negative share for one of the above mentioned ERR groups cannot be argued with differences between overall- and partial-NPV and ERR projects.¹⁹⁵

The following table provides an overview of the source groups and their corresponding counts and ERR arithmetic mean values as well as of the non-financial indicator ERR group for which the minimum and the maximum ERR arithmetic mean should be calculated:

¹⁹⁵ Cf. *Appendices BO* till *BT* present the outcomes of the overall- and partial- NPV and ERR analyses for all projects, for the ones when “high-value” NPVs/ERRs are excluded and for the projects where no “high-values” at all are contained.

	Negative Share		ERR-Arithmetic Mean	
	Negative Counts	%	Min (Ø)	Max (Ø)
ERRs of NPV Projects	19	11.95	26.7	44.93
ERRs of non-NPV Projects	10	16.95	38.8	57.17
ERRs of Negative & Neutral NPV/ERR Projects	8	42.11	2.93	37.52
ERRs of Positive NPV/ERR Projects	17	8.9	34.22	50.73
<u>ERRs</u> of Non-Financial Indicator Projects	117	21.43	?	?

Table 51: Missing ERRs of Non-Financial Indicator Projects

As per the boundaries set by the source groups the ERR arithmetic mean of the non-financial indicator projects should lie somewhere between 2.93 percent and 57.17 percent. In order to further narrow down the boundary percentages an iterative approximation¹⁹⁶ can be used. In principle, this approximation follows a two-step approach. In the first step the “ERR Share Factor” based on the minimum and the maximum arithmetic means of the 4 source groups is calculated. The ERR share factor is the quotient of the sum of the negative shares and the sum of the minimum and maximum (respectively) arithmetic means of the ERRs of the 4 source groups. The minimum and maximum ERR share factors for the table above are as follows:¹⁹⁷

$$ERR\ Share\ Factor_{Min} = \frac{(11.95+16.95+42.11+8.9)}{(26.7+38.8+2.93+34.22)} = 0.78$$

$$ERR\ Share\ Factor_{Max} = \frac{(11.95+16.95+42.11+8.9)}{(44.93+57.17+37.52+50.73)} = 0.42$$

Equation 7: ERR Share Factor

¹⁹⁶ Cf. Freund/Hoppe (2007), unpag.

¹⁹⁷ Author’s Note: To calculate the ERR share factor based on an unweighted approach using the dividend of the sum of the negative shares divided by the sum of the ERR arithmetic means has proven to provide more reasonable results than using a weighted ERR share factor (refer to *Appendices BU* and *BV*).

Each “Share Difference” which is the difference between the negative share of the source group and the non-financial indicator group can then be divided by the minimum respectively maximum ERR-share factor. The result of each group is then subtracted from the corresponding minimum respectively maximum ERR arithmetic mean of the same group. The minimum and maximum values of the calculated minimum and maximum ERRs of all groups then define the outer boundary values of the ERRs of the non-financial indicator projects. In the second iterative step, the arithmetic means of the calculated minimum and maximum ERRs of all groups can be calculated as to further narrow down the results of the searched minimum and maximum ERRs of the non-financial indicator projects:

	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
	%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	11.95	26.7	44.93	-9.48	14.55	22.36
ERRs of non-NPV Projects	16.95	38.8	57.17	-4.48	33.06	46.5
ERRs of Negative & Neutral NPV/ERR Projects	42.11	2.93	37.52	20.68	29.44	86.76
ERRs of Positive NPV/ERR Projects	8.9	34.22	50.73	-12.53	18.16	20.9
<u>Calculated ERRs of Non-Financial Indicator Projects</u>	21.43	Ø =			23.80	44.13

Table 52: Calculated ERRs of Non-Financial Indicator Projects

The minimum value of 14.55 percent and the maximum value of 86.76 percent describe the respective minimum and maximum value of the minimum and maximum ERR arithmetic mean of the non-financial indicator projects group. The calculated arithmetic means of 23.8 and 44.13 percent further narrow down possible minimum and maximum ERR arithmetic means of the non-financial indicator group. Obviously, this calculation follows an unweighted approach since no total counts of the source groups have been considered. However, the results of 23.8 and 44.13 percent are more reasonable than following weighted approaches:

- The calculated minimum ERR of 23.8 percent is lower than the minimum ERR arithmetic means of all groups disclosing lower negative shares, but as well higher than the one of the negative and neutral NPV/ERR projects and thus favors the hypothesis that the lower the ERRs, the higher the negative share of the overall project performance.
- Even though the maximum calculated ERR of 44.13 percent is only slightly lower than the one of the ERRs of the NPV projects, it is still higher than the one of the negative and neutral NPV/ERR projects and thus favors the hypothesis as well.
- For each of the 2 weighted approaches for calculating the minimum and maximum ERR of the non-financial indicator groups (refer to *Appendices BV* and *BW*) the calculated minimum arithmetic mean ERRs appear to be too low with 16.66 and 19.37 percent. Furthermore, both calculated maximum arithmetic mean ERRs are lower than the corresponding maximum ERR arithmetic mean of the negative and neutral NPV/ERR projects, although the negative share of the non-financial indicator projects is nearly twice as low as the share of the negative and neutral NPV/ERR projects.

However, as shown in previous analyses, “high-values” always lead to distorted results. Therefore, the following table provides the outcome of the ERR calculation based on the exclusion of all “high-value” projects:

	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
	%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	11.68	22.98	32.56	-9.75	14.5	17.56
ERRs of non-NPV Projects	16.36	18.91	31.35	-5.07	14.5	23.55
ERRs of Negative & Neutral NPV/ERR Projects	44.44	2.92	26.69	23.91	23.71	63.12
ERRs of Positive NPV/ERR Projects	7.78	24.82	33.37	-13.65	12.95	13.45
<u>Calculated ERRs</u> of Non-Financial Indicator Projects	21.43			Ø =	16.42	29.42

Table 53: Calculated ERRs of Non-Financial Indicator Projects excluding “High-values”

When excluding the “high-value” projects for the unweighted approach the calculated minimum ERR of 16.42 percent and the calculated maximum ERR of 29.42 percent again favor the hypothesis. On the other hand, both weighted approaches outline the same issues –disclosing lower minimum values than the unweighted approach and lower maximum values than the maximum value of the negative and neutral NPV/ERR projects.¹⁹⁸

4.7.4 Verification of Outcomes and Conclusion

Obviously, the question on ‘how realistic are the presented outcomes in terms of reliability and accuracy’ arrives. Not only because of the “simplified” calculation approach itself, rather because of the fact that the ICR department is WB owned and thus data might be positively biased. In order to validate ICR overall project performance ratings the data basis of the IEG can be used.¹⁹⁹ The following table provides an overview of an exemplary comparison of 40 ICR overall project performances and the corresponding IEG evaluation:²⁰⁰

¹⁹⁸ Cf. *Appendices BX, BY and BZ* provide an overview of the second unweighted and the 2 weighted approaches for calculating the ERRs of the non-financial indicator groups when all “high-values” are excluded.

¹⁹⁹ Author’s Note: Access the IEG data evaluation catalog here: <http://data.worldbank.org/data-catalog/IEG> (call on 2014/12/30).

²⁰⁰ Author’s Note: The projects for the comparison have been selected randomly, containing 33 NPV and 7 non-NPV projects. *Appendix CA* provides an overview of the listing of the compared reports.

	Project Overall Performance	
	ICR	IEG
Total Counts	40	40
Total Negative Counts	10	10
“Moderately Unsatisfactory”	7	6
“Unsatisfactory”	3	4
“Highly Unsatisfactory”	n.a.	n.a.
Total Positive Counts	30	30
“Moderately Satisfactory”	6	13
“Satisfactory”	21	15
“Highly Satisfactory”	3	2

Table 54: ICR vs. IEG Overall Project Performance Comparison

What can be noticed at a glance is that the total counts for the negative (10) and positive (30) overall project performance ratings are the same for the ICR and the IEG evaluations. However, there are differences within the sub-rating categories: IEG evaluation is generally worse outlining more “Unsatisfactory” but less “Moderately Unsatisfactory” ratings for the negative ratings and more “Moderately Satisfactory” but less “satisfactory” for the positive overall project performances than the ICR reports.

Finally, it needs to be noted that even though the overall project performance seems to relate to the ERR arithmetic means, the iterative approaches discussed above are only approximations, since no further project details were considered. The principle idea is to provide the reader with a better feeling for potential ERR outcomes of the non-financial indicator projects. Another point to mention is that this analysis did not distinguished between the underlying detailed rating categories of the positive and negative overall project performances. For the negative sub-category ratings there are only negligible differences, but stronger differences for the positive rating outcomes. Last but not least, the actual results of this analysis may deviate from the outcomes presented, since no additional hints on “lower” or “higher” ERRs

for any of the 546 non-financial indicator projects were provided within the corresponding ICR reports.

4.8 NPV-Cost-Ratios and SROI Calculation

This chapter deals with the results of the calculated NPVCRs of the result stage. The first chapter outlines and discusses the results for the different types of NPVCRs identified for this research project (refer to *chapter 3.2.3 Types of NPV-Cost-Ratios*). Hereby, the difference between all LDC projects and the portfolio excluding “high-value” projects as well as the continent comparison are the focus. In the second chapter missing figures and indicators are calculated using standard values based on the LDCs and the continents itself. The last chapter then deals with the evaluation of the proper SROI ratio whereat certain assumptions are made and variations of the results are examined.

4.8.1 NPV-Cost-Ratios of the ICR Reports

This chapter examines the results of the described NPVCRs. In order to do so the evaluation data to calculate the BCRs was taken directly out of ICR reports and was left unmodified. Therefore, no attempt to calculate any standard values such as discount rates, NPV-horizons or the recurring costs was made for this analysis. However, it needs to be mentioned that the discounted values for

- capital costs,
- remaining capital costs as well as
- the recurring costs

are generally not disclosed within the ICR reports (except for few cases only). Usually, only undiscounted costs can be extracted from the ICR reports. As to calculate the discounted values the project respective discount rate, project duration as well as the NPV-horizon need to be obtainable. If all of the mentioned indicators were outlined within the report, all cost values were distributed evenly across the project

duration respective to the NPV-horizon²⁰¹, in case no other information on the real distribution of costs was provided. The outcomes of the analysis focus on the arithmetic means of all available values per NPVCR. It is distinguished between the minimum and maximum value of the weighted and the unweighted NPVCRs. The minimum and maximum values for each of the ratios are based on the standalone- and scenario-NPVs (refer to *chapter 3.2.2.1 Net Present Value*). The weighted ratios are defined as the summation of each cost type and the NPVs across all NPV projects first before their sums are used to construct the NPVCRs. The unweighted NPVCRs are the arithmetic means of each project NPVCR.

4.8.1.1 Overall Results

The following table provides an overview of the results of the discounted and undiscounted NPVCRs of the LDCs. For each of the NPVCRs the counts of projects as well as the minimum and maximum values for the weighted and unweighted ratios are given:

		LDCs				
		Counts	Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	168	2.8	3.15	2.89	3.62
	discounted	136	3.6	4.13	4.11	5.4
Total-Capital-Costs Ratio	undiscounted	168	2.25	2.51	2.39	3.03
	discounted	136	2.82	3.2	3.41	4.55
Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	71	2.31	2.58	1.71	2.48
	discounted	57	3.31	3.78	2.11	3.51
Total-Capital plus Recurring-Costs Ratio	undiscounted	71	2	2.21	1.51	2.16
	discounted	57	2.68	3.02	1.85	3.03

Table 55: Disclosed NPVCRs of the LDCs

²⁰¹ Author's Note: This is the most meaningful approach based on the evaluation of ICR reports disclosing discounted cost values (refer to *chapter 3.2.2.2 Capital and Recurring Costs*).

The “Pro-Rata-Capital Costs Ratio” and the “Total-Capital Costs Ratio” are based on the same project counts for the discounted and the undiscounted ratios, as well as the “Pro-Rata-Capital plus Recurring-Costs Ratio” and the “Total-Capital plus Recurring-Costs Ratio”. The only difference within each of the 2 ratios is that the pro-rata ratio only considers the partial-NPV related capital costs, whereat the total-capital ratio additionally takes the remaining capital costs respectively the total project costs into account. The difference of 32 project counts between each of the discounted version compared to the undiscounted version of the pro-rata-capital costs and the total-capital-costs ratio is due to missing discount rates. In comparison, the difference of 14 project counts for discounted pro-rata-capital plus recurring-costs as well as for the total-capital plus recurring-costs ratio is additionally due to missing NPV-horizons. In total, 54 out of 168 projects disclose recurring costs, whereas only 17 out of the 168 projects are not based on recurring costs. For 13 out of the 54 projects, the recurring costs are part of the capital cost framework, whereat it is not possible to identify the amount of recurring costs separately. That none of the disclosed NPVCRs is below 1 indicates that for each NPVCR the corresponding NPV sum must be positive (refer to *chapter 3.2.2.1 Net Present Value, table 11: Result NPV Counts of the LDCs*: In total there are only 7 negative and 7 neutral NPVs out of 168 NPV projects). Looking at the discounted ratios only the single NPVCR disclosing a value below 2 is the total-capital plus recurring-costs ratio; all other NPVCRs are all above 2. Specifically highlighting the pro-rata-capital plus recurring-costs ratio –which as per the author’s understanding examines the SROI ratios as per the definition of the WB– it needs to be noted that the outcome is highly positive. For both the unweighted and the weighted ratio values are far above 2 and 3. The weighted ratio basically indicates that the monetary input for development aid actually generates an output which is at least 3.31 times higher in the minimum case, with the potential to grow to 3.78 times in the maximum case. On the other hand the unweighted ratio shows the average span of increase per project independent from the monetary amount loaned or granted. Thus, on average each project at least generates 2.11 times higher outputs than inputs given, and in the optimum case even 3.51 times higher outputs. Even when considering the total project costs using the total-capital plus recurring-costs ratio the outcome is highly positive

with 2.68 for the discounted weighted minimum arithmetic mean and still 1.85 for the discounted unweighted minimum arithmetic mean. Here it needs to be pointed out that even though this ratio always considers the full project costs, the NPVs are still based on the overall- and partial-concept (refer to *chapter 3.2.3 Types of NPV-Cost-Ratios*). What needs to be noted at this stage is that the presented results are only based on 57 out of 168 NPV projects.

In order to understand how the NPVCRs change when “high-value” projects are excluded, the following table provides an overview of the summary of 158 NPV projects. This data set excludes the 7 “high-value” NPVs (refer to *Appendix CB*) as well 3 additional projects which disclose cost values above 1,000 Mio USD:

		LDCs				
		Counts	Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	158	1.6	1.75	2.33	2.82
	discounted	126	2	2.26	3.2	4.04
Total-Capital-Costs Ratio	undiscounted	158	1.38	1.48	1.94	2.34
	discounted	126	1.65	1.82	2.68	3.36
Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	66	1.39	1.48	1.75	2.56
	discounted	52	1.85	2.07	2.19	3.7
Total-Capital plus Recurring-Costs Ratio	undiscounted	66	1.28	1.36	1.55	2.23
	discounted	52	1.58	1.73	1.92	3.19

Table 56: Disclosed NPVCRs of the LDCs excluding “High-values”

In general, the NPVCRs decrease with a lot more ratios falling below 2. Specifically, the weighted values fall above-average which is due to the exclusion of the “high-value” NPV projects. Still, all ratios of the weighed minimum arithmetic mean are highly above 1. The unweighted minimum values are generally above 2, with the exception of the total-capital plus recurring-costs ratio which is only slightly below with 1.92. Looking at the pro-rata-capital plus recurring-costs unweighted ratio of 2.19 and 3.7 it can be noted that on average each project generates at least twice, and in the maximum close to 4 times higher outputs, noting that the maximum value of 3.7 is even higher than the one of the total projects of 3.51 (refer to *table 55*:

Disclosed NPVCRs of the LDCs). Furthermore, the total-capital plus recurring-costs ratio also discloses highly positive ratios with 1.58 and 1.73 for the weighted and 1.92 and 3.19 for the unweighted ratios.

4.8.1.2 Continent Comparison

In order to compare Africa and Asia following table provides an overview of the results of the NPVCRs of the 168 NPV projects:

		Africa				
		Counts	Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	109	1.75	2.05	2.24	3.2
	discounted	88	2.2	2.67	3.18	4.96
Total-Capital-Costs Ratio	undiscounted	109	1.36	1.51	1.88	2.71
	discounted	88	1.55	1.76	2.66	4.23
Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	43	1.5	1.7	1.76	2.89
	discounted	36	2.02	2.41	2.16	4.15
Total-Capital plus Recurring-Costs Ratio	undiscounted	43	1.29	1.41	1.58	2.5
	discounted	36	1.51	1.7	1.92	3.56
		Asia				
		Counts	Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	59	3.4	3.79	4.1	4.4
	discounted	48	4.28	4.85	5.82	6.21
Total-Capital-Costs Ratio	undiscounted	59	3.25	3.64	3.34	3.62
	discounted	48	4.15	4.7	4.78	5.14
Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	28	2.86	3.17	1.64	1.86
	discounted	21	4	4.52	2.02	2.43
Total-Capital plus Recurring-Costs Ratio	undiscounted	28	2.77	3.07	1.42	1.65
	discounted	21	3.88	4.39	1.73	2.12

Table 57: Disclosed NPVCRs of Africa and Asia

Generally, all of the weighted NPVCRs are (highly) positive (greater than 1) which indicates that the overall sum of the NPVs of each ratio must be positive.

Out of the African data set 34 projects disclose recurring costs. For 8 of them no differentiation between the capital and recurring cost type is possible. Furthermore, less than 10 percent of the projects (9 out of 109) are not based on recurring costs. Except the minimum arithmetic mean of the total-capital plus recurring-costs NPVCR of 1.92 all unweighted discounted ratios are above 2 and often even above 4 for the maximum values.

Out of the 59 Asian projects 20 disclose recurring and capital costs, whereas for 5 of them no distinction between capital and recurring costs is provided. In total only 8 projects are not based on recurring costs. What can be noticed at a glance when comparing Africa and Asia is that the Asian weighted values are primarily about twice as high as the African ones. This is attributed to the 4 “high-value” Bangladeshi NPV projects. Comparing the continental weighted values however, the picture looks quite different. Even though the capital costs solely based ratios –namely the pro-rata-capital-costs and the total-capital-costs ratio– are at least a factor of 2 higher in Asia, the picture changes once the recurring project costs are taken into account. Both recurring costs ratios for Africa disclose slightly higher minimum values than the ones for Asia. However, the weighted maximum values of 4.15 of the pro-rata-capital plus recurring costs and 3.56 of the total-capital plus recurring costs ratio are far above the maximum values of 2.43 and 2.12 of Asia. In order to depict a more complete picture, the following table shows the outcomes of the NPVCRs per continent after excluding all “high-value” projects:

		Africa				
		Counts	Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	104	1.5	1.63	2.05	2.64
	discounted	83	1.81	2.03	2.91	3.96
Total-Capital-Costs Ratio	undiscounted	104	1.31	1.39	1.76	2.21
	discounted	83	1.51	1.65	2.49	3.33
Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	41	1.36	1.46	1.78	2.96
	discounted	34	1.74	1.93	2.2	4.31
Total-Capital plus Recurring-Costs Ratio	undiscounted	41	1.25	1.31	1.6	2.57
	discounted	34	1.48	1.6	1.98	3.71
		Asia				
		Counts	Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	54	1.93	2.1	2.87	3.18
	discounted	43	2.7	3.15	3.77	4.19
Total-Capital-Costs Ratio	undiscounted	54	1.66	1.84	2.3	2.59
	discounted	43	2.29	2.63	3.04	3.42
Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	25	1.44	1.54	1.69	1.91
	discounted	18	2.18	2.49	2.16	2.56
Total-Capital plus Recurring-Costs Ratio	undiscounted	25	1.37	1.47	1.45	1.67
	discounted	18	1.96	2.22	1.82	2.21

Table 58: NPVCR Comparison of Africa and Asia excluding “high-value” Projects

As indicated in the above table, the weighted values drop across all NPVCRs for both continents, with larger declines in Asia. Additionally, there is a decrease in both unweighted total capital costs ratios in both continents. As outlined in *chapter 4.3.2. Result NPV Continent Comparison, table 23: Result Comparison of Africa and Asia excluding “high-value” NPVs* the higher ratios of the Asian capital costs solely related NPVCRs prove that projects in Asia are generally more capital costs effective than the ones in Africa. Interestingly, there is an increase in both unweighted recur-

ring costs related ratios for each of the continents compared to the total of all projects. This is attributed to the “high-value” projects –specifically “high-value” NPV projects– which account for the (very) high recurring costs. When comparing the continental unweighted recurring costs related ratios it can be noted that Africa discloses higher minimum and maximum values. This basically means that in Asia the recurring costs are percentile higher than the related capital costs compared to Africa. Finally, it can be concluded that even though Africa discloses higher unweighted recurring cost based NPVCRs compared to Asia, and consequently the weighted recurring costs are higher in Asia, the results of both continents can be regarded as highly positive.

4.8.2 Standardized NPV-Cost-Ratios

This chapter deals with the process and results of the application of standard values in order to generate possible outcomes for missing financial indicators and key figures. The goal is to apply standard values based on the arithmetic means of existing values of the same type. As mentioned in the previous chapter, the reason for missing discounted ratios of the capital related NPVCRs is due to undisclosed discount rates only. Secondly, the difference in counts amongst the undiscounted recurring costs related ratios and the undiscounted capital related ratios lies in missing disclosures of recurring costs. And last but not least, the difference within the total counts of the discounted and undiscounted recurring costs related ratios is due to missing discount rates and NPV-horizons. The following listing provides an overview of how the missing financial indicators and figures were calculated:

- **Discount Rate:** The missing discount rates were calculated based on the arithmetic mean of the existing discount rates of the same data set. Therefore, the whole of existing discount rates of the LDCs was used to calculate the missing ones, whereat only continental discount rates were used to calculate each continent’s standard discount rate in order to conduct a continent comparison.

- NPV-horizon: The missing NPV-horizons were calculated based on the arithmetic mean of the existing NPV-horizons of the same data set (same as for the discount rates).
- Recurring operational and maintenance costs: This is definitely the most crucial as well as vague part of the standard indicator calculation. Projects which have not disclosed their recurring costs present a challenge in being able to reliably compare these against other project types where the amount and distribution of recurring costs have been appropriately disclosed. Furthermore, there may be differences related to the staff, environmental or even country specific factors which need to be taken into consideration. Therefore, a standard approach for calculating missing recurring costs is always biased and restricted to the presumed parameters. However, in the absence of disclosed recurring costs and in order to provide a basis for comparative analysis given what information is available, a “recurring costs coefficient” was calculated. The recurring costs coefficient describes the multiplier for the capital costs of a project in order to derive the project recurring costs. The coefficient was calculated based on a weighted as well as on a unweighted approach:
 - The weighted recurring costs coefficient is the quotient of the sum of all available recurring costs divided by the sum of the related capital costs^{202, 203}
 - The unweighted recurring costs coefficient is the arithmetic mean of all available project recurring costs coefficients. A project recurring costs coefficient is the quotient of the recurring project costs divided by the corresponding capital costs. In order to account for exceptionally high project recurring costs coefficients, different versions of the unweighted recurring costs coefficient will be outlined.

²⁰² Author’s Note: In the case of partial-NPVs only the related capital costs will be considered in order to ensure a common basis for both cost types.

²⁰³ Author’s Note: For projects which are scenario-based and thus provide minimum and maximum values for both cost types, these have been averaged by taking the mean value of the corresponding minimum and maximum cost values.

The discounted ratios were again derived via an even distribution and discounting of all cost types in cases where no specific cost distribution and discounted values were mentioned.

4.8.2.1 Overall Results

This chapter is divided into 2 sections. In the first part the results for the capital costs solely related ratios are presented and discussed. The second part then deals with the recurring costs related NPVCRs, where for each of the 2 ratios different outcomes based on varying recurring costs coefficients will be presented. For all ratios the outcome is sensitized by excluding “high-value” projects.

The following table provides an overview of the 168 weighted and unweighted minimum and maximum ratios of the capital costs related NPVCRs only. The average discount rate based on the 136 available discount rates of the LDCs is 11.21 per cent²⁰⁴. In order to calculate the remaining 32 undiscounted capital costs related ratios, this discount rate is used:

		LDCs			
		Ø - weighted		Ø - unweighted	
		Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	2.8	3.15	2.89	3.62
	discounted	3.42	3.89	3.73	4.81
Total-Capital-Costs Ratio	undiscounted	2.25	2.51	2.39	3.03
	discounted	2.66	3.01	3.08	4.03

Table 59: Standardized Capital Costs solely related NPVCRs of the LDCs

The total-capital-costs ratio always discloses lower values than the pro-rata-capital-costs ratio, since the only difference is due to the amount of costs applied. However,

²⁰⁴ Cf. Chapter 4.3.1 Result NPVs of the LDCs, table 12: Result NPV and Cost Values of the LDCs.

for both of the NPVCRs the outcome is highly positive, with discounted values consistently well above 3 for the pro-rata-capital-costs ratio and still above 2.5 in case all project costs are considered.

When excluding the 10 “high-value” projects the weighted and unweighted total-capital-costs ratios are still above 1.5 using the adjusted average discount rate of 11.25 as shown in the following table:

		LDCs			
		Ø - weighted		Ø - unweighted	
		Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	1.6	1.75	2.33	2.82
	discounted	1.83	2.03	2.99	3.68
Total-Capital-Costs Ratio	undiscounted	1.38	1.48	1.94	2.34
	discounted	1.52	1.66	2.48	3.05

Table 60: Standardized Capital Costs solely related NPVCRs of the LDCs excluding “high-value” Projects

On average the per project total-capital-costs ratio ranges between 2.48 at a minimum and 3.05 for those projects which are managed in the most effective way possible.

In order to calculate the remainder of the 97 recurring costs ratios the missing recurring costs per project need to be determined. For discounting those costs the average NPV-horizon of the LDCs (rounded down to 19 from 19.33 years; refer to *chapter 4.3.1 Result NPVs of the LDCs, table 12: Result NPV and Cost Values of the LDCs*) is used in case no NPV-horizon within the project ICR is mentioned. The weighted recurring costs coefficient of all 41 projects outlining this cost type separately is 3.6. The unweighted coefficient of 13.29 is much higher, but also unrealistic since it is driven by 2 unusually high project recurring costs coefficients of above 132 and 214, as well as by a few additional coefficients with values above 10. When excluding all 10 project recurring costs coefficients of above 10, the unweighted coefficient comes down to 1.34. In the case where the additional 3 project recurring

costs coefficients above 5 are excluded, the recurring costs coefficient reduces further to 0.62.²⁰⁵ The following table provides an overview of the outcome of the 2 recurring costs NPVCRs based on different recurring costs coefficients:

Recurring Costs Coefficient			LDCs			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
3.6	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.7	1.85	1.61	2.02
		discounted	2.3	2.56	2.04	2.69
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.6	1.72	1.5	1.85
		discounted	2.04	2.26	1.85	2.42
13.29	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.31	1.38	1.4	1.75
		discounted	1.65	1.78	1.66	2.2
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.29	1.35	1.31	1.61
		discounted	1.58	1.7	1.53	1.99
1.34	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.99	2.19	1.92	2.4
		discounted	2.7	3.04	2.5	3.28
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.8	1.97	1.74	2.17
		discounted	2.29	2.56	2.21	2.9

Table 61: Standardized Recurring Costs related NPVCRs of the LDCs

In this case it is obvious that all of the weighted NPVCRs are positive (greater than 1), since each ratio is based on the 168 NPV projects which are overall positive. Considering the weighted recurring costs coefficient of 3.6 the outcomes are still highly satisfying for both recurring costs based ratios with all ratios around and above 2. This result basically means that based on the data set considered the value generation is at least twice as high in any case for any given input. Making this statement it again needs to be pointed out that the pro-rata-capital plus recurring-costs NPVCR does not consider all project costs in the case of partial-NPVs. Furthermore, the NPV and the costs blocks of the total-capital plus recurring-costs ratio

²⁰⁵ Cf. *Appendices CC* and *CD* provide an overview of the composition of the different unweighted recurring costs coefficients and of the outcome of the recurring costs ratios based on an unweighted recurring costs coefficient of 0.62.

do not share a common basis. The ratio's NPVs are based on the overall- and partial-concept, whereat in both cases the overall-costs are considered. Even for the average weighted recurring costs indicator of 13.29 the minimum average outcome ratio is higher than 1.5 for any of the discounted weighted or unweighted ratios. Using the adjusted coefficient of 1.34 the unweighted minimum arithmetic mean of the discounted total-capital plus recurring-costs ratio is the lowest with a ratio of 2.21.

As indicated by the table below, the ratios show a slight decline but still yield promising results when the 10 "high-value" projects are excluded. The adjusted average NPV-horizon of the remaining 158 projects is 19.44 years (rounded down to 19 years). Exclusion of the "high-value" projects again affects the recurring costs coefficients. Therefore, the weighted recurring costs coefficient comes down to 2.64, whereas the total unweighted recurring costs coefficient of 12.99 remains quite high. In the case where the 9 project coefficients with values above 10 are excluded the weighted coefficient comes down to 1.17. Supposing the remaining 2 coefficients above 5 are additionally excluded, the recurring costs coefficient further reduces to 0.62.²⁰⁶ The following table outlines the results for the first 3 coefficients mentioned:

²⁰⁶ Cf. *Appendices CE* and *CF* provide an overview of the composition of the different unweighted recurring costs coefficients and of the outcome of the recurring costs ratios based on an unweighted recurring costs coefficient of 0.62. For both appendices the 10 "high-value" projects have been excluded.

Recurring Costs Coefficient			LDCs			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
2.64	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.18	1.23	1.55	1.92
		discounted	1.36	1.46	1.89	2.46
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.16	1.2	1.43	1.74
		discounted	1.29	1.37	1.7	2.17
12.99	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.06	1.08	1.38	1.72
		discounted	1.13	1.17	1.59	2.12
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.06	1.07	1.29	1.58
		discounted	1.12	1.15	1.46	1.9
1.17	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.26	1.33	1.71	2.1
		discounted	1.49	1.61	2.13	2.72
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.21	1.26	1.55	1.88
		discounted	1.36	1.46	1.88	2.37

Table 62: Standardized Recurring Costs related NPVCRs of the LDCs excluding “high-value” Projects

Again, each of the ratios shows a positive result for the minimum and maximum arithmetic means. Thus, it can be concluded the overall sum of the 158 NPVs is positive. The outcome based on the weighted recurring costs coefficient of 2.64 shows positive results for both discounted ratios of above 1.29 in the weighted minimum case, whilst the maximum arithmetic mean of the weighted ratios is above 2. Although the values converge towards 1 when applying the unweighted coefficient of 12.99, both weighted minimum arithmetic means are still positive. Furthermore, the unweighted arithmetic means of the pro-rata-capital plus recurring-costs and total-capital plus recurring-costs NPVCRs still disclose average ratios of 1.59 and 1.46 (respectively) in the minimum and 2.12 and 1.9 (respectively) in the maximum case. Using the adjusted coefficient of 1.17 the minimum arithmetic mean is derived by the minimum ratio of 1.36 of the weighted total-capital plus recurring-costs NPVCR. In contrast the maximum arithmetic mean of the unweighted ratio lies at 2.37.

As a conclusion it needs to be captured that even though the ratio outcomes are lower than compared to the whole NPV data set they are still quite positive for both the discounted weighted and the regulated unweighted recurring costs coefficient of 1.17. Furthermore, the unweighted total-capital plus recurring-costs NPVCRs for both coefficients indicate that on average the NPV projects generate between 1.7 to 2.37 times of their input value. Again, this statement is due to the definition of the NPVCRs.

4.8.2.2 Continent Comparison

This chapter is also divided into 2 sections. Firstly, the outcomes for the capital costs solely related NPVCRs will be compared and highlighted before the result for the recurring costs related ratios are examined based on different recurring costs coefficients.

In order to calculate the missing discounted ratios of Africa and Asia, the respective continental average discount rates have been used.²⁰⁷ The following table presents the African and Asian capital costs solely related NPVCRs:

²⁰⁷ Author's Note: The African specific discount rate is 11.18 which is a bit lower than the one in Asia of 11.27 (refer to *chapter 4.3.2 Result NPV Continent Comparison*, *table 20: Result Total NPV and Cost Values Comparison of Africa and Asia*).

		Africa			
		Ø - weighted		Ø - unweighted	
		Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	1.75	2.05	2.24	3.2
	discounted	2.06	2.48	2.99	4.44
Total-Capital-Costs Ratio	undiscounted	1.36	1.51	1.88	2.71
	discounted	1.49	1.68	2.51	3.79

		Asia			
		Ø - weighted		Ø - unweighted	
		Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	3.4	3.79	4.1	4.4
	discounted	4.15	4.65	5.11	5.49
Total-Capital-Costs Ratio	undiscounted	3.25	3.64	3.34	3.62
	discounted	3.94	4.45	4.13	4.48

Table 63: Standardized Capital Costs solely related NPVCRs of Africa and Asia

Both continents generally disclose higher unweighted than weighted NPVCRs. Furthermore, only Africa discloses high differences of above 1 between the minimum and maximum arithmetic means of the unweighted NPVCRs. Those differences are due to broader variations within the scenario-NPVs in Africa. A further point to note is that the weighted ratios in Asia are close to twice as high as the ones in Africa which is again due to the 4 “high-value” Bangladeshi NPV projects. The Bangladeshi “high-value” projects also support that the unweighted arithmetic means for both NPVCRs are generally higher in Asia than in Africa. When the 10 “high-value” projects are excluded, the outcome of the continental comparison changes as shown in following table:

		Africa			
		Ø - weighted		Ø - unweighted	
		Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	1.5	1.63	2.05	2.64
	discounted	1.71	1.9	2.76	3.62
Total-Capital-Costs Ratio	undiscounted	1.31	1.39	1.76	2.21
	discounted	1.43	1.55	2.37	3.04

		Asia			
		Ø - weighted		Ø - unweighted	
		Min	Max	Min	Max
Pro-Rata-Capital-Costs Ratio	undiscounted	1.93	2.1	2.87	3.18
	discounted	2.24	2.44	3.41	3.81
Total-Capital-Costs Ratio	undiscounted	1.66	1.84	2.3	2.59
	discounted	1.86	2.1	2.69	3.05

Table 64: Standardized Capital Costs solely related NPVCRs of Africa and Asia excluding “high-value” Projects

At a glance it is evident that for each ratio –weighted or unweighted– Asia discloses higher values than Africa. The converging difference in the unweighted total-capital costs ratio compared to the weighted ratio can be explained by the fact that the remaining capital costs of the projects in Asia generally result in stronger project NPV/cost ratios than in Africa. Finally, it can be concluded that the projects in Asia operate at a more capital costs effective level than projects in Africa.

In order to calculate the remainder of the continental recurring costs ratios the missing recurring costs per project are determined using the average continental NPV-horizon.²⁰⁸ For both continents the continental specific weighted and unweighted recurring costs coefficients are used. The weighted coefficient taking all continental

²⁰⁸ Author’s Note: The African specific NPV-horizon is 19.76 years (rounded up to 20 years) and the Asian specific NPV-horizon is 18.62 years (rounded up to 19 years; refer to *chapter 4.3.2 Result NPV Continent Comparison, table 20: Result Total NPV and Cost Values Comparison of Africa and Asia*).

projects into account is 2.21 in Africa and 9.58 in Asia. Specifically for the unweighted coefficient the continental differences are quite high. When considering all projects within both continents, the unweighted coefficient is 4.35 in Africa but 28.8 in Asia. The high unweighted coefficient in Asia can be explained to the fact that the 2 unusually high project recurring costs coefficients of 132 and 214 both belong to Asia. When excluding all of the 10 project recurring costs coefficients of above 10, the unweighted African coefficient comes down to 1.24 and the Asian one to 1.6. In the case where the 3 project recurring costs coefficients above 5 are additionally excluded, the recurring costs coefficient further decreases to 0.56 in Africa and to 0.77 in Asia.²⁰⁹ The following table provides an overview of the outcome of the 2 recurring costs NPVCRs based on different recurring costs coefficients for Africa and Asia:

²⁰⁹ Cf. *Appendices CG* and *CH* provide an overview of the recurring costs ratios based on an unweighted recurring costs coefficient of 0.56 in Africa and 0.77 in Asia.

Recurring Costs Coefficient			Africa			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
2.21	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.25	1.35	1.58	2.17
		discounted	1.5	1.7	1.91	2.89
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.19	1.26	1.46	1.97
		discounted	1.32	1.45	1.73	2.58
4.35	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.17	1.24	1.47	2
		discounted	1.36	1.5	1.74	2.6
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.14	1.19	1.37	1.83
		discounted	1.26	1.36	1.6	2.34
1.24	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.32	1.45	1.69	2.36
		discounted	1.62	1.86	2.09	3.18
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.22	1.31	1.55	2.13
		discounted	1.37	1.51	1.86	2.81

Recurring Costs Coefficient			Asia			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
9.58	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.91	2.07	1.54	1.65
		discounted	2.65	2.93	1.95	2.12
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.89	2.05	1.42	1.54
		discounted	2.59	2.87	1.78	1.95
28.8	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.45	1.53	1.39	1.5
		discounted	1.91	2.06	1.63	1.79
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.45	1.52	1.28	1.39
		discounted	1.89	2.05	1.49	1.65
1.6	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	2.59	2.85	2.26	2.41
		discounted	3.5	3.91	3.16	3.39
	Total-Capital plus Recurring-Costs Ratio	undiscounted	2.52	2.78	2.04	2.2
		discounted	3.37	3.78	2.79	3.01

Table 65: Standardized Recurring Costs related NPVCRs of Africa and Asia

Obviously, varying continental recurring costs coefficients make it difficult to compare the 2 continents with each other. However, what can be noticed at a glance is

that even though Asia discloses higher coefficients for each of the coefficient versions, it still outlines higher weighted arithmetic means than Africa. Comparing the unweighted ratios of the initial, continental weighted and unweighted coefficient, Africa reports higher ratio outcomes than Asia. For adjusted continental unweighted coefficient outcomes of 1.24 in Africa and 1.6 in Asia, the comparative result is quite obvious – even though Asia’s missing recurring costs calculation is based on a slightly higher coefficient, the continent still discloses higher unweighted ratios than Africa.

In order to further research this outcome the following table compares the continental ratios based on the adjusted recurring costs coefficients as well as the adjusted average NPV-horizons²¹⁰ due to the exclusion of the 10 “high-value” projects:

²¹⁰ Author’s Note: The African adjusted NPV-horizon is 19.89 years (rounded up to 20 years) and the Asian adjusted NPV-horizon is 18.66 years (rounded up to 19 years).

Recurring Costs Coefficient			Africa			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
1.77	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.2	1.25	1.56	2.05
		discounted	1.38	1.48	1.89	2.65
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.16	1.2	1.44	1.85
		discounted	1.28	1.36	1.71	2.33
4.26	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.12	1.15	1.44	1.92
		discounted	1.25	1.31	1.7	2.44
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.1	1.13	1.35	1.75
		discounted	1.2	1.26	1.56	2.17
0.98	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.25	1.31	1.66	2.17
		discounted	1.46	1.58	2.04	2.82
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.19	1.24	1.52	1.94
		discounted	1.32	1.41	1.82	2.46

Recurring Costs Coefficient			Asia			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
6.5	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.15	1.19	1.48	1.6
		discounted	1.3	1.38	1.67	1.85
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.14	1.18	1.35	1.47
		discounted	1.28	1.35	1.48	1.65
28.59	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.05	1.06	1.36	1.46
		discounted	1.1	1.13	1.57	1.72
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.05	1.06	1.25	1.35
		discounted	1.1	1.13	1.41	1.56
1.6	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.3	1.37	1.79	1.94
		discounted	1.58	1.71	2.28	2.51
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.27	1.34	1.59	1.74
		discounted	1.48	1.61	1.97	2.19

Table 66: Standardized Recurring Costs related NPVCRs of Africa and Asia excluding “high-value” Projects²¹¹

As shown above there are no noteworthy changes to the continental coefficients once the 10 “high-value” projects are excluded. Obviously, the results of the initial unweighted coefficient of 28.59 in Asia are worse than the lowest coefficient of 4.26 in Africa. However, due to the variations in the continental coefficients it is difficult to judge which country actually performs better. The best way to compare the 2 continents is to look at the outcomes of the most similar coefficients. In this case the weighted coefficient of 1.77 in Africa and the adjusted unweighted coefficient of 1.6 in Asia are the ones most similar. For both weighted NPVCRs Asia shows higher minimum and maximum arithmetic mean ratios. This basically means that in Asia the percentile monetary amount generation is higher than in Africa. Looking at the unweighted ratios the picture looks quite mixed. Taking worse case scenarios into consideration, projects in Asia generally perform better with an average minimum ratio comprising of the pro-rata-capital plus recurring costs ratio of 2.28 and the total-rata-capital plus recurring costs ratio of 1.97, compared to the minimum ratio of 1.89 and 1.71 (respectively) in Africa. However, Africa outshines Asia when comparing the continental maximum values, since for both NPVCRs the unweighted maximum ratios in Africa are slightly higher than the ones in Asia. Therefore as a conclusion, whilst NPVCRs of both continents disclose satisfying results, Asia exhibits slightly higher weighted and more stable and predictable unweighted ratios.²¹²

²¹¹ Cf. *Appendices CI* and *CJ* provide an overview of the recurring costs ratios based on an unweighted recurring costs coefficient of 0.56 in Africa and 0.77 in Asia when the 10 “high-value” projects are excluded. Both of those indicators are based on the exclusion of any coefficient above 5.

²¹² Author’s Note: Those results have been verified by using the average coefficient of 1.68. The changes to the original results of the African coefficient of 1.77 and the Asian coefficient of 1.6 are negligible and if at all only affect the second position after the decimal point by a maximum of 1 digit. Verifying the continental differences using the average recurring costs coefficient of both most similar continental recurring costs coefficients has been done for all of the following continental comparisons.

4.8.3 Calculating the Minimum SROI Ratio

This chapter aims to calculate a minimum SROI ratio for the LDCs. As described in *chapter 3.2.4 Calculation of the proper SROI Ratio* the calculation of the minimum SROI ratio will follow a two-step approach.

In the first approximation step the capital SROI ratio will be calculated based on the remaining capital costs adjusted NPVs. In order to do so all of the remaining capital costs are again distributed and discounted evenly across the corresponding project lifetime period and subsequently deducted from the corresponding partial-NPV value. The following formula visualizes the calculation of a partial-NPV project's capital SROI ratio:

$$\text{Capital SROI Ratio} = \frac{(NPV - C) + (CRi + C)}{(CRi + C)} = \frac{NPV + CRi}{CRi + C}$$

where

CRi = discounted initial NPV-related capital plus recurring operational and maintenance costs (including standardized costs)

C = discounted remaining capital costs of partial-NPVs

Equation 8: Capital SROI Ratio

Due to the fact that the total-capital plus recurring-costs ratio already considers the total project costs, the discounted remaining capital costs have to be deducted from the corresponding NPV value only.

In the second step, standardized recurring costs based on the remaining capital costs of the partial-NPVs are additionally considered. In order to calculate the amount of the recurring costs for partial-NPV projects the already presented varying LDC and continent specific recurring costs coefficients are used. Once calculated, the standardized recurring costs are again distributed and discounted evenly across the project specific NPV-horizon. Thereafter, the discounted project recurring costs will be deducted from the corresponding NPV together with the discounted remaining capital costs calculated in the first step. Since those costs are not yet captured within any of the described NPVCRs they also need to be added to the total costs of the minimum SROI ratio. The formula below visualizes this calculation:

$$\text{Minimum SROI Ratio} = \frac{(NPV - C - R) + (CRi + C + R)}{(CRi + C + R)} = \frac{NPV + CRi}{CRi + C + R}$$

where

CRi = discounted initial NPV-related capital plus recurring operational and maintenance costs (including standardized costs)

CRi + C + R = total costs (discounted)

C = discounted remaining capital costs of partial-NPVs

R = discounted remaining recurring operational and maintenance costs of partial-NPVs (based on remaining capital costs)

Equation 9: Minimum SROI Ratio

The portfolio capital and minimum SROI ratio is calculated using the specific weighted and unweighted recurring costs coefficients of the LDCs and continents as presented in the previous chapter. Furthermore, the portfolio disclosure is again based on the minimum and maximum values of the weighted and unweighted arithmetic means. In the following, the results of the capital SROI ratio for the LDCs and as a continental comparison are presented. Thereafter, the outcomes of the minimum SROI ratio calculation are examined and discussed. For both SROI ratios, the adjustment of the NPV values based on an even distribution and discounting of costs is key. Thus, only discounted ratios are part of this analysis.

4.8.3.1 Overall Results of the Capital SROI Ratio

In the following section, results of the capital SROI ratio of the LDCs are outlined using the 168 NPV projects as a basis. As mentioned in *chapter 3.2.3 Types of NPV-Cost-Ratios* the amount of recurring costs was disclosed for only 54 projects within the ICR reports. Further, 17 project NPVs were not based on recurring costs. For the remainder of the 97 projects, the standard approach for calculating the recurring costs based on the LDC specific recurring costs coefficients was applied (refer to previous chapter). Additionally, all of the 108 partial-NPV values (refer to *chapter 4.3.1 Result NPVs of the LDCs, table 11: Result NPV Counts of the LDCs*) needed to be adjusted respectively and deducted by the corresponding remaining capital costs. In order to do so, all of the remaining capital costs were evenly distributed across the corresponding project duration and discounted using the project specific discount rate. In cases where no discount rate was disclosed the average discount rate of 11.21 percent of the LDCs was used. The table below shows the results of the capital SROI ratio of the LDCs using different recurring costs coefficients:

Recurring Costs Coefficient	LDCs			
	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.62	2.13	2.43	2.21	3
1.34	2.05	2.32	1.96	2.66
3.6	1.85	2.07	1.63	2.2
13.29	1.47	1.59	1.34	1.81

Table 67: Capital SROI Ratio of the LDCs

For each of the recurring costs coefficients of the LDCs, the results are positive across the board. Obviously, when comparing the capital SROI ratios to the ones of the solely standardized coefficients of the total-capital plus recurring-costs NPVCR (refer to *chapter 4.8.2.1 Overall Results, table 61: Standardized Recurring Costs related NPVCRs of the LDCs*), they are lower for each coefficient. Still, even for the highest recurring costs coefficient of 13.29 the weighted and unweighted minimum values are 1.47 and 1.34 respectively, spanning up to 1.59 and 1.81 in the maximum case. Looking at the adjusted unweighted coefficient of 1.34 both capital SROI minimum ratios are around 2 and even around 2.5 in the maximum case.

Excluding the 10 “high-value” projects still results in positive ratios for the weighted and unweighted arithmetic means for each of the coefficients as show in the table below:

Recurring Costs Coefficient	LDCs			
	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.62	1.12	1.23	1.74	2.26
1.17	1.11	1.21	1.62	2.12
2.64	1.09	1.17	1.46	1.94
12.99	1.04	1.07	1.27	1.71

Table 68: Capital SROI Ratio of the LDCs excluding “high-value” Projects

Again, when comparing the capital SROI ratios with the ones of the solely standardized coefficients of the total-capital plus recurring-costs ratio (refer to *chapter 4.8.2.1 Overall Results, table 62: Standardized Recurring Costs related NPVCRs of the LDCs excluding “high-value” Projects*), they are again lower for each coefficient. Interestingly, with increasing coefficients the decreases of the unweighted ratios are stronger than the ones for the weighted ratios which was not the case when the data set still contained the “high-value” projects. Furthermore, even for the weighted coefficient of 2.64 the weighted capital SROI ratio still ranges between 1.09 and 1.17 and even higher for the unweighted ratio with arithmetic means of 1.46 in the minimum and 1.94 in the maximum case. So it can be concluded that even when the “high-value” projects are excluded, the ratios remain positive for all coefficients. This essentially means that even through the deduction of the 108 partial-NPV values by the remaining capital costs, the ratio outcomes are positive throughout and thus, a positive value is generated for the LDCs.

4.8.3.2 Continental Comparison of the Capital SROI Ratio

When comparing results of the 168 NPV projects with comparable high coefficients across Africa and Asia, consistently higher outcomes are evident in Asia:

Africa				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.56	0.96	1.12	1.76	2.84
1.24	0.96	1.11	1.6	2.55
2.21	0.97	1.09	1.48	2.33
4.35	0.97	1.08	1.37	2.11

Asia				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.77	3.44	3.88	2.99	3.25
1.6	3.32	3.74	2.56	2.8
9.58	2.56	2.84	1.6	1.79
28.8	1.87	2.03	1.34	1.51

Table 69: Capital SROI Ratio of Africa and Asia

The main reasons for higher arithmetic means in Asia are again attributed to the Bangladeshi “high-value” NPV projects which specifically increase the weighted ratios. But not only are the Asian weighted ratios higher. Looking at the results of similarly high recurring costs coefficients, e.g. 1.24 of Africa and 1.6 of Asia, the unweighted capital SROI ratios in Asia are also higher. Furthermore, the weighted minimum ratios in Africa stay more or less equal within the range of the applied recurring costs coefficients.

However, as seen in previous analyses “high-value” projects always lead to a distorted continental comparison picture. Therefore the following table provides the continental results of the capital SROI ratios based on the exclusion of the 10 “high-value” projects:

Africa				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.56	1.04	1.13	1.65	2.3
0.98	1.03	1.12	1.55	2.19
1.77	1.03	1.1	1.45	2.08
4.26	1.02	1.07	1.33	1.94

Asia				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.77	1.36	1.54	1.91	2.17
1.6	1.31	1.48	1.73	1.96
6.5	1.19	1.29	1.4	1.59
28.59	1.07	1.1	1.25	1.41

Table 70: Capital SROI Ratio of Africa and Asia excluding “high-value” Projects

Logically, the weighted ratios in Asia drop drastically once the “high-value” projects are excluded. As for comparable high recurring costs coefficients, Asia still discloses higher weighted minimum and maximum arithmetic means. The results of the un-weighted ratios however are a little diffused. For example, when considering similarly high recurring costs coefficients of 1.77 for Africa and 1.6 for Asia, Asia discloses higher values (in this specific instance 1.73 compared to 1.45 in Africa). On the other hand, Africa discloses higher maximum values with 2.08 for the mentioned recurring costs coefficient of 1.77 compared to 1.96 for a recurring costs coefficient of 1.6 in Asia. This shows that defining African financial outcomes is less predictable, since the ranges of the expected result are wider. As a conclusion it needs to be captured that both continents disclose satisfying capital SROI ratio results, again

with slightly higher weighted and a more stable and predictable unweighted ratio outcome in Asia.

4.8.3.3 Overall Results of the Minimum SROI Ratio

This chapter presents the results of the minimum SROI ratio of the LDCs. The 168 NPV projects again served as a basis, and the procedure to calculate standard recurring costs was the same as for the previous analyses. As already described at the beginning of this chapter, the difference compared to the calculation of the capital SROI ratio lies therein that additional recurring costs for the already applied remaining capital costs of the partial-NPV projects are assumed. In order to calculate those additional recurring costs, the respective LDC continent specific recurring costs coefficients were again used. Once the additional recurring costs were calculated they were distributed and discounted evenly across the corresponding NPV-horizon using standard NPV-horizons and discount rates where necessary. The table below shows the results of the minimum SROI ratio of the LDCs based on different recurring costs coefficients:

Recurring Costs Coefficient	LDCs			
	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.62	1.92	2.19	2.11	2.88
1.34	1.68	1.91	1.8	2.45
3.6	1.25	1.41	1.38	1.87
13.29	0.74	0.82	0.98	1.25

Table 71: Minimum SROI Ratio of the LDCs

Generally, the outcome of the minimum SROI ratio of all 168 discovered NPVs can be regarded as positive. Even for the weighted coefficient of 3.6 the result ratios are positive with a ratio of 1.25 and 1.41 for the weighted minimum and maximum ratios and 1.38 and 1.87 for the unweighted minimum and maximum ratios. Only when

applying the initial and unadjusted unweighted recurring costs coefficient of 13.29 the weighted ratio outcomes are both negative, whereas the unweighted ratios still yield positive results with 0.98 in the minimum and 1.25 in the maximum case. In order to account for the “high-value” bolters, the data set is cleaned up by excluding the 10 projects again. The table below shows the outcome of the minimum SROI ratio for varying recurring costs coefficients:

Recurring Costs Coefficient	LDCs			
	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.62	1.01	1.12	1.66	2.16
1.17	0.94	1.03	1.5	1.96
2.64	0.82	0.89	1.27	1.68
12.99	0.62	0.66	0.91	1.15

Table 72: Minimum SROI Ratio of the LDCs excluding “high-value” Projects

The unweighted ratios remain positive even for the weighted recurring costs coefficient of 2.64. Only when applying the initial and unadjusted unweighted recurring costs coefficient of 12.99 the minimum value falls to 0.91, whilst the maximum ratio remains at 1.15. The weighted ratios are only positive for the fully embellished unweighted recurring costs coefficient of 0.62. For the adjusted unweighted recurring costs coefficient of 1.17 the minimum value is already down at 0.94, whereas the maximum one still stays positive with a ratio of 1.03. Obviously, the weighted minimum SROI ratios decreases further as the applied recurring costs coefficient increases. In order to shed some more light on the outcome of the minimum SROI ratios, *chapter 4.8.4 Making Meaning of the Results* will deal further on how the results can be interpreted.

4.8.3.4 Continental Comparison of the Minimum SROI Ratio

When comparing the data set pertaining to the 168 NPV projects for the African and Asian continents, the outcome looks as follows:

Africa				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.56	0.82	0.96	1.67	2.73
1.24	0.72	0.83	1.45	2.35
2.21	0.63	0.71	1.29	2.05
4.35	0.53	0.59	1.11	1.71

Asia				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.77	3.34	3.8	2.85	3.11
1.6	3.14	3.59	2.36	2.59
9.58	2.08	2.43	1.27	1.42
28.8	1.35	1.58	0.93	1.04

Table 73: Minimum SROI Ratio of Africa and Asia

What can be realized at a glance is that for none of the African recurring costs coefficients the weighted minimum SROI ratio is greater than 1. On the contrary, all of the Asian weighted ratios are (highly) positive for any recurring costs coefficient. This still refers back to the “high-value” Bangladeshi NPV projects. For both continents the outcome of the unweighted minimum SROI ratio looks highly promising with positive outcomes throughout. The negligible exception is shown for the Asian minimum ratio of 0.93 with a recurring costs coefficient of 28.8. Comparing similar

continental recurring costs coefficients again shows that the unweighted ratios of Asia are also higher than the ones in Africa.

In order to validate the outcomes, the “high-value” projects need to be excluded. The following table provides the continental results of the minimum SROI ratios based on the exclusion of the 10 “high-value” projects:

Africa				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.56	0.94	1.02	1.57	2.19
0.98	0.88	0.96	1.44	2.03
1.77	0.81	0.87	1.29	1.84
4.26	0.7	0.74	1.09	1.55

Asia				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
0.77	1.23	1.44	1.83	2.08
1.6	1.11	1.31	1.59	1.81
6.5	0.82	0.98	1.13	1.28
28.59	0.63	0.74	0.82	0.92

Table 74: Minimum SROI Ratio of Africa and Asia excluding “high-value” Projects

The picture for the embellished data set looks a bit different than the one described above. For each of the recurring costs coefficient the African weighted outcomes are generally higher. Furthermore, for the fully embellished coefficient of 0.56 the maximum arithmetic mean shows the only positive ratio of 1.02. Comparing the results of similar high recurring costs coefficients, e.g. the recurring costs coefficient of 0.56 of Africa and 0.77 of Asia, shows that the weighted SROI ratios in Asia are

still higher than the ones in Africa. Even though the Asian weighted ratios drop drastically, the results of 1.11 in the minimum and 1.31 in the maximum case of the embellished recurring costs coefficient of 1.6 outline that there is still an overall value generation which is percentile higher than the one in Africa. Looking at the unweighted results the outcomes are positive throughout –ignoring the ones of the single high recurring costs coefficient of 28.59 of Asia. Considering comparable recurring costs coefficients, the outcome looks similar to the capital SROI comparison (refer to *table 70: Capital SROI Ratio of Africa and Asia excluding “high-value” Projects*): Asia outlines higher unweighted minimum values, but lower maximum ratios than Africa. Thus the range in expected results for Africa is broader and therefore outcomes are less predictable for this continent.

In summary, the weighted results in Africa appear to be value degenerating from an overall monetary point of view. Asia on the other hand outlined positive outcomes for the weighted minimum SROI ratio in cases where a slightly embellished recurring costs coefficient of 1.6 was applied. On a positive note, regardless of the monetary size respective to the projects’ costs, the projects generated positive outcomes on average (refer to the unweighted ratios) within both continents.

4.8.4 Making Meaning of the Results

This chapter deals with the outcome analysis of the minimum SROI of the LDCs. The idea is to define the most realistic recurring costs coefficient for the LDCs. In order to do so, specifically those projects which outlined the highest recurring costs coefficients were analyzed further. Furthermore, as discussed in the previous chapter the weighted minimum ratio of the LDCs was already below 1 for a recurring costs coefficient of 1.17 (refer to *chapter 4.8.3.3 Overall Results of the Minimum SROI Ratio, table 72: Minimum SROI Ratio of the LDCs excluding “high-value” Projects*). As shown in the continental analysis the African projects were the ones decreasing the weighted arithmetic means of the LDCs. In a second step the projects which were responsible for decreasing the weighted arithmetic means of the minimum SROI of the LDCs were further analyzed and discussed.

The question to be answered in the first step was, ‘which projects actually drive the recurring costs coefficient up and what kind of projects are these?’

Analyzing the recurring costs coefficients of the 5 LDC projects with very high recurring costs coefficients helped to address this question. The following table provides an overview of those projects and the respective recurring costs coefficient:

High-Value Recurring Costs Coefficients			
Continent	Country	Report	Recurring Costs Coefficient
Africa	Benin	21512	33
Africa	Chad	ICR1891	27
Asia	Cambodia	22038	19
Asia	Samoa	ICR1793	132
Asia	Yemen	27440	214

Table 75: “High-value” Recurring Costs Coefficients of the LDCs

The highest recurring costs coefficients range from 19 to 214, with 3 of them belonging to the Asian continent. In the case where those 5 coefficients are excluded when calculating the average unweighted recurring costs coefficient of the LDC data set (excluding the 10 “high-value” projects) the coefficient comes down from 12.99 to 2.4.²¹³ In order to understand what kind of projects the “high-value” recurring costs coefficient projects are, the following table provides an overview of their NPV and costs values:

²¹³ Author’s Note: The exclusion of the 5 “high-value” recurring costs coefficients brings the coefficient of the complete dataset of the 168 NPV projects of 13.29 down to 2.52.

Values in Mio USD		5 High Recurring Costs Coefficient Projects	
		Cum Sum	Ø
NPV	Minimum	42	8
	Maximum	56	11
Minimum NPV (Coefficient 2.64)	Minimum	-122	-24
	Maximum	-108	-22
Total NPV Capital Costs	Minimum	69	14
	Maximum	69	14
Total Project Costs		150	30
Total World Bank Costs		132	26
Discount Rate [%]		10.9	
Project Duration [years]		7	
NPV-horizon [years]		27	

Table 76: NPV and Cost Values of the “high-value” Recurring Costs Coefficient Projects²¹⁴

Comparing the average NPV and cost values with the ones of the LDCs excluding the “high-value” NPVs there are multiple and relevant differences (*chapter 4.3.1 Result NPVs of the LDCs, table 17: Result NPV and Cost Values of the LDCs excluding “high-value” NPVs*):

- The average NPVs of 8 and 11 Mio USD are far below the ones of the LDCs excluding “high-value” NPVs of 38 and 48 Mio USD.
- After adjustment of partial-NPVs via the corresponding project remaining capital and recurring costs the average NPVs come down to -24 and -22 Mio USD using the weighted recurring costs coefficient of 2.64 of the LDCs excluding “high-value” projects.
- The average NPV capital costs of 14 Mio USD is well below the ones of the LDCs excluding “high-value” NPV projects of 65 and 66 Mio USD.
- The average total project costs of 30 Mio USD is well below the ones of the LDCs excluding “high-value” NPV projects of 126 Mio USD.

²¹⁴ Cf. *Appendix CK* shows the sector codes of the WB funds of the 5 “high-value” recurring costs coefficient projects.

- The share of the WB costs of the total project costs is approx. 87 percent and thus a lot higher than the one of the LDCs excluding “high-value” NPV projects of 39 percent (49 out of 126 Mio USD).
- Considering the outcome of the sector types of the WB funds, it can be noted that 50 out of the 132 Mio USD belong to the ‘Distribution and Transmission’ sector code. This amount actually mirrors the total amount for this sector code for the whole of the WB funds of the whole LDCs (refer to *Appendix F*).
- The average NPV-horizon of 27 years is much higher than the 19.74 years of the LDCs excluding “high-value” NPVs.

In summary, the “high-value” recurring costs coefficient projects are characterized by low NPV, low NPV related capital, and low total project costs projects. Furthermore, the WB funded amount of the total project costs is unusually high, particularly for the ‘Distribution and Transmission’ sector code. Therefore, the exclusion of the 5 “high-value” recurring costs coefficients from the remaining coefficient of the LDCs, together with the exclusion of “high-value” projects is reasonable and moreover justifiable.

In the second step, the calculated minimum NPVs of the adjustment via the project specific remaining capital and the corresponding recurring costs were analyzed further. In summary, of the 158 projects which initially contained 7 neutral as well as 7 negative NPV projects, the number of negative NPV projects drastically increased to 76. In total the numerical count of the neutral project remained at 7. This basically means that around half of the projects of –and due to– the minimum SROI calculation are based on SROI ratios of less than 1. In order to understand which kind of projects are the negative SROI drivers, 10 projects with calculated minimum NPVs of less than -200 Mio USD were identified. The following table provides an overview of the 10 highest negative calculated minimum NPV projects:

Values in Mio USD		10 highest negative calculated Minimum NPVs		Detailed Project Information		
		Cum Sum	Ø	Continent	Country	Report
NPV	Minimum	477	48	Africa Africa Africa Africa Africa Africa Africa Africa Africa Asia	Ethiopia Mali Mali Mozambique Mozambique Senegal Senegal Tanzania Zambia Afghanistan	ICR350 34791 32130 ICR586 27511 ICR699 32408 26745 32521 ICR524
	Maximum	756	76			
Minimum NPV (Coefficient 2.64)	Minimum	-5,566	-557			
	Maximum	-4,756	-476			
Total NPV Capital Costs	Minimum	1,249	125			
	Maximum	1,410	141			
Total Project Costs		4,541	454			
Total World Bank Costs		1,308	131			
Discount Rate [%]		11.3				
Project Duration [years]		7				
NPV-horizon [years]		20.27				

Table 77: NPV and Cost Values of the 10 highest negative calculated Minimum NPV Projects²¹⁵

Again there are some differences worth highlighting when comparing the average NPV and cost values with the ones of the LDCs where the “high-value” NPVs were excluded:

- The average NPVs before the adjustment of 48 and 76 Mio USD are slightly higher than the ones of the LDCs excluding “high-value” NPVs of 38 and 48 Mio USD.
- After adjustment of partial-NPVs via the corresponding project remaining capital and recurring costs, the average NPVs come down to -557 and -476 Mio USD using the weighted recurring costs coefficient of 2.64 of the LDCs excluding “high-value” projects.
- The average NPV capital costs of 125 and 141 Mio USD are twice as high as the ones of the LDCs excluding “high-value” NPV projects of 65 and 66 Mio USD.

²¹⁵ Cf. *Appendix KL* shows the sector codes of the WB funds of the 10 highest negative calculated minimum NPV projects.

- The average total project costs of 454 Mio USD are about 3.5 times higher than the ones of the LDCs excluding “high-value” NPV projects of 126 Mio USD.
- The share of the WB costs of the total project costs of 29 percent is 10 percent lower than the one of the LDCs excluding “high-value” NPV projects of 39 percent (49 out of 126 Mio USD).

In regards to the sector codes there are no noteworthy differences compared to the overall sector codes of the LDCs. The 3 largest sector codes are the ‘Roads and highways’, ‘Power’ and the ‘Other social services’ (refer to *Appendix CL*), whereat all of them are core sector codes of the LDCs excluding “high-value” NPV projects (refer to *Appendix M*). However, what can be noted is the fact that the 10 projects outlining the most negative NPVs are projects with NPVs which are slightly higher than the average NPV of the LDCs excluding “high-value” NPVs, but feature high NPV related capital costs and even higher total project costs. In cases where these projects are excluded from the data set of the 158 NPV projects, the weighted recurring costs coefficient of the remaining 148 projects actually increases from 2.64 to 3. The following table outlines the results of the capital SROI ratio for the remainder of the 148 LDC projects:

148 LDC Projects				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
3	1.2	1.28	1.51	2.01

Table 78: Capital SROI Ratio of the Remainder of the 148 LDC Projects

Even though the recurring costs coefficient increased to 3, the results of the capital SROI ratio are generally better throughout than the ones of the LDCs excluding “high-value” projects (refer to *chapter 4.8.3.1 Overall Results of the Capital SROI Ratio, table 68: Capital SROI Ratio of the LDCs excluding “high-value” Projects*). When calculating the minimum SROI ratio for the 148 projects of the LDCs, the results are as follows:

148 LDC Projects				
Recurring Costs Coefficient	Ø - weighted		Ø - unweighted	
	Min	Max	Min	Max
3	1	1.06	1.33	1.72

Table 79: Minimum SROI Ratio of the Remainder of the 148 LDC Projects

The outcome is simple to interpret. Even in the minimum case of the weighted arithmetic mean, the ratio is slightly above 1 (when rounded to the nearest whole number). The weighted maximum ratio of 1.06 shows a possible monetary value creation of 6 percent for the total of 148 projects. Furthermore, the unweighted ratios show that the average project generates a 1.3 to 1.72 times higher output compared to the given input.

In order to understand what kind of projects actually outline minimum SROIs which are at least as high as the unweighted minimum arithmetic mean of 1.3, the following table provides an overview of their NPV and cost values:

Values in Mio USD		Projects with Minimum SROIs ≥ 1.3	
		Cum Sum	Ø
NPV	Minimum	3,132	104
	Maximum	3,354	112
Total NPV Capital Costs	Minimum	1,213	40
	Maximum	1,213	40
Total Project Costs		1,261	42
Total World Bank Costs		846	28
Discount Rate [%]		11.27	
Project Duration [years]		6.4	
NPV-horizon [years]		19.7	

Table 80: Projects with Minimum SROIs greater than or equal to 1.3²¹⁶

²¹⁶ Cf. *Appendix CM* shows the sector codes of the WB funds of the projects with minimum SROI ratios of greater than or equal to 1.3.

When analyzing these projects in detail, it needs to be noted that 16 out of 30 are partial-NPV based. Each project at minimum discloses a SROI ratio of 1.3. The following major differences are noted when comparing the average NPV and cost values of the minimum SROI projects with the ones of the LDCs excluding the “high-value” NPVs:

- The average NPVs of 104 and 112 Mio USD are between 2.3 and 2.7 times higher than the ones of the LDCs excluding “high-value” NPVs of 38 and 48 Mio USD.
- The average NPV capital costs of 40 Mio USD only make up about 60 percent of the ones of the LDCs excluding “high-value” NPV projects of 65 and 66 Mio USD.
- The average total project costs of 42 Mio USD are similarly high as the NPV related capital costs which basically means that the amount of remaining capital costs of the partial-NPVs is negligible.
- The share of the WB costs of the total project costs of 67 percent is about 1.7 times higher than the one of the LDCs excluding “high-value” NPV projects of 39 percent.
- Looking at the outcome of the sector types of the WB funds, ‘Roads and highways’, ‘Power’, ‘Central government administration’ and ‘Water supply’ still remain the core sector codes. However, differences can be found in the ‘Power’ sector code which shows the second largest %-share but the highest WB costs with 189 Mio USD out of 846 Mio USD (22.34 percent; refer to *Appendix CM*) compared to the LDCs excluding “high-value” NPV projects where the ‘Power’ sector code only makes up 783 Mio USD out of 7,845 Mio USD (10 percent). On the other hand the ‘Roads and highways’ sector code only makes up 16.55 percent compared to 29.75 percent of the LDCs excluding “high-value” NPVs.

Summarizing the outcomes above, it can be noted that projects with high minimum SROIs are not necessarily overall-NPV projects, but at least partial-NPVs with negligible remaining capital costs. Furthermore, these projects disclose around 2.5

times higher NPVs and 40 percent lower NPV related capital costs than the average project of the LDCs excluding “high-value” NPV projects. Last but not least, the share of the WB costs of the total project costs is comparably high, particularly in the ‘Power’ sector and less in the ‘Roads and highways’ sector.

Furthermore, the positive overall outcome of the weighted minimum SROI ratio is driven by the still higher Asian weighted ratios. In case the recalculated unweighted recurring costs coefficient of 2.4 was used, the weighted ratios of the 148 LDC projects would increase further. Nonetheless, the most interesting find is the fact that even in the worst case scenario by calculating the minimum SROI ratio of the LDCs and under consideration of the above mentioned restrictions, the outcome for the WB projects is positive for each of the 2 minimum SROI ratios.

4.9 Summary and Conclusion

In the first part of chapter 4 the appraisal NPV outcomes of 91 ICR reports of the LDCs were outlined and discussed. The results were positive with NPVs ranging from 17,667 to 20,067 Mio USD with total project costs of 13,981 Mio USD. Excluding the 3 “high-value” NPV projects which disclosed NPVs greater than 1,000 Mio USD led to NPV values of 4,678 to 4,713 Mio USD (8,021 Mio USD of total project costs). The comparison of outcomes between of the African and Asian continents for the whole data set disclosed 2.5 to 3 times higher NPV values in Asia, even though the NPV related capital costs of both continents were equally high. Once the “high-value” NPVs were excluded, Africa’s total NPV sum of 3,827 Mio USD was more than 4.5 times higher than the 886 Mio USD in Asia. However, the NPV/capital cost ratio of Asia was still slightly higher than the one of Africa. In the subsequent chapter the 68 NPVs of the result stage were evaluated. Aside from the 14 negative and neutral NPVs, the results were (highly) positive with NPV values of 42,059 to 50, 779 Mio USD with 33,506 Mio USD of total project costs, and 6,188 to 7,799 Mio USD with 20,280 Mio USD of total project costs for the LDCs when excluding the 7 “high-value” NPV projects. The comparison between the continental results showed a higher NPV as well as a higher NPV/capital cost ratio in Asia. Even with the exclusion of the “high-value” NPVs, the NPV/capital cost ratio of Asia was still close to

twice as high as the African one. Whether or not the continental NPV and cost value differences were due to differences in project types accordingly to the WB funding could not be evaluated further since the WB funding made up only 27 percent of the total project costs for a data set of 168 NPVs (39 percent of the total project costs for the data set excluding “high-value” NPVs). In order to compare the planned NPV and cost values of the appraisal stage with the ones of the result stage, 67 projects could be used as the comparison basis. Interestingly, the result stage NPVs were 2 times higher even though the cost values were slightly lower. Excluding the 4 “high-value” NPVs, appraisal stage showed a slightly higher minimum NPV compared to the result stage. However, the maximum NPV of the result stage was slightly higher than the one of the appraisal stage. Although the NPV related capital and total project costs increased by roughly 8 and 14 percent from the appraisal to the result stage, project outcomes are generally found to be developing as planned.

The analysis of the ERRs showed that the 159 NPV-intersectional ERRs disclosed slightly lower arithmetic means than the total of the 219 ERRs. Excluding the 20 “high-value” ERRs disclosing values of 100 or higher, the picture shifts towards minimal higher arithmetic means of the intersectional ERRs compared to the ERRs of the non-NPV projects. Still, the minimum and maximum arithmetic means of the ERRs of the non-NPV projects were respectively 18.91 and 31.35 percent and thus much higher than the corresponding discount rate of 10.83 percent. The comparison of the continents based on the entire data set revealed the following facts:

- Africa disclosed higher ERR values than Asia for both groups –the ERRs of the NPV projects and the non-NPV project ERRs.
- Even though the differences for the ERRs of the NPV projects were negligible, the African minimum arithmetic mean of 48.79 percent of the non-NPV projects was more than twice as much as the Asian maximum of 23.97 percent.
- Through the exclusion of the 20 “high-value” ERRs, the picture changed in such a way that Africa still disclosed higher total ERRs for both the minimum and maximum arithmetic means.

- A more in depth look at the 2 groups revealed that Asia's maximum arithmetic mean (34.69 percent) of the ERRs of the NPV projects was slightly higher than the 32.6 percent in Africa, albeit differentiated by just over 1 percent.
- Above all, the 35.57 percent maximum value of the ERRs of the non-NPV projects in Africa was still significantly higher than the 23.97 percent in Asia.

The analysis of the additional economic and financial indicators only produced limited insights. Whilst the analysis was able to show to a limited degree that there are additional projects which outlined positive outcomes in 4 of the economic and financial indicators other than the NPV or ERR, for the most part, the amount of the additional indicators is negligible and there was overall, insufficient financial information to adequately bring the outlined values in line with the NPV or ERRs. In order to establish a concept on how potential outcomes of the remainder of projects which did not disclosed any economic or financial indicators could look like, the overall project performance and the results of the disclosed ERRs were used. Even though the outcomes based on varying ERR share factors were positive and only a little below the results of the ones of the NPV and non-NPV groups, they can only be regarded as approximations, since no further project details were considered.

In the final part of the economic and financial analysis chapter the outcomes of all NPVCRs based on the data set as disclosed within the ICRs showed positive results throughout. Obviously, the more cost types that were considered, the worse the NPVCR. Nonetheless, for the data set which excluded the 10 "high-value" projects, the discounted weighted and unweighted ratios maintained (highly) positive with values between 1.58 in the weighted minimum case and 3.19 in the unweighted maximum case based on the inclusion of all outlined project costs (total-capital plus recurring-costs ratio). The continent comparison based on the exclusion of "high-value" NPVs showed mixed results consisting of higher African ratios for the unweighted but lower weighted NPVCRs. Looking at the standardized outcomes the capital costs solely based ratios of the LDCs disclosed positive results throughout, even with the exclusion of the 10 "high-value" projects. Furthermore, the recurring costs based NPVCRs disclosed high weighted and unweighted ratios. Even for the

unadjusted unweighted recurring costs coefficient of 13.29, all ratios ranged between 1.58 and 1.99 when all project costs were included. When the “high-value” projects were excluded, the ratios dropped but still remained positive with a weighted recurring costs coefficient of 2.64. This translated to values between 1.29 for the weighted minimum and 2.17 for the unweighted maximum case of the total-capital plus recurring costs NPVCR. The continental comparison of the whole data set outlined higher capital costs solely based NPVCRs for the Asian continent, which in the case of weighted ratios were more than twice as high as the African ones. Even with the exclusion of “high-values”, Asia still showed higher values throughout, lending support to the fact that the projects run in Asia are indeed more capital costs effective than projects in Africa. With regards to the recurring costs related NPVCRs, it needs to be noted that the weighted and unweighted recurring costs coefficients were much higher in Asian compared to Africa. However, as a result of the “high-value” NPV projects, Asia still disclosed higher weighted ratios even when comparing the results of the Asian unadjusted unweighted recurring costs coefficient of 28.8 against the African one of 4.35. Further on this, Asia disclosed higher weighted NPVCRs based on comparably high coefficients. Excluding the 10 “high-value” projects demonstrated almost unchanged recurring costs coefficients for both continents, even though there was notable decline in continental differences. Although Asia still disclosed slightly higher weighted ratios for comparable recurring costs coefficients and thus a higher percentile monetary amount generation than Africa, the picture for the unweighted NPVCRs looked quite different. With higher unweighted minimum but lower maximum values, the Asian outcomes appeared to be more stable and predictable. In order to account for the remaining capital costs of the partial-NPVs, the capital SROI ratio was calculated using different recurring costs coefficients. Even for the initial and unadjusted unweighted recurring costs coefficient of 13.29, both weighted and unweighted ratios were positive. The cleansed data set of 158 NPVs also disclosed positive outcomes for each of the recurring costs coefficients. Considering the weighted coefficient of 2.64, the weighted outcomes remained between 1.09 and 1.17, whereas the unweighted capital SROI ratio resulted with higher arithmetic means of 1.46 and 1.94. The continent comparison of the full data set looked one-sided: Higher Asian capital SROI ratios

for any recurring costs coefficient, whereas the African weighted minimum ratios were consistently below 1 for any of the African recurring costs coefficients. When excluding the “high-value” projects and considering comparable high recurring costs coefficients, the continent comparison revealed the same picture as for the standardized values: Higher weighted capital SROI ratios as well as a higher unweighted minimum capital SROI ratio, but a lower unweighted maximum capital SROI ratio in Asia. Thus as an outcome of the continental comparison, it can be summarized that Asia usually disclosed little higher weighted, more stable and predictable unweighted ratio outcomes. In the penultimate chapter, the theoretic minimum SROI ratio was calculated. Therefore, the assumption that the remaining capital costs of the partial-NPVs come along with additional recurring costs was taken. Deducting those –theoretically assumed– recurring costs from the NPV values still disclosed positive weighted and unweighted SROIs for all except 1 recurring costs indicators across all LDCs. For the initial unadjusted unweighted recurring costs indicator of 13.29, both of the weighted ratios were below 1 with arithmetic means of 0.74 in the minimum and 0.82 in the maximum case, whereas the unweighted SROI ratio still lay between 0.98 and 1.25. Excluding the 10 high bolter projects disclosed positive results throughout only for the fully embellished recurring costs coefficient of 0.62. Even for the adjusted recurring costs coefficient of 1.17 the weighted minimum value of the minimum SROI came down to 0.94. Using the weighted recurring costs coefficient of 2.64, both weighted ratios were below 1, with an unweighted minimum SROI ratio range between 1.27 and 1.68. Only when applying the initial unadjusted unweighted recurring costs coefficient of 12.99 the weighted minimum value fell to 0.91. The continental comparison of the total 168 NPV projects showed that Asia discloses about 4 times higher weighted results as well as higher unweighted results for the minimum SROI based on comparably high recurring costs coefficients. Even for the fully embellished unweighted coefficient of 0.56, Africa’s weighted minimum SROI ratio was below 1, even for the maximum case. In comparison, Asia’s minimum SROIs were all positive, except the unweighted minimum ratio of 0.93 of the initial unadjusted unweighted recurring costs coefficient of 28.8. Since the African initial unadjusted unweighted recurring costs of 4.35 was much lower, all unweighted ratios are (highly) above 1. With the exclusion of the “high-values” the

Asian ratios dropped most notably. However, for comparable recurring costs coefficients, Asia disclosed a higher weighted minimum SROI which was still positive in the minimum case for the adjusted recurring costs coefficient of 1.6. Africa only disclosed 1 positive weighted minimum SROI ratio which was the maximum ratio value of 1.02 of the fully embellished unweighted recurring costs of 0.56. The unweighted ratios of both continents were all positive, except for the Asian recurring costs coefficient of 28.59. As a conclusion of the continental minimum SROI comparison, the African weighted results appear to be value degenerating from an overall monetary point of view. Positively noted is the fact that regardless of the monetary size respective to the projects' costs, the projects generate positive outcomes within both continents on average, but with more stable and predictable unweighted ratio outcomes in Asia.

The last chapter of the economic and financial analysis delved further into the NPV projects and the driving factors for high recurring costs coefficients and a low weighted minimum SROI ratio. An outcome of this analysis revealed 5 "high-value" recurring costs coefficient projects. These projects outlined low NPVs, NPV related capital and total project costs. Furthermore, the WB funding amount of the total project costs was unusually high, specifically in the 'Distribution and Transmission' sector code. In the case where the 5 projects and the "high-values" were excluded when calculating the unweighted recurring costs coefficient for the LDCs, the coefficient came down from 12.99 to 2.4.

The analysis also revealed an additional 10 projects with calculated minimum NPVs of less than -200 Mio USD. These projects disclosed NPVs less than the average NPV of the LDCs excluding "high-value" NPVs, but maintained high NPV related capital costs and even higher total project costs. In the case where these projects were excluded from the data set, the remaining 148 projects disclosed a positive capital SROI ratio of 1.2 and 1.28 for the weighted and 1.51 and 2.01 in the unweighted case. Furthermore, the minimum SROI ratio of 1 and 1.06 in the weighted and the 1.3 and 1.72 in the unweighted case were as well positive.

Considering these findings, it needs to be summarized that even in the worst case scenario, the calculated minimum SROI resulted in positive outcomes. Therefore, it can be concluded, that the projects delivered by the WB have a positive effect on

the LDCs for the researched data set. Furthermore, the projects generating the highest minimum SROIs were generally projects with approx. 2.5 times above-average NPVs of 104 and 112 Mio USD, and low NPV related capital and total project costs of 40 and 42 Mio USD respectively. Additionally, those projects were generally more focused on the 'Power' and less on the 'Roads and highways' sector compared to the average NPV project of the LDCs.

5 Qualitative Data Analysis

This chapter focuses on the qualitative data analysis of ICR reports used in this research paper. The qualitative data analysis represents the second of the 2 major parts of this doctoral thesis. The following chapter discusses the type of qualitative data analysis –content analysis– used for this research project. Subsequent chapters then define and further explain the 3 types of content analyses –namely the sustainability evaluation and the bank and borrower performance. In the end, a closing chapter concludes each of the 3 areas summarizing findings and providing recommendations on how to address negative sustainability ratings and negative performance outcomes.

5.1 Content Analysis

Content analysis is a research tool used to determine the presence of certain words or concepts within texts or sets of texts. Researchers quantify and analyze the presence, meanings and relationships of words and concepts before making inferences about the messages within the texts. To conduct a content analysis the text is coded, or broken down into manageable pieces and categories on a variety of levels, such as words, word senses, phrases, sentences, or themes.²¹⁷ In principle, there are 2 complementary types of content analysis:²¹⁸

- Conceptual Analysis (quantitative): Analyzing the existence and frequency of concepts which are used the most within the text.
- Relational Analysis (qualitative): Analyzing the relationship among concepts within the text.

²¹⁷ Author's Note: Please see *Appendix CN* for the process of executing a content analysis.

²¹⁸ Cf. Mayring (2008), unpag.; Krippendorff (2004), unpag.; Hausmann/Rudolph (2014), s. 8.

In order to conduct qualitative data analyses for the 3 mentioned assessment areas of ICR reports (sustainability, bank and borrower performance), content analysis will be used.

5.2 Sustainability

The first section of this chapter provides the definition of sustainability rating of ICR reports and identifies risk factors that are potential drivers behind negative sustainability ratings. In the second part of this chapter, different categories for sustainability ratings are explained.

5.2.1 Sustainability Rating Definition

Looking at the sustainability rating within an ICR document, one must not be confused with the “three-pillar model” of Corporate Social Responsibility (CSR).²¹⁹ A sustainability rating of an ICR describes the ‘Risk to Development Outcome’ as “the risk, at the time of evaluation, that development outcomes (or expected outcomes) will not be maintained (or realized). This refers to outcomes that have actually been achieved (or are expected to be achieved).”²²⁰ In principle, the risk to development outcome has 2 dimensions:²²¹

1. The likelihood that some changes may occur from the operation that are detrimental to the ultimate achievement of the development outcome.
2. The impact from the operation and the development outcomes if some or all of these changes materialize.

²¹⁹ Cf. Lexikon der Nachhaltigkeit (2014); Lexikon der Nachhaltigkeit (2013): In general, the foundation of Corporate Social Responsibility (CSR) concept lies in the “three-pillar model” of sustainability –in particular economic and financial world. Thereafter, economic, environmental and social are *pari passu* and equally weighted, both at the macroeconomic and political level as well as at the global and corporate level.

²²⁰ WB (2013), p. 40.

²²¹ WB (2013), p. 40; WB (2010), p. 3.

There are internal risks primarily related to the operation itself and external risks which arise from factors outside of the project (e.g. at the country or global level). The sustainability rating helps to identify those operations that require a close monitoring and controlling process in managing risks which may affect project outcome and benefits. Therefore, rating ICR sustainability requires an assessment of uncertainties, which the operation might face over its remaining useful lifetime, and whether adequate measures and arrangements are in place to mitigate or even avoid the impact of those uncertainties. Defined by the WB, the ICR sustainability rating is the “evaluator’s judgment of the uncertainties faced by the operation’s development out-comes over its expected remaining useful life, taking account of any risk mitigation measures already in place at the time of evaluation.”^{222 223}

The ICR sustainability rating is 1 of 3 major rating categories that describe the overall project performance of an operation or project as described in *chapter 4.7 Overall Project Performance*. It is worth noting that sustainability rating does not give an indication for the absolute level of project benefits. For example, a project with a positive NPV or a high expected ERR but a negative sustainability rating can still receive a satisfactory overall project performance rating.²²⁴

In order to establish the most adequate and reliable assessment of a sustainability rating, the evaluator and its team (mainly project staff and ICR WB employees) must consider operational, sector, and country-specific related issues by weighing in the relative importance of each individualized criterion of a risk and how it may affect the planned project outcome – these risk factors include:²²⁵

- Technical (e.g. innovative technologies and system)
- Financial (e.g. robustness of financial flows and financial viability)
- Economic (e.g. regional and global)
- Social (e.g. strength of stakeholder support)

²²² WB (2013), p. 40.

²²³ Cf. WB (2013), p. 40; WB (2010), p. 3.

²²⁴ Cf. WB (2013), p. 40 et seq.; WB (2010), p. 3.

²²⁵ Cf. WB (2013), p. 41.

- Political (e.g. volatility of political situation)
- Environmental (including positive and negative impacts)
- Government Ownership and Commitment (e.g. supportive policies, budget)
- Other Stakeholder Ownership (e.g. from private sector/civil society)
- Institutional Support and Capacity (e.g. from project entities; and/or related to legal/legislative framework)
- Governance
- Natural Disasters Exposure

5.2.2 Sustainability Rating Categories

There are 4 rating categories suggested by the WB for the ‘Risk to Development Outcome’ rating category:²²⁶

- Negligible to Low
- Moderate
- Significant
- High

Based on the outcome of the research conducted 2 major findings need to be outlined:

1. In ICR documents disclosed before mid-2005²²⁷, rating category ‘Risk to Development Outcome’ was assessed (and named) as ‘Sustainability’ and thus has controversial rating categories.
2. Besides the above rating categories suggested by the WB, other rating categories have been used for the ‘Risk to Development Outcome’ category, such as substantial, medium, and modest.

²²⁶ Cf. WB (2013), p. 41 et seq.; WB (2013), p. 42: The lack of sufficient information, or other circumstances, makes it impossible to assign one of the above ratings, whereas “non-evaluable” should be recorded.

²²⁷ Author’s Note: The exact date can’t be determined because both situations exist: ICR documents disclosed before 2005 contain ‘Risk to Development Outcome’ as a rating category; ICR documents disclosed after 2005 contain ‘Sustainability’ as a rating category.

To account for these findings and to build the foundation for the content analysis, mapping of the ratings and categories was performed and is illustrated in the table below. Within the context of this research paper only negative rating categories are researched using the content analysis.²²⁸

Rating Category Type	Sustainability	Risk to Development Outcome
positive	Highly Likely	Low or Negligible
	Likely	
neutral	Modest	Medium / Moderate / Modest
negative	Unlikely	Substantial / Significant
	Highly Unlikely	High

Table 81: Mapping of Sustainability Rating Categories

Going forward both categories will be referred to as sustainability.

5.3 Bank Performance

This chapter focuses on the definition of ICR bank performance and its rating categories.

5.3.1 Bank Performance Definition

The performance of the WB (referred as “bank performance”) is defined as the extent to which the services provided ensures quality at entry of the cooperation (project) and provides support throughout the implementation phase through appropriate supervision to achieve planned development outcomes. Supervision also includes defining adequate transition arrangements for operations and tasks to be executed after loan/credit closing.²²⁹

²²⁸ Cf. WB (2006), p. 26.

²²⁹ Cf. WB (2013), p. 42.

Overall, the bank performance rating is assessed by 2 dimensions: ²³⁰

1. Quality at entry²³¹: Quality at entry defines to which extent the WB has identified and facilitated the preparation of the appraised project in such dimensions that the achievement of project outcomes are most likely and consistent with the bank's fiduciary role. Generally, landing performance is rated against following criteria, whereat the evaluator should account for the project environment, such as sector and country context, when weighting the relative importance of each criterion:²³²
 - Strategic relevance and approach
 - Technical, structural, financial and (macro-)economic aspects
 - Poverty, gender and social development aspects
 - Environmental aspects
 - Fiduciary aspects
 - Policy and institutional aspects
 - Implementation arrangements
 - Monitoring and evaluation arrangements
 - Risk assessment
 - Bank inputs and processes
2. Quality of supervision: Quality of supervision defines the extent to which the WB identified, prevented and managed risks to PDOs and the WB's fiduciary role. Quality of supervision is rated against following criteria, whereat as the evaluation should again account for environmental aspects when weighting the relative importance of each criterion:²³³
 - Focus on development impact
 - Supervision of fiduciary and safeguard aspects (where applicable)
 - Adequacy of supervision inputs and processes
 - Candor and quality of performance reporting

²³⁰ Cf. WB (2013), p. 42 et seq.

²³¹ Author's Note: Quality at entry is often referred as "Quality at Landing" by the WB.

²³² Cf. WB (2013), p. 42 et seq.

²³³ Cf. WB (2013), p. 43 et seq.

- Role in ensuring adequate transition arrangements for support after Loan/Credit closing

As shown, each of the 2 dimensions is rated against different criteria whereat the evaluator rates both dimensions separately.²³⁴

5.3.2 Bank Performance Categories

The (overall) bank performance rating is based on the ratings for each of the 2 mentioned dimensions –namely quality at entry and quality of supervision. For each of the 2 dimensions their own rating categories are used²³⁵, whereat in order to provide transparency both rating outcomes are presented separately but are combined to derive the overall bank performance. Ratings for the common combinations of ratings of quality at entry and quality of supervision are presented in the following table:²³⁶

	Rating Category		Definition
Positive Ratings	Highly Satisfactory	●●●	Bank performance was rated Highly Satisfactory on both dimensions.
	Satisfactory	●●	Bank performance was rated Satisfactory on both dimensions, or was rated Satisfactory on one dimension and Highly Satisfactory on the other dimension.
	Moderately Satisfactory	●	Bank performance was rated Moderately Satisfactory on both dimensions, or was rated Moderately Satisfactory on one dimension and Satisfactory or Highly Satisfactory on the other dimension.
Negative Ratings	Moderately Unsatisfactory	●	Bank performance was rated Moderately Unsatisfactory on both dimensions.
	Unsatisfactory	●●	Bank performance was rated Unsatisfactory on both dimensions, or was rated Unsatisfactory on one dimension and Moderately Unsatisfactory on the other dimension.
	Highly Unsatisfactory	●●●	Bank performance was rated Highly Unsatisfactory on both dimensions, or was rated Moderately Unsatisfactory or Unsatisfactory on one dimension and Highly Unsatisfactory on the other dimension.

Table 82: Bank Performance Rating Categories

In order to derive the overall bank performance the ratings of both dimensions need to be considered: In case the rating for 1 dimension is in the satisfactory range

²³⁴ Cf. WB (2013), p. 42.

²³⁵ Author's Note: *Appendices CO* and *CP* provide an overview of the rating categories of quality at entry and quality of supervision.

²³⁶ Cf. WB (2013), p. 45.

(“Moderately Satisfactory” or better) while the rating for the other dimension is in the unsatisfactory range, the rating for the bank performance normally depends on the overall project performance rating. Thus, overall bank performance is rated “Moderately Satisfactory” if overall project performance is rated in the satisfactory range and “Moderately Unsatisfactory” if overall project performance is rated in the unsatisfactory range.²³⁷

5.4 Borrower Performance

This chapter focuses on the definition borrower performance and its rating categories outlined within the ICR reports.

5.4.1 Borrower Performance Definition

Borrower performance is defined as the extent to which the borrower –consisting of the government and implementing agency or agencies– ensures quality of preparation and implementation based on agreements in order to ensure the achievement of the PDOs. Similar to bank performance, borrower performance is rated by assessing the following 2 dimensions:²³⁸

1. Government Performance (central and/or local government): Government performance is rated against the following criteria, whereat the evaluator should account for the project environment, such as sector and country context, when weighting the relative importance of each criterion:²³⁹
 - Government ownership and commitment to achieving PDOs
 - Enabling environment including supportive macroeconomic, sectorial, and institutional policies
 - Adequacy of beneficiary and stakeholder consultations

²³⁷ Cf. WB (2013), p. 45.

²³⁸ Cf. WB (2013), p. 46.

²³⁹ Cf. WB (2013), p. 46.

- Readiness for implementation and its arrangements and capacity, and appointment of staff
 - Timely resolution of implementation issues
 - Fiduciary, such as financial management, governance, provision of counterpart funding, reimbursements and procurement
 - Adequacy of monitoring and evaluation arrangements
 - Relationships and coordination with donors, partners and stakeholders
 - Adequacy of transition arrangements for regularly, supported activities after loan closing
2. Implementing Agency Performance: Under consideration of environmental aspects, implementation agency performance should be rated against the following criteria:²⁴⁰
- Agency commitment to achieving PDOs
 - Adequacy of beneficiary and stakeholder involvement
 - Readiness for implementation, implementation arrangements and appointment of key staff
 - Timely resolution of occurring issues
 - Fiduciary, such as financial management, governance, provision of counterpart funding, reimbursements and procurement
 - Adequacy of monitoring and evaluation arrangements
 - Relationships and coordination with partners and stakeholders
 - Appropriateness of transition arrangements for regular operations of supported activities after project closing

Where the government and implementing agency can't be distinguished, only an overall rating which takes the relevant criteria from both lists into account is necessary.²⁴¹

²⁴⁰ Cf. WB (2013), p. 47.

²⁴¹ Cf. WB (2013), p. 46.

5.4.2 Borrower Performance Categories

The overall borrower performance rating is based on the ratings for each of the 2 mentioned dimensions –namely government and implementation agency performance. For each of the 2 dimensions their own rating categories are used²⁴², whereat for transparency reasons both rating outcomes are presented separately but are combined to derive the overall bank performance. Ratings for the common combinations of ratings are presented in the following table:²⁴³

	Rating Category	Definition
Positive Ratings	Highly Satisfactory 	Performance was rated Highly Satisfactory on both dimensions.
	Satisfactory 	Performance was rated Satisfactory on both dimensions, or was rated Satisfactory on one dimension and Highly Satisfactory on the other dimension.
	Moderately Satisfactory 	Performance was rated Moderately Satisfactory on both dimensions, or was rated Moderately Satisfactory on one dimension and Satisfactory or Highly Satisfactory on the other dimension.
Negative Ratings	Moderately Unsatisfactory 	Performance was rated Moderately Unsatisfactory on both dimensions.
	Unsatisfactory 	Performance was rated Unsatisfactory on both dimensions, or was rated Unsatisfactory on one dimension and Moderately Unsatisfactory on the other dimension.
	Highly Unsatisfactory 	Performance was rated Highly Unsatisfactory on both dimensions or was rated Moderately Unsatisfactory or Unsatisfactory on one dimension and Highly Unsatisfactory on the other dimension.

Table 83: Borrower Performance Rating Categories

In order to derive the overall borrower performance ratings both dimensions need to be considered, whereat the same rules as for the overall bank performance determination apply (refer to previous chapter).²⁴⁴

²⁴² Author's Note: *Appendices CQ* and *CR* provide an overview of the rating categories of government and implementation agency performance.

²⁴³ Cf. WB (2013), p. 48.

²⁴⁴ Cf. WB (2013), p. 49.

6 Results of the Qualitative Data Analysis

In this chapter the approach and outcomes of the analyses conducted are presented and discussed. Therefore, the following chapters firstly look at each assessment area of its own, starting with the sustainability outcome, followed by the bank and borrower performances. For each of the assessment areas a quantitative counting of the negative and positive ratings is examined in the first stage. Thereafter, a classical content analysis is applied in the second stage to identify the reasons for “negative” ratings. Each of the chapters of assessment closes with a short summary and conclusion of findings from the research conducted.

6.1 Sustainability

The following chapters analyze the outcome of the sustainability ratings of the LDCs. In the next chapter the quantitative counts of the negative, neutral and positive sustainability ratings are assessed. Thereafter, the outcome of the content analysis for the negative sustainability projects is presented. For this purpose, results of the LDCs, the ones for Haiti followed by the outcome comparison of the African and Asian continents, are examined. The chapter will continue with an excursus on the results of positive NPV projects, before a summary and conclusion finalize this section.

6.1.1 Quantitative Assessment of Sustainability Ratings

In order to assess the quantitative number of negative, neutral and positive sustainability ratings, each ICR document had to be reviewed individually as the rating categories are not yet available in any other format nor have they been published anywhere else by the WB.²⁴⁵

²⁴⁵ Cf. Email from WB (Jeannette Smith) on 2012/12/27; Author’s Note: Only IEG ratings can be publicly accessed on a portfolio level via <http://data.worldbank.org/data-catalog/IEG>.

Table 84 provides an overview of the neutral and positive sustainability ratings for the LDCs per continent, whereat the ratings related to the risk to development outcome category have been added to the corresponding sustainability rating category in order to provide a comprehensive summary (same mapping approach as for the negative sustainability rating; refer to *chapter 5.2.2 Sustainability Categories, table 81: Mapping of Sustainability Rating Categories*):

	Neutral Sustainability Counts	Positive Sustainability Counts
Total	160	364
Africa	121	276
Asia	36	88
Latin America (Haiti)	3	0

Table 84: Quantitative Assessment of Neutral and Positive Sustainability Ratings²⁴⁶

Approx. 20 percent (160 out of 790) of all ICR documents disclose a neutral rating for sustainability, such as modest or moderate. On a global level, 364 out of 790 (46.1 percent) projects exhibit a positive rating. Interestingly enough, Asia contains approx. the same number of projects with positive and negative sustainability ratings (roughly 80 projects). On the contrary, Haiti does not even disclose 1 project with a positive rating. In Africa – the continent where most projects have been executed – nearly half of the projects (about 48 percent) received a positive rating for sustainability.

In conclusion, it can be adhered that these WB projects are rather sustainable in terms of having a positive likelihood that the project outcome will be sustained after project closing, as there are more positive than negative ratings available.

Table 85 provides an overview of the negative sustainability ratings per continent as well as per category:

²⁴⁶ Author's Note: The mismatch in the sum of the total number of projects per category and the total number of ICR documents (790) is due to the fact that some documents did not provide a rating for sustainability.

	ICR Report Counts	Negative Sustainability Counts Quotations	Sustainability „Unlikely“ Counts Quotations	Sustainability „Highly Unlikely“ Counts Quotations
Total	790	263 1.432	212 1.107	51 325
Africa	574	174 896	145 723	29 173
Asia	206	82 501	64 371	18 130
Latin America (Haiti)	10	7 35	3 13	4 22

Table 85: Quantitative Assessment of Negative Sustainability Ratings²⁴⁷

When broken down by percentages, the “Highly Unlikely” sustainability rating makes up about 20 percent of the total negative sustainability ratings. In Africa, 17 percent of the negative sustainability projects have a “Highly Unlikely” rating. In Asia about 22 percent account for this rating. In Haiti 4 out of 10 projects disclose this negative sustainability rating.

As a summary it must be outlined that in total 33.3 percent of the projects analyzed have an “Unlikely” or “Highly Unlikely” sustainability rating (30.3 percent in Africa, 39.8 percent in Asia).

However, it must be noted that the sustainability rating may always be subject to change due to any reasons which might not have been foreseen during the rating period –more than likely resulting in a positive rating moving towards a more negative rating rather than the other way around.²⁴⁸

²⁴⁷ Author’s Note: The numbers mentioned in the lower right corner for each category represent the number of quotations used.

²⁴⁸ WB (2013), p. 40 et sqq.

6.1.2 Outcome of the Content Analysis

In order to evaluate the reasons why achieved project outcomes might not be sustained after project completion with a “significant” or even “high” negative sustainability rating, classical content analyses were conducted using ATLAS.ti. The following chapters outline the results of the content analyses per continent.

6.1.2.1 Types of Reasons

In order to assess the types of reasons for negative sustainability ratings within the LDCs as a whole, each sustainability ICR report was researched using the qualitative toolset of content analysis. The following table provides an overview of the existing types of reasons which were identified across all LDCs:

Category
Macroeconomic / Country Risks
Governmental Risks Government Commitment Risks Government Action / Task Risks Government Capacity Risks
Political Risks
Security Risks Natural Disasters / Environmental Risks War / Conflict Risks Epidemic Risks Thirst & Hunger
Corruption
Fiduciary Risks Funding / Donor Risks Payment Risks
Implementation Capacity / Institutional Risks
Infrastructure Risks
Project Risks PDO Risks Implementation Agency Risks Bank Risks Technical Risks Other Project Risks

Table 86: Identified Reasons for a Negative Sustainability Rating

Macroeconomic or country risks affect the country as a whole and can be due to external/global reasons, such as a financial crisis. Governmental risks relate to the government itself, its (lack of) commitment, its (lack of) actions and its limited capacity²⁴⁹ (e.g. resources). The third type of reason for a negative rating in sustainability is political risks, such as political instability or uncertainty within the country perhaps due to upcoming elections. Another group of risks identified are security risks – sub-divided by natural disaster and environmental risks, war and conflicts, epidemic risks, and thirst and hunger. The fifth category of reasoning is corruption which is still a major issue in the LDCs. Following this, fiduciary risks, implementation capacity and institutional risks, and infrastructure risks are 3 additional reasons mentioned. Lastly, project specific risks such as PDO risks (which directly have an impact on the project’s PDOs), implementation agency risks, bank risks, technical risks, and other project risks make up the final reasoning in this assessment.²⁵⁰

6.1.2.2 Overall Results

Looking at the LDCs as a whole (including Haiti), 33.3 percent have a negative outcome for the sustainability rating (263 out of 790 projects). Out of the 263 projects, 212 projects have an “Unlikely” rating and 51 have a “Highly Unlikely” rating for sustainability (refer to *chapter 6.1.1 Quantitative Assessment of Sustainability Ratings, table 85: Quantitative Assessment of Negative Sustainability Ratings*). The following table highlights the major reasons for negative sustainability ratings within the LDCs compared against the total number of 790 projects researched within this paper:

²⁴⁹ Author’s Note: In case where government capacity was mentioned in the context of funding the code was counted under point 6 Funding Risks instead of 2.3 Government Capacity Risks.

²⁵⁰ Author’s Note: *Appendix CS* provides an overview of appropriate codes used for the most relevant types of reasons.

Category	% of Projects	% of total Projects
Macroeconomic / Country Risks	27.4	9.1
Governmental Risks	44.9	14.9
Government Commitment Risks	27.4	9.1
Government Action / Task Risks	22.4	7.5
Government Capacity Risks	10.6	3.5
Political Risks	20.2	6.7
Security Risks	28.1	9.4
Natural Disasters / Environmental Risks	6.8	2.3
War / Conflict Risks	23.2	7.7
Epidemic Risks	0.8	0.3
Thirst & Hunger	0.8	0.3
Corruption	4.2	1.4
Fiduciary Risks	54.8	18.2
Funding / Donor Risks	54.4	18.1
Payment Risks	4.6	1.5
Implementation Capacity / Institutional Risks	50.6	16.8
Infrastructure Risks	8.7	2.9
Project Risks	48.3	16.1
PDO Risks	27	9
Implementation Agency Risks	19.8	6.6
Bank Risks	1.9	0.6
Technical Risks	11.4	3.8
Other Project Risks	19	6.3

Table 87: Top Reasons for a Negative Sustainability Rating across LDCs²⁵¹

The “% of Projects” column represents the percentages of the risk category in comparison to the 263 negative sustainability projects, whereat the “% of total projects” shows the share in regards to the total of 790 projects. The majority of reasoning for the LDCs to share a negative sustainability rating lies behind fiduciary risks (54.8 percent), whereat this affects about 18 percent of all WB projects researched in the context of the LDCs. More than 50 percent of the negative sustainability rated projects have implementation capacity and institutional risks, close to 50 percent have project-related risks, specifically in areas of PDO and implementation agency. Fur-

²⁵¹ Author’s Note: See *Appendix CT* for reasons for „Unlikely“ and „Highly Unlikely“ sustainability ratings across all LDCs.

thermore, nearly 45 percent of the negative projects today are assessed with governmental risks, followed closely by security risks at 30 percent (roughly 20 percent of these being due to war and conflict). Interestingly, corruption and political risks are still 2 types of risks which affect more than 20 percent of the negative sustainability projects, whereat infrastructure risks represent the smallest type of reasoning with a share of 8.7 percent only.

Out of 263 negative sustainability projects across the LDCs there are 81 sector codes available in total. Table 88 represents the 10 most common sector codes across the negative sustainability LDC projects, whereat the %-points and the %-share are given based on the project type definition already presented in *chapter 2.5.3 Project Types* (refer to *table 3: Example of a Project Type Definition*):²⁵²

Sector Code	%-Points	%-Share
Central government administration	54.04	20.4
Health	27.32	10.31
Other social services	16	6.04
Power	13.22	4.99
Roads and highways	11.77	4.44
Water supply	10.42	3.93
General public administration sector	9.08	3.43
Primary education	8.98	3.39
Agricultural extension and research	8.08	3.05
Irrigation and drainage	6.82	2.57
Total	165.73	62.55

Table 88: Most common Sector Codes across Negative Sustainability Projects of the LDCs²⁵³

The ‘Central government administration’ sector code counts about 54 %-points and is thus the largest shared among sector codes (about 20 percent). It belongs to 148

²⁵² Author’s Note: Sector codes were assessed using both negative sustainability ratings („Unlikely“ and „Highly Unlikely“).

²⁵³ Author’s Note: The uneven %-points of 165.73 are the result of some projects not disclosing whole %-points in total.

out of the 263 total negative sustainability projects. The 'Health' sector code counts approx. 27 %-points and has a share of roughly 10 percent, accounting for 61 projects. Other major sector codes to be noted are 'Other social services', 'Power', 'Roads and highways', as well as water supply.

In order to better understand for which type of projects the negative sustainability project accounts, following %-shares for the project type categories can be outlined:

Project Type Category	LDCs
	%-Share
Governmental Administration	22.87
Health	12.41
Transportation Development	10.98
Other	9.38
Agriculture	8.52
Education	8.1
Sectorial Development & Reforming	8.06
Power	7.24
Water & Sanitation	5.85
Financial Development	4.17
Natural Resources	1.26
IT & Communication	1.16
Total	100

Table 89: Project Type Categories of the Negative Sustainability Projects of the LDCs²⁵⁴

Other than for the NPV projects of the LDCs (refer to *chapter 4.3.1 Result NPVs of the LDCs, table 14: Project Type Categories of the NPV Projects of the LDCs*) the 'Governmental Administration' project type represents the major project type of the

²⁵⁴ Cf. *Appendix CU* provides an overview of all 81 available sector codes and their %-shares assigned to the corresponding project type categories.

negative sustainability projects of the LDCs with a share of about 23 percent. Additional major project types are 'Health' (12.41 percent) and 'Transportation Development' (10.98 percent).

Out of the 263 projects there are only a few projects which share the same sector codes. In most cases only 2 projects share the same sector code(s), whereat most of the time projects are based on 1 sector code only. In the latter case, the %-point is 1 and the same for both projects. The following table provides an overview of the projects which are based on 1 sector code only:

Number of projects sharing the same sector code	Project Report Numbers	Shared Sector Codes
11	25705; 29215; ICR1458; 36329; ICR1489; 32710; ICR1536; ICR352; ICR1688; ICR2074; ICR1374	• Central government administration
4	25102; ICR152; ICR785; ICR1143	• Power
2	22231; 25624	• Agricultural extension and research
2	20756; ICR1968	• Irrigation and drainage
2	22167; 22804	• Primary education
2	ICR2154; ICR1041	• Railways
2	22416; 21529	• Oil & Gas Transportation
2	37359; ICR666	• General public administration sector
2	ICR2421; ICR1200	• Law and justice

Table 90: Project with common Sector Codes and Percentage Points

11 projects share the 'Central government administration' sector code while additional 4 projects the 'Power' sector code. Furthermore, there are 7 groups which consist of 2 projects each, sharing a different sector code. Another 46 projects can be grouped into smaller project groups sharing at least the same sector codes per group, whereat in none of the project groups the %-points per sector code match.²⁵⁵ Therefore, it can be concluded that besides the 29 projects mentioned in the table above, the rest of the 188 projects do not have matching sector codes and therefore have different project types. In order to increase the probability of finding patterns that can explain relationship between certain sector codes, their %-points and the

²⁵⁵ Author's Note: See *Appendix CV* for an overview of matching sector codes and the corresponding projects. In total, there are 75 projects consisting of 23 groups which share the same sector code(s).

effect on the sustainability rating, a detailed statistical analysis is required. Furthermore, the fundamental issue that the WB costs only making up about 27 percent of the total projects costs for the 168 NPV projects as described in *chapter 4.3.1 Result NPVs of the LDCs* remains for the negative sustainability projects: A sample analysis of roughly about 100 negative sustainability projects has shown that the WB costs account for only less than 40 percent of the total project costs. Therefore, further in-depth statistical analyses can be conducted only after the sector codes of the total project funding are known.

As an outcome of this analysis it needs to be noted that the project types of the negative sustainability projects of the LDCs:

- Are based on 81 different sector codes but not representative for the total funding amount of those projects.
- Vary extremely and are rarely the same across projects/countries.
- Can only serve as “trend-setters” due to the mentioned limitations.

6.1.2.3 Results in Haiti

Haiti is the only country within LA that belongs to the LDCs and therefore not representative for LA as a continent. Nevertheless, as outlined in *chapter 6.1.1 Quantitative Assessment of Sustainability Ratings, table 84: Quantitative Assessment of Neutral and Positive Sustainability Ratings* it contains 10 ICR documents out of which 7 do have a negative sustainability rating.²⁵⁶ The following table is provided to highlight the top reasons for receiving “Unlikely” and “Highly Unlikely” sustainability ratings in Haiti by comparing the percentages of negative projects per reasoning type to all negative sustainability projects (“% of Projects”) and in relation to all (10) projects (“% of total Projects”) within the country:

²⁵⁶ Cf. *Chapter 6.1.1 Quantitative Assessment of Sustainability Ratings, table 85: Quantitative Assessment of Negative Sustainability Ratings*.

Category	% of Projects	% of total Projects
Macroeconomic / Country Risks	57.1	40
Governmental Risks	42.9	30
Government Commitment Risks	42.9	30
Government Action / Task Risks	28.6	20
Government Capacity Risks	-	-
Political Risks	71.4	50
Security Risks	42.9	30
Natural Disasters / Environmental Risks	42.9	30
War / Conflict Risks	-	-
Epidemic Risks	-	-
Thirst & Hunger	-	-
Corruption	-	-
Fiduciary Risks	42.9	30
Funding / Donor Risks	42.9	30
Payment Risks	-	-
Implementation Capacity / Institutional Risks	71.4	50
Infrastructure Risks	-	-
Project Risks	14.3	10
PDO Risks	-	-
Implementation Agency Risks	-	-
Bank Risks	-	-
Technical Risks	-	-
Other Project Risks	14.3	10

Table 91: Top Reasons for a Negative Sustainability Rating in Haiti²⁵⁷

According to the research conducted, the major reasons for a negative sustainability outcome in Haiti are implementation capacity and institutional risks, political risks, and macroeconomic risks, followed by 3 additional types, security, fiduciary and governmental risks, with equal shares.

6.1.2.4 Continent Comparison

In this chapter, the outcome of the content analyses of Africa and Asia are presented and compared.²⁵⁸

²⁵⁷ Author's Note: See *Appendix CW* for reasons for „Unlikely“ and „Highly Unlikely“ sustainability ratings in Haiti.

²⁵⁸ Author's Note: Due to fact that Haiti is the only LDC in LA continent, LA will not be part of the continental comparison.

In Africa, 30 percent of the projects outline a negative outcome for the sustainability rating (174 out of 574). Out of the 174 projects, 145 projects have an “Unlikely” while 29 disclose a “Highly Unlikely” rating for sustainability.²⁵⁹ The following table provides an overview of the major reasons for a negative sustainability rating within Africa (174 to count) compared to the total number of projects researched within the continent (574 in total; refer to column “% of total Projects”). The comparison is based on the combination of both negative sustainability ratings (“Unlikely” and “Highly Unlikely”):

Category	% of Projects	% of total Projects
Macroeconomic / Country Risks	29.9	9.1
Governmental Risks	42.5	12.9
Government Commitment Risks	25.3	7.7
Government Action / Task Risks	20.1	6.1
Government Capacity Risks	7.5	2.3
Political Risks	17.8	5.4
Security Risks	23	7
Natural Disasters / Environmental Risks	6.3	1.9
War / Conflict Risks	18.4	5.6
Epidemic Risks	0.6	0.2
Thirst & Hunger	1.1	0.3
Corruption	1.1	0.3
Fiduciary Risks	60.3	18.3
Funding / Donor Risks	60.3	18.3
Payment Risks	4	1.2
Implementation Capacity / Institutional Risks	49.4	15
Infrastructure Risks	6,9	2.1
Project Risks	50	15.2
PDO Risks	27.6	8.4
Implementation Agency Risks	18.4	5.6
Bank Risks	1.7	0.5
Technical Risks	10.3	3.1
Other Project Risks	19.5	5.9

Table 92: Top Reasons for a Negative Sustainability Rating in Africa²⁶⁰

²⁵⁹ Cf. Chapter 6.1.1 Quantitative Assessment of Sustainability Ratings, table 85: Quantitative Assessment of Negative Sustainability Ratings.

²⁶⁰ Author's Note: See Appendix CX for reasons for „Unlikely“ and „Highly Unlikely“ sustainability ratings in Africa.

The major issues in terms of making project outcomes in Africa sustainable are fiduciary risks (60.3 percent), especially due to lack of donor funding, whereat a donor could be the government or loans or grants from external sources. Project-related risks (50 percent) represent the secondary majority of reasoning in Africa, directly followed by implementation capacity and institutional risks (49.2 percent). Governmental risks (40 percent) and macroeconomic or country risks (30 percent) account for the last majority of reasoning for negative sustainability projects.

In Asia, roughly 40 percent of the projects receive a negative outcome for the sustainability rating (82 out of 206 projects), wherein the bulk of these projects (64 in total) has an “Unlikely” rating for sustainability and only 18 ICR reports receive a “Highly Unlikely” sustainability rating.²⁶¹ The following table is provided to present the major reasons for the negative sustainability rating for projects within Asia (82 to count), showing comparison against the corresponding total percentage of ICR reports of Asia as a basis (206 in total; refer to column “% of total Projects”):

²⁶¹ Cf. *Chapter 6.1.1 Quantitative Assessment of Sustainability Ratings*, table 85: *Quantitative Assessment of Negative Sustainability Ratings*.

Category	% of Projects	% of total Projects
Macroeconomic / Country Risks	20.7	8.3
Governmental Risks	56.1	22.3
Government Commitment Risks	31.7	12.6
Government Action / Task Risks	26.8	10.7
Government Capacity Risks	18.3	7.3
Political Risks	22	8.7
Security Risks	41.5	16.5
Natural Disasters / Environmental Risks	6.1	2.4
War / Conflict Risks	36.6	14.6
Epidemic Risks	1.2	0.5
Thirst & Hunger	-	-
Corruption	11	4.5
Fiduciary Risks	52.4	20.9
Funding / Donor Risks	47.6	18.9
Payment Risks	7.3	2.9
Implementation Capacity / Institutional Risks	62.2	24.8
Infrastructure Risks	13.4	5.3
Project Risks	62.2	24.8
PDO Risks	32.9	13.1
Implementation Agency Risks	25.6	10.2
Bank Risks	2.4	1
Technical Risks	15.9	6.3
Other Project Risks	20.7	8.3

Table 93: Top Reasons for a Negative Sustainability Rating in Asia²⁶²

Interestingly, the 3 leading reasons (implementation capacity/institutional risks at 62.2 percent, project-related risks at 62.2 percent, and governmental risks at 56.1 percent) account for far more than 50 percent of the negative sustainability projects. Furthermore, it can be stated that about every fourth project executed in Asia is facing either implementation capacity/institutional or project risks. Additionally, governmental risks and fiduciary risks account for more than 50 percent of the negative ICR reports, affecting at least every fifth project in Asia. Security risks such as war and conflict make up the third type of reason with about 40 percent of the negative sustainability ratings.

In order to allow for a direct comparison of the 2 continents the results outlined above are summarized in the following table:

²⁶² Author's Note: See *Appendix CY* for reasons for „Unlikely“ and „Highly Unlikely“ sustainability ratings in Asia.

Category	Africa		Asia	
	% of Projects	% of total Projects	% of Projects	% of total Projects
Macroeconomic / Country Risks	5 (29.9)	9.1	20.7	8.3
Governmental Risks	4 (42.5)	12.9	2 (56.1)	22.3
Political Risks	17.8	5.4	22	8.7
Security Risks	23	7	4 (41.5)	16.5
Corruption	1.1	0.3	11	4.5
Fiduciary Risks	1 (60.3)	18.3	3 (52.4)	20.9
Implementation Capacity / Institutional Risks	3 (49.4)	15	1 (62.2)	24.8
Infrastructure Risks	6.9	2.1	13.4	5.3
Project Risks	2 (50)	15.2	1 (62.2)	24.8

Table 94: Direct Risk Comparison of Asia and Africa

The table above reflects 2 major points of interest:

1. Based on the risk assessment for Asia, the percentage numbers for the total number of projects (“% of total Projects” column) within the continent are considerably higher for 8 out of the 9 major risk types than that in Africa. The only risk type where Africa shows a slightly higher percentage number than Asia is macroeconomics. Therefore, it can be concluded that the likelihood for project outcomes to not be maintained after project completion is generally higher in Asia than in Africa.
2. Perhaps contrary to common belief, corruption is indicated to be the lowest risk type for both continents. Therefore, it can be concluded that corruption is not a major risk in affecting the sustainability of the development outcome of a project. A deeper look into the projects reveals that corruption typically affects projects at the beginning when arrangements are first made and money transfers are agreed.²⁶³

²⁶³ Cf. Moyo (2009), unpag.; Ahrens (2005), unpag.

While fiduciary or funding-related risks have the highest percentage of all risks in Africa (60.3 percent), it is only the third- or fourth-ranked risk (being that there are 2 first-ranked risks) in Asia with close to 8 percent less than that in Africa (52 percent). The 2 primary risks for Asia, namely project risks and implementation capacity/institutional risks (both at 62.2 percent) are closely matched by that in Africa (second-ranked project risks at 50 percent and third-ranked implementation capacity/institutional risks at 49.4 percent). Fiduciary risks remain a high risk within both continents. However, risks related to the government (its associated commitment and actions) have been assessed with a 14 percent variance (ranked second in Asia at 56.1 percent and fourth in Africa with about 42.5 percent). A potential explanation could be inferred, i.e. due to the higher occurrence of other risks within Asia (e.g. security risks), the governmental support has suffered (note: this has not been validated within this study). Macroeconomic/country risks are ranked as the top fifth risk in Africa (approx. 30 percent), unlike its ranking in Asia (21 percent). The largest difference between the 2 continents can be found looking at the security risk which is among the top 5 risks for Asia (41.5 percent) and about 19 percent higher than that for Africa (23 percent). An explanation of this variance can be found looking at Afghanistan specifically which accounts for about 27 percent of the negative sustainability projects (22 out of 82 projects received “Unlikely” (12) and “Highly Unlikely” (10) sustainability ratings). Due to the fact that the country has been plagued by conflicts, wars and political instability for many years, almost all of the 22 projects outline that security within the country might potentially affect the project development outcome and thus increase the security risk for Asia when comparing continents at large.

In summary, the 4 highest-ranked risks in Asia are shared differently in Africa (implementation capacity/institutional risks, project risks, governmental risks, and security risks). The 2 types of risk in Africa shown to be (slightly) higher in percentage than that in Asia are macroeconomic and fiduciary.

To conclude those findings, it again needs to be mentioned that the total negative sustainability projects in Asia are in average more often affected by any risk type or by a combination of multiple risk types. In general, this indicates that projects in Asia

run worse in term of sustainability, or vice-versa, projects in Africa have a better adoption of development aid in term of making project outcomes more sustainable. In order to account for the high number of various sector codes and to consolidate findings, the sector codes were grouped by similarity to different project type categories. The following table provides a continental comparison of the %-share per project type category:²⁶⁴

Project Type Category	Africa	Asia
	%-Share	%-Share
Governmental Administration	24.56	22.71
Health	14.46	5.25
Agriculture	12.01	11.83
Other	9.61	6.44
Education	9.09	6.47
Transportation Development	8.41	11.23
Power	8.19	5.33
Sectorial Development & Reforming	5.99	12.71
Water & Sanitation	3.72	10.38
Financial Development	1.75	5.03
Natural Resources	1.39	0.76
IT & Communication	0.82	1.86
Total	100	100

Table 95: Project Type Comparison in Asia and Africa

Even though no research on the total WB costs of the negative sustainability projects was conducted the differences of the project type categories are generally larger as the ones found for the continental NPV projects (refer to *chapter 4.3.2 Result NPV Continent Comparison, table 24: Continental Project Type Category Comparison excluding “high-value” NPVs*). The largest variance in sector codes between the 2 continents can be found in the ‘Health’ project type category with roughly

²⁶⁴ Author’s Note: *Appendix CZ* provides an overview on the detailed sector codes mapping per category based on the %-share on a continental comparison.

10 percent more in Africa. In total, the health-related sector codes grouped under the 'Health' project type category affect 54 African projects compared to 7 Asian projects. Additionally, 2 further but respectively minor differences can be found in the 'Water and Sanitation' and the 'Agriculture' groupings: A variance of about 6.5 percent more for the 'Water and Sanitation' as well as the 'Sectorial Development and Reforming' project type category in Asia. All other categories show differences with percentages of lower than 5 percent and therefore have not been explored any further.

In order to further explain continental differences and to determine if any of the above mentioned project type categories account for an extremity in specific risk types, separate code reviews per project type category –using only project type category relevant projects– were conducted. *Appendix DA* provides a comparison analysis of risk types and their corresponding percentage of distribution over all African negative sustainability projects and specific African health-related projects. Fiduciary risks affected roughly about 75 percent of all health related projects in Africa – which is 15 percent more than compared to all negative sustainability projects in Africa. Besides this, no noteworthy results were found. Controlling vice versa –looking at Asian health-related category projects– only provided “insignificant” differences (lower than 5 percent) when compared to the entirety of negative sustainability projects in Asia. Furthermore, analyses of the 'Water and Sanitation', 'Sectorial Development and Reforming', 'Agriculture' and 'Education' project type categories resulted as well in insignificant differences (lower than 5 percent) for any risk type in either continent. Therefore, it must be summarized that even though some sector codes varied between the 2 continents, the research on varying sector codes in respect to their defined project types did not provide any further insights to the differences of continental risk types in general. The potential reasoning for this might be lying in the reference to the amount of the WB financing which is comparably low to the overall project funding.

6.1.3 Excursus: Positive NPV Projects

In this chapter, negative sustainability projects of the LDCs that outline a positive NPV are analyzed. The goal is to find out if there are specific types of risks which cause a negative sustainability for projects with a positive NPV at project closure. Therefore, each of the 263 negative sustainability reports was searched for positive project NPV values at project completions.²⁶⁵ The following table provides an overview of the negative sustainability projects within the LDCs, categorized per continent and by category rating type:

	Total No of "Implementation Completion & Result Reports"	Total No of „negative“ Sustainability counts	Total No of "negative" Sustainability counts of positive NPV projects	Sustainability „Unlikely“ of positive NPV projects	Sustainability „Highly Unlikely“ of positive NPV projects
TOTAL	790	263	39	34	5
Africa	574	174	19	16	3
Asia	206	82	20	18	2
Latin America (Haiti)	10	7	0	0	0

Table 96: Negative Sustainability Counts of Positive NPV Projects

On a LDC level, about 15 percent of the negative sustainability ICR documents (39 out of 263) expose a positive NPV. The highest share (34 projects) can be found in the “Unlikely” sustainability rating. Interestingly, the share as well as the absolute number of the negative sustainability ratings of the positive NPV projects in Asia (40 percent (20 out of 50 projects)) is higher than that in Africa (18 percent (19 out of 104 projects)) although Africa counts more than double of the negative sustainability

²⁶⁵ Author’s Note: In the first step, it was not distinguished if the outlined NPV relates to a single or multiple main components of the project only (“partial” NPV) or to the project as a whole (“overall” NPV).

projects (174 projects compared to that of Asia with 82 negative sustainability projects in total) as well as double of positive NPV projects (104 compared to that 50 positive NPV projects in Asia; refer to *chapter 4.3.2 Result NPV Continent Comparison, table 19: Result NPV Counts of Africa and Asia*). This outcome basically underlines the results of the previous chapter: Asian projects are generally more affected by any risk type negatively influencing the future maintenance of project outcomes than African projects, whereat this effect appears to be even stronger for positive NPV projects.

The following table compares the overall outcome amongst the LDCs' projects with the evaluated risk types in relationship to the percentage of negative sustainability projects containing a positive NPV:

Category	% of Projects	% of positive NPV Projects
Macroeconomic / Country Risks	27.4	17.9
Governmental Risks	44.9	30.8
Government Commitment Risks	27.4	20.5
Government Action / Task Risks	22.4	17.9
Government Capacity Risks	10.6	7.7
Political Risks	20.2	15.4
Security Risks	28.1	25.6
Natural Disasters / Environmental Risks	6.8	7.7
War / Conflict Risks	23.2	17.9
Epidemic Risks	0.8	-
Thirst & Hunger	0.8	-
Corruption	21.6	2.6
Fiduciary Risks	54.8	61.5
Funding / Donor Risks	54.4	56.4
Payment Risks	4.6	10.3
Implementation Capacity / Institutional Risks	50.6	56.4
Infrastructure Risks	8.7	20.5
Project Risks	48.3	51.3
PDO Risks	27	28.2
Implementation Agency Risks	19.8	20.5
Bank Risks	1.9	2.6
Technical Risks	11.4	10.3
Other Project Risks	19	20.5

Table 97: Results of Negative Sustainability Projects with a Positive NPV

The largest variance relates to corruption and shows that negative sustainability projects with a positive NPV are 19 percent less affected than the total of negative sustainability projects and thus only show a negligible effect of 2.6 percent in total.

Furthermore, negative sustainability projects with a positive NPV are approx. 15 percent less affected by governmental risks. On the other hand negative sustainability projects with a positive NPV seem to be more affected by infrastructure risks at 20.5 percent compared to 8.7 percent of the total of negative sustainability projects. Aside from this, there are 2 more noteworthy variances with less than 10 percent difference: On the one hand negative sustainability projects with a positive NPV are 9.5 percent less affected by macroeconomic risks but roughly 7 percent more affected by fiduciary risks than the total of negative sustainability projects.

6.1.4 Summary and Conclusion

Fiduciary was the primary risk relating to 54.8 percent of the negative sustainability projects, representing roughly 20 percent of all WB projects researched in the context of the LDCs. These were followed by the implementation capacity risks, and thereafter, by project specific risks. Corruption was the single type of risk which affected the least number of projects, followed by infrastructure and political risks.

Comparing the risk types of the Asian and the African continents showed various differences. In general, Asian projects were more often affected by any type of risk. When directly compared, the main 2 differences between the continents were found in the governmental and security risk types. For both risk types, negative sustainability projects in Asia showed 15 or more %-points. An explanation for the governmental differences could be the higher occurrence of other risks causing governments to be overwhelmed and therefore ineffective. The differences in the security risk type found in Afghanistan are due to the fact that the country has been in a war situation for several years.

When controlling for project types the major variance was found in the health-related project types. In this context, fiduciary risks showed to be the only noticeable difference affecting 75 percent of the negative sustainability health-related projects in Africa, whereat no noteworthy detection for Asia was found. Furthermore, it was concluded that – based on the research conducted – no other significant relationship was found amongst the minor differences within the continental project types and their associated risk types.

The excursus on the negative sustainability projects outlining a positive NPV showed that those projects are in principle not affected by corruption which was 19 percent lower comparing to that of the total of negative sustainability projects (refer to *chapter 6.1.3 Excursus: Positive NPV Projects*). Furthermore, negative sustainability projects with a positive NPV were also approx. 15 percent less affected by governmental risks, whereat on the other hand they seemed to be more affected by infrastructure risks at 20.5 percent comparing to 8.7 percent of the total of negative sustainability projects.

To ensure a higher sustainability after project completion and to cope for the discovered risks and their respective underlying factors various recommendations need to be considered. With fiduciary being the major risk affecting sustainability, this paper is in line with the generally stated need that far more aid assistance from rich countries as well as debt forgiveness and better trade terms are needed. Looking at implementation capacity as the second major risk to sustainability, donor countries and their respective development aid institutions need to ensure access to new and appropriate technologies to the LDCs. There is as well a need to provide more guidance, support and training to the respective implementation agencies and the staff on the ground. In regards to governmental risks the UN needs to get a better understanding of how to provide a more adequate assistance to support the governmental processes and tasks, such as the establishment of regulatory rules and laws, project prioritization, money distribution, and community support. With respect to security risks it is difficult to make a judgment. As outlined earlier, war and conflicts are actually “burning” money, since governmental priorities for allocating donor aid change dramatically. Especially in such conflict-environments, help and support of external sources is more than ever required to stabilize the situation within the country. Therefore, the WB might need to establish a framework which allows prioritizing, transferring and handling development aid better within the context of war- and conflict-affected countries. A start could be the re-evaluation of the country’s PRS together with the government itself.

6.2 Bank Performance

The following chapters look at the outcome of the bank performance ratings of the LDCs. First, the negative, neutral and positive sustainability ratings are assessed. Thereafter, the outcome of the content analysis conducted for the negative bank performance projects is presented. For this purpose, results of the LDCs, the ones for Haiti followed by the outcome comparison of the African and Asian continent, are examined. The chapter closes with a short summary and its findings.

6.2.1 Quantitative Assessment of Bank Performance Ratings

Similar to what is needed for sustainability assessment, the numbers of positive and negative ratings within the ICR documents had to be researched individually. Table 98 provides an overview of the positive and negative bank performance ratings for the LDCs per continent:

	ICR Report Counts	Positive Bank Performance Counts	Negative Bank Performance Counts
Total	790	652	134
Africa	574	477	94
Asia	206	168	37
Latin America (Haiti)	10	7	3

Table 98: Quantitative Assessment of Positive and Negative Bank Performance Ratings²⁶⁶

On a global level, 652 out of 790 (83 percent) projects show a positive bank performance rating. Africa discloses nearly 3 times as many positive bank performance ratings (477) than Asia (168) which corresponds to the share of the continental ICR report counts (574 African vs. 206 Asian projects). Thus, both continents disclose

²⁶⁶ Author's Note: The mismatch between the total number of ratings per category and the total number of ICR documents (790) is due to missing bank performance ratings in some documents.

roughly about 80 percent positive compared to 20 percent negative bank performance ratings. Haiti –being handled separately again– discloses 7 positive out of a total of 10 project ratings.

In summary it can be concluded that the researched WB projects are generally well supported in terms of quality at entry and quality of supervision by WB.

Table 99 provides an overview of the negative bank performance ratings per continent as well as per category:

	ICR Report Counts	Negative Bank Performance Counts Quotations	„Moderately Unsatisfactory“ Counts Quotations	„Unsatisfactory“ Counts Quotations
Total	790	134 1776	63 906	71 870
Africa	574	94 1384	44 732	50 652
Asia	206	37 365	18 166	19 199
LA (Haiti)	10	3 27	1 8	2 19

Table 99: Quantitative Assessment of Negative Bank Performance Ratings²⁶⁷

When broken down by percentages, the “Unsatisfactory” bank performance rating makes up about 53 percent of the total negative bank performance ratings. In Africa 53 percent (50 out of 94) of the projects disclose an “Unsatisfactory” rating, whereat the percentage declines slightly to 51 percent (19 out of 37) in Asia. Out of the 3 negative bank performance ratings in Haiti, 2 projects disclose “Unsatisfactory” ratings. None of the 134 negative bank performance ratings is categorized as “Highly Unsatisfactory”.

²⁶⁷ Author’s Note: The numbers mentioned in the lower right corner for each category represent the number of quotations used.

6.2.2 Outcome of the Content Analysis

In order to evaluate the reasons behind negative bank performance ratings classical content analyses using ATLAS.ti were conducted. After discussing the types of reason results of the content analyses for the LDCs as well as that per continent are presented.

6.2.2.1 Types of Reasons

In order to assess the types of reasons for negative bank performance ratings within the LDCs as a whole, each relevant ICR report was researched using the qualitative toolset of content analysis. The following table provides an overview of the existing types of reasons which have been identified across all LDCs:

Category
Bank Competence Decision Making Process Management, Coordination, Administration & Evaluation Communication
Implementation Framework General & Other – e.g. Systems, Processes, Technology Data Quality Government, Institutions & Policy Schedule/Timing Project Design Environment & Macroeconomic Restructuring (<i>only applicable for Supervision</i>)
Monitoring & Controlling General Reporting Risks & Issues KPIs PDOs
Resources & Capacity
Budget & Financing

Table 100: Identified Reasons for a Negative Bank Performance

Aspects of bank competence that negatively affect the performance rating can generally be grouped into weak or slow decision processes, poor management, coordination, administration and evaluation of the tasks and actions and lack of quality of communication. As a second main category the project implementation framework has been identified. Within the implementation framework there are numerous sub-categories, such as the provided data quality, the government, its institutions and policy, schedule and timing of project and its deliverables, project design, environment and macroeconomic, and general or other issues which cannot be grouped under one of the already mentioned categories (e.g. systems, processes and technology). The last sub-category is purely referring to issues during the restructuring of the project, whereat this group only applies to the supervision dimension. The third main category is referred to as the monitoring and controlling of the project. The sub-categories are general aspects, issues with the reporting, KPIs and PDOs as well as project relevant risks and issues which might have been missed. Last but not least the fourth and the fifth main groups are project resources and capacity issues and problems with project budget and financing.

6.2.2.2 Overall Results

As already mentioned in the previous chapter the overall results for the LDCs look quite promising with only 17 percent negative bank performance ratings. Out of those 134 projects, 63 projects disclose a “Moderately Unsatisfactory” rating and 71 have a “Unsatisfactory” rating for bank performance (refer to *chapter 6.2.1 Quantitative Assessment of Bank Performance Ratings, table 99: Quantitative Assessment of Negative Bank Performance Ratings*). The table highlights the major reasons for negative bank performance ratings within the LDCs compared against the total number of 790 projects researched within this paper. For this comparison both available performance ratings (“Moderately Unsatisfactory” and “Unsatisfactory”) and the 2 dimensions for the bank performance rating (quality at entry and quality of supervision) have been comprehensively assessed:²⁶⁸

²⁶⁸ Author’s Note: See *Appendices DB and DC* for reasons for negative bank performances across the LDCs separated by the landing and the supervision dimension.

Category	% of Projects	% of total Projects
Bank Competence	84.3	14.3
Decision Making Process	3.7	0.6
Management, Coordination, Administration & Evaluation	84.3	14.3
Communication	8.2	1.4
Implementation Framework	96	16.2
General & Other – e.g. Systems, Processes, Technology	53	9
Data Quality	22.4	3.8
Government, Institutions & Policy	47	8
Schedule/Timing	38.1	6.5
Project Design	51.5	8.7
Environment & Macroeconomic	17.2	2.9
Restructuring (<i>only applicable for Supervision</i>)	25.4	4.3
Monitoring & Controlling	87.3	14.8
General	49.3	8.4
Reporting	26.9	4.6
Risks & Issues	35.8	6.1
KPIs	44.8	7.6
PDOs	43.5	7.4
Resources & Capacity	68	11.5
Budget & Financing	35.8	6.1

Table 101: Top Reasons for a Negative Bank Performance Rating across LDCs²⁶⁹

The “% of Projects” column represents percentages of projects, out of total of 134 with a negative bank performance, with respectively identified reasons, whereat the “% of total projects” shows the share in regards to the total of 790 projects. The major reasons which affect 96 percent of all negative bank performance projects lie within the implementation framework. Main drivers are issues such as project design, systems, processes and technologies and the scheduling changes of project components. Reasons under implementation framework category affect 16.2 percent of all 790 ICR reports. Another major category affecting about 87 percent of negative bank performance projects is monitoring and controlling. Specifically, lack of clear definition and unavailability of KPIs as well as overambitious PDOs make it difficult to have an effective monitoring and controlling process. Bank’s competence in managing, coordinating and evaluating bank activities re-presents a major driver affecting about 84 percent of all negative bank performance ratings. Last but not

²⁶⁹ Author’s Note: *Appendices DD and DE* provide an overview of the reasons of negative bank performances across the LDCs for the “Moderately Unsatisfactory” and the “Unsatisfactory” rating categories.

least 68 percent of the negative bank performance projects are affected by resource and capacity constraints, e.g. unskilled workers and missing knowledge.

6.2.2.3 Results in Haiti

Out of the 10 ICR reports for Haiti only 3 disclose a negative bank performance rating.²⁷⁰ The following table highlights the top reasons for both available bank performance rating categories and dimensions in Haiti by comparing the percentages of negative projects per reason category to all negative bank performance projects (“% of Projects”) and in relation to all (10) projects (“% of total Projects”) within the country:

Category	% of Projects	% of total Projects
Bank Competence	100	30
Decision Making Process	-	-
Management, Coordination, Administration & Evaluation	100	30
Communication	-	-
Implementation Framework	100	30
General & Other – e.g. Systems, Processes, Technology	33.3	10
Data Quality	33.3	10
Government, Institutions & Policy	66.7	20
Schedule/Timing	33.3	10
Project Design	66.7	20
Environment & Macroeconomic	33.3	10
Restructuring (<i>only applicable for Supervision</i>)	-	-
Monitoring & Controlling	100	30
General	66.7	20
Reporting	33.3	10
Risks & Issues	33.3	10
KPIs	33.3	10
PDOs	33.3	10
Resources & Capacity	66.7	20
Budget & Financing	-	-

Table 102: Top Reasons for a Negative Bank Performance Rating in Haiti²⁷¹

²⁷⁰ Cf. Chapter 6.2.1 Quantitative Assessment of Bank Performance Ratings, table 99: Quantitative Assessment of Negative Bank Performance Ratings.

²⁷¹ Author’s Note: Appendices DF, DG, DH and DI provide an overview of the reasons for negative bank performances in Haiti separated by the entry and the supervision dimension and for each of the 2 rating categories (“Moderately Unsatisfactory” and “Unsatisfactory”).

As highlighted in the above table all of the 3 negative bank performance projects outline issues based on the bank competence, the implementation framework (specifically challenges with the government and issues related to project design) and with the monitoring and controlling process.

6.2.2.4 Continent Comparison

In this chapter the outcome of the content analyses of Africa and Asia is presented and compared against each other.²⁷²

About 16 percent of all African projects receive a negative outcome for the bank performance rating. Out of the 94 projects, 44 projects have a “Moderately Unsatisfactory” and 50 receive an “Unsatisfactory” rating for bank performance rating.²⁷³ The following table provides an overview of the major reasons for a negative bank performance within Africa compared against the total number of projects researched within the continent (refer to column “% of total Projects”), whereat the percentages are based on both available bank performance rating categories (“Moderately Unsatisfactory” and “Unsatisfactory”) and dimensions (quality at entry and quality of supervision):

²⁷² Author's Note: Due to fact that Haiti is the only LDC in LA, LA will not be part of the continental comparison.

²⁷³ Cf. Chapter 6.2.1 Quantitative Assessment of Bank Performance Ratings, table 99: Quantitative Assessment of Negative Bank Performance Ratings.

Category	% of Projects	% of total Projects
Bank Competence	84	13.8
Decision Making Process	5.3	0.9
Management, Coordination, Administration & Evaluation	84	13.8
Communication	7.4	1.2
Implementation Framework	94.7	15.5
General & Other – e.g. Systems, Processes, Technology	56.4	9.2
Data Quality	22.3	3.7
Government, Institutions & Policy	46.8	7.7
Schedule/Timing	39.4	6.4
Project Design	45.7	7.5
Environment & Macroeconomic	14.9	2.4
Restructuring (<i>only applicable for Supervision</i>)	27.7	4.5
Monitoring & Controlling	88.3	14.5
General	52.1	8.5
Reporting	28.7	4.7
Risks & Issues	33	5.4
KPIs	47.9	7.8
PDOs	43.6	7.1
Resources & Capacity	63.8	10.5
Budget & Financing	40.4	6.6

Table 103: Top Reasons for a Negative Bank Performance Rating in Africa²⁷⁴

As highlighted in the table above the implementation framework affects 94.7 percent of all negative bank performance projects in Africa. Looking at the total of 574 African projects researched in this paper, about every sixth project faces implementation framework category issues, most noticeably general issues with the related technology and processes, followed by governmental and institutional challenges and issues with the project design. The monitoring and controlling category accounts for 88.3 percent followed by the bank competence category affecting 84 percent of the negative bank performance projects in Africa.

In Asia, roughly 18 percent of the projects receive a negative bank performance rating (37 out of 206 projects), whereat both rating categories are equally present with 18 ratings for the “Moderately Unsatisfactory” and 19 for the “Unsatisfactory” category.²⁷⁵ The following table presents the major reasons for the negative bank

²⁷⁴ Author's Note: *Appendices DJ, DK, DL and DM* provide an overview of the reasons for negative bank performances in Africa separated by the entry and the supervision dimension and for each of the 2 available rating categories (“Moderately Unsatisfactory” and “Unsatisfactory”).

²⁷⁵ Cf. *Chapter 6.2.1 Quantitative Assessment of Bank Performance Ratings, table 99: Quantitative Assessment of Negative Bank Performance Ratings.*

performance ratings of projects within Asia (37 to count) and shows 2 types of percentage calculations driven by different comparison basis, namely counts of projects with negative bank performance ratings (“% of Projects”) and total number of ICR reports in Asia (“% of total Projects”):

Category	% of Projects	% of total Projects
Bank Competence	83.8	15
Decision Making Process	-	-
Management, Coordination, Administration & Evaluation	83.8	15
Communication	-	-
Implementation Framework	97.3	17.5
General & Other – e.g. Systems, Processes, Technology	45.9	8.3
Data Quality	21.6	3.9
Government, Institutions & Policy	45.9	8.3
Schedule/Timing	32.4	5.8
Project Design	62.1	11.2
Environment & Macroeconomic	21.6	3.9
Restructuring (<i>only applicable for Supervision</i>)	22.7	5.8
Monitoring & Controlling	83.8	15
General	37.8	6.8
Reporting	21.6	3.9
Risks & Issues	43.4	7.8
KPIs	37.8	6.8
PDOs	43.4	7.8
Resources & Capacity	78.4	14.1
Budget & Financing	27	4.9

Table 104: Top Reasons for a Negative Bank Performance Rating in Asia²⁷⁶

Interestingly, the 3 leading rating categories in Asia (implementation framework at 97.3 percent, monitoring and controlling and bank competence both at 83.8 percent) are the same ones seen in Africa with only minor differences. Project design issues in the implementation framework category affect 62.1 percent of the projects with negative bank performance rating, whereat the resources and capacity category 78.4 percent.

²⁷⁶ Author's Note: *Appendices DN, DO, DP and DQ* provide an overview of the reasons for negative bank performances in Asia separated by the entry and the supervision dimension and for each of the 2 available rating categories (“Moderately Unsatisfactory” and “Unsatisfactory”).

In order to compare the reasons for the negative bank performance ratings in Africa and Asia, the most relevant results differing the 2 continents are summarized in the following table:²⁷⁷

Category	Africa		Asia	
	% of Projects	% of total Projects	% of Projects	% of total Projects
Bank Competence	84	13.8	83.8	15
Implementation Framework	94.7	15.5	97.3	17.5
Project Design	45.7	7.5	62.1	11.2
General & Other	56.4	9.2	45.9	8.3
Monitoring & Controlling	88.3	14.5	83.8	15
General	52.1	8.5	37.8	6.8
Risks & Issues	33	5.4	43.4	7.8
KPIs	47.9	7.8	37.8	6.8
Resources & Capacity	63.8	10.5	78.4	14.1
Budget & Financing	40.4	6.6	27	4.9

Table 105: Direct Comparison of African and Asian Bank Performance

The direct comparison of the continents reflects some major points of interest. First of all, the 3 major reason categories in both continents (implementation framework, monitoring and controlling and bank competence) show similar percentages, whereat differences still lie in the sub-categories. For instance, project design issues (implementation framework category) affect about 16 percent fewer of the negative bank performance projects in Africa than in Asia. On the other hand, general issues with technologies and processes affect more than 10 percent more African than Asian negative bank performance projects. In the monitoring and controlling category, risks and issues affect about 10 percent fewer negative bank performance projects in Africa than in Asia, whereat the picture is reversed for the KPIs with 47.9 percent in Africa and 37.8 percent in Asia. Also worth noticing is that resources and capacity category affects 63.8 percent of the negative bank performance projects in Africa, about 15 percent lower than the 78.4 percent in Asia. Furthermore, budget and financing category –although accounting for 40.4 percent in Africa– does not have major influence on negative bank performance projects in either of the conti-

²⁷⁷ Author's Note: Sector codes with respect to project types have not been researched for this comparison, since the WB financing of the negative bank performance projects is less than 50 percent in both continents.

nents. The main categories are easy to identify when looking at the “% of total Projects” column, although they show only minimal continental differences. One exception is the budget and financing category which affects only 4.9 percent of all 206 Asian projects and 6.6 percent of the 574 African projects. In summary, Asian projects are on average more affected by any of the major categories than the African projects (same as for the sustainability risk categories; refer to *chapter 6.1.2.4 Continent Comparison, table 94: Direct Risk Comparison of Asia and Africa*).

6.2.3 Summary and Conclusion

Across the LDCs the implementation framework category was the major category affecting 96 percent of all negative bank performance projects. Specifically, issues with unrealistic project designs and poor support of institutions and the government were causing negative bank performance ratings, followed by the monitoring and controlling category at 87.3 percent and the bank competence category at 84.3 percent.

The outcome in terms of category relevance was also shared by Haiti and the African and Asian continents. Continental comparison between Asia and Africa showed similarities for the main categories but with differences in the sub-categories. Besides higher percentages for general sub-categories of the implementation framework and the monitoring and controlling category as well as for the KPIs sub-category and the budget and financing category in Africa, challenges according to the risks and issues and the project design sub-categories as well as the resources and capacity category were affecting more projects in Asia. Furthermore, it had been found that Asian projects were on average more often affected by any main category than the African projects –except for the budget and financing category.

Since both continents disclosed a high percent for the management, coordination, administration and evaluation sub-category respectively the bank competence category (84 percent in Africa and 83.8 percent in Asia), the need for the WB to take a closer look at its own project management team in order improve performance should be highlighted. Additionally, ensuring a more realistic project design through proper consideration for the framework of resources, support of government and its

institutions, planned project deliverables and timeframes is a step into the right direction. Furthermore, defining KPIs and PDOs more properly considering input factors such as resources and capacity and the project environment based on a corporate approach between the WB and the borrower could further improve the bank performance.

6.3 Borrower Performance

The following chapters take a detailed look at the outcome of the borrower performance ratings of the LDCs. The first chapter examines the quantity of positive and negative borrower performance ratings. Thereafter, the results of the content analysis conducted for the negative borrower performance projects are presented. For this purpose, the outcomes of the LDCs, the ones for Haiti followed by the outcome comparison of the African and Asian continents, are examined. The chapter closes with a short summary and its findings.

6.3.1 Quantitative Assessment of Borrower Performance Ratings

Similar to the assessment performed for sustainability and banking performance ratings, borrower performance assessment required a quantitative analysis of the positive and negative ratings across all ICR documents. Table 106 provides an overview of the positive and negative borrower performance ratings for the LDCs per continent:

	ICR Report Counts	Positive Borrower Performance Counts	Negative Borrower Performance Counts
Total	790	616	167
Africa	574	451	120
Asia	206	160	42
Latin America (Haiti)	10	5	5

Table 106: Quantitative Assessment of Positive and Negative Borrower Performance Ratings²⁷⁸

On a global level, 616 out of 790 (78 percent) projects show a positive and 167 (21 percent) a negative borrower performance rating. Africa discloses nearly 3 times as many positive borrower performance ratings (451) than Asia (120), whereat both continents have much higher positive rating percentages (approx. 80 percent) than negative ones. Haiti –being handled separately again– discloses equal numbers of positive and negative ratings.

In summary it can be concluded that the researched WB projects are generally well supported by the fund receiving countries –specifically by their governments and the implementation agencies.

Table 107 provides an overview of the negative borrower performance ratings per continent and per category:

	ICR Report Counts	Negative Borrower Performance Counts Quotations	„Moderately Unsatisfactory“ Counts Quotations	„Unsatisfactory“ Counts Quotations	„Highly Unsatisfactory“ Counts Quotations
Total	790	167 869	72 405	93 443	2 21
Africa	574	120 621	44 257	75 354	1 10
Asia	206	42 225	27 144	15 81	0 0
LA (Haiti)	10	5 23	1 4	3 8	1 11

Table 107: Quantitative Assessment of Negative Borrower Performance Ratings²⁷⁹

²⁷⁸ Author's Note: The mismatch in the sum of the total number per category in comparison to the total number of ICR documents (790) is due to missing borrower performance ratings in some documents.

²⁷⁹ Author's Note: The numbers mentioned in the lower right corner for each category represent the number of quotations used.

When broken down by percentages, the “Unsatisfactory” borrower performance rating makes up about 56 percent of the total negative borrower performance ratings of the LDCs. In Africa 63 percent (75 out of 120) of the projects disclose an “Unsatisfactory” rating, whereat the percentage drastically declines to 36 percent (15 out of 42) for the Asian continent. In Haiti, out of 5 negative borrower performances, 3 are rated “Unsatisfactory”. In total there are only 2 “Highly Unsatisfactory” ratings – 1 in Africa and 1 in Haiti.

6.3.2 Outcome of the Content Analysis

In order to evaluate the reasons behind negative borrower performance ratings classical content analyses using ATLAS.ti were conducted. After discussing the types of reasons, results of the content analyses for the LDCs as well as that per continent are presented.

6.3.2.1 Types of Reasons

As for the sustainability and the bank performance ratings each relevant ICR report was researched for negative borrower performance reasons using the qualitative toolset of content analysis. The following table provides an overview of the existing types of reason which have been identified across all LDCs:

Category
Borrower Competence Commitment Decision Making Process "Management" of Things Leadership, Coordination & Administration Communication Cooperation
Implementation Framework General & Other – e.g. Systems, Processes, Technology Data Quality Project Design Institutional Setup Policy Corruption Schedule/Timing Environment & Macroeconomic
Monitoring & Controlling General Reporting PDOs
Resources & Capacity
Budget & Financing

Table 108: Identified Reasons for a Negative Borrower Performance

Overall, reasons in the borrower competence category can be grouped into several sub-categories, such as missing commitment, weak or slow decision processes, “management” of things²⁸⁰ as well as leadership, coordination and administration of the resources and tasks, quality of communication and borrower co-operation. The second main group concerns with the implementation framework, similar to what was already presented for the bank performance. Differences are: Bank performance sub-category government, institutions and policy can be split into institutional setup and policy; 1 additional sub-category –namely corruption– was added while the restructuring sub-category was removed. Monitoring and controlling of the project was identified as the third main group, with general aspects, reporting of the project and PDOs as its sub-categories. Last but not least the fourth and fifth main

²⁸⁰ Author’s Note: The “management” of things sub-category contains all quotations where “management” as a noun has been used.

groups were defined to focus on challenges around project resources and capacity as well as project budget and financing.

6.3.2.2 Overall Results

This chapter focuses on the results of the negative borrower performance ratings of the LDCs which make up approx. 17 percent of the total of 790 ICR reports. Out of the 167 projects with negative borrower performance ratings, 72 disclose a “Moderately Unsatisfactory” and 93 an “Unsatisfactory” rating (refer to *chapter 6.3.1 Quantitative Assessment of Borrower Performance Ratings*, *table 107: Quantitative Assessment of Negative Borrower Performance Ratings*). The following table highlights the major reasons for negative borrower performance ratings within the LDCs (“% of Projects” column) compared against the total number of 790 projects (“% of total Projects” column). For this comparison both available performance ratings (“Moderately Unsatisfactory”, “Unsatisfactory” and “Highly Unsatisfactory”) and the 2 dimensions for the borrower performance rating (government and implementation agency) were comprehensively assessed:²⁸¹

²⁸¹ Author’s Note: See *Appendices DR* and *DS* for reasons for negative bank performances across the LDCs separated by the government and implementation agency dimension.

Category	% of Projects	% of total Projects
Borrower Competence	91.6	19.5
Commitment	47.3	10
Decision Making Process	13.2	2.8
"Management" of Things	44.9	9.5
Leadership, Coordination & Administration	74.3	15.7
Communication	12.6	2.7
Cooperation	6.6	1.4
Implementation Framework	50.9	10.9
General & Other – e.g. Systems, Processes, Technology	19.8	4.2
Data Quality	3.6	0.8
Project Design	7.8	1.6
Institutional Setup	6.6	1.4
Policy	8.4	1.8
Corruption	1.2	0.3
Schedule/Timing	18	3.8
Environment & Macroeconomic	13.2	2.8
Monitoring & Controlling	37.7	8
General	19.2	4.1
Reporting	10.8	2.3
PDOs	24.6	5.2
Resources & Capacity	51.5	11
Budget & Financing	22.8	4.8

Table 109: Top Reasons for a Negative Borrower Performance Rating across LDCs²⁸²

Different than for the bank performance the major reasons behind negative borrower performance ratings are not lying within the implementation framework category. Instead it is the borrower competence category which affects 91.6 percent of the total of 167 negative borrower performance projects. For the whole data set of 790 ICR reports, roughly 20 percent (every fifth project) suffer from weak borrower competence. The main drivers within this category are leadership, coordination and administration, commitment of the government and its institutions as well as the “management” of things. The resources and capacity category affects 51.5 percent of the negative borrower performance projects. Causes collected under the implementation framework category, e.g. challenges or issues affecting project systems and processes or schedule interference by macroeconomic circumstance, affect 50.9 percent of all negative borrower performance projects and about 11 percent of all

²⁸² Author's Note: *Appendices DT, DU and DV* provide an overview of the reasons of negative bank performances across the LDCs for the “Moderately Unsatisfactory”, “Unsatisfactory” and “Highly Unsatisfactory” rating categories.

790 ICR reports researched within this paper. Interestingly, corruption sub-category, affecting the least number of projects (1.2 percent), is not a relevant reason for negative borrower performance ratings. Finally, categories monitoring and controlling as well as budget and financing have a noticeable influence on borrower performance, namely 37.7 and 22.8 percent respective.

6.3.2.3 Results in Haiti

Out of the 10 Haiti ICR reports 5 disclose a negative borrower performance rating.²⁸³ The following table highlights the top reasons for both the 3 available borrower performance rating categories and their 2 dimensions in Haiti by comparing the percentages of negative projects per reason category to all negative borrower performance projects (“% of Projects”) and in relation to all (10) projects (“% of total Projects”) within the country:

Category	% of Projects	% of total Projects
Borrower Competence	80	40
Commitment	20	10
Decision Making Process	40	20
"Management" of Things	20	10
Leadership, Coordination & Administration	40	20
Communication	-	-
Cooperation	-	-
Implementation Framework	80	40
General & Other – e.g. Systems, Processes, Technology	20	10
Data Quality	-	-
Project Design	-	-
Institutional Setup	20	10
Policy	-	-
Corruption	-	-
Schedule/Timing	20	10
Environment & Macroeconomic	40	20
Monitoring & Controlling	20	10
General	-	-
Reporting	-	-
PDOs	20	10
Resources & Capacity	20	10
Budget & Financing	20	10

Table 110: Top Reasons for a Negative Borrower Performance Rating in Haiti²⁸⁴

²⁸³ Cf. Chapter 6.3.1 Quantitative Assessment of Borrower Performance Ratings, table 107: Quantitative Assessment of Negative Borrower Performance Ratings.

²⁸⁴ Author's Note: Appendices DW till EA provide an overview of the reasons for negative bank performances in Haiti separated by the landing and the supervision dimension and for each of the 3 rating categories ("Moderately Unsatisfactory", "Unsatisfactory" and "Highly Unsatisfactory").

As highlighted in the above table reasons in borrower competence and implementation framework categories account for 80 percent (or 4 out of 5 negative borrower performance projects) of the negative borrower performance ratings in Haiti. The remaining 3 categories affect only 1 project.

6.3.2.4 Continent Comparison

In this chapter the outcome of the content analyses of Africa and Asia are presented and compared to one another.²⁸⁵

About 21 percent of all African projects received a negative outcome for the borrower performance rating. Out of the 120 projects, 44 projects have a “Moderately Unsatisfactory” and 75 received an “Unsatisfactory” rating for borrower performance. Furthermore, 1 project received a “Highly Unsatisfactory” rating.²⁸⁶ The following table provides an overview of the major reasons for a negative borrower performance within Africa compared against the total number of projects researched within the continent, whereat the percentages are based on all 3 available borrower performance rating categories (“Moderately Unsatisfactory”, “Unsatisfactory” and “Highly Unsatisfactory”) and dimensions (government and implementation agency):

²⁸⁵ Author’s Note: Due to fact that Haiti is the only LDC in LA, it will not be part of the continental comparison.

²⁸⁶ Cf. *Chapter 6.3.1 Quantitative Assessment of Borrower Performance Ratings*, table 107: *Quantitative Assessment of Negative Borrower Performance Ratings*.

Category	% of Projects	% of total Projects
Borrower Competence	92.5	19.3
Commitment	46.7	9.8
Decision Making Process	16.7	3.5
"Management" of Things	47.5	9.9
Leadership, Coordination & Administration	75	15.7
Communication	9.2	1.9
Cooperation	8.3	1.7
Implementation Framework	46.7	9.8
General & Other – e.g. Systems, Processes, Technology	17.5	3.7
Data Quality	2.5	0.5
Project Design	6.7	1.4
Institutional Setup	7.5	1.6
Policy	10.8	2.3
Corruption	1.7	0.3
Schedule/Timing	19.2	4
Environment & Macroeconomic	8.3	1.7
Monitoring & Controlling	39.2	8.2
General	18.3	3.8
Reporting	11.7	2.4
PDOs	25	5.2
Resources & Capacity	50.8	10.6
Budget & Financing	29.2	6.1

Table 111: Top Reasons for a Negative Borrower Performance Rating in Africa²⁸⁷

As highlighted in the table above borrower competence affects 92.5 percent of all negative borrower performance projects in Africa, a share of roughly 20 percent of all 574 African projects. Within this category, leadership, coordination and administration subcategory is the driving force at 75 percent, followed by “management” of things at 47.5 percent and commitment at 46.7 percent. The resources and capacity category discloses the second highest share at 50.8 percent. Implementation framework, at 46.7 percent, ranks third driven mainly by project schedule and timing issues as well as technology and process challenges. The monitoring and controlling category accounts for 39.2 percent, whereat 29.2 percent of the negative borrower performance projects in Africa are affected by budget and financing issues. In Asia, roughly 20 percent of the projects received a negative outcome for the borrower performance rating, whereat the “Moderately Unsatisfactory” rating category

²⁸⁷ Author's Note: *Appendices EB* till *EG* provide an overview of the reasons for negative bank performances in Africa separated by the landing and the supervision dimension and for each of the 3 available rating categories (“Moderately Unsatisfactory”, “Unsatisfactory” and “Highly Unsatisfactory”).

acclaims 27 projects and the “Unsatisfactory” category 15 projects.²⁸⁸ The following table presents the major reasons for the negative borrower performance ratings within Asia (42 to count) and shows 2 types of percentage calculations driven by different comparison basis, namely counts of projects with negative bank performance ratings (“% of Projects”) and total number of ICR reports in Asia (“% of total Projects”):

Category	% of Projects	% of total Projects
Borrower Competence	90.5	18.4
Commitment	52.4	10.7
Decision Making Process	-	-
"Management" of Things	40.5	8.3
Leadership, Coordination & Administration	76.2	15.5
Communication	19	3.9
Cooperation	2.4	0.5
Implementation Framework	61.9	12.6
General & Other – e.g. Systems, Processes, Technology	26.2	5.3
Data Quality	7.1	1.5
Project Design	11.9	2.4
Institutional Setup	2.4	0.5
Policy	2.4	0.5
Corruption	-	-
Schedule/Timing	14.3	2.9
Environment & Macroeconomic	26.2	5.3
Monitoring & Controlling	38.1	7.8
General	23.8	4.9
Reporting	9.5	1.8
PDOs	23.8	4.9
Resources & Capacity	54.8	11.2
Budget & Financing	4.8	1

Table 112: Top Reasons for a Negative Borrower Performance Rating in Asia²⁸⁹

In Asia, the leading 3 rating categories (borrower competence at 90.5 percent, implementation framework at 61.9 percent and resources and capacity at 54.8 percent) are the same as the ones in Africa, except the ranking ordering of the second

²⁸⁸ Cf. *Chapter 6.3.1 Quantitative Assessment of Borrower Performance Ratings, table 107: Quantitative Assessment of Negative Borrower Performance Ratings.*

²⁸⁹ Author's Note: *Appendices EH, EI, EJ and EK* provide an overview of the reasons for negative bank performances in Asia separated by the landing and the supervision dimension and for each of the 3 available rating categories (“Moderately Unsatisfactory”, “Unsatisfactory” and “Highly Unsatisfactory”).

and third categories. The borrower competence is again driven by leadership, coordination and administration sub-category at 76.2 percent, followed by commitment at 52.4 percent. Within the implementation framework category the environment and macroeconomic as well as the general and other sub-categories both at 26.2 percent are the main drivers. The resources and capacity category affects 54.8 percent of the projects rated negatively for borrower performance, whereat the budget and financing category is only at 4.8 percent in the Asian continent.

In order to compare the reasons for the negative borrower performance ratings between Africa and Asia, the most relevant results differing between the 2 continents are summarized in the following table:²⁹⁰

Category	Africa		Asia	
	% of Projects	% of total Projects	% of Projects	% of total Projects
Borrower Competence	92.5	19.3	90.5	18.4
Decision Making Process	16.7	3.5	-	-
"Management" of Things	47.5	9.9	40.5	8.3
Communication	9.2	1.9	19	3.9
Implementation Framework	46.7	9.8	61.9	12.6
General & Other	17.5	3.7	26.2	5.3
Policy	10.8	2.3	2.4	0.5
Environment & Macroeconomic	8.3	1.7	26.2	5.3
Monitoring & Controlling	39.2	8.2	38.1	7.8
Resources & Capacity	50.8	10.6	54.8	11.2
Budget & Financing	29.2	6.1	4.8	1

Table 113: Direct Comparison of African and Asian Borrower Performance

The direct comparison of the continents shows various differences in multiple categories. Except for 2 categories –namely the implementation framework and the budget and financing– the overall percentages are more or less equal. Still, looking at that the borrower competence sub-categories, a minor difference of 7 percent for the “management” of things and a larger one of about 10 percent for the communication framework can be recognized. Additionally, the decision making process which affects 16.7 percent of the African projects does not have any influence for Asian projects. In regards to the implementation framework there are 2 noteworthy

²⁹⁰ Author's Note: Sector codes respectively project types have not been researched for this comparison, since WB financing of the negative bank performance projects is less than 50 percent for both continents.

continental variances with percentage difference greater than 5 but smaller than 10 percent –namely the general and the policy sub-category. The environment and macroeconomic sub-category differs about 20 percent at 8.3 percent in Africa and 26.2 percent in Asia. Yet, the budget and financing category shows the largest difference with 29.2 percent affected projects in Africa and 4.8 percent in Asia.

6.3.3 Summary and Conclusion

Looking at the LDCs as a whole the borrower performance suffered primarily from reasons in the borrower competence category. Overall, out of the total of 167 negative borrower performance projects 91.6 percent were affected by reasons belonging to this category. More specifically, leadership, coordination and administration by the government and its implementation agencies were the leading causes within the borrower competence affecting 74.3 percent of the projects. Furthermore, the resources and capacity and the implementation framework categories affected about 50 percent of all negative borrower performance projects.

In Haiti, 4 out of 5 negative borrower performance projects were affected by lower borrower competence and the negative causes within the implementation framework.

The comparison of the African and Asian continents showed that the borrower competence category captured the major reasons behind negative borrower performance ratings in both Africa (92.5 percent) and Asia (90.5 percent). Slight differences were found within the implementation framework which was at 46.7 percent in Africa and 61.9 percent in Asia. However, the largest difference was found within the budget and financing category. Here the difference was tremendous, since roughly one-third (29.2 percent) of all negative borrower performance projects in Africa were affected by such causes, whereat only 4.8 percent of the Asian projects outlined budget or financing constraints.

In order to ensure a decrease in negative borrower performance ratings similar recommendations as for the bank performance can be given. First of all, the borrower consisting of the government, its institutions and implementation agencies needs to

be better supported by professional and educated WB staff in order to improve borrower competence. Specifically the leadership and management on the borrower's side need to be guided and supervised to ensure a better understanding throughout the implementation process and to increase borrower commitment. Considering the resources and capacity challenges more time and effort need to be invested for the up-skilling and training of planned project resources. Another aspect is to ensure a more accurate and cooperated monitoring and controlling process, by defining more realistic PDOs and workable approaches for a reliable project reporting. Furthermore, the implementation framework, its (planned) systems, order and delivery processes, potential technologies as well as the country's political environment and macroeconomic situation should be evaluated more precisely –preferably before any arrangements are made.

7 Overall Summary and Conclusion

This doctoral thesis focused on the evaluation of the WB performance in delivering development aid to the LDCs. Even though a positive outcome is already evidenced by the minimum SROI ratio, the research analyses and findings support the general demand to provide even more development assistance to LDCs in order to overcome remaining hurdles.

In the first chapter the challenges of today's global poverty were outlined and a basic understanding of the MDGs provided. The second chapter discussed the research approach, the paper's objective as well as the structure. Additionally, the definition of the LDCs was given, the WB and its institutions presented and the fundamentals of the researched data set explained. The study and research questions concluded the second chapter. The third chapter brought insights into the theoretical foundation of the economic and financial analysis, where the SROI and the CBA were defined and the identified NPVCRs illustrated. Furthermore, additional economic and financial indicators of the ICR reports were also briefly discussed.

In the fourth chapter the results of the economic and financial analysis were presented. The outcome of the appraisal NPV values were presented with positive results of 17,667 to 20,067 Mio USD (with total project costs of 13,981 Mio USD) for the 91 NPVs at the appraisal stage and 4,678 to 4,713 Mio USD (8,021 Mio USD of total project costs) for the remainder of 88 projects excluding the 3 "high-value" NPVs (NPVs greater 1,000 Mio USD). The continental comparison of Africa and Asia based on the exclusion of "high-value" NPVs disclosed a higher NPV/capital cost ratio for the Asian continent.

The same analyses for the 168 result NPVs of the LDCs showed NPV values of 42,059 to 50,779 Mio USD (33,506 Mio USD of total project costs) and 6,188 to 7,799 Mio USD (20,280 Mio USD of total project costs) with 7 "high-value" NPV projects being excluded. Similarly, the continental comparison excluding "high-value" NPVs depicted a nearly twice as high NPV/capital cost ratio in Asia than in Africa. The research on project types did not lead to any further conclusions due to missing data for the total project funding.

When comparing appraisal and result values of a data set of 67 projects where “high-value” NPVs were excluded a slightly decreasing minimum NPV but an increasing maximum NPV by as well slightly increasing capital and total project costs at the result stage was shown.

Chapter 4.5 outlined the analysis on project ERRs. It demonstrated that the ERRs, excluding the 20 “high-value” ERRs (ERRs equal or greater than 100 percent), from the non-NPV projects were between 18.91 and 31.35 percent, comparable to those of the total ERR group but noticeably higher than the corresponding discount rate of 10.83 percent. Furthermore, the continental comparison of the “high-values” excluded data set showed similar values for the ERRs of the NPV projects, but a higher maximum ERR value for non-NPV projects of 35.57 percent in Africa vs. 23.97 percent in Asia.

Using the overall project performance and the results of the disclosed ERRs to calculate potential ERR outcomes for the remainder of non-financial indicator projects gave result of positive outcomes of a calculated minimum ERR of 16.42 percent and a maximum of 29.42 percent which were considerably higher than the standard discount rate of 11 percent. However, those results can only be regarded as approximations due to limitations in the calculation approach.

In the next step various NPVCRs were calculated and compared. For both –the initial and the standardized NPVCRs– the outcomes were (highly) positive. Specifically looking at the standardized values, the total-capital plus recurring costs NPVCR indicators were between 1.29 for the weighted minimum and 2.17 for the unweighted maximum case when excluding “high-value” NPVs and using the weighted recurring costs coefficient of 2.64.

The calculation of the capital SROI ratio also showed positive results. Even when using the initial and unadjusted unweighted recurring costs coefficient of 13.29, both weighted and unweighted ratios were positive. Excluding all “high-value” projects, the remaining 158 projects showed positive results of 1.09 and 1.17 for the weighted and 1.46 and 1.94 for the unweighted ratio using the weighted coefficient of 2.64. The continent comparison based on the whole data set showed higher capital SROI ratios in Asia for any recurring costs coefficient. For the cleaned-up data set it can

be noted that Asia disclosed slightly higher weighted and more stable and predictable unweighted ratios.

Calculating the theoretic minimum SROI disclosed positive results for all 168 projects: Except when using the initial unadjusted, unweighted recurring costs indicator of 13.29, the weighted ratios were below 1 with arithmetic means of 0.74 in the minimum and 0.82 in the maximum case, whereas the unweighted SROI ratio lay between 0.98 and 1.25. Excluding the “high-value” projects positive results were only displayed for the fully embellished recurring costs coefficient of 0.62. For the adjusted recurring costs coefficient of 1.17 the weighted minimum ratio was already at 0.94. Using the weighted recurring costs coefficient of 2.64 only the unweighted ratio stayed positive, whereas both weighted ratios lowered to 0.82 and 0.89. The continental comparison of the complete and the “high-value” embellished data set showed higher ratios for the Asian continent for both ratios using comparable high recurring costs coefficients. The African weighted results seemed to indicate value degeneration from a monetary point of view. On the other hand the unweighted ratios showed that on average a positive outcome was generated by projects regardless of the monetary size with respect to project costs.

In the first step of the final chapter projects responsible for “high-value” recurring costs coefficients were identified. Excluding those projects caused the unweighted recurring costs coefficient to drop from 12.99 to 2.4 for the LDC “high-value” embellished data set. In the second step 10 projects with calculated minimum NPVs of less than -200 Mio USD with high total project costs were identified and excluded from the data set. The remaining 148 projects disclosed positive weighted and unweighted capital SROI ratios based on the weighted recurring costs coefficient of 3 corresponding to the data set. Furthermore, the minimum SROI ratio disclosed values of 1 and 1.06 in the weighted and the 1.3 and 1.72 in the unweighted case. In the last step projects disclosing the highest minimum SROIs were further analyzed. Those were projects with about 2.5 times above the average NPVs of 104 and 112 Mio USD and lower NPV related capital and total project costs of 40 and 42 Mio USD as well as a specific focus on the ‘Power’ sector code.

As a result from the performed economic and financial analysis, it can be concluded the outcome of the theoretic calculated minimum SROI does signal a positive effect

on the LDCs. For both types of the ratio –weighted and unweighted– the outcomes were all above 1, with a minimum weighted ratio of 1 (rounded down). This means that the WB delivers development aid projects successfully and in general increases the welfare of the recipient country.

Looking at the qualitative analysis of the ICR sustainability ratings, it is obvious that fiduciary risk was the primary risk affecting 54.8 percent of the negative sustainability projects representing approx. 20 percent of the researched projects of the LDCs. The fiduciary risk was followed by the implementation capacity, and thereafter, by project specific risks, whereas risks in lowest ranking were infrastructure and political risks. The continental risks comparison showed that Asian projects were more often affected by all types of risks. The 2 main differences between continents were found in the governmental and security risk types. In Asia, projects with negative sustainability showed 15 or higher percentages for both risk types. The differences in the governmental risk type could potentially be explained by Asian governments being overburdened whereat the differences in the security risk type were contributed by Afghanistan undergoing a war period for several years. When controlling for project types the major variance was found in the health-related project types. In this context, fiduciary risks showed to be the only noticeable difference affecting 75 percent of the negative sustainability health-related projects in Africa, whereat no noteworthy detection for Asia was found.

Even though there were fewer negative bank than borrower performance ratings (134 negative bank performances vs. 167 negative borrower performance ratings), the qualitative data analyses –following the same approach for both performance rating categories– showed similarities in the setup of the main categories. At a LDC level, the implementation framework category was the major category affecting 96 percent of all negative bank performance projects (or 16.2 percent of all 790 researched ICR reports). Furthermore, major effects on the bank performance were shown from the monitoring and controlling category at 87.3 percent and the bank competence category at 84.3 percent. In comparison the borrower performance mostly suffered from the borrower competence category affecting 91.6 percent of all

167 projects (equivalent to every fifth project researched within this paper). The second and third ranked categories –resources and capacity and the implementation framework– affected only half of the negative borrower performance projects.

The outcome in terms of category relevance to the LDCs was shared by the African and Asian continents for both rating categories. The continental comparison of the bank performance showed similarities among the most relevant categories, whereas differences were found in the sub-categories. Higher percentages were found for the general sub-categories of the implementation framework as well as for the KPIs sub-category in Africa. On the other hand Asia disclosed higher percentage values for the risks and issues sub-category but as well for the resources and capacity category. In general, Asian projects were on average more often affected by any main category than the African projects –except for the budget and financing category. In terms of the continental borrower performance the comparison showed that the borrower competence category was the major reasoning group accounting for over 90 percent in both continents. Differences were found in the implementation framework category (46.7 percent in Africa and 61.9 percent in Asia) and in the budget and financing category which showed nearly no relevance for the Asian projects (4.8 percent) but affected 29.2 percent of the negative borrower performance projects in Africa.

In conclusion, findings of this paper support the general demand to provide even more development assistance to the LDCs. As shown, fiduciary risks are the major reason affecting the sustainability of most projects delivered to the LDCs. A move into the right direction would be the outcome-oriented continuation of the debt forgiveness discussion and the establishment of better global trade terms between donor states and the LDCs. Specifically the access to better technology standards and improved channels of resources and knowledge transfers could further improve the situation. Customized trainings on project processes and the surrounding project conditions for the planned resources need to be ensured. Further, preventing planned staff of the implementation agencies heading for other opportunities by implementing appropriate incentive systems is another possibility to keep the knowledge within the project environment. Specifically the development and correct

treatment of resources and project capacity is the key, not only for future sustainability, but as well for an improved bank and borrower performance. In order to minimize governmental risks project responsible from the donor side should get a better understanding and feeling on how to support the governments, e.g. through the establishment of regulatory rules and laws. Recommendations in terms of country risks are difficult to make. Understandably, countries in a war situation do have an even higher need for external support and aid. However, credits and funds are often misused by various parties for conflict-related self-interests. In order to start overcoming such conflict and war related hurdles actions such as carefully selecting cooperation partners within the country and the establishment of a trustful future negotiation basis need to be undertaken. In terms of the bank performance the action of the WB to take a closer look at its own project management team in order to improve related tasks, coordination and administration will help to further decrease issues with the bank competence. Issues related to project design, project KPIs and PDOs also require an advanced and more precise evaluation and setup of the project framework –including resources and the project environment– from the bank's side. Looking at the borrower performance there is a specific need to support the government, its institutions and agencies in order to increase the competence, management and leadership of the borrower. Here, the WB can take a leading role in order to support and guide the borrower, strengthen the implementation process and increase the commitment towards the PDOs. As stated above, skilled, knowledgeable and committed resources are all key challenges in order to improve the performance of the Borrower. Last but not least the evaluation and consideration of planned systems, technologies as well as the political and macroeconomic context play a key role in order to improve performances and successfully deliver development aid.

8 Critical Acclaim and Recommendations

There are some critical points to the approach and the data set itself which need to be noted in regards to this dissertation project. First of all, the portfolio values generated under this research project are based on different ICR disclosure dates and very often on local currencies. Even though many reports already outline the economic and financial outcomes based on USD, those outcomes are tied to the currency conversion rate of the corresponding ICR specific calculation date. Thus it was not possible to account for currency fluctuations. Another critical point to note is the fact that many ICR reports outlining an overall-NPV simply provide this information at an 'overall project' level. Whether or not all project costs were considered is, however, often not part of the financial description. Even though the ERRs of the 546 non-financial indicator projects (projects with no financial indicators disclosed) based on the overall project performance were positive, it is difficult to conclusively determine if NPVs respectively SORIs are also positive. As no other economic or financial related indicator was available, evaluation and making final conclusions based on ERR outcomes alone would most likely lead to distorted statements. Furthermore, data quality issues associated with total project costs by project type made it nearly impossible to undertake the continental comparison of any of the economic and financial indicators. Another dimension to be considered is the general question on how biased are the outlined economic and financial results as well as the rating category outcomes. Even though an exemplary comparison of 40 project ICR rating category outcomes and the IEG evaluation of the same projects showed that the rating outcomes are very similar²⁹¹, it is a known fact that the ICR reports are established by a WB owned department. Although minor differences can be found within the sub-rating categories of the project overall, bank and borrower performances and the sustainability rating (noting that the IEG ratings are generally more negative), it cannot be assumed that the outcomes of rating categories directly correlate to the financial outcomes. That the result NPVs and any other economic

²⁹¹ Cf. *Appendices CN* and *EL* provide an overview of the comparison of the ICR rating category outcomes and the IEG ratings as well as the listing of the compared reports.

and financial indicators are only based on projections into the future and whether the outlined results have been or will be achieved would need to be evaluated further. In order to do so post-evaluations monitoring post-project developments by including independent evaluation institutions are necessary. In regards to the interpretation of the minimum SROI calculation it needs to be noted that although the 148 LDC projects outline positive outcomes for the weighted and unweighted minimum SROI ratio, the approach of “excluding projects from the data set” can be argued. On the other hand it can also be argued that “high-value” NPV projects should be part of the data set. Obviously, single extremely high or low runaway values can always distort and bias a data set. Therefore, excluding “high-value” NPV projects which would actually ensure positive minimum SROI ratios, justifies the exclusion of the most negative minimum SROI ratios on the other side. The one thing which both –the “high-value” NPV and the most negative calculated minimum NPV projects– have in common are above-average high costs. First of all, the NPV capital costs are at least twice as high as the LDC average of about 66 Mio USD. Secondly, the total project costs are even 3.5 times higher than the LDC average of 126 Mio USD. Therefore, as a recommendation for future project execution the evaluation process must ensure that all potential risks and issues are taken into account. Additionally, the execution of such process should be guided by a well-tailored monitoring process, such as using phase-specific quality gates which ensure that deliverables can be met in the right quality and at the right time. Furthermore, specifically projects disclosing minimum SROIs within the unweighted range of 1.3 and 1.72 (refer to *chapter 4.8.4 Making Meaning of the Results*) could serve as best practices of processes, setup and delivery. Obviously, the project type needs to be considered when reviewing the ICR documents. However, the lessons learned provided as part of each ICR document provide valuable information which could lay the foundation for the establishment of “best practices for development aid delivery”. Last but not least it needs to be noted that even for negative NPV projects a value for the society could still possibly be created since delivering development aid is not only related to monetary aspects, but also to supporting and bringing communities and states to cooperate together to ensure knowledge transfers in the global community.

9 Outlook and Future Research

Based on the initial data load in September 2012, 2,000 ICR reports were identified out of the 30,500 documents referenced within the country spreadsheets (refer to *chapter 2.5.1 Data Basis*). Even without detailed statistical probability calculations the 168 result NPVs of the 790 ICR reports researched in this paper can be regarded as significant. Nevertheless, further research and evaluation is warranted to determine how realistic the disclosed result NPV and ERR outcomes are. Such crucial analyses need critical project data and access to personal staff within the country and extended project materials and thus require the participation of the WB and the involvement of governments. However, engaging in post-project evaluations to validate the outcomes gathered can provide valuable insights. Another valuable research further upon this research paper would be to expand the data basis by evaluating the remaining ICR reports before the passing date of the MDGs (9/8/2000). Then, a comparison of the outcomes of this paper with the ones of the “pre-MDG passing” could be conducted to find out if project delivery improved after the MDG agreement. Additionally, any newer reports disclosed after the data extract for this research paper in September 2012 could also be considered to further validate the outcomes.²⁹² Furthermore, a drill-down into the country results could be worthwhile to get a better understanding for country specific results, e.g. highly positive project outcomes in Bangladesh.

In terms of the qualitative data analysis causes for positive performance could be further researched. The outcomes could then be aggregated and used to establish best practices and recommendations for project deliveries. Additionally, aside from researching the positive outcomes, a further drill-down into the negative ones could also be conducted. Here cross-analyses considering specific indicator groups, such as negative or neutral NPV projects and the corresponding project rating outcomes could lead to additional findings helping to improve the delivery of development aid.

²⁹² Author’s Note: A suggestion at this stage would be to make use of the already created data basis and VBA analysis programs. Please reach out to the author (schaefer.dominik@yahoo.de) for further information.

Last but not least, non-financial indicator projects could be qualitatively researched to get a better understanding why no economic or financial indicators have been provided. This would further help to validate if development aid had indeed a positive effect on those projects.

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Appendix

Appendix A²⁹³

Millennium Development Goals

Goals and Targets (from the Millennium Declaration)	Indicators for monitoring progress
Goal 1: Eradicate extreme poverty and hunger	
Target 1.A: Halve, between 1990 and 2015, the proportion of people whose income is less than one dollar a day	1.1 Proportion of population below \$1 per day 1.2 Poverty gap ratio 1.3 Share of poorest quintile in national consumption
Target 1.B: Achieve full and productive employment and decent work for all, including women and young people	1.4 Growth rate of GDP per person employed 1.5 Employment-to-population ratio 1.6 Proportion of employed people living below \$1 per day 1.7 Proportion of own-account and contributing family workers in total employment
Target 1.C: Halve, between 1990 and 2015, the proportion of people who suffer from hunger	1.8 Prevalence of underweight children under-five years of age 1.9 Proportion of population below minimum level of dietary energy consumption
Goal 2: Achieve universal primary education	
Target 2.A: Ensure that, by 2015, children everywhere, boys and girls alike, will be able to complete a full course of primary schooling	2.1 Net enrolment ratio in primary education 2.2 Proportion of pupils starting grade 1 who reach last grade of primary 2.3 Literacy rate of 15-24 year-olds, women and men
Goal 3: Promote gender equality and empower women	
Target 3.A: Eliminate gender disparity in primary and secondary education, preferably by 2005, and in all levels of education no later than 2015	3.1 Ratios of girls to boys in primary, secondary and tertiary education 3.2 Share of women in wage employment in the non-agricultural sector 3.3 Proportion of seats held by women in national parliament
Goal 4: Reduce child mortality	
Target 4.A: Reduce by two-thirds, between 1990 and 2015, the under-five mortality rate	4.1 Under-five mortality rate 4.2 Infant mortality rate 4.3 Proportion of 1 year-old children immunized against measles
Goal 5: Improve maternal health	
Target 5.A: Reduce by three quarters, between 1990 and 2015, the maternal mortality ratio	5.1 Maternal mortality ratio 5.2 Proportion of births attended by skilled health personnel
Target 5.B: Achieve, by 2015, universal access to reproductive health	5.3 Contraceptive prevalence rate 5.4 Adolescent birth rate 5.5 Antenatal care coverage (at least one visit and at least four visits) 5.6 Unmet need for family planning
Goal 6: Combat HIV/AIDS, malaria and other diseases	
Target 6.A: Have halted by 2015 and begun to reverse the spread of HIV/AIDS	6.1 HIV prevalence among population aged 15-24 years 6.2 Condom use at last high-risk sex 6.3 Proportion of population aged 15-24 years with comprehensive correct knowledge of HIV/AIDS 6.4 Ratio of school attendance of orphans to school attendance of non-orphans aged 10-14 years
Target 6.B: Achieve, by 2010, universal access to treatment for HIV/AIDS for all those who need it	6.5 Proportion of population with advanced HIV infection with access to antiretroviral drugs
Target 6.C: Have halted by 2015 and begun to reverse the incidence of malaria and other major diseases	6.6 Incidence and death rates associated with malaria 6.7 Proportion of children under 5 sleeping under insecticide-treated bednets 6.8 Proportion of children under 5 with fever who are treated with appropriate anti-malarial drugs 6.9 Incidence, prevalence and death rates associated with tuberculosis 6.10 Proportion of tuberculosis cases detected and cured under directly observed treatment short course

²⁹³ Source: Millennium Development Goals Indicators – The Official United Nations Site for the MDG Indicators (2011).

Goal 7: Ensure environmental sustainability	
Target 7.A: Integrate the principles of sustainable development into country policies and programmes and reverse the loss of environmental resources	7.1 Proportion of land area covered by forest 7.2 CO ₂ emissions, total, per capita and per \$1 GDP 7.3 Consumption of ozone-depleting substances 7.4 Proportion of fish stocks within safe biological limits 7.5 Proportion of total water resources used 7.6 Proportion of terrestrial and marine areas protected 7.7 Proportion of species threatened with extinction
Target 7.B: Reduce biodiversity loss, achieving, by 2010, a significant reduction in the rate of loss	
Target 7.C: Halve, by 2015, the proportion of people without sustainable access to safe drinking water and basic sanitation	7.8 Proportion of population using an improved drinking water source 7.9 Proportion of population using an improved sanitation facility
Target 7.D: By 2020, to have achieved a significant improvement in the lives of at least 100 million slum dwellers	7.10 Proportion of urban population living in slums
Goal 8: Develop a global partnership for development	
Target 8.A: Develop further an open, rule-based, predictable, non-discriminatory trading and financial system Includes a commitment to good governance, development and poverty reduction – both nationally and internationally	Some of the indicators listed below are monitored separately for the least developed countries (LDCs), Africa, landlocked developing countries and small island developing States. <u>Official development assistance (ODA)</u> 8.1 Net ODA, total and to the least developed countries, as percentage of OECD/DAC donors' gross national income 8.2 Proportion of total bilateral, sector-allocable ODA of OECD/DAC donors to basic social services (basic education, primary health care, nutrition, safe water and sanitation) 8.3 Proportion of bilateral official development assistance of OECD/DAC donors that is untied 8.4 ODA received in landlocked developing countries as a proportion of their gross national incomes 8.5 ODA received in small island developing States as a proportion of their gross national incomes
Target 8.B: Address the special needs of the least developed countries Includes: tariff and quota free access for the least developed countries' exports; enhanced programme of debt relief for heavily indebted poor countries (HIPC) and cancellation of official bilateral debt; and more generous ODA for countries committed to poverty reduction	<u>Market access</u> 8.6 Proportion of total developed country imports (by value and excluding arms) from developing countries and least developed countries, admitted free of duty 8.7 Average tariffs imposed by developed countries on agricultural products and textiles and clothing from developing countries 8.8 Agricultural support estimate for OECD countries as a percentage of their gross domestic product 8.9 Proportion of ODA provided to help build trade capacity
Target 8.C: Address the special needs of landlocked developing countries and small island developing States (through the Programme of Action for the Sustainable Development of Small Island Developing States and the outcome of the twenty-second special session of the General Assembly)	<u>Debt sustainability</u> 8.10 Total number of countries that have reached their HIPC decision points and number that have reached their HIPC completion points (cumulative) 8.11 Debt relief committed under HIPC and MDRI Initiatives 8.12 Debt service as a percentage of exports of goods and services
Target 8.D: Deal comprehensively with the debt problems of developing countries through national and international measures in order to make debt sustainable in the long term	
Target 8.E: In cooperation with pharmaceutical companies, provide access to affordable essential drugs in developing countries	8.13 Proportion of population with access to affordable essential drugs on a sustainable basis
Target 8.F: In cooperation with the private sector, make available the benefits of new technologies, especially information and communications	8.14 Telephone lines per 100 population 8.15 Cellular subscribers per 100 population 8.16 Internet users per 100 population

Current Progress towards the MDGs

MDG	Current Status
Goal 1: Eradicate extreme poverty and hunger	- The proportion of people living in extreme poverty declined by half at the global level. - In developing regions, the proportion of people living on less than \$1.25 a day fell from 47 per cent in 1990 to 22 per cent in 2010, five years ahead of schedule. - While the proportion of undernourished people globally decreased from 23.2 per cent in 1990–1992 to 14.9 per cent in 2010–2012, this still leaves 870 million people—one in eight worldwide—going hungry.
Goal 2: Achieve universal primary education	- Literacy rates among adults and youths are on the rise and gender gaps are narrowing. - New national data show the number of out-of-school children dropped from 102 million to 57 million from 2000 to 2011. - Primary education enrolment in developing countries reached 90 per cent in 2010.
Goal 3: Promote gender equality and empower women	- Steady progress has been made towards equal access of girls and boys to education, though disparities remain between regions and education levels. - Globally, the share of women employed outside of agriculture rose to 40 per cent, but rose to only 20 per cent in Southern Asia, Western Asia and Northern Africa. - The global share of women in parliament continues to rise slowly and reached 20 per cent in 2012—far short of gender parity, though an increase of one percentage point was seen during 2012.
Goal 4: Reduce child mortality	- Since 1990, the under-five mortality rate has dropped by 47 per cent. - While around 17,000 fewer children are dying each day, 6.6 million children under five died in 2012—mostly from preventable diseases. - More than 10 million lives have been saved through measles vaccines since 2000. - In sub-Saharan Africa, one in ten children dies before age five, more than 15 times the average for developed regions.
Goal 5: Improve maternal health	- Globally, maternal mortality declined by 47 per cent over the last two decades. - Maternal mortality has declined by about two-thirds in Eastern Asia, Northern Africa and Southern Asia. - Only half of pregnant women in developing regions receive the recommended minimum of four antenatal care visits. - Complications during pregnancy or childbirth are one of the leading causes of death for adolescent girls. - Some 140 million women worldwide who are married or in union say they would like to delay or avoid pregnancy, but do not have access to voluntary family planning. - Most maternal deaths in developing countries are preventable through adequate nutrition, proper health care, including access to family planning, the presence of a skilled birth attendant during delivery and emergency obstetric care.
Goal 6: Combat HIV/AIDS, malaria and other diseases	- Worldwide, the number of people newly infected with HIV continues to fall, dropping 33 per cent from 2001 to 2011. - In 2012, 290,000 fewer children under age 15 were infected with HIV than in 2001. - A record 9.7 million people were receiving antiretroviral therapy for HIV in 2012. - In the decade since 2000, 1.1 million deaths from malaria were averted. - Treatment for tuberculosis has saved some 20 million lives between 1995 and 2011.
Goal 7: Ensure environmental sustainability	- More than 2.1 billion people have gained access to improved drinking water sources since 1990, exceeding the MDG target. - While almost 2 billion more people now have access to proper sanitation than in 1990, 2.5 billion still do not have access to toilets or latrines. - An estimated 863 million people reside in slums in developing countries. - Global carbon dioxide emissions have increased by more than 46 per cent since 1990. - Nearly one-third of marine fish stocks have been overexploited and the world's fisheries can no longer produce maximum sustainable yields. - More species are at risk of extinction despite an increase in protected areas. - Forests, particularly in South America and Africa, are disappearing at an alarming rate.
Goal 8: Develop a Global Partnership for Development	- Official development assistance stood at \$126 billion in 2012. - A total of 83 per cent of least developed country exports enter developed countries duty-free. - In 2012, trade of developing countries and transition economies outpaced the world average. - In the developing world, 31 per cent of the population use the Internet, compared with 77 per cent of the developed world.

²⁹⁴ Source: UN (2015): Fact Sheets.

Appendix C

Appraisal Project NPVs greater than 1,000 Mio USD

Continent	Country	Report	Project Name	NPV Type	NPV	NPV - Min	NPV - Max	Capital Costs	Capital Costs - Min	Capital Costs - Max
Africa	Lesotho	ICR168	Lesotho Highlands Water Project – Phase1B	overall	1,200	-	-	1,132	-	-
Asia	Bangladesh	ICR317	Health, Nutrition and Population Sector Program (HNPS)	overall	-	610	2,975	-	4,300	4,300
Asia	Bangladesh	35449	Third Road Rehabilitation and Maintenance Project (RRMP III)	partial	11,179	-	-	359	-	-

This table shows the projects which disclose NPV values of more than 1,000 Mio USD at the appraisal stage. All values are expressed in Mio USD. The Bangladeshi ICR report 35449 discloses a NPV of 11,179 Mio USD and is thus the project with the highest disclosed NPV value at the appraisal stage.

Appendix D

Appraisal Overall-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Overall-NPVs		Asia Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	2,458	95	1,081	90
	Maximum	2,475	95	3,445	287
Total NPV Capital Costs (planned)	Minimum	2,483	95	4,609	384
	Maximum	2,483	95	4,609	384
Total Project Costs (planned)		2,483	96	4,609	384
Total World Bank Costs (planned)		997	38	539	45
Discount Rate [%]		11.73		10.67	
Planned Project Duration [years]		5.38		5.5	
NPV-horizon [years]		16.97		17.17	

This table compares the outcomes of the NPVs and the corresponding costs for the overall-NPV projects in Africa and Asia at the appraisal stage.²⁹⁵

²⁹⁵ Author's Note: Taking a closer look at the overall-NPV projects in Africa, one notices slightly lower overall-NPV related capital costs (2,482,842,000 USD) than the total project costs (2,483,162,000 USD). The difference is either due to projects where funding is used for operational and maintenance

Appendix E

Appraisal Partial-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Partial-NPVs		Asia Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	2,551	71	11,577	681
	Maximum	2,551	71	11,594	682
Total NPV Capital Costs (planned)	Minimum	3,326	92	1,120	66
	Maximum	3,335	93	1,120	66
Total Project Costs (planned)		5,010	142	1,789	105
Total World Bank Costs (planned)		2,519	70	1,182	70
Discount Rate [%]		11.27		10.58	
Planned Project Duration [years]		5.25		5.29	
NPV-horizon [years]		21.56		20.32	

This table compares the outcomes of the NPVs and the corresponding costs for the partial-NPV projects in Africa and Asia at the appraisal stage.

works or mismatches between the outlined total project costs in the Appendices of the ICR documents (refer to *table 2: ICR Report Chapter Structure: Annex 1 - Project Costs and Financing*) and the costs disclosed in the detailed NPV calculation tables (if available).

Appendix F

Sector Codes of the NPV Projects of the LDCs

Sector Code	%-Points	%-Share	World Bank Costs
Roads and highways	34.47	20.52	2,848,686,000
Central government administration	16.83	10.01	848,680,200
Water supply	16.21	9.65	703,396,900
Power	13.31	7.92	875,771,900
General agriculture, fishing and forestry sector	6.28	3.74	178,222,300
Irrigation and drainage	5.95	3.54	166,654,200
Sub-national government administration	5.62	3.35	238,821,300
Other social services	5.57	3.32	320,312,800
Agricultural extension and research	4.94	2.94	157,014,000
Health	4.94	2.94	381,252,100
General water, sanitation and flood protection sector	3.16	1.88	182,377,300
General public administration sector	3.14	1.87	156,978,200
Mining and other extractive	3.00	1.79	54,056,300
Agricultural marketing and trade	2.83	1.68	57,444,330
Secondary education	2.28	1.36	44,925,050
General transportation sector	2.21	1.32	98,818,980
Ports, waterways and shipping	2.17	1.29	127,326,000
Crops	2.17	1.29	71,158,430
Distribution & Transmission	2.00	1.19	50,030,000
Forestry	1.87	1.11	21,354,400
Railways	1.80	1.07	130,051,800
Sanitation	1.76	1.05	85,626,720
General industry and trade sector	1.75	1.04	50,442,300
Flood protection	1.73	1.03	72,951,200
Sewerage	1.67	0.99	71,856,000
Transportation Adjustment	1.50	0.89	57,400,000
General finance sector	1.43	0.85	115,579,400
Primary education	1.29	0.77	127,965,100
Aviation	1.12	0.67	37,996,180
Hydro	1.00	0.60	152,970,000
Urban Management	1.00	0.60	29,300,000
Natural Resources Management	1.00	0.60	78,520,000
Other Transportation	1.00	0.60	35,920,000
Other industry	0.94	0.56	26,409,760
Petrochemicals and fertilizers	0.92	0.55	150,972,000
Oil and gas	0.88	0.52	84,500,270
Capital markets	0.81	0.48	25,171,800
Animal production	0.79	0.47	13,630,720
Other domestic and international trade	0.74	0.44	4,610,940
General education sector	0.63	0.38	29,210,760
Primary Health, Including Reproductive Health	0.50	0.30	5,900,000
Urban Development Adjustment	0.50	0.30	12,625,000
Telecommunications	0.45	0.27	1,576,800
Law and justice	0.44	0.26	4,111,350
Renewable Energy	0.41	0.24	39,249,050
Solid waste management	0.38	0.23	15,406,970
Tertiary education	0.32	0.19	5,587,010
Rural services and infrastructure	0.29	0.17	8,346,200
General information and communications sector	0.28	0.17	4,270,932
Pollution management and environmental health	0.28	0.17	8,058,400
Agency Reform	0.25	0.15	2,525,000
Other Agriculture	0.25	0.15	2,525,000
Micro- and SME finance	0.21	0.13	4,741,556
Postal services	0.20	0.12	700,800
Agro-industry	0.15	0.09	4,726,800
General energy sector	0.10	0.06	623,100
Non-compulsory health finance	0.10	0.06	16,965,150
Information technology	0.10	0.06	2,274,000
Adult literacy/non-formal education	0.05	0.03	2,425,091
Housing finance and real estate markets	0.02	0.01	1,137,200
Vocational training	0.02	0.01	1,384,000

This table provides an overview of all available sector codes of the 168 NPV projects of the LDCs. The total %-points are 168 with a %-share of 100 percent. The total WB costs are approx. 9,110 Mio USD.

Appendix G

Shared Sector Codes of the NPV Projects

Number of projects sharing the sector code(s)	Project Report Numbers	Shared Sector Codes
6	20987; 22254; 22364; 26258; 24937; 29653	• Roads and highways
2	ICR705; ICR168	• Water supply
2	24401; ICR1904	• Mining and other extractive
2	25102; ICR785	• Power
2	21512; 22038	• Distribution & Transmission
7	ICR1221; 26930; ICR907; ICR688; 35449; 27285; ICR128	• Central government administration • Roads and highways
4	25255; ICR1832; ICR760; 34608	• Power • Central government administration • Roads and highways
3	4485; 27511; ICR2229	• Central government administration • Sub-national government administration • Agricultural extension and research
3	ICR1415; ICR1421; 29813	• Agricultural marketing and trade • Central government administration
3	ICR502; 32900; 25199	• Aviation • Central government administration • Roads and highways
2	27539; ICR700	• Water supply • Sub-national government administration • Central government administration
2	ICR790; ICR586	• Central government administration • Health • Roads and highways
2	ICR2000; 30003	• Central government administration • Water supply
2	ICR2265; 38161	• Central government administration • General agriculture fishing and forestry sector • Sub-national government administration
2	ICR1178; 36380	• Agricultural extension and research • Central government administration • Irrigation and drainage • Other social services
2	ICR1755; 32004	• Central government administration • Power • Renewable Energy
2	ICR750; 30800	• Sanitation • Water supply
2	ICR1352; 25247	• Sanitation • Sewerage • Water supply
2	ICR730; ICR1844	• Central government administration • Health • Sanitation • Sub-national government administration • Water supply
2	30930; ICR951	• Central government administration • Agricultural extension and research
2	26745; ICR2152	• Power • Oil and gas • Central government administration
2	ICR916; 27813	• Central government administration • General transportation sector • Roads and highways

This table shows the projects and their shared sector codes. Except those with 1 sector code only, none of the projects sharing 2 or more sector codes discloses matching %-points for the sector codes.

Appendix H

Sector Code Category Mapping of the NPV Projects

Category Mapping	LDCs	
	%-Share	WB Costs [Mio USD]
Water & Sanitation	13.57	1,043
• Water Supply • General water, sanitation and flood protection sector • Sanitation • Sewerage	9.65 1.88 1.05 0.99	703 182 86 72
Power	10.53	1,204
• Power • General energy sector • Renewable energy • Hydro • Distribution & transmission • Oil and gas	7.92 0.06 0.24 0.6 1.19 0.52	876 1 39 153 50 85
Transportation Development	27.43	3,386
• Roads and highways • Railways • Transportation Adjustment • Urban development adjustment • Urban management • Ports, waterways and shipping • General transportation sector • Other transportation • Aviation • Rural services and infrastructure	20.52 1.07 0.89 0.3 0.6 1.29 1.32 0.6 0.67 0.17	2,849 130 57 13 29 127 99 36 38 8
Health	3.47	412
• Health • Primary health, including reproductive health • Non-compulsory health finance • Pollution management and environmental health	2.94 0.3 0.06 0.17	381 6 17 8
Education	2.73	211
• Primary education • Secondary education • Tertiary education • General education sector • Vocational training • Adult literacy/non-formal education	0.77 1.36 0.19 0.38 0.01 0.03	128 45 6 29 1 2

Agriculture	15.09	810
• Agricultural extension and research	2.94	157
• Agricultural marketing and trade	1.68	57
• General agriculture, fishing and forestry sector	3.74	178
• Crops	1.29	71
• Irrigation and drainage	3.54	167
• Forestry	1.11	21
• Agro-Industry	0.09	5
• Petrochemicals and fertilizers	0.55	151
• Other Agriculture	0.15	3
Financial Development	1.46	147
• Capital markets	0.48	25
• General finance sector	0.85	116
• Micro- and SME finance	0.13	5
• Housing finance and real estate markets	0.01	1
Governmental Administration	13.36	1,088
• Central government administration	10.01	849
• Sub-national government administration	3.35	239
Sectorial Development & Reform	3.75	219
• General public administration sector	1.87	157
• General industry and trade sector	1.04	50
• Other domestic and international trade	0.44	5
• Law and justice	0.26	4
• Agency reform	0.15	3
IT & Communication	0.5	8
• General information and communications sector	0.17	4
• Telecommunications	0.27	2
• Information technology	0.06	2
Natural Resources	2.38	133
• Mining and other extractive	1.79	54
• Natural Resources Management	0.6	79
Other	5.73	449
• Other social services	3.32	320
• Other industry	0.56	26
• Flood protection	1.03	73
• Animal production	0.47	14
• Solid waste management	0.23	15
• Postal services	0.12	1

This table shows the project type mapping of the 61 sector codes of the 168 result NPV projects and their corresponding %-share as well as the WB costs in Mio USD.

Appendix I

Neutral Result NPV and Cost Values of the LDCs

Values in Mio USD		LDCs					
		Total Neutral NPVs		Neutral Overall-NPVs		Neutral Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	-23	-3	-7	-2	-15	-5
	Maximum	47	7	35	9	12	4
Total NPV Capital Costs (actual)	Minimum	296	42	67	17	228	76
	Maximum	285	41	57	14	228	76
Total Project Costs (actual)		533	76	67	17	231	77
Total World Bank Costs (actual)		294	42	61	15	465	155
Discount Rate [%]		11.17		12		10.33	
Actual Project Duration [years]		7.43		7.5		7.33	
NPV-horizon [years]		22.6		19.33		27.5	

This table compares the outcomes of the neutral total, overall- and partial-NPVs and the corresponding cost values of the LDCs.

Appendix J

Negative Result NPV and Cost Values of the LDCs

Values in Mio USD		LDCs					
		Total Negative NPVs		Negative Overall-NPVs		Negative Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	-27	-4	-7	-2	-20	-7
	Maximum	-27	-4	-7	-2	-20	-7
Total NPV Capital Costs (actual)	Minimum	202	29	80	20	122	41
	Maximum	202	29	80	20	122	41
Total Project Costs (actual)		344	49	74	18	269	90
Total World Bank Costs (actual)		200	29	63	16	137	46
Discount Rate [%]		11.2		11.33		11	
Actual Project Duration [years]		6.86		6.75		7	
NPV-horizon [years]		23.75		25		22.5	

This table compares the outcomes of the negative total, overall- and partial-NPVs and the corresponding cost values of the LDCs.

Appendix K

Result Project NPVs greater than 1,000 Mio USD

Continent	Country	Report	Project Name	NPV Type	NPV	NPV - Min	NPV - Max	Capital Costs	Capital Costs - Min	Capital Costs - Max
Africa	Ethiopia	34485	Road Sector Development Program Support Project	partial	1,211	-	-	368	-	-
Africa	Tanzania	ICR667	Participatory Agriculture Development and Empowerment Project	partial	1,190	-	-	56	-	-
Africa	Uganda	ICR688	Health, Nutrition and Population Sector Program (HNPSP)	overall	-	3	1,490	-	34	34
Asia	Bangladesh	ICR317	Health, Nutrition and Population Sector Program (HNPSP)	overall	-	610	2,975	-	4,300	4,300
Asia	Bangladesh	ICR358	Private Sector Infrastructure Development Project (PSIDP)	overall	1,724	-	-	300	-	-
Asia	Bangladesh	ICR2036	Primary Education Development Project II	overall	-	2,216	5,472	-	7,508	7,508
Asia	Bangladesh	35449	Third Road Rehabilitation and Maintenance Project (RRMP III)	partial	28,918	-	-	384	-	-

This table shows projects that disclose NPV values of more than 1,000 Mio USD at the result stage. All values are expressed in Mio USD. Interestingly, all 4 Asian “high-value” projects are Bangladeshi projects, with ICR report 35449 disclosing a NPV of 28,918 Mio USD.

Appendix L

Sector Codes of the 7 “high-value” NPV Projects

Sector Code	Project Counts	%-Points	%-Share	World Bank Costs [Mio USD]
Roads and highways	3	2.15	30.71	514
Central government administration	7	2.07	29.5	313
Crops	1	0.7	10	38
Primary education	1	0.65	9.29	89
Power	1	0.5	7.07	93
Health	1	0.4	5.71	117
Sub-national government administration	3	0.2	2.86	23
Other social services	1	0.11	1.57	32
General public administration sector	1	0.09	1.29	26
Agricultural extension and research	1	0.05	0.71	8
Agricultural marketing and trade	1	0.05	0.71	8
General education sector	1	0.04	0.57	12
Total		7	100	1,273

This table shows the sector codes of the 7 “high-value” NPV projects with WB financing. Note that the ‘Water Supply’ sector code is absent from the top 12 sector codes listing, but ‘Crops’, ‘Primary education’, and ‘General education sector’ sector codes are included in the list.

Appendix M

Sector Codes of the NPV projects of the LDCs excluding “high-value” NPVs

Sector Code	%-Points	%-Share	World Bank Costs
Roads and highways	32.32	20.07	2,334,341,000
Water supply	16.21	10.07	703,396,300
Central government administration	14.76	9.17	535,348,400
Power	12.81	7.96	783,070,800
General agriculture, fishing and forestry sector	6.28	3.90	178,222,300
Irrigation and drainage	5.95	3.70	166,654,200
Other social services	5.46	3.39	288,038,800
Sub-national government administration	5.42	3.37	215,665,500
Agricultural extension and research	4.89	3.04	154,273,600
Health	4.54	2.82	263,892,100
General water, sanitation and flood protection sector	3.16	1.96	182,377,300
General public administration sector	3.05	1.89	130,572,200
Mining and other extractive	3.00	1.86	54,056,300
Agricultural marketing and trade	2.78	1.73	54,703,950
Secondary education	2.28	1.42	44,925,050
General transportation sector	2.21	1.37	98,818,980
Ports, waterways and shipping	2.17	1.35	127,326,000
Distribution & Transmission	2.00	1.24	50,030,000
Forestry	1.87	1.16	21,354,400
Railways	1.80	1.12	130,051,800
Sanitation	1.76	1.09	85,626,720
General industry and trade sector	1.75	1.09	50,442,300
Flood protection	1.73	1.07	72,951,200
Sewerage	1.67	1.04	71,856,000
Transportation Adjustment	1.50	0.93	57,400,000
Crops	1.47	0.91	32,793,100
General finance sector	1.43	0.89	115,579,400
Aviation	1.12	0.70	37,996,180
Hydro	1.00	0.62	152,970,000
Urban Management	1.00	0.62	29,300,000
Natural Resources Management	1.00	0.62	78,520,000
Other Transportation	1.00	0.62	35,920,000
Other industry	0.94	0.58	26,409,760
Petrochemicals and fertilizers	0.92	0.57	150,972,000
Oil and gas	0.88	0.55	84,500,270
Capital markets	0.81	0.50	25,171,800
Animal production	0.79	0.49	13,630,720
Other domestic and international trade	0.74	0.46	4,610,940
Primary education	0.64	0.40	38,527,720
General education sector	0.59	0.37	17,474,760
Primary Health, Including Reproductive Health	0.50	0.31	5,900,000
Urban Development Adjustment	0.50	0.31	12,625,000
Telecommunications	0.45	0.28	1,576,800
Law and justice	0.44	0.27	4,111,350
Renewable Energy	0.41	0.25	39,249,050
Solid waste management	0.38	0.24	15,406,970
Tertiary education	0.32	0.20	5,587,010
Rural services and infrastructure	0.29	0.18	8,346,200
General information and communications sector	0.28	0.17	4,270,932
Pollution management and environmental health	0.28	0.17	8,058,400
Agency Reform	0.25	0.16	2,525,000
Other Agriculture	0.25	0.16	2,525,000
Micro- and SME finance	0.21	0.13	4,741,556
Postal services	0.20	0.12	700,800
Agro-industry	0.15	0.09	4,726,800
General energy sector	0.10	0.06	623,100
Non-compulsory health finance	0.10	0.06	16,965,150
Information technology	0.10	0.06	2,274,000
Adult literacy/non-formal education	0.05	0.03	2,425,091
Housing finance and real estate markets	0.02	0.01	1,137,200
Vocational training	0.02	0.01	1,384,000

This table provides an overview of all available sector codes of the 161 NPV projects of the LDCs when “high-value” NPV projects are excluded.

Appendix N

Sector Code Category Mapping excluding “high-value” NPVs

Category Mapping	LDCs	
	%-Share	WB Costs [Mio USD]
Water & Sanitation	14.69	1,043
• Water Supply • General water, sanitation and flood protection sector • Sanitation • Sewerage	10.7 1.96 1.09 1.04	703 182 86 72
Power	10.58	1,111
• Power • General energy sector • Renewable energy • Hydro • Distribution & transmission • Oil and gas	7.96 0.06 0.25 0.62 1.24 0.55	783 1 39 153 50 85
Transportation Development	27.8	2,871
• Roads and highways • Railways • Transportation Adjustment • Urban development adjustment • Urban management • Ports, waterways and shipping • General transportation sector • Other transportation • Aviation • Rural services and infrastructure	20.7 1.12 0.93 0.31 0.62 1.35 1.37 0.62 0.7 0.18	2,334 130 57 13 29 127 99 36 38 8
Health	3.26	295
• Health • Primary health, including reproductive health • Non-compulsory health finance • Pollution management and environmental health	2.82 0.31 0.06 0.17	264 6 17 8
Education	2.33	110
• Primary education • Secondary education • Tertiary education • General education sector • Vocational training • Adult literacy/non-formal education	0.4 1.42 0.2 0.37 0.01 0.03	39 45 6 17 1 2

Agriculture	15.16	767
• Agricultural extension and research	3.04	154
• Agricultural marketing and trade	1.73	55
• General agriculture, fishing and forestry sector	3.9	178
• Crops	0.91	33
• Irrigation and drainage	3.7	167
• Forestry	1.16	21
• Agro-Industry	0.09	5
• Petrochemicals and fertilizers	0.57	151
• Other Agriculture	0.16	3
Financial Development	1.42	8
• Capital markets	0.5	25
• General finance sector	0.89	116
• Micro- and SME finance	0.13	5
• Housing finance and real estate markets	0.01	1
Governmental Administration	12.4	751
• Central government administration	9.17	535
• Sub-national government administration	3.37	216
Sectorial Development & Reform	3.77	193
• General public administration sector	1.89	131
• General industry and trade sector	1.09	50
• Other domestic and international trade	0.46	5
• Law and justice	0.27	4
• Agency reform	0.16	3
IT & Communication	0.42	8
• General information and communications sector	0.17	4
• Telecommunications	0.28	2
• Information technology	0.06	2
Natural Resources	2.38	132
• Mining and other extractive	1.86	54
• Natural Resources Management	0.62	79
Other	5.79	417
• Other social services	3.39	288
• Other industry	0.58	26
• Flood protection	1.07	73
• Animal production	0.49	14
• Solid waste management	0.24	15
• Postal services	0.12	1

This table shows the project type mapping of the 61 sector codes of the 161 result NPV projects and their corresponding %-share as well as the WB costs in Mio USD when the 7 “high-value” NPVs are excluded.

Appendix O

Result Overall-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Overall-NPVs		Asia Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,317	36	5,463	238
	Maximum	3,529	95	11,402	496
Total NPV Capital Costs (actual)	Minimum	2,715	73	12,933	562
	Maximum	2,718	73	12,923	562
Total Project Costs (actual)		2,894	78	12,915	562
Total World Bank Costs (actual)		1,469	40	1,113	48
Discount Rate [%]		11.53		11.16	
Actual Project Duration [years]		7.49		6.48	
NPV-horizon [years]		18.08		18.24	

This table compares the outcomes of the NPVs and the corresponding costs for the overall-NPV projects in Africa and Asia at the result stage.

Appendix P

Result Partial-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Partial-NPVs		Asia Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	5,141	71	30,139	837
	Maximum	5,452	76	30,395	844
Total NPV Capital Costs (actual)	Minimum	5,860	81	1,900	53
	Maximum	5,860	81	2,077	58
Total Project Costs (actual)		14,771	205	2,925	81
Total World Bank Costs (actual)		4,658	65	1,870	52
Discount Rate [%]		10.99		11.34	
Actual Project Duration [years]		7.26		6.42	
NPV-horizon [years]		20.55		18.86	

This table compares the outcomes of the NPVs and the corresponding costs for the partial-NPV projects in Africa and Asia at the result stage.

Appendix Q

Sector Codes of the African NPV Projects

Sector Code	%-Points	%-Share	World Bank Costs
Roads and highways	21.42	19.65	1,858,115,000
Water supply	11.19	10.27	509,955,000
Central government administration	10.74	9.85	457,457,600
Power	9.44	8.66	707,837,800
General agriculture, fishing and forestry sector	4.47	4.10	134,015,800
Sub-national government administration	4.05	3.72	181,103,100
Agricultural extension and research	3.62	3.32	135,697,500
Other social services	3.32	3.05	219,041,000
Mining and other extractive	3.00	2.75	54,056,300
Health	2.85	2.61	219,644,100
General water, sanitation and flood protection sector	2.44	2.24	161,430,400
Ports, waterways and shipping	2.16	1.98	127,046,600
Crops	2.12	1.94	69,958,430
Agricultural marketing and trade	2.12	1.94	35,945,530
General public administration sector	1.91	1.75	109,200,500
Railways	1.80	1.65	130,051,800
General industry and trade sector	1.75	1.61	50,442,300
Irrigation and drainage	1.69	1.55	59,111,290
Transportation Adjustment	1.50	1.38	57,400,000
Sanitation	1.34	1.23	74,000,160
Sewerage	1.28	1.17	60,593,100
General transportation sector	1.26	1.16	81,538,580
Secondary education	1.23	1.13	13,411,700
Distribution & Transmission	1.00	0.92	14,870,000
Hydro	1.00	0.92	152,970,000
Urban Management	1.00	0.92	29,300,000
Petrochemicals and fertilizers	0.92	0.84	150,972,000
Other industry	0.91	0.83	25,699,160
Forestry	0.87	0.80	4,454,400
Flood protection	0.79	0.72	17,514,300
Other domestic and international trade	0.74	0.68	4,610,940
Oil and gas	0.58	0.53	84,398,270
Primary Health, Including Reproductive Health	0.50	0.46	5,900,000
Urban Development Adjustment	0.50	0.46	12,625,000
General finance sector	0.44	0.40	16,598,380
Solid waste management	0.37	0.34	15,244,960
General education sector	0.36	0.33	6,788,956
Capital markets	0.33	0.30	2,659,800
Agency Reform	0.25	0.23	2,525,000
Other Agriculture	0.25	0.23	2,525,000
Micro- and SME finance	0.21	0.19	4,741,556
Aviation	0.20	0.18	16,959,480
Primary education	0.19	0.17	9,995,900
Renewable Energy	0.18	0.17	4,957,052
Tertiary education	0.17	0.16	1,749,300
Agro-industry	0.15	0.14	4,726,800
General energy sector	0.10	0.09	623,100
Non-compulsory health finance	0.10	0.09	16,965,150
Animal production	0.05	0.05	2,525,500
General information and communications sector	0.05	0.05	2,001,932
Adult literacy/non-formal education	0.05	0.05	2,425,091
Housing finance and real estate markets	0.02	0.02	1,137,200
Vocational training	0.02	0.02	1,384,000

This table provides an overview of all available sector codes of the African NPV projects.

Appendix R

Sector Codes of the Asian NPV Projects

Sector Code	%-Points	%-Share	World Bank Costs
Roads and highways	20.13	18.99	1,540,572,000
Water supply	11.19	10.56	509,955,000
Central government administration	9.95	9.39	413,512,200
Power	9.44	8.91	707,837,800
General agriculture, fishing and forestry sector	4.47	4.22	134,015,800
Sub-national government administration	3.93	3.71	168,954,900
Agricultural extension and research	3.57	3.37	132,957,100
Other social services	3.32	3.13	219,041,000
Mining and other extractive	3.00	2.83	54,056,300
Health	2.85	2.69	219,644,100
General water, sanitation and flood protection sector	2.44	2.30	161,430,400
Ports, waterways and shipping	2.16	2.04	127,046,600
Agricultural marketing and trade	2.07	1.95	33,205,150
General public administration sector	1.91	1.80	109,200,500
Railways	1.80	1.70	130,051,800
General industry and trade sector	1.75	1.65	50,442,300
Irrigation and drainage	1.69	1.59	59,111,290
Transportation Adjustment	1.50	1.42	57,400,000
Crops	1.42	1.34	31,593,100
Sanitation	1.34	1.26	74,000,160
Sewerage	1.28	1.21	60,593,100
General transportation sector	1.26	1.19	81,538,580
Secondary education	1.23	1.16	13,411,700
Distribution & Transmission	1.00	0.94	14,870,000
Hydro	1.00	0.94	152,970,000
Urban Management	1.00	0.94	29,300,000
Petrochemicals and fertilizers	0.92	0.87	150,972,000
Other industry	0.91	0.86	25,699,160
Forestry	0.87	0.82	4,454,400
Flood protection	0.79	0.75	17,514,300
Other domestic and international trade	0.74	0.70	4,610,940
Oil and gas	0.58	0.55	84,398,270
Primary Health, Including Reproductive Health	0.50	0.47	5,900,000
Urban Development Adjustment	0.50	0.47	12,625,000
General finance sector	0.44	0.42	16,598,380
Solid waste management	0.37	0.35	15,244,960
General education sector	0.36	0.34	6,788,956
Capital markets	0.33	0.31	2,659,800
Agency Reform	0.25	0.24	2,525,000
Other Agriculture	0.25	0.24	2,525,000
Micro- and SME finance	0.21	0.20	4,741,556
Aviation	0.20	0.19	16,959,480
Primary education	0.19	0.18	9,995,900
Renewable Energy	0.18	0.17	4,957,052
Tertiary education	0.17	0.16	1,749,300
Agro-industry	0.15	0.14	4,726,800
General energy sector	0.10	0.09	623,100
Non-compulsory health finance	0.10	0.09	16,965,150
Animal production	0.05	0.05	2,525,500
General information and communications sector	0.05	0.05	2,001,932
Adult literacy/non-formal education	0.05	0.05	2,425,091
Housing finance and real estate markets	0.02	0.02	1,137,200
Vocational training	0.02	0.02	1,384,000

This table provides an overview of all available sector codes of the Asian NPV projects.

Appendix S

Sector Code Category Mapping of Africa and Asia

Category Mapping	Africa		Asia	
	%-Share	WB Costs [Mio USD]	%-Share	WB Costs [Mio USD]
Water & Sanitation	14.91	806	11.09	237
• Water Supply	10.27	510	8.51	193
• General water, sanitation and flood protection sector	2.24	161	1.22	21
• Sanitation	1.23	74	0.71	12
• Sewerage	1.17	61	0.66	11
Power	11.29	966	9.13	237
• Power	8.66	708	6.55	168
• General energy sector	0.09	1	-	-
• Renewable energy	0.17	5	0.39	34
• Hydro	0.92	153	-	-
• Distribution & transmission	0.92	15	1.69	35
• Oil and gas	0.53	84	0.51	0
Transportation Development	27.38	2,313	27.48	1,073
• Roads and highways	19.65	1,858	22.12	991
• Railways	1.65	130	-	-
• Transportation Adjustment	1.38	57	-	-
• Urban development adjustment	0.46	13	-	-
• Urban management	0.92	29	-	-
• Ports, waterways and shipping	1.98	127	0.02	0
• General transportation sector	1.16	82	1.61	17
• Other transportation	-	-	1.69	36
• Aviation	0.18	17	1.56	21
• Rural services and infrastructure	-	-	0.49	8
Health	3.16	242	4.01	170
• Health	2.61	220	3.54	162
• Primary health, including reproductive health	0.46	6	-	-
• Non-compulsory health finance	0.09	17	-	-
• Pollution management and environmental health	-	-	0.47	8
Education	1.87	34	4.32	176
• Primary education	0.17	10	1.86	118
• Secondary education	1.13	13	1.78	32
• Tertiary education	0.16	2	0.25	4
• General education sector	0.33	7	0.46	22
• Vocational training	0.02	1	-	-
• Adult literacy/non-formal education	0.05	2	-	-

Agriculture	14.63	598	15.68	212
• Agricultural extension and research	3.32	136	2.44	21
• Agricultural marketing and trade	1.94	36	1.2	21
• General agriculture, fishing and forestry sector	4.10	134	3.07	44
• Crops	1.94	70	0.08	1
• Irrigation and drainage	1.55	59	7.22	108
• Forestry	0.8	4	1.69	17
• Agro-Industry	0.14	5	-	-
• Petrochemicals and fertilizers	0.84	151	-	-
• Other Agriculture	0.23	3	-	-
Financial Development	0.92	26	2.47	123
• Capital markets	0.3	3	0.81	23
• General finance sector	0.4	17	1.68	99
• Micro- and SME finance	0.19	5	-	-
• Housing finance and real estate markets	0.02	1	-	-
Governmental Administration	12.6	639	12.96	449
• Central government administration	9.85	458	10.31	391
• Sub-national government administration	2.75	181	2.66	58
Sectorial Development & Reform	4.27	167	2.82	52
• General public administration sector	1.75	109	2.08	48
• General industry and trade sector	1.61	50	-	-
• Other domestic and international trade	0.68	5	-	-
• Law and justice	-	-	0.75	4
• Agency reform	0.23	3	-	-
IT & Communication	0.05	2	1.31	6
• General information and communications sector	0.05	2	0.39	2
• Telecommunications	-	-	0.76	2
• Information technology	-	-	0.17	2
Natural Resources	2.75	53	1.68	79
• Mining and other extractive	2.75	54	-	-
• Natural Resources Management	-	-	1.69	79
Other	4.99	281	7.05	169
• Other social services	3.05	219	3.81	101
• Other industry	0.83	26	0.05	1
• Flood protection	0.72	18	1.59	55
• Animal production	0.05	3	1.25	11
• Solid waste management	0.34	15	0.02	0
• Postal services	-	-	0.34	1

This table shows the project type mapping of the 61 sector codes of the 168 African and Asian NPV projects and their corresponding %-share as well as the WB costs in Mio USD.

Appendix T

Positive Result Total NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Total Positive NPVs		Asia Total Positive NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	6,488	62	35,620	712
	Maximum	8,991	86	41,767	835
Total NPV Capital Costs (actual)	Minimum	8,274	80	14,636	293
	Maximum	8,277	80	14,813	296
Total Project Costs (actual)		17,041	164	15,589	312
Total World Bank Costs (actual)		5,774	56	2,843	57
Discount Rate [%]		11.17		11.29	
Actual Project Duration [years]		7.31		6.4	
NPV-horizon [years]		19.51		18.84	

This table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for the positive total NPV projects of the result stage.

Appendix U

Positive Result Overall-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Positive Overall-NPVs		Asia Positive Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,323	38	5,471	322
	Maximum	3,535	101	11,368	669
Total NPV Capital Costs (actual)	Minimum	2,649	76	12,851	756
	Maximum	2,653	76	12,851	756
Total Project Costs (actual)		2,835	81	12,834	755
Total World Bank Costs (actual)		1,418	41	1,039	61
Discount Rate [%]		11.57		10.93	
Actual Project Duration [years]		7.43		6.41	
NPV-horizon [years]		17.37		19.79	

This table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for the positive overall-NPV projects of the result stage.

Appendix V

Positive Result Partial-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Positive Partial-NPVs		Asia Positive Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	5,166	75	30,150	914
	Maximum	5,456	79	30,399	921
Total NPV Capital Costs (actual)	Minimum	5,624	82	1,785	54
	Maximum	5,624	82	1,962	59
Total Project Costs (actual)		14,206	206	2,755	83
Total World Bank Costs (actual)		4,356	63	1,803	55
Discount Rate [%]		10.97		11.5	
Actual Project Duration [years]		7.25		6.39	
NPV-horizon [years]		20.5		18.35	

This table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for the positive partial-NPV projects of the result stage.

Appendix W

Neutral Result Overall-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Neutral Overall-NPVs		Asia Neutral Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	n.a.	n.a.	-7	-2
	Maximum	n.a.	n.a.	35	9
Total NPV Capital Costs (actual)	Minimum	n.a.	n.a.	67	17
	Maximum	n.a.	n.a.	57	14
Total Project Costs (actual)		n.a.	n.a.	67	17
Total World Bank Costs (actual)		n.a.	n.a.	61	15
Discount Rate [%]		n.a.		12	
Actual Project Duration [years]		n.a.		7.5	
NPV-horizon [years]		n.a.		19.33	

This table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for the neutral overall-NPV projects of the result stage.

Appendix X

Neutral Result Partial-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Neutral Partial-NPVs		Asia Neutral Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	-13	-13	-2	-1
	Maximum	7	7	5	3
Total NPV Capital Costs (actual)	Minimum	210	210	18	9
	Maximum	210	210	18	9
Total Project Costs (actual)		392	392	74	37
Total World Bank Costs (actual)		196	196	36	18
Discount Rate [%]		11		10	
Actual Project Duration [years]		8		7	
NPV-horizon [years]		40		15	

This table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for the neutral partial-NPV projects of the result stage.

Appendix Y

Negative Result Overall-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Negative Overall-NPVs		Asia Negative Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	-6	-3	-1	0
	Maximum	-6	-3	-1	0
Total NPV Capital Costs (actual)	Minimum	66	33	15	7
	Maximum	66	33	15	7
Total Project Costs (actual)		60	30	15	7
Total World Bank Costs (actual)		51	25	12	6
Discount Rate [%]		11		12	
Actual Project Duration [years]		8.5		5	
NPV-horizon [years]		25		n.a.	

This table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for the negative overall-NPV projects of the result stage.

Appendix Z

Negative Result Partial-NPV and Cost Values Comparison of Africa and Asia

Values in Mio USD		Africa Negative Partial-NPVs		Asia Negative Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	-11	-5	-9	-9
	Maximum	-11	-5	-9	-9
Total NPV Capital Costs (actual)	Minimum	25	13	97	97
	Maximum	25	13	97	97
Total Project Costs (actual)		173	86	97	97
Total World Bank Costs (actual)		107	53	30	30
Discount Rate [%]		12		10	
Actual Project Duration [years]		7.5		6	
NPV-horizon [years]		20		25	

This table compares the outcomes of the NPVs and the corresponding cost values of Africa and Asia for the negative partial-NPV projects of the result stage.

Appendix AA

Result Overall-NPVs of Africa and Asia excluding “high-value” NPV Projects

Values in Mio USD		Africa Overall-NPVs		Asia Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,313	36	9,13	46
	Maximum	2,039	57	1,231	62
Total NPV Capital Costs (actual)	Minimum	2,681	74	825	41
	Maximum	2,684	75	815	41
Total Project Costs (actual)		2,860	79	807	40
Total World Bank Costs (actual)		1,440	40	495	25
Discount Rate [%]		11.52		11.25	
Actual Project Duration [years]		7.42		6.35	
NPV-horizon [years]		17.90		20.6	

This table shows the results of the overall-NPV projects of Africa and Asia.

Appendix AB

Result Partial-NPVs of Africa and Asia excluding “high-value” NPV Projects

Values in Mio USD		Africa Partial-NPVs		Asia Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	2,740	39	1,221	35
	Maximum	3,052	44	1,477	42
Total NPV Capital Costs (actual)	Minimum	5,436	78	1,516	43
	Maximum	5,436	78	1,693	48
Total Project Costs (actual)		14,155	202	2,457	70
Total World Bank Costs (actual)		4,270	61	1,641	47
Discount Rate [%]		10.96		11.32	
Actual Project Duration [years]		7.27		6.4	
NPV-horizon [years]		21.1		18.41	

This table shows the results of the partial-NPV projects of Africa and Asia.

Appendix AC

Sector Codes of Africa excluding “high-value” NPVs

Sector Code	%-Points	%-Share	World Bank Costs
Roads and highways	20.13	18.99	1,540,572,000
Water supply	11.19	10.56	509,955,000
Central government administration	9.95	9.39	413,512,200
Power	9.44	8.91	707,837,800
General agriculture, fishing and forestry sector	4.47	4.22	134,015,800
Sub-national government administration	3.93	3.71	168,954,900
Agricultural extension and research	3.57	3.37	132,957,100
Other social services	3.32	3.13	219,041,000
Mining and other extractive	3.00	2.83	54,056,300
Health	2.85	2.69	219,644,100
General water, sanitation and flood protection sector	2.44	2.30	161,430,400
Ports, waterways and shipping	2.16	2.04	127,046,600
Agricultural marketing and trade	2.07	1.95	33,205,150
General public administration sector	1.91	1.80	109,200,500
Railways	1.80	1.70	130,051,800
General industry and trade sector	1.75	1.65	50,442,300
Irrigation and drainage	1.69	1.59	59,111,290
Transportation Adjustment	1.50	1.42	57,400,000
Crops	1.42	1.34	31,593,100
Sanitation	1.34	1.26	74,000,160
Sewerage	1.28	1.21	60,593,100
General transportation sector	1.26	1.19	81,538,580
Secondary education	1.23	1.16	13,411,700
Distribution & Transmission	1.00	0.94	14,870,000
Hydro	1.00	0.94	152,970,000
Urban Management	1.00	0.94	29,300,000
Petrochemicals and fertilizers	0.92	0.87	150,972,000
Other industry	0.91	0.86	25,699,160
Forestry	0.87	0.82	4,454,400
Flood protection	0.79	0.75	17,514,300
Other domestic and international trade	0.74	0.70	4,610,940
Oil and gas	0.58	0.55	84,398,270
Primary Health, Including Reproductive Health	0.50	0.47	5,900,000
Urban Development Adjustment	0.50	0.47	12,625,000
General finance sector	0.44	0.42	16,598,380
Solid waste management	0.37	0.35	15,244,960
General education sector	0.36	0.34	6,788,956
Capital markets	0.33	0.31	2,659,800
Agency Reform	0.25	0.24	2,525,000
Other Agriculture	0.25	0.24	2,525,000
Micro- and SME finance	0.21	0.20	4,741,556
Aviation	0.20	0.19	16,959,480
Primary education	0.19	0.18	9,995,900
Renewable Energy	0.18	0.17	4,957,052
Tertiary education	0.17	0.16	1,749,300
Agro-industry	0.15	0.14	4,726,800
General energy sector	0.10	0.09	623,100
Non-compulsory health finance	0.10	0.09	16,965,150
Animal production	0.05	0.05	2,525,500
General information and communications sector	0.05	0.05	2,001,932
Adult literacy/non-formal education	0.05	0.05	2,425,091
Housing finance and real estate markets	0.02	0.02	1,137,200
Vocational training	0.02	0.02	1,384,000

This table provides an overview of all available sector codes of the African NPV projects when the 3 “high-value” NPV projects are excluded.

Appendix AD

Sector Codes of Asia excluding “high-value” NPVs

Sector Code	%-Points	%-Share	World Bank Costs
Roads and highways	12.19	22.16	793,768,700
Water supply	5.02	9.13	193,441,300
Central government administration	4.81	8.75	121,836,300
Irrigation and drainage	4.26	7.75	107,542,900
Power	3.37	6.13	75,233,000
Other social services	2.14	3.89	68,997,840
General agriculture, fishing and forestry sector	1.81	3.29	44,206,500
Health	1.69	3.07	44,248,030
Sub-national government administration	1.49	2.71	46,710,540
Agricultural extension and research	1.32	2.40	21,316,480
General public administration sector	1.14	2.07	21,371,730
Secondary education	1.05	1.91	31,513,350
Natural Resources Management	1.00	1.82	78,520,000
Distribution & Transmission	1.00	1.82	35,160,000
Forestry	1.00	1.82	16,900,000
Other Transportation	1.00	1.82	35,920,000
General finance sector	0.99	1.80	98,981,000
General transportation sector	0.95	1.73	17,280,400
Flood protection	0.94	1.71	55,436,900
Aviation	0.92	1.67	21,036,700
Animal production	0.74	1.35	11,105,220
General water, sanitation and flood protection sector	0.72	1.31	20,946,980
Agricultural marketing and trade	0.71	1.29	21,498,800
Capital markets	0.48	0.87	22,512,000
Primary education	0.45	0.82	28,531,820
Telecommunications	0.45	0.82	1,576,800
Law and justice	0.44	0.80	4,111,350
Sanitation	0.42	0.76	11,626,560
Sewerage	0.39	0.71	11,262,900
Oil and gas	0.30	0.55	102,000
Rural services and infrastructure	0.29	0.53	8,346,200
Pollution management and environmental health	0.28	0.51	8,058,400
Renewable Energy	0.23	0.42	34,292,000
General information and communications sector	0.23	0.42	2,269,000
General education sector	0.23	0.42	10,685,800
Postal services	0.20	0.36	700,800
Tertiary education	0.15	0.27	3,837,710
Information technology	0.10	0.18	2,274,000
Crops	0.05	0.09	1,200,000
Other industry	0.03	0.05	710,600
Ports, waterways and shipping	0.01	0.02	279,400
Solid waste management	0.01	0.02	162,010

This table provides an overview of all available sector codes of the Asian NPV projects when the 4 “high-value” NPV projects are excluded.

Appendix AE

Sector Code Category Mapping of Africa and Asia excluding “high-value” NPVs

Category Mapping	Africa		Asia	
	%-Share	WB Costs [Mio USD]	%-Share	WB Costs [Mio USD]
Water & Sanitation	15.33	806	11.91	237
• Water Supply	10.56	510	9.13	193
• General water, sanitation and flood protection sector	2.3	161	1.31	21
• Sanitation	1.26	74	0.76	12
• Sewerage	1.21	61	0.71	11
Power	11.6	966	8.92	144
• Power	8.91	708	6.13	75
• General energy sector	0.09	1	-	-
• Renewable energy	0.17	5	0.42	34
• Hydro	0.94	153	-	-
• Distribution & transmission	0.94	15	1.82	35
• Oil and gas	0.55	84	0.55	0
Transportation Development	26.94	1,996	27.93	876
• Roads and highways	18.99	1,541	22.16	794
• Railways	1.7	130	-	-
• Transportation Adjustment	1.42	57	-	-
• Urban development adjustment	0.47	13	-	-
• Urban management	0.94	29	-	-
• Ports, waterways and shipping	2.04	127	0.02	0
• General transportation sector	1.19	82	1.73	17
• Other transportation	-	-	1.82	36
• Aviation	0.19	17	1.67	21
• Rural services and infrastructure	-	-	0.53	8
Health	3.24	242	3.57	52
• Health	2.69	220	3.07	44
• Primary health, including reproductive health	0.47	6	-	-
• Non-compulsory health finance	0.09	17	-	-
• Pollution management and environmental health	-	-	0.51	8
Education	1.91	35	3.42	76
• Primary education	0.18	10	0.82	29
• Secondary education	1.16	13	1.91	32
• Tertiary education	0.16	2	0.27	4
• General education sector	0.34	7	0.42	11
• Vocational training	0.02	1	-	-
• Adult literacy/non-formal education	0.05	2	-	-

Agriculture	14.54	554	16.64	212
• Agricultural extension and research	3.37	133	2.4	21
• Agricultural marketing and trade	1.95	33	1.29	21
• General agriculture, fishing and forestry sector	4.22	134	3.29	44
• Crops	1.34	32	0.09	1
• Irrigation and drainage	1.59	59	7.75	108
• Forestry	0.82	4	1.82	17
• Agro-Industry	0.14	5	-	-
• Petrochemicals and fertilizers	0.87	151	-	-
• Other Agriculture	0.24	3	-	-
Financial Development	0.94	25	2.66	123
• Capital markets	0.31	3	0.87	23
• General finance sector	0.42	17	1.8	99
• Micro- and SME finance	0.2	5	-	-
• Housing finance and real estate markets	0.02	1	-	-
Governmental Administration	13.1	583	11.46	169
• Central government administration	9.39	414	8.75	122
• Sub-national government administration	3.71	169	2.71	47
Sectorial Development & Reform	4.39	166	2.87	25
• General public administration sector	1.8	109	2.07	21
• General industry and trade sector	1.65	50	-	-
• Other domestic and international trade	0.7	5	-	-
• Law and justice	-	-	0.8	4
• Agency reform	0.24	3	-	-
IT & Communication	0.05	2	1.42	6
• General information and communications sector	0.05	2	0.42	2
• Telecommunications	-	-	0.82	2
• Information technology	-	-	0.18	2
Natural Resources	2.82	53	1.82	79
• Mining and other extractive	2.83	54	-	-
• Natural Resources Management	-	-	1.82	79
Other	5.14	281	7.38	137
• Other social services	3.13	219	3.89	69
• Other industry	0.86	26	0.05	1
• Flood protection	0.75	18	1.71	55
• Animal production	0.05	3	1.35	11
• Solid waste management	0.35	15	0.02	0
• Postal services	-	-	0.36	1

This table shows the project type mapping of the 61 sector codes of the 161 African and Asian NPV projects and their corresponding %-share as well as the WB costs in Mio USD when the 7 “high-value” NPV projects are excluded

Appendix AF

Overall-NPVs of the LDCs at Appraisal and Result Stage

Values in Mio USD		LDCs Appraisal – Overall-NPVs		LDCs Result – Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	2,992	103	1,658	57
	Maximum	5,357	185	4,329	149
Total NPV Capital Costs	Minimum	6,679	230	6,092	210
	Maximum	6,679	230	6,082	210
Total Project Costs		6,679	230	6,122	211
Total World Bank Costs		1,213	42	1,169	40
Discount Rate [%]		11.53		11.55	
Project Duration [years]		5.59		6.86	
NPV-horizon [years]		16.52		18.19	

This table shows the results for the overall-NPV and cost values of the LDCs at the appraisal and result stage in comparison.

Appendix AG

Partial-NPVs of the LDCs at the Appraisal and Result Stage

Values in Mio USD		LDCs Appraisal – Partial-NPVs		LDCs Result – Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	13,054	344	31,990	842
	Maximum	13,072	344	32,014	842
Total NPV Capital Costs	Minimum	3,490	92	3,909	103
	Maximum	3,499	92	3,909	103
Total Project Costs		5,257	138	6,019	158
Total World Bank Costs		2,710	71	2,688	71
Discount Rate [%]		10.89		11.10	
Project Duration [years]		5.16		6.87	
NPV-horizon [years]		21.13		22.86	

This table shows the results for the partial-NPV and cost values of the LDCs at the appraisal and result stage in comparison.

Appendix AH

Appraisal vs. Result Project NPVs greater than 1,000 Mio USD

Continent	Country	Report	Project Name	NPV Type	Appraisal			Result		
					NPV	NPV - Min	NPV - Max	NPV	NPV - Min	NPV - Max
Africa	Ethiopia	34485	Road Sector Development Program Support Project	partial	253	-	-	1,211	-	-
Africa	Lesotho	ICR168	Lesotho Highlands Water Project – Phase1B	overall	1,200	-	-	230	-	-
Asia	Bangladesh	35449	Third Road Rehabilitation and Maintenance Project (RRMP III)	partial	11,179	-	-	28,918	-	-
Asia	Bangladesh	ICR317	Health, Nutrition and Population Sector Program (HNPS)	overall	-	610	2,975	-	610	2,975

This table shows the NPV projects which do disclose “high-value” NPVs for the appraisal and/or the result stage. The Bangladeshi project “35449” is the project disclosing the highest NPV values for the appraisal and the result stage.

Appendix AI

Overall-NPVs of the LDCs at the Appraisal and Result Stage excluding “high-value” NPVs

Values in Mio USD		LDCs Appraisal – Overall-NPVs		LDCs Result – Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,182	43	818	30
	Maximum	1,182	43	1,124	42
Total NPV Capital Costs	Minimum	1,247	46	1,164	43
	Maximum	1,247	46	1,153	43
Total Project Costs		1,247	46	1,194	44
Total World Bank Costs		868	32	854	32
Discount Rate [%]		11.73		11.7	
Project Duration [years]		5.56		6.81	
NPV-horizon [years]		16.07		17.5	

This table shows the results of the overall-NPVs and the related capital cost values of the LDCs at the appraisal and the result stage when excluding the “high-value” projects.

Appendix AJ

Partial-NPVs of the LDCs at the Appraisal and Result Stage excluding “high-value” NPVs

Values in Mio USD		LDCs Appraisal – Partial-NPVs		LDCs Result – Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,623	45	1,862	52
	Maximum	1,640	46	1,886	52
Total NPV Capital Costs	Minimum	2,743	76	3,157	88
	Maximum	2,751	76	3,157	88
Total Project Costs		4,190	116	5,016	139
Total World Bank Costs		2,128	59	2,125	59
Discount Rate [%]		10.8		11.03	
Project Duration [years]		5.11		6.86	
NPV-horizon [years]		21.19		23.11	

This table shows the results of the partial-NPVs and the related capital cost values of the LDCs at the appraisal and the result stage when excluding the “high-value” projects.

Appendix AK

Overall-NPVs of Africa and Asia at the Appraisal and Result Stage

Values in Mio USD		Africa				Asia			
		Appraisal Overall-NPVs		Result Overall-NPVs		Appraisal Overall-NPVs		Result Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,927	96	961	48	1,066	118	696	77
	Maximum	1,927	96	1,037	52	3,431	381	3,293	366
Total NPV Capital Costs	Minimum	2,155	108	1,578	79	4,524	503	4,515	502
	Maximum	2,155	108	1,578	79	4,524	503	4,505	501
Total Project Costs		2,155	108	1,616	81	4,524	503	4,507	501
Total World Bank Costs		739	37	708	35	474	53	461	51
Discount Rate [%]		11.69		11.73		11		11.14	
Actual Project Duration [years]		5.65		7.1		5.44		6.33	
NPV-horizon [years]		16.67		18.04		15.75		18.6	

This table shows the results of the overall-NPVs and the related capital cost values of Africa and Asia at the appraisal and the result stage.

Appendix AL

Partial-NPVs of Africa and Asia at the Appraisal and Result Stage

Values in Mio USD		Africa				Asia			
		Appraisal Partial-NPVs		Result Partial-NPVs		Appraisal Partial-NPVs		Result Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,599	64	2,617	105	11,456	881	29,374	2,260
	Maximum	1,599	64	2,628	105	11,473	883	29,387	2,261
Total NPV Capital Costs	Minimum	2,633	105	3,041	122	858	66	868	67
	Maximum	2,641	106	3,041	122	858	66	868	67
Total Project Costs		3,982	159	4,819	193	1,275	98	1,200	92
Total World Bank Costs		1,873	75	1,904	76	837	64	784	60
Discount Rate [%]		11.06		11.15		10.56		11	
Actual Project Duration [years]		5.24		7.28		5		6.08	
NPV-horizon [years]		21.64		24.25		20.44		20.39	

This table shows the results of the partial-NPVs and the related capital cost values of Africa and Asia at the appraisal and the result stage.

Appendix AM

Overall-NPVs of Africa and Asia at the Appraisal and Result Stage excluding “high-value” NPVs

Values in Mio USD		Africa				Asia			
		Appraisal Overall-NPVs		Result Overall-NPVs		Appraisal Overall-NPVs		Result Overall-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	727	38	731	38	456	57	86	11
	Maximum	727	38	807	42	456	57	318	40
Total NPV Capital Costs	Minimum	1,023	54	949	50	224	28	215	27
	Maximum	1,023	54	949	50	224	28	205	26
Total Project Costs		1,023	54	987	52	224	28	207	26
Total World Bank Costs		694	37	686	36	174	22	168	21
Discount Rate [%]		11.83		11.86		11.33		11.33	
Actual Project Duration [years]		5.63		7.05		5.38		6.25	
NPV-horizon [years]		15.19		17.04		24		18.6	

This table shows the results of the overall-NPVs and the related capital cost values of Africa and Asia at the appraisal and the result stage when excluding the “high-value” projects.

Appendix AN

Partial-NPVs of Africa and Asia at the Appraisal and Result Stage excluding “high-value” NPVs

Values in Mio USD		Africa				Asia			
		Appraisal Partial-NPVs		Result Partial-NPVs		Appraisal Partial-NPVs		Result Partial-NPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
NPV	Minimum	1,346	56	1,406	59	277	23	456	38
	Maximum	1,346	56	1,417	59	294	24	469	39
Total NPV Capital Costs	Minimum	2,244	93	2,673	111	499	42	484	40
	Maximum	2,252	94	2,673	111	499	42	484	40
Total Project Costs		3,443	143	4,284	179	746	62	732	61
Total World Bank Costs		1,564	65	1,571	65	564	47	555	46
Discount Rate [%]		11		11.12		10.38		10.9	
Actual Project Duration [years]		5.21		7.29		4.92		6	
NPV-horizon [years]		21.64		24.53		20.5		20.44	

This table shows the results of the partial-NPVs and the related capital cost values of Africa and Asia at the appraisal and the result stage when excluding the “high-value” projects.

Appendix AO

Result Overall-ERRs of the LDCs

Values in Percentage		Overall-ERRs		Overall-ERRs of NPV Projects		Overall-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		40.27	50.74	29.61	39.76	62.80	73.96
ERR-Range	Min Scenario	3.3	1,105.65	3.3	137.55	6.7	1,105.65
	Max Scenario	3.3	1,105.65	3.3	311	6.7	1,105.65
Actual Total Project Costs [Mio USD]		17,670		16,127		1,543	
Actual Total World Bank Costs [Mio USD]		3,151		2,450		701	
Discount Rate [%]		11.29		11.38		10.67	
Actual Project Duration [years]		6.96		7.07		6.73	
ERR-horizon [years]		18.42		18.6		17.7	

This table shows the results of the overall-ERRs and the corresponding cost values of the LDCs at the result stage. A distinction has been provided between the total of the overall-ERRs, the overall-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AP

Result Partial-ERRs of the LDCs

Values in Percentage		Partial-ERRs		Partial-ERRs of NPV Projects		Partial-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		23.89	46.77	25.17	47.66	19.88	43.94
ERR-Range	Min Scenario	-8	117	-4	103	-8	117
	Max Scenario	-8	202	8	202	-8	200
Actual Total Project Costs [Mio USD]		19,522		16,605		2,917	
Actual Total World Bank Costs [Mio USD]		7,661		6,320		1,342	
Discount Rate [%]		11.13		11.19		10.75	
Actual Project Duration [years]		6.88		6.93		6.73	
ERR-horizon [years]		19.81		20.2		17.42	

This table shows the results of the partial-ERRs and the corresponding cost values of the LDCs at the result stage. A distinction has been provided between the total of the partial-ERRs, the partial-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AQ

Result Project ERRs greater than 100 %

Continent	Country	Report	ERR Type	ERR	ERR - Min	ERR - Max
Africa	Guinea-Bissau	ICR1347	overall	1,105.65	-	-
Africa	Malawi	ICR764	partial	117	-	-
Africa	Mozambique	ICR2273	partial	103	-	-
Africa	Senegal	32102	overall	137.55	-	-
Africa	Burkina Faso	ICR316	partial	-	3	200
Africa	Central African Republic	ICR1990	overall	-	23	226
Africa	Ethiopia	ICR350	partial	-	34.7	153.4
Africa	Madagascar	29309	partial	-	12	202
Africa	Madagascar	26161	overall	-	78	109
Africa	Malawi	ICR252	partial	-	96	103
Africa	Niger	26930	partial	-	31	146
Africa	Tanzania	ICR667	partial	-	8.87	118.6
Africa	Uganda	ICR951	partial	-	38	175
Africa	Uganda	30285	partial	-	23.8	120
Africa	Uganda	29653	partial	-	51.4	143.4
Africa	Uganda	21418	partial	-	4.3	180
Asia	Bangladesh	ICR23	overall	120	-	-
Asia	Cambodia	ICR2229	partial	-	21	131
Asia	East Timor	ICR785	partial	-	9	120
Asia	Nepal	ICR2029	overall	-	29	311

This table shows the ERR projects which do disclose percentages of above 100. All values are expressed in percentage [%].

Appendix AR

Result Overall-ERRs of the LDCs excluding “high-value” Projects

Values in Percentage		Overall-ERRs		Overall-ERRs of NPV Projects		Overall-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		23.58	28.01	24.79	29.6	21.01	24.63
ERR-Range	Min Scenario	3.3	84	3.3	84	6.7	51.5
	Max Scenario	3.3	84	3.3	84	6.7	51.5
Actual Total Project Costs [Mio USD]		17,543		16,044		1,499	
Actual Total World Bank Costs [Mio USD]		3,078		2,408		670	
Discount Rate [%]		11.30		11.32		11.2	
Actual Project Duration [years]		7.05		7.2		6.73	
ERR-horizon [years]		18.51		18.78		17.44	

This table shows the results of the overall-ERRs and the corresponding cost values of the LDCs at the result stage when “high-value” ERR projects are excluded. A distinction has been provided between the total of the overall-ERRs, the overall-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AS

Result Partial-ERRs of the LDCs excluding “high-value” Projects

Values in Percentage		Partial-ERRs		Partial-ERRs of NPV Projects		Partial-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		21.38	35.73	22.76	35.45	17.29	36.55
ERR-Range	Min Scenario	-8	99	8	95	-8	99
	Max Scenario	-8	99	-4	76	-8	99
Actual Total Project Costs [Mio USD]		18,450		15,679		2,771	
Actual Total World Bank Costs [Mio USD]		7,054		5,813		1,240	
Discount Rate [%]		11.1		11.16		10.65	
Actual Project Duration [years]		6.93		6.99		6.74	
ERR-horizon [years]		20.22		20.82		16.73	

This table shows the results of the partial-ERRs and the corresponding cost values of the LDCs at the result stage when “high-value” ERR projects are excluded. A distinction has been provided between the total of the partial-ERRs, the partial-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AT

Result Overall-ERRs of Africa and Asia

Africa Values in Percentage		Overall-ERRs		Overall-ERRs of NPV Projects		Overall-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean ($\emptyset$)		53.62	62.29	31.6	36.17	121.67	143.03
ERR-Range	Min Scenario	3.3	1,105.65	3.3	137.55	6.7	1,105.65
	Max Scenario	3.3	1,105.65	3.3	137.55	6.7	1,105.65
Actual Total Project Costs [Mio USD]		4,225		3,219		1,006	
Actual Total World Bank Costs [Mio USD]		1,661		1,341		319	
Discount Rate [%]		11.38		11.54		10	
Actual Project Duration [years]		7.33		7.44		7	
ERR-horizon [years]		18.27		18.31		18	
Asia Values in Percentage		Overall-ERRs		Overall-ERRs of NPV Projects		Overall-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean ($\emptyset$)		23.58	36.3	26.39	45.59	19.63	23.3
ERR-Range	Min Scenario	9	120	9	120	11	51.5
	Max Scenario	9	311	9	311	11	51.5
Actual Total Project Costs [Mio USD]		13,445		12,908		537	
Actual Total World Bank Costs [Mio USD]		1,490		1,109		381	
Discount Rate [%]		11.18		11.16		11.33	
Actual Project Duration [years]		6.5		6.48		6.53	
ERR-horizon [years]		18.61		19.03		17.5	

This table shows the results of the overall-ERRs and the corresponding cost values of the African and Asian continents at the result stage. A distinction has been provided between the total of the overall-ERRs, the overall-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AU

Result Partial-ERRs of Africa and Asia

Africa Values in Percentage		Partial-ERRs		Partial-ERRs of NPV Projects		Partial-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean ($\emptyset$)		23.44	48.95	24.77	49.69	20.16	47.15
ERR-Range	Min Scenario	-8	117	-4	103	-8	117
	Max Scenario	-8	202	8	202	-8	200
Actual Total Project Costs [Mio USD]		16,412		13,694		2,718	
Actual Total World Bank Costs [Mio USD]		5,645		4,464		1,181	
Discount Rate [%]		11.04		11.12		10.64	
Actual Project Duration [years]		7.1		7.2		6.86	
ERR-horizon [years]		19.86		20.48		16.9	
Asia Values in Percentage		Partial-ERRs		Partial-ERRs of NPV Projects		Partial-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean ($\emptyset$)		24.99	41.46	25.94	43.67	18.34	25.98
ERR-Range	Min Scenario	3	99	5	99	3	30
	Max Scenario	11	131	11	131	5	30
Actual Total Project Costs [Mio USD]		3,110		2,911		199	
Actual Total World Bank Costs [Mio USD]		2,017		1,855		161	
Discount Rate [%]		11.34		11.32		12	
Actual Project Duration [years]		6.35		6.4		6	
ERR-horizon [years]		19.7		19.67		20	

This table shows the results of the partial-ERRs and the corresponding cost values of the African and Asian continents at the result stage. A distinction has been provided between the total of the partial-ERRs, the partial-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AV

Result Overall-ERRs of Africa and Asia excluding “high-value” ERRs

Africa Values in Percentage		Overall-ERRs		Overall-ERRs of NPV Projects		Overall-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		26.06	29.87	26.84	30.72	23.3	26.86
ERR-Range	Min Scenario	3.3	84	3.3	84	6.7	42
	Max Scenario	3.3	84	3.3	84	6.7	42
Actual Total Project Costs [Mio USD]		4,157		3,195		962	
Actual Total World Bank Costs [Mio USD]		1,621		1,331		289	
Discount Rate [%]		11.46		11.5		11	
Actual Project Duration [years]		7.48		7.56		7.13	
ERR-horizon [years]		18.34		18.48		17.33	
Asia Values in Percentage		Overall-ERRs		Overall-ERRs of NPV Projects		Overall-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		20.58	25.76	21.33	27.7	19.63	23.3
ERR-Range	Min Scenario	9	65	9	65	11	51.5
	Max Scenario	9	65	9	65	11	51.5
Actual Total Project Costs [Mio USD]		13,386		12,850		537	
Actual Total World Bank Costs [Mio USD]		1,458		1,077		381	
Discount Rate [%]		11.1		11.06		11.33	
Actual Project Duration [years]		6.56		6.58		6.53	
ERR-horizon [years]		18.73		19.25		17.5	

This table shows the results of the overall-ERRs and the corresponding cost values of the African and Asian continents at the result stage based on the exclusion of “high-value” overall-ERRs. A distinction has been provided between the total of the overall-ERRs, the overall-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AW

Result Partial-ERRs of Africa and Asia excluding “high-value” ERRs

Africa Values in Percentage		Partial-ERRs		Partial-ERRs of NPV Projects		Partial-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		20.6	35.14	22.14	33.63	17.09	38.58
ERR-Range	Min Scenario	-8	99	8	85	-8	99
	Max Scenario	-8	99	-4	76	-8	99
Actual Total Project Costs [Mio USD]		15,367		12,795		2,572	
Actual Total World Bank Costs [Mio USD]		5,055		3,975		1,079	
Discount Rate [%]		10.96		11.05		10.5	
Actual Project Duration [years]		7.18		7.31		6.88	
ERR-horizon [years]		20.20		21.15		16	

Asia Values in Percentage		Partial-ERRs		Partial-ERRs of NPV Projects		Partial-ERRs of Non-NPV Projects	
		Min	Max	Min	Max	Min	Max
ERR-Arithmetic Mean (Ø)		23.15	37.04	23.88	38.71	18.34	25.98
ERR-Range	Min Scenario	3	67.34	5	67.34	3	30
	Max Scenario	11	95	11	95	22.3	30
Actual Total Project Costs [Mio USD]		3,084		2,884		199	
Actual Total World Bank Costs [Mio USD]		1,999		1,838		161	
Discount Rate [%]		11.37		11.35		12	
Actual Project Duration [years]		6		6		6	
ERR-horizon [years]		20.25		20.27		20	

This table shows the results of the partial-ERRs and the corresponding cost values of the African and Asian continents at the result stage based on the exclusion of “high-value” partial-ERRs. A distinction has been provided between the total of the partial-ERRs, the partial-ERRs of projects disclosing a NPV and the ones for non-NPV projects.

Appendix AX

Result Net Benefits of the LDCs – 1

Values in Mio USD		Total Net Benefits		Net Benefits of NPV/ERR Projects	
		Cum Sum (Ø)		Cum Sum (Ø)	
Net Benefit	Overall	17,581 (419)	12,232 (680)	16,893 (433)	11,564 (680)
	Partial		5,349 (223)		5,328 (242)
Total Net Benefit Costs	Overall	6,721 (160)	3,303 (183)	6,450 (165)	3,059 (180)
	Partial		3,418 (142)		3,391 (154)
Actual Total Project Costs	Overall	4,685 (112)	2,070 (115)	4,399 (113)	1,826 (107)
	Partial		2,615 (109)		2,573 (117)
Actual Total World Bank Costs	Overall	1,875 (45)	764 (42)	1,601 (41)	530 (31)
	Partial		1,111 (46)		1,072 (49)
Discount Rate [%]	Overall	10.44	11.2	10.73	11.2
	Partial		10		10.56
Actual Project Duration [years]	Overall	7.1	6.94	7.33	7.24
	Partial		7.21		7.41
Net Benefit-horizon [years]	Overall	24.87	24.08	24.87	24.08
	Partial		25.39		25.39

This table shows the first 2 groups of the result Net Benefits of the LDCs, providing a view of the overall and partial-Net Benefits and corresponding cost values.

Appendix AY

Result Net Benefits of the LDCs – 2

Values in Mio USD		Net Benefits of Non-NPV/ERR Projects		Net Benefits of Non-NPV/ERR Projects – BCR Intersection		Net Benefits of Non-NPV/ERR Projects – Single	
		Cum Sum (Ø)		Cum Sum (Ø)		Cum Sum (Ø)	
Net Benefit	Overall	688	668 (668)	672	668 (668)	16	n.a.
	Partial	(229)	20 (10)	(336)	4 (4)	(16)	16 (16)
Total Net Benefit Costs	Overall	271	244 (244)	269	244 (244)	2	n.a.
	Partial	(90)	27 (14)	(135)	25 (25)	(2)	2 (2)
Actual Total Project Costs	Overall	286	244 (244)	269	244 (244)	17	n.a.
	Partial	(95)	42 (21)	(135)	25 (25)	(17)	17 (17)
Actual Total World Bank Costs	Overall	273	234 (234)	260	234 (234)	13	n.a.
	Partial	(91)	39 (19)	(130)	25 (25)	(13)	13 (13)
Discount Rate [%]	Overall	3	n.a.	3	n.a.	n.a.	n.a.
	Partial		3		3		n.a.
Actual Project Duration [years]	Overall	4	2	3.5	2	5	n.a.
	Partial		5		5		5
Net Benefit-horizon [years]	Overall	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Partial		n.a.		n.a.		n.a.

This table shows the remaining 3 groups of the result Net Benefits of the LDCs, providing the breakdown by overall- and partial-Net Benefits and corresponding cost values.

Appendix AZ

Result Project Net Benefits greater than 1,000 Mio USD

Continent	Country	Report	Type	Net Benefit	Total Net Benefit Costs
Africa	Ethiopia	35573	partial	1,927	385
Africa	Mali	30931	overall	2,808	463
Africa	Mauritania	27481	overall	2,808	463
Africa	Senegal	27482	overall	2,135	450
Africa	Uganda	24406	overall	1,120	218

This table shows the 5 identified “high-value” Net Benefit projects.

Appendix BA

Result Net Benefits of the LDCs excluding “high-value” Net Benefits – 1

Values in Mio USD		Total Net Benefits		Net Benefits of NPV/ERR Projects	
		Cum Sum (Ø)		Cum Sum (Ø)	
Net Benefit	Overall	6,782	3,361 (240)	6,094	2,693 (207)
	Partial	(183)	3,422 (149)	(179)	3,401 (162)
Total Net Benefit Costs	Overall	4,742	1,709 (122)	4,471	1,465 (113)
	Partial	(128)	3,033 (132)	(132)	3,006 (143)
Actual Total Project Costs	Overall	2,981	722 (52)	2,695	478 (37)
	Partial	(81)	2,259 (98)	(79)	2,217 (106)
Actual Total World Bank Costs	Overall	1,485	572 (41)	1,212	338 (26)
	Partial	(40)	913 (40)	(36)	874 (42)
Discount Rate [%]	Overall	10.4	11.11	10.79	11.11
	Partial		10		10.6
Actual Project Duration [years]	Overall	7.05	6.86	7.32	7.23
	Partial		7.17		7.38
Net Benefit-horizon [years]	Overall	23.37	22.7	23.37	22.7
	Partial		23.76		23.76

This table shows the first 2 groups of the result Net Benefits of the LDCs when “high-value” Net Benefits are excluded, with a breakdown by overall- and partial-Net Benefits and corresponding cost values.

Appendix BB

Result Net Benefits of the LDCs excluding “high-value” Net Benefits – 2

Values in Mio USD		Net Benefits of Non-NPV/ERR Projects		Net Benefits of Non-NPV/ERR Projects – Net Benefit Intersection		Net Benefits of Non-NPV/ERR Projects – Single	
		Cum Sum (Ø)		Cum Sum (Ø)		Cum Sum (Ø)	
Net Benefit	Overall	688	668 (668)	672	668 (668)	16	n.a.
	Partial	(229)	20 (10)	(336)	4 (4)	(16)	16 (16)
Total Net Benefit Costs	Overall	271	244 (244)	269	244 (244)	2	n.a.
	Partial	(90)	27 (14)	(135)	25 (25)	(2)	2 (2)
Actual Total Project Costs	Overall	286	244 (244)	269	244 (244)	17	n.a.
	Partial	(95)	42 (21)	(135)	25 (25)	(17)	17 (17)
Actual Total World Bank Costs	Overall	273	234 (234)	260	234 (234)	13	n.a.
	Partial	(91)	39 (19)	(130)	25 (25)	(13)	13 (13)
Discount Rate [%]	Overall	3	n.a.	3	n.a.	n.a.	n.a.
	Partial		3		3		n.a.
Actual Project Duration [years]	Overall	4	2	3.5	2	5	n.a.
	Partial		5		5		5
Net Benefit-horizon [years]	Overall	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Partial		n.a.		n.a.		n.a.

This table shows the remaining 3 groups of the result Net Benefits of the LDCs when “high-value” Net Benefits are excluded, providing a breakdown by overall- and partial-Net Benefits and corresponding cost values.

Appendix BC

Result Project FNPVs greater than 1,000 Mio USD

Continent	Country	Report	FNPV	Total Project Costs	WB Costs	Object of Consideration
Africa	Chad	36560	3,721	4,817	93	cross-country project: 2 countries (Cameroon, Chad, operator)

This table shows the FNPV project which does disclose a FNPV value above 1,000 Mio USD.

Appendix BD

Result FNPVs

Values in Mio USD		Total FNPVs		FNPVs of NPV/ERR Projects		FNPVs of Non-NPV/ERR Projects		FNPVs of Non-NPV/ERR Projects – FRR Intersection		FNPVs of Non-NPV/ERR Projects – Single	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
FNPV	Minimum	4,190	110	461	14	3,729	622	8	2	3,721	3,721
	Maximum	4,231	111	502	16	3,729	622	8	2	3,721	3,721
Actual Total Project Costs		7,195	189	2,280	71	4,916	819	99	20	4,817	4,817
Actual Total World Bank Costs		1,641	43	1,455	45	186	31	93	19	93	93
Discount Rate [%]		10.86		10.87		10.8		11		10	
Actual Project Duration [years]		6.97		7.19		5.83		6.2		4	
FNPV-horizon [years]		16.7		17.93		8.5		8.5		n.a.	

This table shows the total FNPVs based on the 5 grouping differentiation of the result stage.

Appendix BE

Result FNPV Comparison

Values in Mio USD		Total FNPVs		Positive FNPVs		Neutral FNPVs		Negative FNPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
FNPV	Minimum	4,190	110	4,285	134	-14	-14	-80	-80
	Maximum	4,231	111	4,308	135	3	3	-80	-80
Actual Total Project Costs		7,195	189	6,498	203	135	135	564	113
Actual Total World Bank Costs		1,641	43	1,298	41	98	98	245	49
Discount Rate [%]		10.86		11.33		5		9	
Actual Project Duration [years]		6.97		6.63		8		9	
FNPV-horizon [years]		16.7		14.89		n.a.		23.2	

This table shows the total, positive, neutral and negative FNPVs of the result stage.

Appendix BF

Result FNPV and FRR Objects of Consideration

Report	Financial Indicator	Object of consideration
ICR705	FRR	project as a whole
21422	FRR	revenue generating activities
ICR2172	FRR&FNPV	specific sub-project
36560	FNPV	cross-country project: 2 countries (Cameroon, Chad, operator)
30711	FRR&FNPV	project as a whole
25283	FRR&FNPV	different models, sub projects
27537	FRR&FNPV	typical subproject financed under the project
35573	FRR	Ethiopian electric power corporation
33784	FRR	revenue-earning infrastructure (e.g. markets, abattoirs)
ICR2265	FRR&FNPV	whole project which is relevant for the analysis (not all cost associated!)
ICR764	FRR	Privatization Component
ICR243	FRR&FNPV	project as a whole
32130	FRR	railway component
25255	FRR	project as a whole
ICR2294	FRR&FNPV	overall micro-projects > pooled
34787	FRR&FNPV	Rice product & diversification (Sample)
27558	FRR&FNPV	Beneficiaries' Contribution on various activities such as, small dams, bunds, Arabic gum
ICR1118	FRR&FNPV	port and railway system
ICR337	FRR	overall crop production
ICR1231	FRR&FNPV	urban water supply component
ICR1178	FRR&FNPV	project (PIP2) as a whole
26677	FRR	income-generating sub-projects
25669	FNPV	one income-generating activity
24288	FRR	cultivated surface area
ICR675	FNPV	gravity irrigation (project as a whole)
ICR1415	FRR&FNPV	project (PSAOP2) as a whole
ICR106	FRR	profitability of MFI (Micro-Finance Institution) activities
ICR1819	FRR&FNPV	main project component A of the project (Songas)
32225	FRR&FNPV	UWAS (Urban Water and Sewerage Authority)
26232	FRR&FNPV	TRC (Tanzania Railways Corporation)
22256	FRR	Evaluation of 3 institutions
21559	FRR&FNPV	THA (Tanzania Harbours Authority)
27529	FRR&FNPV	the corresponding cities
24406	FRR	revalued fixed assets
35722	FRR&FNPV	asset holding company – mining and municipal services
ICR507	FRR	rural and urban piped water schemes
ICR777	FRR&FNPV	microcredit disbursed to the LIL beneficiaries
ICR358	FRR&FNPV	450 MW power plant
ICR23	FRR	all 4 project components: inland open-water, coastal shrimp aquaculture, freshwater aquaculture, aquatic resource
ICR90	FRR&FNPV	IDA credit
27415	FRR&FNPV	per farmer as a pilot
24350	FRR	Evaluation of the Borrower
ICR312	FRR	water supply component (without tourism component)
25207	FRR	the (forestry) project
ICR1491	FRR&FNPV	Government level
30003	FRR	component PPWSA= Phnom Penh Water Supply Authority
ICR2152	FNPV	seepage gas was available for a power generation respectively as Seep Harvesting Project (project as a whole)
ICR1270	FRR	FRR dealt with the financial benefits and costs to the government
32004	FRR&FNPV	EDL (Electricite de Laos)
ICR1348	FRR&FNPV	Nepal Bank Limited (NBL)
ICR1197	FRR	selected activities of sub-projects
31484	FRR	Farmer-Managed Irrigation Scheme (FMIS) Sites
27813	FRR&FNPV	Biratnagar and Bhairahawa Inland Container Depots (ICDs)
ICR1125	FRR&FNPV	farm-level
36380	FRR	for the land development, livestock and fisheries components
29813	FNPV	benefits for farmers from the fertilizer
27483	FRR&FNPV	3 different water networks
27440	FRR&FNPV	water supply scheme by the water utility (NWSA)
22117	FRR	Highland areas (Tihama, Lahej and Dhamar)

This table shows the FRR and FNPV projects and the corresponding objects of consideration. The “Financial Indicator” column indicates if the report discloses a FNPV and/or FRR.

Appendix BG

Result FNPV Comparison excluding the single “high-value” FNPV

Values in Mio USD		Total FNPVs		Positive FNPVs		Neutral FNPVs		Negative FNPVs	
		Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø	Cum Sum	Ø
FNPV	Minimum	469	13	564	18	-14	-14	-80	-80
	Maximum	510	14	587	19	3	3	-80	-80
Actual Total Project Costs		2,378	64	1,681	54	135	135	564	113
Actual Total World Bank Costs		1,548	42	1,205	39	98	98	245	49
Discount Rate [%]		10.89		11.39		5		9	
Actual Project Duration [years]		7.05		6.71		8		9	
FNPV-horizon [years]		16.7		14.89		n.a.		23.2	

This table shows the total, positive, neutral and negative FNPVs of the result stage when the single “high-value” FNPV is excluded.

Appendix BH

Result Project FRRs greater than 100 %

Continent	Country	Report	FRR	FRR- Min	FRR - Max	Object of Consideration
Africa	Mauritania	27558	-	21	500	Rice product & diversification (Sample): Contribution of beneficiaries on various activities, such as small dams, bunds, Arabic gum
Asia	Bangladesh	ICR23	116	-	-	all 4 project components: Inland open-water, coastal shrimp aquaculture, freshwater aquaculture, aquatic resource
Asia	Nepal	ICR1197	-	21	847	selected activities of sub-projects

This table shows the FRR projects which disclose percentages above 100. All values are expressed in percentage [%].

Appendix BI

Result Project FRRs excluding “high-value” FRRs

Values in Percentage		Total FRRs		FRRs of NPV/ERR Projects		FRRs of Non-NPV/ERR Projects		FRRs of Non-NPV/ERR Projects – FNPV Intersection		FRRs of Non-NPV/ERR Projects – Single	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
FRR-Arithmetic Mean ($\emptyset$)		21.11	29.58	18.44	26.4	36.5	47.88	34	51.8	40.67	41.33
FRR-Range	Min Scenario	-7.66	78	-7.66	78	12	77	12	77	13	66
	Max Scenario	-7.66	100	-7.66	100	13	100	14	100	13	68
Actual Total Project Costs [Mio USD]		4,668		4,513		155		99		56	
Actual Total World Bank Costs [Mio USD]		2,407		2,271		136		93		43	
Discount Rate [%]		10.96		11		10.8		11		10	
Actual Project Duration [years]		7.1		7.38		6.13		6.2		6	
FRR-horizon [years]		18.38		19.79		8.5		8.5		n.a.	

This table shows the outcomes of the 5 groups for the result FRRs when the 3 “high-value” FRRs are excluded.

Appendix BJ

Negative Project Shares – Groups Comparison 1 excluding all “High-values”

		All Projects		All Financial Indicator Projects		Non-Financial Indicator Projects	
		Counts: 755		Counts: 209		Counts: 546	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	19.34	146	13.88	29	21.43	117
“Moderately Unsatisfactory” Share	negative	7.28	55	5.26	11	8.06	44
“Unsatisfactory” Share	negative	11.79	89	8.61	18	13	71
“Highly Unsatisfactory” Share	negative	0.26	2	0	0	0.37	2

This table shows the results for the groups comparison 1 when all “high-values” of all financial indicators are excluded. There are no noteworthy changes compared to the group comparison of all values.

Appendix BK

Negative Project Shares – Groups Comparison 2 excluding all “High-values”

		Total NPV Projects		Total ERR Projects		Other Financial Indicator Projects	
		Counts: 142		Counts: 185		Counts: 97	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	12.68	18	13.51	25	14.43	14
“Moderately Unsatisfactory” Share	negative	4.93	7	5.41	10	5.15	5
“Unsatisfactory” Share	negative	7.75	11	8.11	15	9.28	9
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

This table shows the results for the groups comparison 2 when all “high-values” of all financial indicators are excluded. There are no noteworthy changes compared to the group comparison of all values.

Appendix BL

Negative Project Shares – Groups Comparison 3 excluding all “High-values”

		ERRs of NPV Projects		ERRs of non-NPV Projects	
		Counts: 133		Counts: 52	
		%	Counts	%	Counts
Total Negative Share	negative	12.03	16	17.31	9
“Moderately Unsatisfactory” Share	negative	5.26	7	5.77	3
“Unsatisfactory” Share	negative	6.77	9	11.54	6
“Highly Unsatisfactory” Share	negative	0	0	0	0

This table shows the results for the groups comparison 3 when all “high-values” of all financial indicators are excluded.

Appendix BM

Negative Project Shares – Groups Comparison 3 excluding “high-value” ERRs

		ERRs of NPV Projects (excluding 20 “high value” ERRs)		ERRs of non-NPV Projects (excluding 20 “high value” ERRs)	
		Counts: 143		Counts: 55	
		%	Counts	%	Counts
Total Negative Share	negative	11.89	17	16.36	9
“Moderately Unsatisfactory” Share	negative	5.59	8	5.45	3
“Unsatisfactory” Share	negative	6.29	9	10.91	6
“Highly Unsatisfactory” Share	negative	0	0	0	0

This table shows the results for the groups comparison 3 when “high-value” ERRs are excluded.

Appendix BN

Negative Project Shares – Groups Comparison 4 excluding all “High-values”

		Negative & Neutral NPV/ERR Projects		Positive NPV/ERR Projects	
		Counts: 17		Counts: 161	
		%	Counts	%	Counts
Total Negative Share	negative	47.06	8	8.07	13
“Moderately Unsatisfactory” Share	negative	17.65	3	2.48	4
“Unsatisfactory” Share	negative	29.41	5	5.59	9
“Highly Unsatisfactory” Share	negative	0	0	0	0

This table shows the results for the groups comparison 4 when “high-value” ERRs are excluded.

Appendix BO

Negative Project Shares of the NPV Projects

		Total NPV Projects		Overall-NPV Projects		Partial-NPV Projects	
		Counts: 168		Counts: 60		Counts: 108	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	12.5	21	15	9	11.11	12
“Moderately Unsatisfactory” Share	negative	5.36	9	8.33	5	3.7	4
“Unsatisfactory” Share	negative	7.14	12	6.67	4	7.41	8
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

This table shows the results of the negative project shares of the NPV projects.

Appendix BP

Negative Project Shares of the NPV Projects excluding “high-value” NPVs

		Total NPVs (excluding 7 “high value” NPVs)		Overall-NPVs (excluding 7 “high value” NPVs)		Partial-NPVs (excluding 7 “high value” NPVs)	
		Counts: 161		Counts: 56		Counts: 105	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	12.42	20	14.29	8	11.43	12
“Moderately Unsatisfactory” Share	negative	4.97	8	7.14	4	3.81	4
“Unsatisfactory” Share	negative	7.45	12	7.14	4	7.62	8
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

This table shows the results of the negative project shares of the NPV projects when the 7 “high-value” NPV projects are excluded.

Appendix BQ

Negative Project Shares of the NPV Projects excluding all “High-values”

		Total NPV Projects		Overall-NPV Projects		Partial-NPV Projects	
		Counts: 142		Counts: 50		Counts: 92	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	12.68	18	16	8	10.87	10
“Moderately Unsatisfactory” Share	negative	4.93	7	8	4	3.26	3
“Unsatisfactory” Share	negative	7.75	11	8	4	7.61	7
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

This table shows the results of the negative project shares of the NPV projects when all “high-value” projects of all financial indicators are excluded.

Appendix BR

Negative Project Shares of the ERR Projects

		Total ERR Projects		Overall-ERR Projects		Partial-ERR Projects	
		Counts: 218		Counts: 81		Counts: 137	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	13.3	29	16.05	13	11.68	16
“Moderately Unsatisfactory” Share	negative	5.96	13	7.41	6	5.11	7
“Unsatisfactory” Share	negative	7.34	16	8.64	7	6.57	9
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

This table shows the results of the negative project shares of the ERR projects.

Appendix BS

Negative Project Shares of the ERR Projects excluding “high-value” ERRs

		Total ERR Projects (excluding 20 “high value” ERRs)		Overall-ERR Projects (excluding 20 “high value” ERRs)		Partial-ERR Projects (excluding 20 “high value” ERRs)	
		Counts: 198		Counts: 75		Counts: 123	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	13.13	26	16	12	11.38	14
“Moderately Unsatisfactory” Share	negative	5.56	11	6.67	5	4.88	6
“Unsatisfactory” Share	negative	7.58	15	9.33	7	6.5	8
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

This table shows the results of the negative project shares of the ERR projects when the 20 “high-value” ERR projects are excluded.

Appendix BT

Negative Project Shares of the ERR Projects excluding all “High-values”

		Total ERR Projects		Overall-ERR Projects		Partial-ERR Projects	
		Counts: 185		Counts: 69		Counts: 116	
		%	Counts	%	Counts	%	Counts
Total Negative Share	negative	13.51	25	15.94	11	12.07	14
“Moderately Unsatisfactory” Share	negative	5.41	10	5.8	4	5.17	6
“Unsatisfactory” Share	negative	8.11	15	10.14	7	6.9	8
“Highly Unsatisfactory” Share	negative	0	0	0	0	0	0

This table shows the results of the negative project shares of the ERR projects when all “high-value” projects of all financial indicators are excluded.

Appendix BU

Calculated ERRs of Non-Financial Indicator Projects – Unweighted ERR

Share Factor

	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
	%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	11.95	26.7	44.93	-9.48	24.26	24.76
ERRs of non-NPV Projects	16.95	38.8	57.17	-4.48	37.65	47.64
ERRs of Negative & Neutral NPV/ERR Projects	42.11	2.93	37.52	20.68	8.26	82.52
ERRs of Positive NPV/ERR Projects	8.9	34.22	50.73	-12.53	30.99	24.07
Calculated ERRs of Non-Financial Indicator Projects	21.43	Ø =			25.27	44.75

$$ERR \text{ Share Factor}_{Min} = \frac{\frac{11.95}{26.7} + \frac{16.95}{38.8} + \frac{42.11}{2.93} + \frac{8.9}{34.22}}{4} = 3.88$$

$$ERR \text{ Share Factor}_{Max} = \frac{\frac{11.95}{44.93} + \frac{16.95}{57.17} + \frac{42.11}{37.52} + \frac{8.9}{50.73}}{4} = 0.47$$

This table shows the calculated ERRs of the non-financial indicator projects based on the unweighted calculation of the ERR share factor.

Appendix BV

Calculated ERRs of Non-Financial Indicator Projects – Weighted Option 1

	Total Counts	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
		%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	159	11.95	26.7	44.93	-9.48	3.58	8.47
ERRs of non-NPV Projects	59	16.95	38.8	57.17	-4.48	6.02	21.57
ERRs of Negative & Neutral NPV/ERR Projects	19	42.11	2.93	37.52	20.68	53.37	82.47
ERRs of Positive NPV/ERR Projects	191	8.9	34.22	50.73	-12.53	3.66	2.54
Calculated ERRs of Non-Financial Indicator Projects	546	21.43	Ø =			16.66	28.76

$$ERR \text{ Share Factor}_{Min} = \frac{[(11.95*159)+(16.95*59)+(42.11*19)+(8.9*191)]}{[(26.7*159)+(38.8*59)+(2.93*19)+(34.22*191)]} = 0.41$$

$$ERR \text{ Share Factor}_{Max} = \frac{[(11.95*159)+(16.95*59)+(42.11*19)+(8.9*191)]}{[(44.93*159)+(57.17*59)+(37.52*19)+(50.73*191)]} = 0.26$$

This table shows the calculated ERRs of the non-financial indicator projects based on the weighted option 1, whereat the ERR share factor is already weighted.

Appendix BW

Calculated ERRs of Non-Financial Indicator Projects – Weighted Option 2

	Total Counts	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
		%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	159	11.95	26.7	44.93	-9.48	14.55	22.36
ERRs of non-NPV Projects	59	16.95	38.8	57.17	-4.48	33.06	46.5
ERRs of Negative & Neutral NPV/ERR Projects	19	42.11	2.93	37.52	20.68	29.44	86.76
ERRs of Positive NPV/ERR Projects	191	8.9	34.22	50.73	-12.53	18.16	20.9
Calculated ERRs of Non-Financial Indicator Projects	546	21.43	W-Ø =			19.37	27.9

$$ERR\ Share\ Factor_{Min} = \frac{(11.95+16.95+42.11+8.9)}{(26.7+38.8+2.93+34.22)} = 0.78$$

$$ERR\ Share\ Factor_{Max} = \frac{(11.95+16.95+42.11+8.9)}{(44.93+57.17+37.52+50.73)} = 0.42$$

This table shows the calculated ERRs of the non-financial indicator projects based on the weighted option 2 using an unweighted ERR share factor. In this option the weighted-arithmetic mean (W-Ø) of 19.37 and 27.9 percent is calculated based on the weighted minimum and maximum calculated ERRs of the source groups.

Appendix BX

Calculated ERRs of Non-Financial Indicator Projects excluding “High-values” – Unweighted ERR Share Factor

	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
	%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	11.68	22.98	32.56	-9.75	20.68	18.43
ERRs of non-NPV Projects	16.36	18.91	31.35	-5.07	17.71	24
ERRs of Negative & Neutral NPV/ERR Projects	44.44	2.92	26.69	23.91	8.57	39.11
ERRs of Positive NPV/ERR Projects	7.78	24.82	33.37	-13.65	21.59	2.08
Calculated ERRs of Non-Financial Indicator Projects	21.43	Ø =			17.14	20.91

$$ERR\ Share\ Factor_{Min} = \frac{\frac{11.68}{22.98} + \frac{16.36}{18.91} + \frac{44.44}{2.92} + \frac{7.78}{24.82}}{4} = 4.23$$

$$ERR\ Share\ Factor_{Max} = \frac{\frac{11.68}{32.56} + \frac{16.36}{31.35} + \frac{44.44}{26.69} + \frac{7.78}{33.37}}{4} = 0.69$$

This table shows the calculated ERRs of the non-financial indicator projects based on the unweighted calculation of the ERR share factor. “High-value” projects have been excluded.

Appendix BY

Calculated ERRs of Non-Financial Indicator Projects excluding “High-values” – Weighted Option 1

	Total Counts	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
		%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	137	11.68	22.98	32.56	-9.75	3.86	5.48
ERRs of non-NPV Projects	55	16.36	18.91	31.35	-5.07	8.97	6.43
ERRs of Negative & Neutral NPV/ERR Projects	18	44.44	2.92	26.69	23.91	49.8	93.11
ERRs of Positive NPV/ERR Projects	167	7.78	24.82	33.37	-13.65	-1.94	-4.55
Calculated ERRs of Non-Financial Indicator Projects	546	21.43	Ø =			15.17	25.12

$$ERR\ Share\ Factor_{Min} = \frac{[(11.68*137)+(16.36*55)+(44.44*18)+(7.78*167)]}{[(22.98*137)+(18.91*55)+(2.92*18)+(24.82*167)]} = 0.51$$

$$ERR\ Share\ Factor_{Max} = \frac{[(11.68*137)+(16.36*55)+(44.44*18)+(7.78*167)]}{[(32.56*137)+(31.35*55)+(26.69*18)+(33.37*167)]} = 0.36$$

This table shows the calculated ERRs of the non-financial indicator projects based on the weighted option 1 when “high-values” are excluded, whereat the ERR share factor is already weighted.

Appendix BZ

Calculated ERRs of Non-Financial Indicator Projects excluding “High-values” – Weighted Option 2

	Total Counts	Negative Share	ERR-Arithmetic Mean		Share Difference	Calculated ERR	
		%	Min (Ø)	Max (Ø)	-	Min	Max
ERRs of NPV Projects	137	11.68	22.98	32.56	-9.75	14.5	17.56
ERRs of non-NPV Projects	55	16.36	18.91	31.35	-5.07	14.5	23.55
ERRs of Negative & Neutral NPV/ERR Projects	18	44.44	2.92	26.69	23.91	23.71	63.12
ERRs of Positive NPV/ERR Projects	167	7.78	24.82	33.37	-13.65	12.95	13.45
Calculated ERRs of Non-Financial Indicator Projects	546	21.43	W-Ø =			14.25	18.79

$$ERR\ Share\ Factor_{Min} = \frac{(11.68+16.36+44.44+7.78)}{(22.98+18.91+2.92+24.82)} = 1.15$$

$$ERR\ Share\ Factor_{Max} = \frac{(11.68+16.36+44.44+7.78)}{(32.56+31.35+26.69+33.37)} = 0.65$$

This table shows the calculated ERRs of the non-financial indicator projects based on the weighted option 2 when “high-values” are excluded and using an unweighted ERR share factor. In this option the weighted-arithmetic mean (W-Ø) of 14.25 and 18.79 percent is calculated based on the weighted min respectively max calculated ERRs of the source groups.

Appendix CA

ICR and IEG Project Rating Comparison Projects

ICR vs. IEG – Sample Reports		
Continent	Country-ID	Report
Africa	AN	ICR2094
Africa	AN	24825
Africa	BE	ICR962
Africa	BE	24379
Africa	BF	ICR650
Africa	BF	23520
Africa	BU	ICR1106
Africa	ZR	ICR1321
Africa	TS	ICR790
Africa	AT	ICR1221
Africa	GU	24401
Africa	LE	ICR2000
Africa	MK	26161
Africa	MA	25255
Africa	MU	34787
Africa	MO	ICR700
Africa	NI	ICR1178
Africa	RU	ICR750
Africa	SE	ICR1763
Africa	SL	25102
Africa	TA	26232
Africa	UG	ICR907
Africa	UG	ICR688
Africa	UG	30284
Africa	UG	27529
Africa	UG	22254
Africa	SB	32520
Asia	AF	ICR1968
Asia	BA	ICR2036
Asia	BA	ICR358
Asia	BA	ICR90
Asia	BA	24323
Asia	KA	ICR2229
Asia	KA	22038
Asia	OT	ICR1191
Asia	LA	32004
Asia	LA	22325
Asia	NE	ICR2029
Asia	SA	32900
Asia	JE	36380

This table shows the projects for which the ICR rating category outcomes and the IEG project ratings have been compared.

Appendix CB

Result “high-value” Projects greater than 1,000 Mio USD

Continent	Country	Report	Project Name	NPV Type	NPV	Total Project Costs
Africa	Tanzania	ICR1511	Second Health Sector Development Project	partial	25	3,947
Continent	Country	Report	Project Name	NPV Type	NPV	Recurring Costs
Africa	Tanzania	ICR1299	Central Transport Corridor Project	overall	46	1,132
Asia	Yemen	34608	Sana 'a Emergency Power	overall	4	1,733

This table shows 3 “high-value” projects. One of them discloses total project costs of 3,947 Mio USD and 2 recurring costs of above 1,000 Mio USD.

Appendix CC

Unweighted Recurring Costs Coefficients of the LDCs

LDCs – Unweighted Recurring Costs Coefficients		
13.29	1.34	0.62
Unweighted – all coefficients	Unweighted – excluding coefficients above 10	Unweighted – excluding coefficients between 5 and 10
32,97819623		
0,233253589		
27,05328431		
0,224992839		
1,151785714		
0,35410832	0,233253589	
1,458855824	0,224992839	
0,1171875	1,151785714	0,233253589
0,242214533	0,35410832	0,224992839
0,392118611	1,458855824	1,151785714
0,868992684	0,1171875	0,35410832
9,413924051	0,242214533	1,458855824
0,91546976	0,392118611	0,1171875
12,405	0,868992684	0,242214533
0,211959589	9,413924051	0,392118611
0,059352153	0,91546976	0,868992684
0,175067274	0,211959589	0,91546976
1,411283763	0,059352153	0,211959589
13,34448161	0,175067274	0,059352153
6,568047337	1,411283763	0,175067274
1,387716303	6,568047337	1,411283763
0,553285594	1,387716303	1,387716303
0,628906594	0,553285594	0,553285594
0,455333333	0,628906594	0,628906594
-0,176844432	0,455333333	0,455333333
0,548495964	-0,176844432	-0,176844432
1,130569948	0,548495964	0,548495964
19,45846154	1,130569948	1,130569948
8,226352461	8,226352461	0,640977444
0,640977444	0,640977444	0,552377358
0,552377358	0,552377358	1,133516997
1,133516997	1,133516997	0,493353028
0,493353028	0,493353028	1,340595507
131,5166667	1,340595507	0,783095499
1,340595507	0,783095499	0,082804637
0,783095499	0,082804637	
10,63468232		
10,3232226		
31,78899083		
213,9535714		
0,082804637		

This table shows the 3 unweighted recurring costs coefficients of the LDCs. In order to arrive at a mean coefficient of 1.34 the 10 above 10 coefficients need to be excluded from the whole of 41 coefficients. In order to arrive at a coefficient of 0.62 additional 3 coefficients above 5 need to be excluded.

Appendix CD

Recurring Costs Coefficient 0.62 of the LDCs

Recurring Costs Coefficient			LDCs			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
0.62	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	2.14	2.38	2.19	2.74
		discounted	2.89	3.26	2.86	3.74
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.9	2.08	1.95	2.44
		discounted	2.39	2.68	2.48	3.26

This table shows the outcome of the recurring costs ratios of the LDCs based on a coefficient of 0.62.

Appendix CE

Recurring Costs Coefficients of the LDCs excluding “high-value” Projects

LDCs – Unweighted Recurring Costs Coefficients		
12.99	1.17	0.62
Unweighted – all coefficients	Unweighted – excluding coefficients above 10	Unweighted – excluding coefficients between 5 and 10
32,97819623		
0,233253589		
27,05328431		
0,224992839		
1,151785714		
0,35410832	0,233253589	
1,458855824	0,224992839	0,233253589
0,1171875	1,151785714	0,224992839
0,242214533	0,35410832	1,151785714
0,392118611	1,458855824	0,35410832
0,868992684	0,1171875	1,458855824
9,413924051	0,242214533	0,1171875
0,91546976	0,392118611	0,242214533
12,405	0,868992684	0,392118611
0,211959589	9,413924051	0,868992684
0,059352153	0,91546976	0,91546976
0,175067274	0,211959589	0,211959589
1,411283763	0,059352153	0,059352153
13,34448161	0,175067274	0,175067274
1,387716303	1,411283763	1,411283763
0,553285594	1,387716303	1,387716303
0,628906594	0,553285594	0,553285594
0,455333333	0,628906594	0,628906594
-0,176844432	0,455333333	0,455333333
0,548495964	-0,176844432	-0,176844432
1,130569948	0,548495964	0,548495964
19,45846154	1,130569948	1,130569948
8,226352461	8,226352461	0,640977444
0,640977444	0,640977444	0,552377358
0,552377358	0,552377358	1,133516997
1,133516997	1,133516997	0,493353028
0,493353028	0,493353028	1,340595507
131,5166667	1,340595507	0,783095499
1,340595507	0,783095499	0,082804637
0,783095499	0,082804637	
10,63468232		
10,3232226		
213,9535714		
0,082804637		

This table shows the 3 unweighted recurring costs coefficients of the LDCs when the 10 “high-value” projects are excluded from the whole of 39 coefficients. In order to arrive at a mean coefficient of 1.17 the 9 above 10 coefficients need to be excluded. In order to arrive at a coefficient of 0.62 additional 2 coefficients above 5 need to be excluded.

Appendix CF

Recurring Costs Coefficient 0.62 of the LDCs excluding “high-value” Projects

Recurring Costs Coefficient			LDCs			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
0.62	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.31	1.38	1.85	2.25
		discounted	1.56	1.7	2.31	2.93
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.24	1.3	1.65	1.99
		discounted	1.4	1.51	2.01	2.52

This table shows the outcome of the recurring costs ratios of the LDCs based on a coefficient of 0.62 and the exclusion of the 10 “high-value” projects.

Appendix CG

Recurring Costs Coefficient 0.56 of Africa

Recurring Costs Coefficient			Africa			
			Ø - weighted		Ø - unweighted	
			Min	Max	Min	Max
0.56	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.4	1.56	1.87	2.62
		discounted	1.73	2.02	2.34	3.56
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.26	1.36	1.67	2.33
		discounted	1.41	1.56	2.04	3.12

This table shows the outcome of the recurring costs ratios of Africa based on a coefficient of 0.56.

Appendix CH

Recurring Costs Coefficient 0.77 of Asia

			Asia			
			Ø - weighted		Ø - unweighted	
Recurring Costs Coefficient			Min	Max	Min	Max
0.77	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	2.72	3	2.7	2.88
		discounted	3.65	4.07	3.74	3.99
	Total-Capital plus Recurring-Costs Ratio	undiscounted	2.64	2.92	2.4	2.57
		discounted	3.5	3.93	3.23	3.47

This table shows the outcome of the recurring costs ratios of Asia based on a coefficient of 0.77.

Appendix CI

Recurring Costs Coefficient 0.56 of Africa excluding “high-value” Projects

			Africa			
			Ø - weighted		Ø - unweighted	
Recurring Costs Coefficient			Min	Max	Min	Max
0.56	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.29	1.36	1.75	2.27
		discounted	1.51	1.65	2.18	2.98
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.21	1.27	1.58	2.01
		discounted	1.35	1.44	1.92	2.57

This table shows the outcome of the recurring costs ratios of Africa based on a coefficient of 0.56 and the exclusion of the 5 African “high-value” projects.

Appendix CJ

Recurring Costs Coefficient 0.77 of Asia excluding “high-value” Projects

			Asia			
			Ø - weighted		Ø - unweighted	
Recurring Costs Coefficient			Min	Max	Min	Max
0.77	Pro-Rata-Capital plus Recurring-Costs Ratio	undiscounted	1.36	1.44	2.01	2.18
		discounted	1.68	1.82	2.55	2.81
	Total-Capital plus Recurring-Costs Ratio	undiscounted	1.31	1.39	1.76	1.93
		discounted	1.55	1.69	2.17	2.42

This table shows the outcome of the recurring costs ratios of Asia based on a coefficient of 0.77 and the exclusion of the 5 Asian “high-value” projects.

Appendix CK

Sector Codes of the 5 „high-value” Recurring Costs Coefficient Projects

Sector Code	Project Counts	%-Points	%-Share	World Bank Costs [Mio USD]
Distribution & Transmission	2	2	40	50
Power	1	0.64	12.8	20
Telecommunications	1	0.45	9	3
Water supply	1	0.36	7.2	12
Law and justice	1	0.35	7	1
Other social services	1	0.23	0.46	11
General water, sanitation and flood protection sector	1	0.23	0.46	11
Roads and highways	1	0.23	0.46	11
General education sector	1	0.23	0.46	11
Postal services	1	0.2	4	1
Central government administration	1	0.08	0.16	1
Total		5	100	132

This table shows the sector codes of the WB funds of the “high-value” recurring costs coefficient projects.

Appendix CL

Sector Codes of the 10 „high-value” negative Minimum NPV Projects

Sector Code	Project Counts	%-Points	%-Share	World Bank Costs [Mio USD]
Roads and highways	7	4.63	46.3	602
Power	2	0.93	9.3	183
Other social services	3	0.74	7.4	136
General transportation sector	4	0.68	6.8	53
General public administration sector	2	0.61	6.1	50
General water, sanitation and flood protection sector	2	0.44	4.4	35
Central government administration	6	0.43	4.3	53
Railways	2	0.34	3.4	24
Water supply	1	0.25	2.5	42
Renewable Energy	1	0.2	2	33
Ports, waterways and shipping	1	0.18	1.8	15
Irrigation and drainage	1	0.17	1.7	28
Health	2	0.17	1.7	15
Primary education	1	0.11	1.1	18
Oil and gas	1	0.07	0.7	14
Sub-national government administration	1	0.03	0.3	6
Vocational training	1	0.02	0.2	1
Total		10	100	1,308

This table shows the sector codes of the WB funds of the “high-value” negative minimum NPV projects after adjustment of the NPVs.

Appendix CM

Sector Codes of LDC Projects with Minimum SROIs greater than or equal to

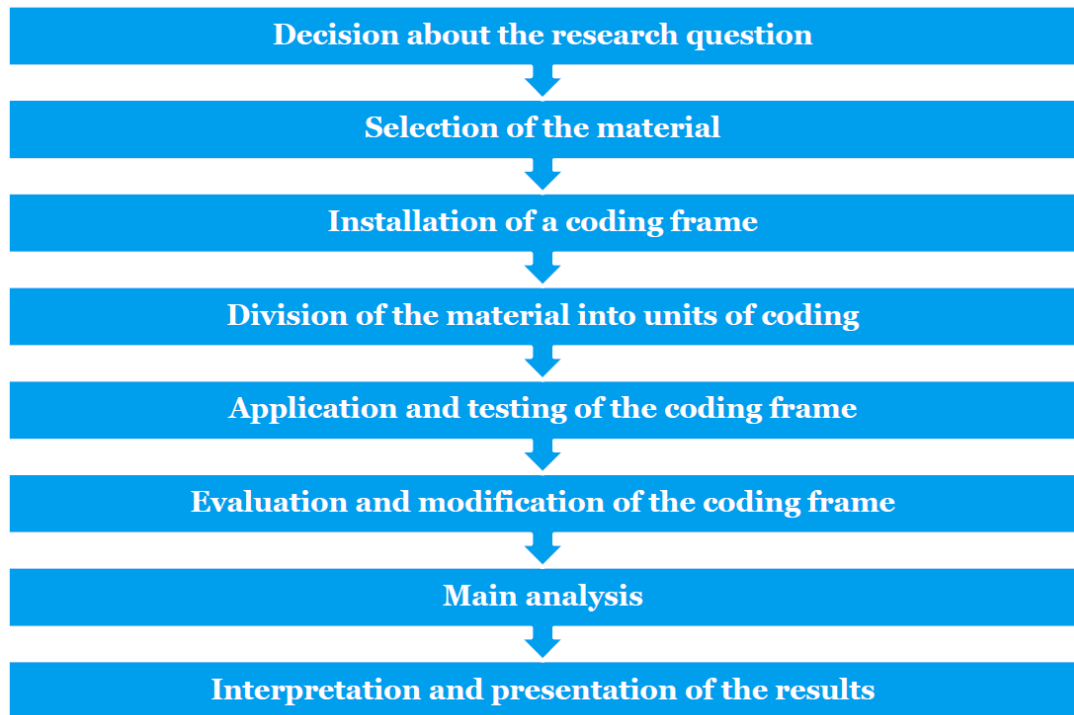
1.3

Sector Code	Project Counts	%-Points	%-Share	World Bank Costs [Mio USD]
Roads and highways	8	4.7	15.67	140
Power	5	4.22	14.07	189
Central government administration	19	3.05	10.17	72
Water supply	4	2.43	8.10	80
Mining and other extractive	4	2	6.67	21
Agricultural extension and research	4	1.36	4.53	37
Other social services	6	1.22	4.07	36
Forestry	2	1.37	4.57	7
Agricultural marketing and trade	2	1.03	3.43	11
Other industry	3	0.88	2.93	24
Flood protection	1	0.79	2.63	18
Other domestic and international trade	1	0.74	2.47	5
Animal production	1	0.52	1.73	8
Oil and gas	2	0.51	1.70	71
Natural Resources Management	1	0.5	1.67	3
General agriculture, fishing and forestry sector	2	0.49	1.63	26
Sub-national government administration	6	0.48	1.60	10
Sewerage	2	0.45	1.50	14
Secondary education	1	0.45	1.50	12
Health	2	0.43	1.43	8
General education sector	1	0.36	1.20	7
Sanitation	2	0.33	1.10	8
Crops	1	0.26	0.87	4
Railways	1	0.25	0.83	7
Irrigation and drainage	1	0.22	0.73	12
General water, sanitation and flood protection sector	2	0.21	0.70	4
Renewable Energy	2	0.19	0.63	2
Agro-industry	1	0.15	0.50	5
Primary education	1	0.14	0.47	4
General energy sector	1	0.1	0.33	1
General public administration sector	1	0.09	0.30	2
Tertiary education	2	0.08	0.27	2
Total		30	100	846

This table shows the sector codes of the WB funds of the LDC Projects with Minimum SROIs greater than or equal to 1.3.

Appendix CN²⁹⁶

Process of Execution of the Content Analysis



Appendix CO²⁹⁷

Quality at Entry Rating Categories

	Rating Category	Definition
Positive Ratings	Highly Satisfactory 	There were no shortcomings in identification, preparation, or appraisal.
	Satisfactory 	There were minor shortcomings in identification, preparation, or appraisal.
	Moderately Satisfactory 	There were moderate shortcomings in identification, preparation, or appraisal.
Negative Ratings	Moderately Unsatisfactory 	There were significant shortcomings in identification, preparation, or appraisal.
	Unsatisfactory 	There were major shortcomings in identification, preparation, or appraisal.
	Highly Unsatisfactory 	There were severe shortcomings in identification, preparation, or appraisal.

This table shows the rating categories of the quality at entry dimension of the bank performance.

²⁹⁶ Source: Hausmann/Rudolph (2014), s. 8.

²⁹⁷ Source: WB (2013), p. 43.

Appendix CP²⁹⁸

Quality of Supervision Rating Categories

	Rating Category		Definition
Positive Ratings	Highly Satisfactory	●●●	There were no shortcomings in the proactive identification of opportunities and resolution of threats.
	Satisfactory	●●	There were minor shortcomings in the proactive identification of opportunities and resolution of threats.
	Moderately Satisfactory	●	There were moderate shortcomings in the proactive identification of opportunities and resolution of threats.
Negative Ratings	Moderately Unsatisfactory	●	There were significant shortcomings in the proactive identification of opportunities and resolution of threats.
	Unsatisfactory	●●	There were major shortcomings in the proactive identification of opportunities and resolution of threats.
	Highly Unsatisfactory	●●●	There were severe shortcomings in the proactive identification of opportunities and resolution of threats.

This table shows the rating categories of the quality of supervision dimension of the bank performance.

Appendix CQ²⁹⁹

Government Performance Rating Categories

	Rating Category		Definition
Positive Ratings	Highly Satisfactory	●●●	There were no shortcomings in government performance.
	Satisfactory	●●	There were minor shortcomings in government performance.
	Moderately Satisfactory	●	There were moderate shortcomings in government performance.
Negative Ratings	Moderately Unsatisfactory	●	There were significant shortcomings in government performance.
	Unsatisfactory	●●	There were major shortcomings in government performance.
	Highly Unsatisfactory	●●●	There were severe shortcomings in government performance.

This table shows the rating categories of the government performance dimension of the borrower performance.

²⁹⁸ Source: WB (2013), p. 44.

²⁹⁹ Source: WB (2013), p. 47.

Appendix CR³⁰⁰

Implementation Agency Performance Rating Categories

	Rating Category		Definition
Positive Ratings	Highly Satisfactory	●●●	There were no shortcomings in implementing agency or agencies' performance.
	Satisfactory	●●	There were minor shortcomings in implementing agency or agencies' performance.
	Moderately Satisfactory	●	There were moderate shortcomings in implementing agency or agencies' performance.
Negative Ratings	Moderately Unsatisfactory	●	There were significant shortcomings in implementing agency or agencies' performance.
	Unsatisfactory	●●	There were major shortcomings in implementing agency or agencies' performance.
	Highly Unsatisfactory	●●●	There were severe shortcomings in implementing agency or agencies' performance.

This table shows the rating categories of the implementation agency performance dimension of the borrower performance.

Appendix CS

Appropriate Codes for the most common Types of Reasons

Fiduciary Risks	▪ "fiduciary accountability of the country/project remains fragile" ▪ "missing funds for project future support" ▪ "reliance on external donor funding"
Implementation Capacity / Institutional Risks	▪ "high turn-over of staff will remain a problem for the national HIV/AIDS response" ▪ "missing availability of trained resources" ▪ "missing consistent implementation of current tariff policy"
Project Risks	▪ "demand for project outcome is far beyond its capacity" ▪ "missing a follow-up project" ▪ "lack of improvement by the project"
Governmental Risks	▪ "missing overall rural infrastructure strategy of the government" ▪ "missing Government commitment to liberalize the transport sector" ▪ "government missed to ensure resource replacement of project staff by project closing"
Security Risks	▪ "natural disasters including earthquakes, hurricanes and floods" ▪ "existing armed conflicts are still going on" ▪ "remaining escalation of the insurgency is one of the major risks"
Macroeconomic / Country Risks	▪ "existing global economic challenges" ▪ "high current debt level of the country" ▪ "the global Financial crisis has negatively affected the country as a whole"

³⁰⁰ Source: WB (2013), p. 49.

Appendix CT

Reasons for “Unlikely” and “Highly Unlikely” Sustainability across all LDCs

Category	% of Projects	
	<i>Unlikely</i>	<i>Highly Unlikely</i>
Macroeconomic / Country Risks	28.3	25.5
Governmental Risks	46.2	66.7
Government Commitment Risks	27.4	35.3
Government Action / Task Risks	21.2	29.4
Government Capacity Risks	9	17.6
Political Risks	21.2	21.6
Security Risks	25.5	47.1
Natural Disasters / Environmental Risks	5.2	15.7
War / Conflict Risks	21.2	33.3
Epidemic Risks	0.5	2
Thirst & Hunger	0.9	-
Corruption	3.8	5.9
Fiduciary Risks	63.2	52.9
Funding / Donor Risks	62.7	47.1
Payment Risks	3.8	9.8
Implementation Capacity / Institutional Risks	54.7	70.6
Infrastructure Risks	7.1	15.7
Project Risks	53.8	64.7
PDO Risks	27.8	37.3
Implementation Agency Risks	20.3	21.6
Bank Risks	2.4	-
Technical Risks	12.3	13.7
Other Project Risks	20.8	19.6

This table shows the reasons for “Unlikely” and “Highly Unlikely” sustainability ratings across all LDCs.

Appendix CU

Project Type Category Mapping of the Negative Sustainability Projects of the LDCs

Category Mapping	LDCs
	%-Share
Water & Sanitation	5.85
• Water Supply • General water, sanitation and flood protection sector • Sanitation • Sewerage	3.93 1.01 0.72 0.19
Power	7.24
• Power • General Energy Sector • Renewable Energy • Hydro • Other Power & Energy Conversion • Energy efficiency in Power Sector • Oil and gas • Oil & Gas Exploration & Development • District heating and energy efficiency services	4.99 0.43 0.28 0.19 0.19 0.01 0.75 0.38 0.02
Transportation Development	10.98
• Roads and highways • Railways • Urban Environment • Other Urban Development • Urban Management • Ports, waterways and shipping • General Transportation Sector • Public Administration- Transportation • Other Transportation • Transportation Adjustment • Oil & Gas Transportation • Aviation	4.54 1.28 0.38 0.38 0.57 0.4 0.93 0.08 0.38 1.21 0.75 0.08
Education	8.1
• (Pre-)Primary education • General Education Sector • Vocational training • Tertiary education • Secondary education • Adult Literacy/non-formal education	3.51 1.54 1.27 0.97 0.47 0.34
Financial Development	4.17
• Banking • General finance sector • Micro- and SME finance • Housing finance and real estate markets • Social Funds & Social Assistance • Public Financial Management • Reform and financing • Financial Sector Development	1.12 0.67 0.32 0.09 0.19 0.19 0.38 1.21
IT & Communication	1.16
• General information and communications sector • Telecommunications • Media	0.43 0.69 0.04

Health	12.41
• Health	10.83
• Population, Health & Nutrition Adjustment	0.07
• Non-compulsory health finance	0.01
• Compulsory health finance	0.01
• Non-compulsory health finance	1.18
• Compulsory health finance	0.31
Governmental Administration	22.87
• Central Government Administration	20.35
• Sub-national Government Administration	2.52
Agriculture	8.52
• Agricultural Extension and Research	3
• Agricultural Marketing and Trade	1.64
• General agriculture, fishing and forestry sector	1.51
• Other Agriculture	0.07
• Crops	0.98
• Irrigation and drainage	2.52
• Forestry	1.32
• Petrochemicals and fertilizers	0.3
• Agro-Industry	0.18
Sectorial Development & Reform	8.06
• General Public Administration Sector	3.43
• General industry and trade sector	1.99
• Law and justice	1.66
• Institutional Development	0.38
• Agency reform	0.35
• Other domestic and international trade	0.25
Other	9.38
• Other social services	6.22
• Other industry	0.38
• Flood protection	0.15
• Animal production	0.44
• Solid waste management	0.03
• Postal services	0.19
• Fisheries & Aquaculture	0.19
• Economic management	0.19
• Compulsory pension and unemployment insurance	0.27
• Research	0.12
• Non-compulsory pensions and insurance	0.4
• Other economic	0.38
• Macro/Non-Trade	0.38
• Housing construction	0.04
Natural Resources	1.26
• Mining and other extractive	0.88
• Natural Resources Management	0.38

This illustration shows the outcome of the sector codes mapped to project type categories. For each of the sector codes respectively project types the %-share based on the corresponding %-points is given.

Appendix CV

Negative Sustainability Projects with matching Sector Codes

Number of Projects sharing the same Sector Code	Project Report Numbers	Shared Sector Codes
11	25705; 29215; ICR1458; 36329; ICR1489; 32710; ICR1536; ICR352; ICR1688; ICR2074; ICR1374	Central government administration
8	26250; 35610; 28022; 30277; 27387; ICR769; 32159; ICR1298	- Central government administration - Health
8	34153; ICR72; ICR1045; ICR1187; ICR1325; ICR604; ICR1324; ICR1984	- Central government administration - Sub-national government administration
7	26251; ICR158; ICR1131; ICR1497; ICR2023; ICR2146; ICR1275	- Other social services - Health - Central government administration - Sub-national government administration
4	25102; ICR152; ICR785; ICR1143	Power
3	27570; ICR2037; ICR1014	- Central government administration - General industry and trade sector
2	36464; ICR8	- Sub-national government administration - Central government administration - Other social services
2	ICR992; ICR1716	- Central government administration - Crops - General industry and trade sector
2	25283; ICR1415	- Agricultural extension and research - Central government administration - Agricultural marketing and trade
2	ICR2056; ICR1129	- Health - Other social services - Sub-national government administration
2	ICR1432; 26249	- Health - Other social services - Primary education - Roads and highways
2	26510; 25247	- Water supply - Sewerage - Sanitation
2	26745; ICR2152	- Power - Oil and gas - Central government administration
2	ICR1779; ICR1365	- Sanitation - Water supply
2	ICR134; 27527	- General public administration sector - Other social services
2	26930; 24252	- Roads and highways - Central government administration
2	22231; 25624	Agricultural extension and research
2	20756; ICR1968	Irrigation and drainage
2	22167; 22804	Primary education
2	ICR2154; ICR1041	Railways
2	22416; 21529	Oil & Gas Transportation
2	37359; ICR666	General public administration sector
2	ICR2421; ICR1200	Law and justice

This table shows the result of any negative sustainability projects within the LDCs with common sector codes. It needs to be noted that only where there is only 1

sector code available the %-points for this sector code are the same for the projects sharing this sector code (namely 1 %-point).

Appendix CW

Reasons for “Unlikely” and “Highly Unlikely” Sustainability in Haiti

Category	% of Projects	
	<i>Unlikely</i>	<i>Highly Unlikely</i>
Macroeconomic / Country Risks	33.3	75
Governmental Risks	66.7	25
Government Commitment Risks	66.7	25
Government Action / Task Risks	33.3	25
Government Capacity Risks	-	-
Political Risks	66.7	75
Security Risks	-	75
Natural Disasters / Environmental Risks	-	75
War / Conflict Risks	-	-
Epidemic Risks	-	-
Thirst & Hunger	-	-
Corruption	-	-
Fiduciary Risks	66.7	25
Funding / Donor Risks	66.7	25
Payment Risks	-	-
Implementation Capacity / Institutional Risks	33.3	100
Infrastructure Risks	-	-
Project Risks	-	25
PDO Risks	-	-
Implementation Agency Risks	-	-
Bank Risks	-	-
Technical Risks	-	-
Other Project Risks	-	25

This table shows the reasons for “Unlikely” and “Highly Unlikely” sustainability ratings based on 3 respectively 4 projects.

Appendix CX

Reasons for “Unlikely” and “Highly Unlikely” Sustainability in Africa

Category	% of Projects	
	<i>Unlikely</i>	<i>Highly Unlikely</i>
Macroeconomic / Country Risks	30.3	27.6
Governmental Risks	44.1	65.5
Government Commitment Risks	24.8	37.9
Government Action / Task Risks	20.7	20.7
Government Capacity Risks	7.6	6.9
Political Risks	19.3	17.2
Security Risks	21.4	34.5
Natural Disasters / Environmental Risks	5.5	10.3
War / Conflict Risks	16.5	27.6
Epidemic Risks	-	3.4
Thirst & Hunger	1.4	-
Corruption	1.4	-
Fiduciary Risks	68.3	55.2
Funding / Donor Risks	68.3	55.2
Payment Risks	3.4	6.9
Implementation Capacity / Institutional Risks	53.1	65.5
Infrastructure Risks	4.8	17.2
Project Risks	52.4	65.5
PDO Risks	27.6	37.9
Implementation Agency Risks	17.2	27.6
Bank Risks	2.1	-
Technical Risks	11	13.8
Other Project Risks	20.7	20.7

This table shows the reasons for “Unlikely” and “Highly Unlikely” sustainability ratings in Africa.

Appendix CY

Reasons for “Unlikely” and “Highly Unlikely” Sustainability in Asia

Category	% of Projects	
	<i>Unlikely</i>	<i>Highly Unlikely</i>
Macroeconomic / Country Risks	23.4	11.1
Governmental Risks	50	77.8
Government Commitment Risks	31.3	33.3
Government Action / Task Risks	21.9	44.4
Government Capacity Risks	12.5	38.9
Political Risks	23.4	16.7
Security Risks	35.9	61.1
Natural Disasters / Environmental Risks	4.7	11.1
War / Conflict Risks	32.8	50
Epidemic Risks	1.6	-
Thirst & Hunger	-	-
Corruption	9.4	16.7
Fiduciary Risks	51.6	55.6
Funding / Donor Risks	50	38.9
Payment Risks	4.7	16.7
Implementation Capacity / Institutional Risks	59.4	72.2
Infrastructure Risks	12.5	16.7
Project Risks	59.4	72.2
PDO Risks	29.7	44.4
Implementation Agency Risks	28.1	16.7
Bank Risks	3.1	-
Technical Risks	15.6	16.7
Other Project Risks	21.9	16.7

This table shows the reasons for “Unlikely” and “Highly Unlikely” sustainability ratings in Asia.

Appendix CZ

Project Type Category Mapping of the Negative Sustainability Projects of Africa and Asia

Category Mapping	Africa	Asia
	%-Share	%-Share
Water & Sanitation	3.72	10.38
- Water Supply - General water, sanitation and flood protection sector - Sanitation - Sewerage	2.27 0.87 0.52 0.06	7.43 1.32 1.16 0.47
Power	8.19	5.33
- Power - General Energy Sector - Renewable Energy - Hydro - Other Power & Energy Conversion - Energy efficiency in Power Sector - Oil and gas - Oil & Gas Exploration & Development - District heating and energy efficiency services	5.32 0.57 0.27 0.26 0.26 0.01 0.94 0.54 0.02	4.72 - 0.25 - - - 0.36 - -
Transportation Development	8.41	11.23
- Roads and highways - Railways - Urban Environment - Other Urban Development - Urban Management - Ports, waterways and shipping - General Transportation Sector - Public Administration- Transportation - Other Transportation - Transportation Adjustment - Oil & Gas Transportation - Aviation	3.7 1.34 0.54 0.54 0.54 0.01 0.66 - - 0.54 0.54 -	5.59 1.21 - - - 1.22 0.31 0.24 1.21 - 1.21 0.24
Education	9.09	6.47
- Primary education - General Education Sector - Vocational training - Tertiary education - Secondary education - Adult Literacy/non-formal education	4.13 1.73 1.84 1 0.36 -	2.33 1.22 0.12 0.92 0.8 1.08
Financial Development	1.75	5.03
- Banking - General finance sector - Micro- and SME finance - Housing finance and real estate markets - Social Funds & Social Assistance - Public Financial Management - Reform and financing - Financial Sector Development	0.47 0.33 0.35 0.08 0.26 0.26 - -	2.53 0.18 0.05 0.05 - - 1.21 1.21
IT & Communication	0.82	1.86
- General information and communications sector - Telecommunications - Media	0.41 0.41 -	0.45 1.29 0.12

Health	14.46	5.25
• Health	13.89	4.14
• Population, Health & Nutrition Adjustment	0.16	-
• Non-compulsory health finance	0.02	-
• Compulsory health finance	0.01	-
• Non-compulsory health finance	0.02	1.21
• Compulsory health finance	0.36	-
Governmental Administration	24.56	22.71
• Central Government Administration	18.97	21.68
• Sub-national Government Administration	3.38	1.03
Agriculture	12.01	11.83
• Agricultural Extension and Research	3.29	2.29
• Agricultural Marketing and Trade	2.17	0.64
• General agriculture, fishing and forestry sector	1.51	1.73
• Other Agriculture	0.3	-
• Crops	1.5	0.06
• Irrigation and drainage	2.36	5.31
• Forestry	1	1.8
• Petrochemicals and fertilizers	0.5	-
• Agro-Industry	0.38	-
Sectorial Development & Reform	5.99	12.71
• General Public Administration Sector	2.18	6.31
• General industry and trade sector	1.82	2.45
• Law and justice	0.61	3.96
• Institutional Development	0.54	-
• Agency reform	0.49	-
• Other domestic and international trade	0.35	-
Other	9.61	6.44
• Other social services	7.8	2.77
• Other industry	-	0.15
• Flood protection	0.11	0.18
• Animal production	0.52	0.29
• Solid waste management	0.01	0.12
• Postal services	0.26	-
• Fisheries & Aquaculture	0.26	-
• Economic management	0.26	-
• Compulsory pension and unemployment insurance	0.2	-
• Research	0.16	0.51
• Non-compulsory pensions and insurance	0.03	-
• Other economic	-	1.21
• Macro/Non-Trade	1.21	1.21
Natural Resources	1.39	0.76
• Mining and other extractive	1.13	0.16
• Natural Resources Management	0.26	0.6

This illustration shows the outcome of the sector codes mapped to project type categories. For each of the sector codes respectively project types the %-share based on the corresponding %-points is given.

Appendix DA

Risk Percentage Outcome of African 'Health' related Projects

Category	Africa	
	% of Projects	% of total Projects
Macroeconomic / Country Risks	5 (29.9)	5 (29.7)
Governmental Risks	4 (42.5)	4 (42.6)
Political Risks	17.8	18.5
Security Risks	23	25.9
Corruption	1.1	1.9
Fiduciary Risks	1 (60.3)	1 (74.1)
Implementation Capacity / Institutional Risks	3 (49.4)	2 (57.4)
Infrastructure Risks	6.9	3.7
Project Risks	2 (50)	3 (50)

This table shows a comparison of risk types and their corresponding percentage of distribution over all African negative sustainability projects (“% of Projects” with relationship to 174 total projects) and specific African health-related projects (“% of Health Projects” with relationship to 54 projects). Fiduciary risks do affect roughly about 75 percent of all health related projects in Africa – which is 15 percent more than compared to all negative sustainability projects in Africa – and thus mainly contributes to the higher percentage in the African country. Following this, there is a slight increase of variance amongst the implementation capacity/ institutional risks. Other than that there are no noteworthy differences.

Appendix DB

Reasons for Negative Bank Performance of the LDCs – Landing

Category	% of Projects	% of total Projects
Bank Competence	54.5	9.2
Decision Making Process	1.5	0.3
Management, Coordination, Administration & Evaluation	54.5	9.2
Communication	0	0
Implementation Framework	75.4	12.8
Data Quality	11.9	2
Government, Institutions & Policy	27.6	4.7
Schedule/Timing	15.7	2.7
Project Design	44	7.5
Environment & Macroeconomic	8.2	1.4
General & Other – e.g. Systems, Processes, Technology	31.3	5.3
Restructuring (<i>only applicable for Supervision</i>)	-	-
Monitoring & Controlling	66.4	11.3
General	20.1	3.4
Reporting	6.7	1.1
Risks & Issues	29.9	5.1
KPIs	22.4	3.8
PDOs	34.3	5.8
Resources & Capacity	32.1	5.3
Budget & Financing	11.9	2

This table shows the reasons for a negative bank performance across the LDCs for the landing dimension.

Appendix DC

Reasons for Negative Bank Performance of the LDCs – Supervision

Category	% of Projects	% of total Projects
Bank Competence	63.2	10.7
Decision Making Process	1.5	0.3
Management, Coordination, Administration & Evaluation	63.4	10.7
Communication	8.2	1.4
Implementation Framework	73.1	12.4
Data Quality	3.7	0.6
Government, Institutions & Policy	25.4	4.3
Schedule/Timing	29.5	5.1
Project Design	1.8	0.3
Environment & Macroeconomic	7.5	1.3
General & Other – e.g. Systems, Processes, Technology	23.1	3.9
Restructuring (<i>only applicable for Supervision</i>)	23.1	3.9
Monitoring & Controlling	65.7	11.1
General	30.6	5.2
Reporting	20.9	3.5
Risks & Issues	7.5	1.3
KPIs	30.6	5.2
PDOs	14.2	2.4
Resources & Capacity	44	7.5
Budget & Financing	25.4	4.3

This table shows the reasons for a negative bank performance across the LDCs for the quality of supervision dimension.

Appendix DD

Reasons for “Moderately Unsatisfactory” Bank Performance across all LDCs

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	82.5	46	42.9
Decision Making Process	3.2	-	1.6
Management, Coordination, Administration & Evaluation	82.5	46	42.9
Communication	1.6	-	1.6
Implementation Framework	95.2	73	68.3
Data Quality	25.4	11.1	1.6
Government, Institutions & Policy	38.1	23.8	17.5
Schedule/Timing	44.4	17.5	34.9
Project Design	50.8	44.4	1.6
Environment & Macroeconomic	15.9	7.9	4.8
General & Other – e.g. Systems, Processes, Technology	54	28.6	23.8
Restructuring (<i>only applicable for Supervision</i>)	19	-	15.9
Monitoring & Controlling	84.1	69.8	55.6
General	42.9	23.8	19
Reporting	19	4.8	14.3
Risks & Issues	36.5	31.7	3.2
KPIs	50.8	28.6	31.7
PDOs	46	34.9	12.7
Resources & Capacity	50.8	28.6	31.7
Budget & Financing	30.2	6.3	25.4

This table shows the reasons for “Moderately Unsatisfactory” bank performance across the LDCs and separated by the 2 dimensions.

Appendix DE

Reasons for “Unsatisfactory” Bank Performance across all LDCs

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	85.9	62	67.6
Decision Making Process	4.2	2.8	2.8
Management, Coordination, Administration & Evaluation	85.9	62	66.2
Communication	14.1	-	14.1
Implementation Framework	95.8	77.5	77.5
Data Quality	19.7	12.7	5.6
Government, Institutions & Policy	54.9	31	32.4
Schedule/Timing	31	14.1	21.1
Project Design	50.7	43.7	1.4
Environment & Macroeconomic	18.3	8.5	9.9
General & Other – e.g. Systems, Processes, Technology	52.1	33.8	18.3
Restructuring (<i>only applicable for Supervision</i>)	31	-	29.6
Monitoring & Controlling	90.1	63.4	73.2
General	53.5	16.9	40.8
Reporting	33.8	8.5	26.8
Risks & Issues	35.2	28.2	11.3
KPIs	39.4	16.9	29.6
PDOs	40.8	33.8	14.1
Resources & Capacity	71.8	35.2	54.9
Budget & Financing	40.8	16.9	25.4

This table shows the reasons for “Unsatisfactory” bank performance across the LDCs and separated by the 2 dimensions.

Appendix DF

Reasons for Negative Bank Performance in Haiti – Landing

Category	% of Projects	% of total Projects
Bank Competence	33.3	10
Decision Making Process	-	-
Management, Coordination, Administration & Evaluation	33.3	10
Communication	-	-
Implementation Framework	66.7	20
Data Quality	33.3	10
Government, Institutions & Policy	33.3	10
Schedule/Timing	-	-
Project Design	66.7	20
Environment & Macroeconomic	-	-
General & Other – e.g. Systems, Processes, Technology	33.3	10
Restructuring (<i>only applicable for Supervision</i>)	-	-
Monitoring & Controlling	66.7	20
General	-	-
Reporting	-	-
Risks & Issues	33.3	10
KPIs	-	-
PDOs	33.3	10
Resources & Capacity	33.3	10
Budget & Financing	-	-

This table shows the reasons for a negative bank performance in Haiti for the landing dimension.

Appendix DG

Reasons for Negative Bank Performance in Haiti – Supervision

Category	% of Projects	% of total Projects
Bank Competence	66.7	20
Decision Making Process	-	-
Management, Coordination, Administration & Evaluation	66.7	20
Communication	-	-
Implementation Framework	100	30
Data Quality	-	-
Government, Institutions & Policy	66.7	20
Schedule/Timing	33.3	10
Project Design	-	-
Environment & Macroeconomic	33.3	10
General & Other – e.g. Systems, Processes, Technology	-	-
Restructuring (<i>only applicable for Supervision</i>)	-	-
Monitoring & Controlling	100	30
General	66.7	20
Reporting	33.3	10
Risks & Issues	-	-
KPIs	33.3	10
PDOs	33.3	10
Resources & Capacity	66.7	20
Budget & Financing	-	-

This table shows the reasons for a negative bank performance in Haiti for the quality of supervision dimension.

Appendix DH

Reasons for “Moderately Unsatisfactory” Bank Performance in Haiti

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	100	100	-
Decision Making Process	-	-	-
Management, Coordination, Administration & Evaluation	100	100	-
Communication	-	-	-
Implementation Framework	100	100	100
Data Quality	100	100	-
Government, Institutions & Policy	-	-	-
Schedule/Timing	-	-	-
Project Design	100	100	-
Environment & Macroeconomic	100	-	100
General & Other – e.g. Systems, Processes, Technology	100	100	-
Restructuring (<i>only applicable for Supervision</i>)	-	-	-
Monitoring & Controlling	100	100	100
General	-	-	-
Reporting	-	-	-
Risks & Issues	-	-	-
KPIs	100	-	100
PDOs	100	100	100
Resources & Capacity	-	-	-
Budget & Financing	-	-	-

This table shows the reasons for “Moderately Unsatisfactory” bank performance in Haiti and separated by the 2 dimensions.

Appendix DI

Reasons for “Unsatisfactory” Bank Performance in Haiti

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	100	-	100
Decision Making Process	-	-	-
Management, Coordination, Administration & Evaluation	100	-	100
Communication	-	-	-
Implementation Framework	100	50	100
Data Quality	-	-	-
Government, Institutions & Policy	100	-	100
Schedule/Timing	100	-	50
Project Design	50	50	-
Environment & Macroeconomic	-	-	-
General & Other – e.g. Systems, Processes, Technology	-	-	-
Restructuring (<i>only applicable for Supervision</i>)	-	-	-
Monitoring & Controlling	100	50	100
General	100	-	100
Reporting	50	-	50
Risks & Issues	50	50	-
KPIs	-	-	-
PDOs	-	-	-
Resources & Capacity	100	50	100
Budget & Financing	-	-	-

This table shows the reasons for “Unsatisfactory” bank performance in Haiti and separated by the 2 dimensions.

Appendix DJ

Reasons for Negative Bank Performance in Africa – Landing

Category	% of Projects	% of total Projects
Bank Competence	47.9	7.8
Decision Making Process	2.1	0.3
Management, Coordination, Administration & Evaluation	47.9	7.8
Communication	-	-
Implementation Framework	70.2	11.5
Data Quality	11.7	1.9
Government, Institutions & Policy	28.7	4.7
Schedule/Timing	19.1	3.1
Project Design	36.2	5.9
Environment & Macroeconomic	4.3	0.7
General & Other – e.g. Systems, Processes, Technology	29.8	4.9
Restructuring (<i>only applicable for Supervision</i>)	-	-
Monitoring & Controlling	66.7	10.8
General	21.3	3.5
Reporting	7.4	1.2
Risks & Issues	29.8	4.9
KPIs	24.5	4
PDOs	33.9	5.4
Resources & Capacity	27.7	4.5
Budget & Financing	13.8	2.3

This table shows the reasons for a negative bank performance in Africa for the landing dimension.

Appendix DK

Reasons for Negative Bank Performance in Africa – Supervision

Category	% of Projects	% of total Projects
Bank Competence	63.8	10.5
Decision Making Process	2.1	0.3
Management, Coordination, Administration & Evaluation	62.8	10.3
Communication	7.4	1.2
Implementation Framework	73.4	12
Data Quality	1.1	0.2
Government, Institutions & Policy	25.5	4.2
Schedule/Timing	26.6	4.4
Project Design	2.1	0.4
Environment & Macroeconomic	8.5	1.4
General & Other – e.g. Systems, Processes, Technology	20.2	3.3
Restructuring (<i>only applicable for Supervision</i>)	24.5	4
Monitoring & Controlling	65.6	10.8
General	30.9	5.1
Reporting	22.3	3.7
Risks & Issues	5.3	0.9
KPIs	33.9	5.4
PDOs	11.7	1.9
Resources & Capacity	40.4	6.6
Budget & Financing	27.7	4.5

This table shows the reasons for a negative bank performance in Africa for the quality of supervision dimension.

Appendix DL

Reasons for “Moderately Unsatisfactory” Bank Performance in Africa

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	81.8	36.4	61.4
Decision Making Process	4.5	-	2.3
Management, Coordination, Administration & Evaluation	81.8	36.4	61.4
Communication	2.3	-	2.3
Implementation Framework	93.2	63.6	68.2
Data Quality	27.3	9.1	-
Government, Institutions & Policy	40.9	25	20.5
Schedule/Timing	50	22.7	36.4
Project Design	40.9	31.8	2.3
Environment & Macroeconomic	11.4	45.5	2.3
General & Other – e.g. Systems, Processes, Technology	54.5	22.7	22.7
Restructuring (<i>only applicable for Supervision</i>)	20.5	-	15.9
Monitoring & Controlling	84.1	68.2	56.8
General	50	27.3	20.5
Reporting	20.5	2.3	18.2
Risks & Issues	29.5	27.3	-
KPIs	52.3	29.5	34.1
PDOs	45.5	31.8	11.4
Resources & Capacity	63.6	22.7	29.5
Budget & Financing	36.4	6.8	29.5

This table shows the reasons for “Moderately Unsatisfactory” bank performance in Africa and separated by the 2 dimensions.

Appendix DM

Reasons for “Unsatisfactory” Bank Performance in Africa

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	86	58	66
Decision Making Process	6	4	2
Management, Coordination, Administration & Evaluation	86	58	64
Communication	12	-	12
Implementation Framework	96	76	78
Data Quality	18	14	2
Government, Institutions & Policy	52	32	33
Schedule/Timing	34	16	18
Project Design	50	40	2
Environment & Macroeconomic	18	4	14
General & Other – e.g. Systems, Processes, Technology	58	36	18
Restructuring (<i>only applicable for Supervision</i>)	34	-	32
Monitoring & Controlling	92	64	74
General	54	16	40
Reporting	36	12	26
Risks & Issues	36	32	10
KPIs	44	20	32
PDOs	42	34	12
Resources & Capacity	64	32	50
Budget & Financing	44	20	26

This table shows the reasons for “Unsatisfactory” bank performance in Africa and separated by the 2 dimensions.

Appendix DN

Reasons for Negative Bank Performance in Asia – Landing

Category	% of Projects	% of total Projects
Bank Competence	72.9	13.1
Decision Making Process	-	-
Management, Coordination, Administration & Evaluation	72.9	13.1
Communication	-	-
Implementation Framework	89.1	16
Data Quality	10.8	1.9
Government, Institutions & Policy	27	4.9
Schedule/Timing	8.1	1.5
Project Design	62.1	11.2
Environment & Macroeconomic	18.9	3.4
General & Other – e.g. Systems, Processes, Technology	35.1	6.3
Restructuring (<i>only applicable for Supervision</i>)	-	-
Monitoring & Controlling	67.6	12.1
General	18.9	3.4
Reporting	5.4	1
Risks & Issues	29.7	5.3
KPIs	18.9	3.4
PDOs	37.8	6.8
Resources & Capacity	43.2	7.8
Budget & Financing	8.1	1.5

This table shows the reasons for a negative bank performance in Asia for the landing dimension.

Appendix DO

Reasons for Negative Bank Performance in Asia – Supervision

Category	% of Projects	% of total Projects
Bank Competence	62.1	11.2
Decision Making Process	-	-
Management, Coordination, Administration & Evaluation	62.1	11.2
Communication	10.8	1.9
Implementation Framework	70.2	12.6
Data Quality	10.8	1.9
Government, Institutions & Policy	21.6	3.9
Schedule/Timing	29.7	5.3
Project Design	-	-
Environment & Macroeconomic	2.7	0.5
General & Other – e.g. Systems, Processes, Technology	24.3	4.4
Restructuring (<i>only applicable for Supervision</i>)	21.6	3.9
Monitoring & Controlling	59.5	10.7
General	27	4.9
Reporting	16.2	2.9
Risks & Issues	13.5	2.4
KPIs	24.3	4.4
PDOs	16.2	2.9
Resources & Capacity	51.3	9.2
Budget & Financing	21.6	3.9

This table shows the reasons for a negative bank performance in Asia for the quality of supervision dimension.

Appendix DP

Reasons for “Moderately Unsatisfactory” Bank Performance in Asia

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	83.3	66.7	55.6
Decision Making Process	-	-	-
Management, Coordination, Administration & Evaluation	83.3	66.7	55.6
Communication	-	-	-
Implementation Framework	100	94.4	66.7
Data Quality	16.7	11.1	5.6
Government, Institutions & Policy	33.3	22.2	11.1
Schedule/Timing	33.3	5.6	33.3
Project Design	72.2	72.2	-
Environment & Macroeconomic	22.2	16.7	5.6
General & Other – e.g. Systems, Processes, Technology	50	38.9	27.8
Restructuring (<i>only applicable for Supervision</i>)	16.7	-	16.7
Monitoring & Controlling	83.3	72.2	50
General	27.8	16.7	16.7
Reporting	16.7	11.1	5.6
Risks & Issues	55.6	44.4	11.1
KPIs	44.4	27.8	22.2
PDOs	44.4	38.9	11.1
Resources & Capacity	66.7	44.4	38.9
Budget & Financing	16.7	5.6	16.7

This table shows the reasons for “Moderately Unsatisfactory” bank performance in Asia and separated by the 2 dimensions.

Appendix DQ

Reasons for “Unsatisfactory” Bank Performance in Asia

Category	% of Projects		
	Total	Landing	Supervision
Bank Competence	84.2	78.9	68.4
Decision Making Process	-	-	-
Management, Coordination, Administration & Evaluation	84.2	78.9	68.4
Communication	21.1	-	21.1
Implementation Framework	94.7	84.2	73.7
Data Quality	26.3	10.5	15.8
Government, Institutions & Policy	57.9	31.6	31.6
Schedule/Timing	31.6	10.5	26.3
Project Design	52.6	52.6	-
Environment & Macroeconomic	21.1	21.1	-
General & Other – e.g. Systems, Processes, Technology	42.1	31.6	21.1
Restructuring (<i>only applicable for Supervision</i>)	26.3	-	26.3
Monitoring & Controlling	84.2	63.2	68.4
General	47.4	21.1	36.8
Reporting	26.3	-	26.3
Risks & Issues	31.6	15.8	15.8
KPIs	31.6	10.5	26.3
PDOs	42.1	36.8	21.1
Resources & Capacity	89.5	41.1	63.2
Budget & Financing	36.8	10.5	26.3

This table shows the reasons for “Unsatisfactory” bank performance in Asia and separated by the 2 dimensions.

Appendix DR

Reasons for Negative Borrower Performance of the LDCs – Government

Category	% of Projects	% of total Projects
Borrower Competence	80.2	17
Commitment	31.7	6.7
Decision Making Process	11.4	2.4
"Management" of Things	22.2	4.7
Leadership, Coordination, Administration	52.7	11.1
Communication	7.8	1.6
Cooperation	4.8	1
Implementation Framework	33.5	7.1
General & Other – e.g. Systems, Processes, Technology	15	3.2
Data Quality	1.8	0.4
Project Design	4.2	0.9
Institutional Setup	3	0.6
Policy	6.6	1.4
Corruption	1.2	0.3
Schedule/Timing	7.8	1.6
Environment & Macroeconomic	8.4	1.8
Monitoring & Controlling	24.6	5.2
General	12	2.5
Reporting	3	0.6
PDOs	12.6	2.7
Resources & Capacity	26.3	5.6
Budget & Financing	19.2	4.1

This table shows the reasons for a negative borrower performance across the LDCs for the government dimension.

Appendix DS

Reasons for Negative Borrower Performance of the LDCs – Implementation

Agency

Category	% of Projects	% of total Projects
Borrower Competence	53.9	11.4
Commitment	17.4	3.7
Decision Making Process	3.6	0.8
"Management" of Things	25.1	5.3
Leadership, Coordination, Administration	41.3	8.7
Communication	4.8	1
Cooperation	1.8	0.4
Implementation Framework	22.2	4.7
General & Other – e.g. Systems, Processes, Technology	6	1.3
Data Quality	2.4	0.5
Project Design	4.8	1
Institutional Setup	3	0.6
Policy	0.6	0.1
Corruption	-	-
Schedule/Timing	8.4	1.8
Environment & Macroeconomic	5.4	1.1
Monitoring & Controlling	22.2	4.7
General	9	1.9
Reporting	9	1.9
PDOs	12	2.5
Resources & Capacity	31.1	6.6
Budget & Financing	1.6	0.3

This table shows the reasons for a negative borrower performance across the LDCs for the implementation agency.

Appendix DT

Reasons for “Moderately Unsatisfactory” Borrower Performance across all LDCs

Category	% of Projects		
	Total	Government	IA
Borrower Competence	95.8	75	61.1
Commitment	50	29.2	19.4
Decision Making Process	9.7	9.7	2.8
"Management" of Things	47.2	22.2	27.8
Leadership, Coordination, Administration	76.4	48.6	44.4
Communication	15.3	9.7	2.8
Cooperation	5.6	5.6	-
Implementation Framework	69.4	44.4	27.8
General & Other – e.g. Systems, Processes, Technology	27.8	20.8	6.9
Data Quality	4.2	1.4	1.4
Project Design	6.9	2.8	1.4
Institutional Setup	5.6	2.8	1.4
Policy	9.7	6.9	-
Corruption	-	-	-
Schedule/Timing	26.4	6.9	16.7
Environment & Macroeconomic	18.1	11.1	5.6
Monitoring & Controlling	40.3	29.2	19.4
General	23.6	18.1	6.9
Reporting	11.1	2.8	8.3
PDOs	13.9	8.3	6.9
Resources & Capacity	63.9	26.4	41.7
Budget & Financing	22.2	18.1	1.4

This table shows the reasons for “Moderately Unsatisfactory” borrower performance across the LDCs and separated by the 2 dimensions.

Appendix DU

Reasons for “Unsatisfactory” Borrower Performance across all LDCs

Category	% of Projects		
	Total	Government	IA
Borrower Competence	88.2	83.9	47.3
Commitment	45.2	33.3	16.1
Decision Making Process	14	12.9	2.2
"Management" of Things	44.1	22.6	23.7
Leadership, Coordination, Administration	72	55.9	39.8
Communication	10.8	6.5	6.5
Cooperation	7.5	4.3	3.2
Implementation Framework	33.3	23.7	17.2
General & Other – e.g. Systems, Processes, Technology	10.8	8.6	4.3
Data Quality	3.2	2.2	3.2
Project Design	8.6	5.4	7.5
Institutional Setup	6.5	3.2	3.2
Policy	7.5	6.5	1.1
Corruption	2.2	2.2	-
Schedule/Timing	11.8	8.6	2.2
Environment & Macroeconomic	9.7	6.5	5.4
Monitoring & Controlling	34.4	20.4	22.6
General	16.1	7.5	10.8
Reporting	9.7	3.2	8.6
PDOs	31.2	15.1	14
Resources & Capacity	41.9	26.9	21.5
Budget & Financing	21.5	19.4	1.1

This table shows the reasons for “Unsatisfactory” borrower performance across the LDCs and separated by the 2 dimensions.

Appendix DV

Reasons for “Highly Unsatisfactory” Borrower Performance across all LDCs

Category	% of Projects		
	Total	Government	IA
Borrower Competence	100	100	100
Commitment	50	50	-
Decision Making Process	100	-	100
"Management" of Things	-	-	-
Leadership, Coordination, Administration	100	50	-
Communication	-	-	-
Cooperation	-	-	-
Implementation Framework	100	100	50
General & Other – e.g. Systems, Processes, Technology	100	100	50
Data Quality	-	-	-
Project Design	-	-	-
Institutional Setup	-	-	-
Policy	-	-	-
Corruption	-	-	-
Schedule/Timing	-	-	-
Environment & Macroeconomic	-	-	-
Monitoring & Controlling	100	50	100
General	-	-	-
Reporting	50	-	50
PDOs	100	50	100
Resources & Capacity	100	-	100
Budget & Financing	100	50	50

This table shows the reasons for “Highly Unsatisfactory” borrower performance across the LDCs and separated by the 2 dimensions.

Appendix DW

Reasons for Negative Borrower Performance in Haiti – Government

Category	% of Projects	% of total Projects
Borrower Competence	60	30
Commitment	20	10
Decision Making Process	-	-
"Management" of Things	-	-
Leadership, Coordination, Administration	40	20
Communication	-	-
Cooperation	-	-
Implementation Framework	40	20
General & Other – e.g. Systems, Processes, Technology	20	10
Data Quality	-	-
Project Design	-	-
Institutional Setup	-	-
Policy	-	-
Corruption	-	-
Schedule/Timing	-	-
Environment & Macroeconomic	20	10
Monitoring & Controlling	20	10
General	-	-
Reporting	-	-
PDOs	20	10
Resources & Capacity	-	-
Budget & Financing	-	-

This table shows the reasons for a negative borrower performance in Haiti for the government dimension.

Appendix DX

Reasons for Negative Borrower Performance in Haiti – Implementation

Agency

Category	% of Projects	% of total Projects
Borrower Competence	40	20
Commitment	-	-
Decision Making Process	20	10
"Management" of Things	20	10
Leadership, Coordination, Administration	-	-
Communication	-	-
Cooperation	-	-
Implementation Framework	60	30
General & Other – e.g. Systems, Processes, Technology	20	10
Data Quality	-	-
Project Design	-	-
Institutional Setup	20	10
Policy	-	-
Corruption	-	-
Schedule/Timing	20	10
Environment & Macroeconomic	20	10
Monitoring & Controlling	20	10
General	-	-
Reporting	-	-
PDOs	20	10
Resources & Capacity	20	10
Budget & Financing	20	10

This table shows the reasons for a negative borrower performance in Haiti for the implementation agency dimension.

Appendix DY

Reasons for “Moderately Unsatisfactory” Borrower Performance in Haiti

Category	% of Projects		
	Total	Government	IA
Borrower Competence	100	100	100
Commitment	-	-	-
Decision Making Process	-	-	-
"Management" of Things	100	-	100
Leadership, Coordination, Administration	100	100	-
Communication	-	-	-
Cooperation	-	-	-
Implementation Framework	100	-	100
General & Other – e.g. Systems, Processes, Technology	-	-	-
Data Quality	-	-	-
Project Design	-	-	-
Institutional Setup	-	-	-
Policy	-	-	-
Corruption	-	-	-
Schedule/Timing	100	-	100
Environment & Macroeconomic	100	-	100
Monitoring & Controlling	-	-	-
General	-	-	-
Reporting	-	-	-
PDOs	-	-	-
Resources & Capacity	-	-	-
Budget & Financing	-	-	-

This table shows the reasons for “Moderately Unsatisfactory” borrower performance in Haiti and separated by the 2 dimensions.

Appendix DZ

Reasons for “Unsatisfactory” Borrower Performance in Haiti

Category	% of Projects		
	Total	Government	IA
Borrower Competence	66.7	33.3	-
Commitment	33.3	33.3	-
Decision Making Process	33.3	-	-
"Management" of Things	-	-	-
Leadership, Coordination, Administration	-	-	-
Communication	-	-	-
Cooperation	-	-	-
Implementation Framework	66.7	33.3	33.3
General & Other – e.g. Systems, Processes, Technology	-	-	-
Data Quality	-	-	-
Project Design	-	-	-
Institutional Setup	33.3	-	-
Policy	-	-	-
Corruption	-	-	-
Schedule/Timing	-	-	-
Environment & Macroeconomic	66.7	33.3	33.3
Monitoring & Controlling	-	-	-
General	-	-	-
Reporting	-	-	-
PDOs	-	-	-
Resources & Capacity	-	-	-
Budget & Financing	-	-	-

This table shows the reasons for “Unsatisfactory” borrower performance in Haiti and separated by the 2 dimensions.

Appendix EA

Reasons for “Highly Unsatisfactory” Borrower Performance in Haiti

Category	% of Projects		
	Total	Government	IA
Borrower Competence	100	100	100
Commitment	-	-	-
Decision Making Process	100	-	100
"Management" of Things	-	-	-
Leadership, Coordination, Administration	100	100	-
Communication	-	-	-
Cooperation	-	-	-
Implementation Framework	100	100	100
General & Other – e.g. Systems, Processes, Technology	100	100	100
Data Quality	-	-	-
Project Design	-	-	-
Institutional Setup	-	-	-
Policy	-	-	-
Corruption	-	-	-
Schedule/Timing	-	-	-
Environment & Macroeconomic	-	-	-
Monitoring & Controlling	100	100	100
General	-	-	-
Reporting	-	-	-
PDOs	100	100	100
Resources & Capacity	100	-	100
Budget & Financing	100	-	100

This table shows the reasons for “Highly Unsatisfactory” borrower performance in Haiti and separated by the 2 dimensions.

Appendix EB

Reasons for Negative Borrower Performance in Africa – Government

Category	% of Projects	% of total Projects
Borrower Competence	83.3	17.4
Commitment	31.7	6.6
Decision Making Process	15.8	3.3
"Management" of Things	22.5	4.7
Leadership, Coordination, Administration	55	11.5
Communication	5.8	1.2
Cooperation	6.7	1.4
Implementation Framework	31.7	6.6
General & Other – e.g. Systems, Processes, Technology	14.2	3
Data Quality	1.7	0.3
Project Design	4.2	0.9
Institutional Setup	3.3	0.7
Policy	9.2	1.9
Corruption	1.7	0.3
Schedule/Timing	7.5	1.6
Environment & Macroeconomic	7.5	1.6
Monitoring & Controlling	23.3	4.9
General	11.7	2.4
Reporting	1.7	0.3
PDOs	13.3	2.8
Resources & Capacity	29.2	6.1
Budget & Financing	25	5.2

This table shows the reasons for a negative borrower performance in Africa for the government dimension.

Appendix EC

Reasons for Negative Borrower Performance in Africa – Implementation Agency

Category	% of Projects	% of total Projects
Borrower Competence	51.7	10.8
Commitment	14.2	3
Decision Making Process	4.2	0.9
"Management" of Things	26.7	5.6
Leadership, Coordination, Administration	41.7	8.7
Communication	4.2	0.9
Cooperation	1.7	0.3
Implementation Framework	17.5	3.7
General & Other – e.g. Systems, Processes, Technology	3.3	0.7
Data Quality	1.7	0.3
Project Design	5	1
Institutional Setup	3.3	0.7
Policy	0.8	0.2
Corruption	-	-
Schedule/Timing	7.5	1.6
Environment & Macroeconomic	1.7	0.3
Monitoring & Controlling	22.5	4.7
General	8.3	1.7
Reporting	10	2.1
PDOs	10	2.1
Resources & Capacity	25	5.2
Budget & Financing	1.7	0.3

This table shows the reasons for a negative borrower performance in Africa for the implementation agency dimension.

Appendix ED

Reasons for “Moderately Unsatisfactory” Borrower Performance in Africa

Category	% of Projects		
	Total	Government	IA
Borrower Competence	97.7	79.5	79.5
Commitment	50	31.8	31.8
Decision Making Process	15.9	15.9	15.9
"Management" of Things	50	22.7	22.7
Leadership, Coordination, Administration	75	50	50
Communication	11.4	6.8	6.8
Cooperation	9.1	9.1	9.1
Implementation Framework	72.7	47.7	47.7
General & Other – e.g. Systems, Processes, Technology	27.3	22.7	22.7
Data Quality	2.3	2.3	2.3
Project Design	4.5	2.3	2.3
Institutional Setup	6.8	2.3	2.3
Policy	13.6	11.4	11.4
Corruption	-	-	-
Schedule/Timing	29.5	4.5	4.5
Environment & Macroeconomic	13.6	11.4	11.4
Monitoring & Controlling	43.2	29.5	29.5
General	25	18.2	18.2
Reporting	15.9	2.3	2.3
PDOs	11.4	11.4	11.4
Resources & Capacity	65.9	29.5	29.5
Budget & Financing	34.1	27.3	27.3

This table shows the reasons for “Moderately Unsatisfactory” borrower performance in Africa and separated by the 2 dimensions.

Appendix EE

Reasons for “Unsatisfactory” Borrower Performance in Africa

Category	% of Projects		
	Total	Government	IA
Borrower Competence	89.3	85.3	46.7
Commitment	44	30.7	14.7
Decision Making Process	16	16	2.7
"Management" of Things	46.7	22.7	25.3
Leadership, Coordination, Administration	74.7	58.7	40
Communication	10.7	5.3	6.7
Cooperation	8	5.3	2.7
Implementation Framework	30.7	21.3	12
General & Other – e.g. Systems, Processes, Technology	10.7	8	2.7
Data Quality	2.7	1.3	2.7
Project Design	8	5.3	6.7
Institutional Setup	8	4	4
Policy	9.3	8	1.3
Corruption	2.7	2.7	-
Schedule/Timing	13.3	9.3	2.7
Environment & Macroeconomic	5.3	5.3	1.3
Monitoring & Controlling	36	20	22.7
General	14.7	8	9.3
Reporting	8	1.3	6.7
PDOs	32	14.7	13.3
Resources & Capacity	41.3	29.3	17.3
Budget & Financing	25.3	22.7	1.3

This table shows the reasons for “Unsatisfactory” borrower performance in Africa and separated by the 2 dimensions.

Appendix EF

Reasons for “Highly Unsatisfactory” Borrower Performance in Africa

Category	% of Projects		
	Total	Government	IA
Borrower Competence	100	100	100
Commitment	100	100	-
Decision Making Process	100	-	100
"Management" of Things	-	-	-
Leadership, Coordination, Administration	100	-	-
Communication	-	-	-
Cooperation	-	-	-
Implementation Framework	100	100	-
General & Other – e.g. Systems, Processes, Technology	100	100	-
Data Quality	-	-	-
Project Design	-	-	-
Institutional Setup	-	-	-
Policy	-	-	-
Corruption	-	-	-
Schedule/Timing	-	-	-
Environment & Macroeconomic	-	-	-
Monitoring & Controlling	100	-	100
General	-	-	-
Reporting	100	-	100
PDOs	100	-	100
Resources & Capacity	100	-	100
Budget & Financing	100	100	-

This table shows the reasons for “Highly Unsatisfactory” borrower performance in Africa and separated by the 2 dimensions.

Appendix EG

Reasons for Negative Borrower Performance in Asia – Government

Category	% of Projects	% of total Projects
Borrower Competence	73.8	15
Commitment	33.3	6.8
Decision Making Process	-	-
"Management" of Things	23.8	4.9
Leadership, Coordination, Administration	47.6	9.7
Communication	14.3	2.9
Cooperation	-	-
Implementation Framework	40.5	8.3
General & Other – e.g. Systems, Processes, Technology	16.7	3.4
Data Quality	2.4	0.5
Project Design	4.8	1
Institutional Setup	2.4	0.5
Policy	-	-
Corruption	-	-
Schedule/Timing	9.5	1.9
Environment & Macroeconomic	11.9	2.4
Monitoring & Controlling	31	6.3
General	14.3	2.9
Reporting	7.1	1.5
PDOs	9.5	1.9
Resources & Capacity	21.4	4.4
Budget & Financing	4.8	1

This table shows the reasons for a negative borrower performance in Asia for the government dimension.

Appendix EH

Reasons for Negative Borrower Performance in Asia – Implementation Agency

Category	% of Projects	% of total Projects
Borrower Competence	61.9	12.6
Commitment	28.6	5.8
Decision Making Process	-	-
"Management" of Things	21.4	4.4
Leadership, Coordination, Administration	42.9	8.7
Communication	7.1	1.5
Cooperation	2.4	0.5
Implementation Framework	31	6.3
General & Other – e.g. Systems, Processes, Technology	11.9	2.4
Data Quality	4.8	1
Project Design	4.8	1
Institutional Setup	-	-
Policy	-	-
Corruption	-	-
Schedule/Timing	7.1	1.5
Environment & Macroeconomic	14.3	2.9
Monitoring & Controlling	23.8	4.9
General	11.9	2.4
Reporting	7.1	1.5
PDOs	16.7	3.4
Resources & Capacity	47.6	9.7
Budget & Financing	-	-

This table shows the reasons for a negative borrower performance in Asia for the implementation agency dimension.

Appendix EI

Reasons for “Moderately Unsatisfactory” Borrower Performance in Asia

Category	% of Projects		
	Total	Government	IA
Borrower Competence	92.6	66.7	63
Commitment	51.9	25.9	29.6
Decision Making Process	-	-	-
"Management" of Things	40.7	22.2	22.2
Leadership, Coordination, Administration	77.8	44.4	44.4
Communication	22.2	14.8	7.4
Cooperation	-	-	-
Implementation Framework	63	40.7	25.9
General & Other – e.g. Systems, Processes, Technology	29.6	18.5	11.1
Data Quality	7.4	-	3.7
Project Design	11.1	3.7	-
Institutional Setup	3.7	3.7	-
Policy	3.7	-	-
Corruption	-	-	-
Schedule/Timing	18.5	11.1	11.1
Environment & Macroeconomic	22.2	11.1	7.4
Monitoring & Controlling	37	29.6	18.5
General	22.2	18.5	7.4
Reporting	3.7	3.7	-
PDOs	18.5	3.7	14.8
Resources & Capacity	59.3	22.2	48.1
Budget & Financing	3.7	3.7	-

This table shows the reasons for “Moderately Unsatisfactory” borrower performance in Asia and separated by the 2 dimensions.

Appendix EJ

Reasons for “Unsatisfactory” Borrower Performance in Asia

Category	% of Projects		
	Total	Government	IA
Borrower Competence	86.7	86.7	60
Commitment	53.3	46.7	26.7
Decision Making Process	-	-	-
"Management" of Things	40	26.7	20
Leadership, Coordination, Administration	73.3	53.3	40
Communication	13.3	13.3	6.7
Cooperation	6.7	-	6.7
Implementation Framework	60	40	40
General & Other – e.g. Systems, Processes, Technology	20	13.3	13.3
Data Quality	6.7	6.7	6.7
Project Design	13.3	6.7	13.3
Institutional Setup	-	-	-
Policy	-	-	-
Corruption	-	-	-
Schedule/Timing	6.7	6.7	-
Environment & Macroeconomic	33.3	13.3	26.7
Monitoring & Controlling	40	33.3	33.3
General	26.7	6.7	20
Reporting	20	13.3	20
PDOs	33.3	20	20
Resources & Capacity	46.7	20	46.7
Budget & Financing	6.7	6.7	-

This table shows the reasons for “Unsatisfactory” borrower performance in Asia and separated by the 2 dimensions.

Appendix EK

ICR and IEG Project Rating Comparison Outcome

	Project Overall Performance		Bank Performance		Borrower Performance	
	ICR	IEG	ICR	IEG	ICR	IEG
Total Counts	40	40	40	40	40	40
Total Negative Counts	10	10	6	7	7	9
“Moderately Unsatisfactory”	7	6	4	3	4	5
“Unsatisfactory”	3	4	2	4	3	4
“Highly Unsatisfactory”	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Total Positive Counts	30	30	30	30	30	30
“Moderately Satisfactory”	6	13	11	12	12	13
“Satisfactory”	21	15	20	19	21	18
“Highly Satisfactory”	3	2	3	2	n.a.	n.a.

	Sustainability	
	ICR	IEG
	40	39
Total Negative Counts	13	11
“Unlikely”	12	9
Highly Unlikely	1	2
“Modest”	7	10
Total Positive Counts	20	18
“Likely”	19	13
“Highly Likely”	1	5

This table shows the outcome of the exemplary project performance ratings of the ICR reports and the evaluation through the IEG. Out of the 40 projects only 7 projects were not NPV based.

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